

Quarterly portfolio construction with uncertain alpha and factor premia

Abstract

Draft abstract. We study an investor who can adjust active mutual funds and exchange-traded funds (ETFs) only at quarterly reviews, under a funded, long-only budget. An exchange-traded fund can replace an active fund's factor exposure only if the required holdings also satisfy position limits and funding. We give an exact feasibility test separating a factor direction missing from the exchange-traded-fund span from a direction present in the span but blocked by those constraints. Two assumed one-quarter examples show why exposure substitution alone does not settle the trading decision: an active-fund advantage can remain despite complete feasible exposure replacement, while adding an ETF can induce an active sale even when the larger fund menu cannot feasibly match the full optimum's exposure. A zero-alpha non-participation result, for a fund with no expected return beyond its specified factor exposures, holds under a feasible, costless replacement and a stated zero cross-moment; costly incumbents, position limits and independent tracking noise give precise limits. The one-quarter belief-average score depends on beliefs about alpha and factor premia only through their means. In a separate finite two-review model, we prove existence of optimal policies and an attained continuation representation under public learning. An assumed example with known zero alpha shows that future ETF adjustment can raise today's active-trading advantage while future active trading lowers it. A second assumed family retains both signs with small conditional risk and uncertain alpha that is not always revealed. Neither family gives a general dynamic trading boundary or calibrated performance. In every instance of the two-review model, the effect of future ETF adjustment lies between differences of that adjustment's premium at the two root optima. The premium is capped by the cash and ETF holdings left after today's trade, and a third assumed family has the opposite sign, so no sign holds across the model. Caps on that premium built only from wealth and return ranges cannot be sharp. Caps using risk-adjusted mean returns vanish exactly when no favorable ETF trade has wealth to trade. In a separate one-quarter statistical example, identical funded portfolio geometry and covariances conceal different full-history certification power. For a class of known bounded finite laws, a second example gives matching-order worst-case history-length bounds for certifying an unspanned-factor trade against the optimized ETF-only class. In a third example, the ETF-only optimizer moves between cash and the ETF; the worst-case rate depends on both joint alpha/premium error directions. A risk-averse example with a continuously moving ETF optimum has a smaller exponent in the *economic advantage* margin because the active trade shrinks at entry; in mean-signal units the testing rate is unchanged. With positive definite return risk, a curvature certificate extends the sufficient rate to directional costs and binding funding or position constraints. Its matching lower bound is class-wide and uses the zero-cost example. When the shock law is unknown but bounded, certifying each ETF-only comparison separately from its range already attains the known-law worst-case order in the moving-comparator example, and no joint or variance-adaptive rule improves that order. Finally, a two-stage procedure that first sets a target factor exposure and then chooses funds, charging for distance from the target, is the joint problem at the premia the target implies. Its loss is at most the first stage's constraint multiplier applied to the exposure difference, and it vanishes when the unconstrained target is admissible. If instead the funds must deliver the target exposure exactly, the loss equals the gain in the best residual outcome from moving the exposure, less the factor objective's loss. Separation is exact when that residual value is maximized at the factor target. A funded long-only menu with manager-borne exposure makes that value depend on the exposure, which a feasibility check on the target does not detect. Back in the statistical model, the optimized ETF-only comparator is exactly a finite set of fixed holdings when the score is linear, and under a curved score exactly when the ETF-only optimizer does not move on the parameter domain. Appendix A records the machine-checked scope.

1. Introduction

An investor may want to change an active mutual fund because its expected manager contribution, factor exposure or risk has changed. Manager alpha is the fund's expected return beyond its specified economic factor exposures, net of internal fund costs. An exchange-traded fund can offer another way to adjust exposure. The relevant comparison is with the *best feasible* exchange-traded-fund trade while leaving active holdings fixed. A spanning calculation alone can miss an unaffordable or prohibited replacement.

The analysis below first fixes the accounting language for a quarterly review, then studies a one-quarter model with one active fund, one or two exchange-traded funds and two economic factors. Both trading classes share one cash budget, long-only holdings and explicit position limits. Purchase and sale costs may differ, and exchange-traded-fund fee and tracking drag may have either sign. The investor has a joint belief about conditional factor premia and manager alpha. The model keeps realized-return risk, uncertainty about those conditional means, and ambiguity aversion distinct.

The accounting identities, compact-optimum arguments and exposure test below apply standard book-keeping, linearity and convex geometry. The no-trade band, continuation examples and statistical certificates also use familiar budget-multiplier, future-flexibility, testing and curvature mechanisms. The contribution here is their explicit scope and limits under a common funded ETF-only comparator, including exact examples; none establishes a general new trading or testing principle. In particular, the statistical rates do not supersede general bounds for identifying the best of several alternatives, known as best-arm identification.

The present draft establishes accounting identities, attained action-class optima, a test for feasible exposure replacement and exact one-quarter comparisons. The assumed examples show a strict full-trading score advantage under complete substitution, a change in active trading after an exchange-traded fund is added, and the conditions under which zero alpha permits non-participation. The score averages conditional mean-variance evaluations over the belief; dispersion of that belief has no effect when its mean stays fixed. The examples are mathematical benchmarks, not estimated portfolio gains. A separate two-review model lets the investor learn from public returns. We prove finite policy attainment and an exact continuation representation there. A second exact example shows opposite effects of future ETF and active trading on today's active advantage when the active fund alone supplies one factor. The signs persist in an explicit family with small conditional risk and partially learned alpha, although the effect still runs through the unspanned factor. These examples do not give a general dynamic trade boundary. A later result gives the exact comparison that decides the sign in every instance. The value of future ETF adjustment is bounded by the cash and ETF holdings that today's position leaves to redeploy. In an example where spending all cash on the active fund leaves nothing to redeploy, the sign is reversed. A further result shows how far such bounds can be sharpened, and where they stop deciding the sign. A statistical rule for deciding when an active trade is justified requires the public history law: an exact example below holds portfolio economics and covariances fixed while changing the information available from ETF returns. It does not give a calibrated rule for observed portfolios. A second statistical result gives a worst-case history-length rate for learning one factor premium when the active fund is its only investment vehicle. The rate is scalar mean-testing behavior in factor-premium units. A joint-error example then shows why checking only one ETF comparison can miss the sampling requirement when the best ETF-only action changes with the unknown parameter. A fourth statistical example shows that a smoothly shrinking active trade changes how a mean-signal error translates into an economic advantage margin. A curvature bound then controls the

optimized ETF comparator under costs and binding constraints, while retaining the same worst-case rate over a broader class of designs. A last statistical result drops knowledge of the shock law, keeping only a bound on its range: comparing with the whole ETF-only class then gives no gain in order over checking each ETF-only comparison separately. Returning to one quarter, two results compare a factor-then-fund procedure with joint optimization. With a soft exposure target, the procedure loses only through the funded budget's effect on the target, and the loss is bounded. With the target imposed exactly, the loss is identified exactly, and separation fails whenever funding makes the best residual choice depend on the exposure. A last result returns to certification and shows when the whole ETF-only comparator reduces to finitely many fixed comparison holdings.

2. Setting and standing assumptions

General review notation

An investor reviews a portfolio at quarterly dates. In the general notation there are m active funds, n ETFs and K economic factors. The loading matrices $B^A \in \mathbb{R}^{m \times K}$ and $B^E \in \mathbb{R}^{n \times K}$ have instruments in rows. Conditional on a parameter θ_t , next-quarter excess returns, measured relative to cash, have the form

$$r_{t+1}^A = B^A f_{t+1} + \alpha_t + \epsilon_{t+1}^A, \quad r_{t+1}^E = B^E f_{t+1} - c^E + \epsilon_{t+1}^E, \quad \mathbb{E}[f_{t+1} | \theta_t] = \lambda_t.$$

The residual shocks have conditional mean zero. The active-fund alpha α_t is relative to the economic factor model and net of internal fund expenses and manager trading effects. The economic factors need not be tradable securities. ETF drag c^E enters ETF returns once; zero ETF manager alpha is assumed, not perfect tracking. The investor has a joint belief about alpha and factor premia at the review. Public fund returns are observed whether or not the investor owns the fund; ownership creates no information. The eligible menu is fixed without a positive-alpha screen and does not depend on realized future alpha.

With pre-trade dollar holdings x^- , cash h^- , trades u and shareholder switching cost $C_t(u)$, accounting at a review is $x^+ = x^- + u$ and $h^+ = h^- - \sum_i u_i - C_t(u)$. Funded long-only trading requires $x^+, h^+ \geq 0$. For holdings $w = (a, p)$ expressed in a common pre-trade normalization, let w^- be the initial normalized holding. Factor exposure is $b(w) = (B^A)^\top a + (B^E)^\top p$. The general one-quarter score used for the paired mean-error identity has the form

$$Q_t(w; \theta) = b(w)^\top \lambda + a^\top \alpha - p^\top c^E - \frac{\gamma}{2} w^\top \Sigma(\theta) w - C_t(w - w^-).$$

Here γ is a fixed risk coefficient and $\Sigma(\theta)$ is the conditional return-covariance input. The cost function in the score is finite and evaluated in the same normalized units at each compared holding. For the funding identity, $C_t(u)$ is a finite nonnegative amount in dollars. The two general identities below use only this fixed algebraic form or the dollar funding equation; they assert no optimizer or particular return law.

Funded one-quarter model

For the optimization results, specialize to one active fund, one or two ETFs, two factors and one investor trade at review $t = 0$. Positions are then held until terminal marking one quarter later, without interim trading or a terminal liquidation charge. Cash has gross return one, and all returns and values are in cash-account units. The loading rows are known and fixed. The parameter $\theta = (\lambda_1, \lambda_2, \alpha)$ belongs to a nonempty finite set Θ ; the decision-date joint belief π has nonnegative masses summing to one. Alpha

and factor premia may be dependent in that belief, and neither has a sign restriction. The known ETF fee-and-tracking term c^E may have either sign. The investor knows the menu, initial holdings, limits, costs, loadings, scenario law and belief before orders are submitted, but not the true parameter or next return scenario.

A nonempty finite scenario set S has masses $q_s \geq 0$ summing to one, fixed across parameter values and actions. Joint factor, active-residual and ETF-residual shocks (z_s^f, z_s^A, z_s^E) each have zero q -weighted mean; their coordinates need not be independent. For every $\theta \in \Theta$ and $s \in S$, set

$$\begin{aligned} f_s(\theta) &= \lambda + z_s^f, \\ r_s^A(\theta) &= B^A f_s(\theta) + \alpha + z_s^A, \\ r_s^E(\theta) &= B^E f_s(\theta) - c^E + z_s^E, \\ \mu(\theta) &= (B^A \lambda + \alpha, B^E \lambda - c^E), \\ \xi_s &= (B^A z_s^f + z_s^A, B^E z_s^f + z_s^E), \quad \Sigma = \sum_s q_s \xi_s \xi_s^\top. \end{aligned}$$

All instrument gross returns $1 + r_{i,s}(\theta)$ are strictly positive at every parameter-scenario pair. The finite scenario law, and hence Σ , is fixed across θ ; Σ may be singular. It describes conditional realized-return risk. The belief describes estimation uncertainty about conditional means and is not a confidence region. No ambiguity-aversion penalty is imposed. For any scenario quantity X_s , write $\mathbb{E}_q X = \sum_s q_s X_s$ and $\text{Var}_q X = \sum_s q_s (X_s - \mathbb{E}_q X)^2$, holding θ fixed.

Pre-trade wealth $W^- = h^- + \sum_i x_i^- > 0$ has nonnegative holdings and cash. Write $w^- = x^-/W^- = (a^-, p^-)$, $k^- = h^-/W^-$, $v = u/W^-$ and $w = x^+/W^- = w^- + v = (a, p)$. Thus $\sum_i w_i^- + k^- = 1$. These are dollars divided by *pre-trade* wealth; they are never renormalized after review costs. Each known position limit $\bar{w}_i \in [0, 1]$ is in the same units, and the initial holding satisfies it.

Each instrument has known purchase and sale rates $\kappa_i^+, \kappa_i^- \in [0, 1]$, with no imposed ranking between the active fund and ETFs. For a signed trade, $v_i^+ = \max(v_i, 0)$ and $v_i^- = \max(-v_i, 0)$ denote its nonnegative purchase and sale parts, not before- and after-review holdings; $u_i^\pm$ use the same convention. The shareholder costs, post-review cash and terminal marked wealth are

$$\begin{aligned} C_0(u) &= \sum_i (\kappa_i^+ u_i^+ + \kappa_i^- u_i^-), \quad \tau(v) = \sum_i (\kappa_i^+ v_i^+ + \kappa_i^- v_i^-), \\ k(w) &= k^- - \sum_i v_i - \tau(v), \quad W_1(w; \theta, s) = W^- \left[k(w) + \sum_i w_i (1 + r_{i,s}(\theta)) \right]. \end{aligned}$$

A purchase rate applies to the quoted value of holdings delivered; a sale rate applies to the gross value removed. Costs are paid at the review from cash and sale proceeds. Active NAV returns already include internal fund costs; c^E is subtracted from ETF returns once and neither is charged again in τ . Execution and settlement at the review quote are immediate. Fixed platform fees, load schedules, taxes, investor flows and nonlinear impact are outside this model.

All three action classes face the same funded cash constraint: $F = \{w : 0 \leq w_i \leq \bar{w}_i, k(w) \geq 0\}$ permits full trading; $E = \{w \in F : a = a^-\}$ permits ETF-only adjustment; and $N = \{w^-\}$ is no trade. Cash is $W^- k(w)$, so neither borrowing nor shorting is hidden in F or E . The initial holding is compliant, and no additional mandate forces an active trade. The inequalities involving holding vectors are componentwise. The ETF-only feasible exposure set is $B_E = \{b(w) : w \in E\}$; its changes from the incumbent are $D_E =$

$\{b(w) - b(w^-) : w \in E\}$. The ETF linear span $L_E = \{(B^E)^\top d : d \in \mathbb{R}^n\}$ ignores funding, limits and signs. An exposure in L_E need not be attainable in E .

For a known risk coefficient $\gamma \geq 0$, the conditional one-quarter score and the implementable belief-average criterion are

$$Q_0(w; \theta) = b(w)^\top \lambda + a\alpha - p^\top c^E - \frac{\gamma}{2} w^\top \Sigma w - \tau(w - w^-), \quad \bar{Q}_0(w) = \sum_{\theta \in \Theta} \pi_\theta Q_0(w; \theta).$$

The score deducts shareholder review cost once and penalizes conditional return variance. It does not use predictive variance over the belief, define a statistical certificate, or solve a multiperiod control problem. Fund returns remain public regardless of holdings. The one-quarter results below therefore have no continuation-value or learning conclusion.

Funded two-review model

For the finite continuation result, keep one active fund, one or two ETFs, two factors, the fixed loading and return formulas above, and the same known directional shareholder rates. Reviews occur at $t = 0$ and $t = 1$, one quarter apart, with terminal marking at $t = 2$. There is no investor trade between reviews or terminal liquidation charge. The unknown parameter $\theta = (\lambda_1, \lambda_2, \alpha)$ is fixed for both quarters and has a finite initial belief π_0 . Conditional on θ , the two scenarios s_0, s_1 are independent draws from the same finite law q . Every specified instrument gross return $1 + r_i(\theta, s)$ is strictly positive, including at zero-mass parameter-scenario pairs. The menu, loadings and possibly signed ETF drag are fixed. All-fund returns are public whether or not the investor holds the funds, and holdings create no information.

At each review use risky *dollar* holdings x_t^- , cash h_t^- and dollar trades u_t , with the funding equations in the general review notation. Initial holdings and cash are nonnegative and $W_0^- = h_0^- + \sum_i x_{0,i}^- > 0$. The same known purchase and sale rates in $[0, 1)$ apply at both reviews: $C(u) = \sum_i [\kappa_i^+ u_i^+ + \kappa_i^- u_i^-]$. The funded full trade class F_t requires $x_t^- + u_t \geq 0$ and $h_t^- - \sum_i u_{t,i} - C(u_t) \geq 0$. The ETF-only class E_t also requires $u_{t,A} = 0$; $N_t = \{0\}$ is no trade. There are no additional concentration limits at either review. In particular, an unchanged dollar active holding remains allowed after return drift. All classes face the same funding equations, with no borrowing or shorting.

Before the second order, the investor observes the first-quarter factor realization and *all* fund returns, $Y_1 = (f(\theta, s_0), r^A(\theta, s_0), r^E(\theta, s_0))$. Let Y be its finite range. For $y \in Y$, define $L_\theta(y) = \sum_{s: Y_1(\theta, s) = y} q_s$ and $P_0(y) = \sum_\theta \pi_0(\theta) L_\theta(y)$. At $P_0(y) > 0$ the posterior is

$$\pi_1(\theta | y) = \frac{\pi_0(\theta) L_\theta(y)}{P_0(y)}.$$

At a zero-probability observation set the posterior to π_0 as a bookkeeping convention. The public observation does not reveal θ in general. If the root trade is u_0 , its post-review cash is h_0^+ ; marking at the observed gross returns $d_i(y) = 1 + r_i(\theta, s_0)$ gives $x_{1,i}^-(y; u_0) = d_i(y)(x_{0,i}^- + u_{0,i})$ and $h_1^-(u_0) = h_0^+$. After a second trade $u_1(y)$, terminal wealth on the path (θ, s_0, s_1) is

$$W_2 = h_1^+ + \sum_i x_{1,i}^+ [1 + r_i(\theta, s_1)].$$

The investor maximizes expected terminal utility $U_\rho(W_2/W_0^-) = -\exp(-\rho W_2/W_0^-)$, with $\rho > 0$ and the *initial* wealth denominator fixed across both reviews. This is not a sum of one-quarter mean-variance

scores. The scenario law describes realized-return risk; the initial belief and posterior describe uncertainty about fixed conditional means. Bayesian integration of terminal utility does not add that uncertainty to the scenario covariance or impose ambiguity aversion.

For a root class $D \in \{F, E, N\}$ and continuation class $R \in \{F, E, N\}$, let $\Pi_{D,R}$ consist of a root trade $u_0 \in D_0$ and one feasible trade $u_1(y) \in R_1$ at each $y \in Y$. Feasibility includes zero-probability nodes. One action per public observation prevents a policy from distinguishing hidden histories with the same observed data. Let Φ be the explicit expected-utility sum

$$\Phi(u_0, u_1) = \sum_{\theta, s_0, s_1} \pi_0(\theta) q_{s_0} q_{s_1} U_\rho \left(\frac{W_2(\theta, s_0, s_1; u_0, u_1)}{W_0^-} \right), \quad V_{D,R} = \sup_{\Pi_{D,R}} \Phi.$$

For a nonnegative marked state (x, h) and posterior π , write $V_1^R(x, h, \pi)$ for the supremum of conditional expected terminal utility over $R_1(x, h)$, retaining the same W_0^- , ρ , return law and costs. For a feasible root trade define

$$H_0^R(u_0) = \sum_{y: P_0(y) > 0} P_0(y) V_1^R(x_1^-(y; u_0), h_1^-(u_0), \pi_1(\cdot | y)).$$

The equality of the policy value to an attained maximum of H_0^R is a result below, rather than part of this definition. Where $V_{D,R} < 0$, express values in units of initial wealth by $CE_{D,R} = -\rho^{-1} \log(-V_{D,R})$. For fixed continuation class, set $\Delta_R = CE_{F,R} - CE_{E,R}$. The continuation classes compare future full trading, ETF-only adjustment and no second trade under identical observations and return laws. Restricting today's active trade does not ban an active trade at the second review when $R = F$.

Repeated-sampling certification model

For a separate one-quarter statistical comparison, retain the funded classes F, E and conditional score $Q(w; \theta)$ above. The factor premia and alpha form a fixed unknown parameter $\theta_* = (\lambda_*, \alpha_*)$ in a known domain Θ , the convex hull of finitely many vertices. Loadings, ETF drag, the centered finite shock law, realized-return covariance Σ , initial portfolio, limits, costs and score coefficient are known. Before one quarterly decision, observe $N \geq 1$ independent public records, each containing the factor realizations and every fund return. The same θ_* governs all records and the following holding quarter; observations do not depend on ownership.

Let $X_l = (f_l, r_l^A - B^A f_l)$, write $\hat{\theta}_N = N^{-1} \sum_l X_l$, and let $\zeta_s = (z_s^f, z_s^A)$ and $\Omega = \sum_s q_s \zeta_s \zeta_s^\top$. ETF returns are public but do not enter this sample-mean estimator. Ω describes mean-estimation error; it is not added to the realized-return covariance Σ in Q . For an error allowance $\eta \in (0, 1)$, the prescribed confidence set uses the finite-law $(1 - \eta)$ -quantile $t_{N, \eta}$ of $T_N = N e_N^\top \Omega^\dagger e_N$, where $e_N = N^{-1} \sum_l \zeta_{s_l}$ and $\Omega^\dagger$ is the Moore–Penrose inverse:

$$C_N = \Theta \cap \{\hat{\theta}_N - e : e \in \text{Im}(\Omega), \quad N e^\top \Omega^\dagger e \leq t_{N, \eta}\}.$$

For a feasible candidate w , compare it with the *optimized* ETF-only class at the same parameter:

$$\text{Adv}(w; \theta) = Q(w; \theta) - \sup_{v \in E} Q(v; \theta), \quad L_N(w) = \inf_{\theta \in C_N} \text{Adv}(w; \theta).$$

If C_N is empty, set $L_N = -\infty$ and fall back to an ETF-only action. The prescribed gate selects full and ETF-only maximizers at $\hat{\theta}_N$ and certifies the selected active trade only when its valid lower bound on L_N exceeds

the stated threshold. The example below uses threshold zero and the exact value L_N . A feasible point in the minimization gives an upper bound on L_N , not a certificate. This is a frequentist repeated-sampling decision at one review, with no Bayesian prior or ambiguity penalty and no multiperiod guarantee.

Multi-review tracking model

For the last result, iterate the one-quarter score over many quarterly reviews and study how the best holding follows a moving target. There are $n \geq 1$ instruments, active funds and ETFs indexed together, reviewed at $t = 0, 1, \dots, T - 1$ and marked at T without a terminal trade. A public state z_t in a finite set is known at review t . Given $z_t = z$, the next state and the vector g of instrument gross returns over the quarter have a known finite law $q_t(z', g \mid z)$, with every $g_i > 0$. The law does not depend on holdings or trades. A known belief-mean function $\mu(t, z)$ gives next quarter's expected excess returns, built from loadings, premia, alphas and ETF drag as in (1). The model takes this function as given and does not specify how beliefs are formed. The return covariance Σ is constant and positive definite, $\gamma > 0$, and future quarters are discounted by $\beta \in (0, 1]$.

Holdings are in dollars. Before review t the investor holds $x_t^- \geq 0$ and cash h_t^- , trades u_t to $x_t^+ = x_t^- + u_t$, and pays the directional cost $C(u) = \sum_i [\kappa_i^+ u_i^+ + \kappa_i^- u_i^-]$ with rates in $[0, 1]$. Holdings are then marked, $x_{t+1}^- = x_t^+ \circ g_{t+1}$ coordinatewise. Post-trade holdings must satisfy dollar caps $0 \leq x_t^+ \leq \bar{x}$ and, in general, keep cash nonnegative. In a *slack-budget* instance, initial cash is large enough, $h_0^- \geq T \sum_i (1 + \kappa_i^+) \bar{x}_i$, that the cash constraint never binds. The investor maximizes the expected discounted sum of one-quarter belief-mean scores, $\mathbb{E} \sum_t \beta^t [\mu(t, z_t)^\top x_t^+ - (\gamma/2) x_t^{+\top} \Sigma x_t^+ - C(u_t)]$. This is a sum of quarterly scores, not a terminal-utility problem.

The *frictionless target* is $x^*(t, z) = \Sigma^{-1} \mu(t, z) / \gamma$, the one-quarter optimum without costs or caps; it may be negative or exceed a cap. Since $\mu^\top x - (\gamma/2) x^\top \Sigma x$ equals a term free of x minus $(\gamma/2)(x - x^*)^\top \Sigma (x - x^*)$, maximizing the objective is the same as minimizing the expected discounted tracking loss $(\gamma/2)(x_t^+ - x_t^*)^\top \Sigma (x_t^+ - x_t^*)$ plus costs. With $X = \prod_i [0, \bar{x}_i]$, $V_T = 0$ and, for $t < T$, the **stay objective**

$$G_t(x, z) = \frac{\gamma}{2} (x - x^*(t, z))^\top \Sigma (x - x^*(t, z)) + \beta \sum_{(z', g')} q_t(z', g' \mid z) V_{t+1}(x \circ g', z'),$$

the minimal remaining loss from a pre-trade holding x is $V_t(x, z) = \min_{x' \in X} [C(x' - x) + G_t(x', z)]$, and the optimal post-trade holding $x_t^+(x, z)$ is the minimizer. The *no-trade set* $NT_t(z)$ collects the holdings $x \in X$ with $x_t^+(x, z) = x$.

Fund-of-funds learning model

The last result studies a larger, stylized problem with learning. It drops the funded long-only budget on purpose, as a declared counterfactual, so its optimal policy is not implementable by a long-only fund of funds. There are N active funds, M ETFs and K factors, reviewed quarterly at $t = 0, \dots, T - 1$. Excess returns over a quarter are

$$f = \lambda + z^f, \quad r^A = B^A f + \alpha + z^A, \quad r^E = B^E f - c^E + z^E,$$

with known loadings B^A , B^E and ETF drag c^E , and shocks $z = (z^f, z^A, z^E)$ that are Gaussian with known positive definite covariance Σ_z , independent across quarters. The premia λ and alphas α are fixed and unknown. The prior on $\theta = (\lambda, \alpha)$ is Gaussian. In the *reference case* the premium and alpha blocks of the prior are independent, Σ_z is block diagonal, and fund residuals are independent. All fund returns are

public, so beliefs $N(m_t, P_t)$ are updated by the Kalman filter from public returns whatever the investor holds. With $G = [[B^A, I_N]; [B^E, 0]]$, the predictive mean and covariance of next quarter's instrument returns are $\mu_t = Gm_t - (0, c^E)$ and $\Sigma_t = GP_tG^\top + \Sigma_r$, where Σ_r is the return covariance implied by Σ_z . Σ_t adds estimation risk to return risk, so this model, unlike the one-quarter model, charges estimation uncertainty through predictive variance.

Positions $x_t \in \mathbb{R}^n$, $n = N + M$, are dollars after the review and may be negative. A trade $u_t = x_t - x_{t-1}$ costs $\frac{1}{2}u_t^\top \Lambda u_t$ with Λ symmetric positive definite and instrument-specific; the reference case is $\Lambda = \text{diag}(\lambda_A I_N, \lambda_E I_M)$. For $\gamma > 0$ and a discount $\rho \in (0, 1]$, the investor maximizes

$$\mathbb{E}_0 \sum_{t=0}^{T-1} \rho^t \left[x_t^\top \mu_t - \frac{\gamma}{2} x_t^\top \Sigma_t x_t - \frac{1}{2} u_t^\top \Lambda u_t \right],$$

a sum of quarterly certainty equivalents under the predictive moments, net of trading costs. The Markowitz portfolio $\text{Mk}_t = (\gamma \Sigma_t)^{-1} \mu_t$ maximizes one quarter's certainty equivalent without costs.

Learning-driven tracking model

The multi-review tracking model takes its target as given. Its *learning-driven* version specifies the target: it is the Markowitz portfolio of the fund-of-funds learning model, $x_t^* = (\gamma \Sigma_t)^{-1} \mu_t$, with that model's instruments, prior, filter and predictive moments, and the fixed-means case. Costs, long-only caps and the objective are those of the multi-review tracking model, with the constant covariance replaced by the predictive Σ_t . In its *finite-law variant*, the premia and alphas are drawn once from a finite law with the prior's mean and covariance, and each quarter's shock from a finite centered law with the shocks' covariance, so all gross returns are positive and the public state is the finite history of observations. The manager runs the same linear filter, which is then the best linear predictor rather than the exact posterior. This variant is an exact evaluation device, not a claim that returns have finite support.

The target moves for two reasons:

$$x_{t+1}^* - x_t^* = (\gamma \Sigma_{t+1})^{-1} G \epsilon_{t+1} + [(\gamma \Sigma_{t+1})^{-1} - (\gamma \Sigma_t)^{-1}] \mu_t,$$

a belief-mean innovation $\epsilon_{t+1} = m_{t+1} - m_t$, with covariance $V_t = P_t - P_{t+1}$, and a *learning drift*, known at t , from the falling risk charge.

3. Main results

Estimated factor premia can affect a comparison between two fixed portfolios differently from an estimate of manager alpha.

Proposition 1 (Paired mean-error identity; general review notation; machine checked). Consider two fixed holdings $w = (a, p)$ and $v = (a_v, p_v)$, expressed in the same pre-trade normalization. Evaluate the one-quarter score Q_t at both holdings, then evaluate it again as $\hat{Q}_t$ after replacing only the factor premia λ and active-fund alpha α by $\hat{\lambda}$ and $\hat{\alpha}$. Keep the loadings, ETF drag, risk coefficient, covariance matrix, pre-trade holdings and switching-cost function identical in all four evaluations, and assume the evaluated costs are finite. Then

$$\begin{aligned} & [\hat{Q}_t(w) - \hat{Q}_t(v)] - [Q_t(w) - Q_t(v)] \\ &= [b(w) - b(v)]^\top (\hat{\lambda} - \lambda) + (a - a_v)^\top (\hat{\alpha} - \alpha). \end{aligned} \tag{1}$$

The factor-premium term is zero for every real factor-premium error if and only if $b(w) = b(v)$. Under that exposure match, the paired error is exactly the alpha-error term; if $a = a_v$ too, it is zero. For one particular factor-premium error, the term can vanish even when exposures differ; for a restricted set of errors, the converse need not hold.

The identity follows by subtracting the two scores at each fixed holding: ETF drag, return risk and switching costs cancel, leaving the two mean-error terms. For the universal cancellation condition, take the factor-premium error equal to $b(w) - b(v)$; its squared length is zero only when the exposure difference is zero.

This is an algebraic result. It does not assert that a feasible exposure-matching pair exists or that the comparator remains optimal when means change. It gives no probability bound or multiperiod decision rule. See Appendix A for the machine-checked scope.

For an investor, matching factor exposure removes one source of mean-estimation error from a fixed portfolio comparison. It does not show that such a match is affordable or that either portfolio should be traded. The cancellation is bilinearity of a fixed score, not a new estimation theorem.

Every comparison of trading choices must first charge shareholder costs to the same pool of available cash.

Proposition 2 (Self-financing at a quarterly review; general review notation; machine checked). Let x^- be nonnegative initial dollar holdings in finitely many active funds and ETFs, let $h^- \geq 0$ be cash, and let u be any trade vector. For a finite, nonnegative shareholder switching cost $C_t(u)$, define $x^+ = x^- + u$ and $h^+ = h^- - \sum_i u_i - C_t(u)$. Then

$$\sum_i x_i^+ + h^+ + C_t(u) = \sum_i x_i^- + h^-. \quad (2)$$

The funded long-only restrictions $x_i^+ \geq 0$ for every instrument and $h^+ \geq 0$ hold exactly when

$$x_i^- + u_i \geq 0 \quad \text{for every } i, \quad \sum_i (x_i^- + u_i) + C_t(u) \leq \sum_i x_i^- + h^-. \quad (3)$$

For a trade satisfying these restrictions, $0 \leq C_t(u) \leq \sum_i x_i^- + h^-$. If $C_t(0) = 0$, the zero trade is feasible under these funding restrictions and leaves holdings and cash unchanged. Additional mandates can still restrict trading.

To see (2), sum $x_i^+ = x_i^- + u_i$ and substitute the post-trade cash definition; the trade and cost terms cancel. Rearranging the equality gives the budget inequality in (3), while the holdings definition gives its coordinate restriction. Nonnegative post-trade holdings and cash give the cost bound. This result concerns accounting at one review. It does not establish a return, a trading score, an optimal policy, or a transfer to a later model version.

The identity says that a trading cost reduces the resources left for holdings and cash by exactly the amount paid. It gives no reason to prefer an active-fund trade over an ETF adjustment. It is the usual self-financing bookkeeping identity specialized to this review.

In the one-quarter model, the score must agree with the wealth and cost accounting before it can be optimized.

Proposition 3 (Score accounting and belief mean; funded one-quarter model; machine checked). Use the funded one-quarter model of Section 2: positive pre-trade wealth W^- , separate purchase and sale switching-cost rates, a finite centered return-scenario law that is fixed across parameter

values, known loadings and signed ETF drag, and a finite joint belief π about factor premia and active-fund alpha. For a feasible holding $w = (a, p)$, set $v = w - w^-$ and $G_s(w; \theta) = W_1(w; \theta, s)/W^- - 1$. The dollar and normalized cost definitions, funding equation and terminal marking give

$$\begin{aligned} C_0(W^- v)/W^- &= \tau(v), \\ \sum_i w_i + k(w) + \tau(v) &= 1, \\ G_s(w; \theta) &= w^\top r_s(\theta) - \tau(v). \end{aligned} \tag{4}$$

With E_q and Var_q taken over scenarios at a fixed θ , the conditional moments and one-quarter score satisfy

$$\begin{aligned} E_q G(w; \theta) &= w^\top \mu(\theta) - \tau(v), \\ \text{Var}_q G(w; \theta) &= w^\top \Sigma w, \\ Q_0(w; \theta) &= E_q G(w; \theta) - \frac{\gamma}{2} \text{Var}_q G(w; \theta). \end{aligned} \tag{5}$$

Thus the review cost enters normalized conditional expected gain once. If $\bar{\theta} = (\bar{\lambda}, \bar{\alpha})$ is the belief mean, finite averaging gives

$$\bar{Q}_0(w) = b(w)^\top \bar{\lambda} + a\bar{\alpha} - p^\top c^E - \frac{\gamma}{2} w^\top \Sigma w - \tau(v) = Q_0(w; \bar{\theta}). \tag{6}$$

Evaluation at $\bar{\theta}$ uses the same affine formula; $\bar{\theta}$ need not belong to the finite parameter support. Its scenario gross returns remain positive. Two beliefs on that support with the same mean and all other inputs fixed therefore give identical scores and rankings at every holding, and identical sets of maximizers on the full-trading, ETF-only and no-trade action classes, whether or not a maximum is attained. The result follows from positive homogeneity of the trade parts, the self-financing equation, centered scenario shocks and linearity of finite belief averages.

This score uses conditional return risk, not predictive variance over the belief. Belief dispersion or dependence at a fixed mean cannot change its rankings; the result adds no ambiguity-aversion penalty, optimizer-existence claim or multiperiod policy. Its directional cost rates do not represent an actual load schedule or redemption window without further assumptions.

Economically, this score responds to a changed belief *mean*, but not to greater uncertainty or a changed dependence pattern at fixed means. A decision effect from those features needs another criterion or a model with learning; this proposition provides neither. The dependence on the mean follows from finite averaging of an affine score, not a general result about learning.

The best ETF-only and full-trading scores are meaningful only if feasible optimal holdings exist in both classes.

Proposition 4 (Attained optimal actions in nested classes; funded one-quarter model; machine checked). Suppose the nonnegative initial holdings in Section 2's funded one-quarter model satisfy the position limits, initial cash is nonnegative, and pre-trade wealth is positive. The purchase and sale cost rates, scenario masses, belief masses and risk coefficient have the stated nonnegative values. With the same funded cash function $k(w)$ in every class, define

$$F = \{w : 0 \leq w_i \leq \bar{w}_i \text{ for every } i, k(w) \geq 0\}, \quad E = \{w \in F : a = a^-\}, \quad N = \{w^-\}. \tag{7}$$

Then $N \subseteq E \subseteq F$, and all three classes are nonempty, closed, bounded and convex. The belief-average one-quarter score $\bar{Q}_0$ is continuous and concave in holdings. It attains a finite maximum on each class; writing $V_D = \max_{w \in D} \bar{Q}_0(w)$ for $D \in \{N, E, F\}$,

$$V_N \leq V_E \leq V_F. \quad (8)$$

The initial holding belongs to every class because $\tau(0) = 0$. Nonnegative purchase and sale rates make the directional cost convex and the post-trade cash function concave; these facts preserve the funded constraint under convex combinations. Nonnegative scenario masses make the return-risk quadratic convex, and nonnegative belief masses preserve concavity under averaging. Its negative risk and cost contributions therefore make the score concave. The classes are closed subsets of a finite position-limit box, and continuity gives attainment. Inclusion then gives (8).

The inequalities may be equalities. The result gives neither a unique optimal action nor a strict advantage from active-fund trading. It covers the one-quarter score under Section 2's fixed menu and compliant initial position, with no forced active trade or additional mandate; it is not a multiperiod policy or a performance claim.

This guarantees that the investor can compare attained one-quarter benchmarks under the same budget. The weak ordering alone says nothing about the size, or even the existence, of an active-trading advantage. Attainment is the compact-set extreme-value argument; nesting gives the ordering.

Before comparing scores, we need to know which factor-exposure changes an ETF-only trade can actually finance.

Proposition 5 (ETF-only exposure geometry; funded one-quarter model; machine checked). Fix the funded one-quarter model's one active fund, one or two ETFs and two factors, with a compliant nonnegative initial holding, nonnegative initial cash, position limits and purchase and sale rates in $[0, 1)$. All actions use the same funded cash constraint. Write p^- for initial ETF holdings, $\bar{p}$ for their limits, k^- for initial cash, and $d = p - p^-$ for an ETF trade. Its net cash outlay and feasible trade set are

$$\begin{aligned} \Psi_E(d) &= \sum_j [d_j + \kappa_{E,j}^+ \max(d_j, 0) + \kappa_{E,j}^- \max(-d_j, 0)], \\ P_E &= \{d : -p^- \leq d \leq \bar{p} - p^-, \Psi_E(d) \leq k^-\}. \end{aligned} \quad (9)$$

For each purchase-or-sale assignment σ across the ETFs, let $c_{\sigma,j} = 1 + \kappa_{E,j}^+$ on a purchase and $c_{\sigma,j} = 1 - \kappa_{E,j}^-$ on a sale. Then $\Psi_E(d) = \max_{\sigma} c_{\sigma}^\top d$, so the last condition in (9) is exactly the finite system $c_{\sigma}^\top d \leq k^-$ for every σ . In particular, sales may have negative net outlay and finance another ETF purchase. With $b(w) = (B^A)^\top a + (B^E)^\top p$,

$$E = \{(a^-, p^- + d) : d \in P_E\}, \quad D_E = (B^E)^\top P_E, \quad B_E = b(w^-) + D_E. \quad (10)$$

The sets P_E , D_E and B_E are nonempty, closed, bounded and convex. The zero trade lies in P_E and D_E , the incumbent exposure lies in B_E , and $D_E \subseteq L_E = \{(B^E)^\top d : d \in \mathbb{R}^n\}$. A target exposure change δ has an ETF-only match if and only if there is a d satisfying

$$(B^E)^\top d = \delta, \quad -p^- \leq d \leq \bar{p} - p^-, \quad c_{\sigma}^\top d \leq k^- \quad \text{for every } \sigma. \quad (11)$$

If $\delta \notin L_E$, the direction is missing from the ETF span. Within that span, the position bounds can rule out every solution; if some bounded solution exists but none meets all cash inequalities, funding rules out a match, possibly together with the bounds. The test examines every solution of the exposure equation.

For one ETF, the entire trade set is the interval

$$P_E = \left[-p^-, \min \left\{ \bar{p} - p^-, \frac{k^-}{1 + \kappa_E^+} \right\} \right]. \quad (12)$$

Its exposure image is this interval times the ETF loading vector, including $\{0\}$ for a zero loading. For two ETFs with invertible $(B^E)^\top$, $L_E = \mathbb{R}^2$; the unique candidate $d = ((B^E)^\top)^{-1}\delta$ matches exactly when it passes the bounds and cash inequalities in (11). Thus full linear span alone gives no feasible-substitution conclusion. For any full action $w = (a, p)$ with $a \neq a^-$, its exposure change is outside L_E exactly when the active loading $(B^A)^\top$ is outside L_E ; if $a = a^-$, the action itself is an ETF-only match.

For each ETF, net outlay is the maximum of $(1 + \kappa_{E,j}^+)d_j$ and $(1 - \kappa_{E,j}^-)d_j$. Summing these maxima gives the finite cash inequalities. Substituting $a = a^-$ in Section 2's cash equation gives $k(a^-, p^- + d) = k^- - \Psi_E(d)$ and proves (10)–(11). The position box and finite inequalities make P_E compact and convex; its linear image and translation have the same properties. The special cases follow by solving the one-dimensional cash bound or the invertible exposure equation.

This is a one-quarter feasibility result. Exposure matching does not establish equal returns, residual risk, costs, cash or score, and an unmatched feasible full action need not outperform ETF-only trading. The result gives no value gap, economic magnitude, optimal trade direction or extension to another model version.

For portfolio construction, an ETF loading matrix can span the desired factors while the required trade still violates holdings limits or the cash budget. The test identifies that obstruction, but an exposure match alone does not establish equal investment value. This is a bounded linear-image feasibility test from standard finite-dimensional convex geometry.

The first exact example asks whether complete feasible exposure replacement removes every reason to hold the active fund.

Proposition 6 (Complete feasible exposure substitution with active-trading advantage; funded one-quarter model; machine checked). The following is one assumed instance of the funded one-quarter model. There is one active fund and one ETF, two factors and one quarterly review. Pre-trade wealth and cash are one, both initial fund holdings are zero, and both position limits are one. The two instruments have the same loading $(1, 0)$, ETF drag is zero, every purchase and sale rate is $1/100$, and $\gamma = 1$. The belief assigns mass $1/2$ to each parameter point $(\lambda, \alpha) = ((1/100, 0), 0)$ and $((3/100, 0), 1/50)$. Two equiprobable scenarios have factor shocks $(1/10, 0)$ and $(-1/10, 0)$, with zero residual shocks. All instrument gross returns are positive.

Write $t = a + p$ for total risky holdings. The identical funded budget in both action classes gives

$$\begin{aligned} F &= \{(a, p) : a, p \geq 0, a + p \leq 100/101\}, \\ E &= \{(0, p) : 0 \leq p \leq 100/101\}, \quad N = \{(0, 0)\}, \\ B_E &= D_E = \{(t, 0) : 0 \leq t \leq 100/101\}. \end{aligned} \quad (13)$$

Every $(a, p) \in F$ has the feasible ETF-only replacement $(0, t) \in E$. The pair has the same exposure $(t, 0)$, conditional variance $t^2/100$, review cost $t/100$, and post-review cash $1 - (101/100)t$. Thus exposure substitution is complete and feasible here, although the second factor is outside the span of both instruments. The belief-average scores satisfy

$$\bar{Q}_0(a, p) = \frac{t}{100} + \frac{a}{100} - \frac{t^2}{200}, \quad \bar{Q}_0(a, p) - \bar{Q}_0(0, t) = \frac{a}{100}. \quad (14)$$

Class	Unique maximizing holding (a, p)	Cash	Review cost	One-quarter score
N	$(0, 0)$	1	0	0
E	$(0, 100/101)$	0	$1/101$	$51/10201$
F	$(100/101, 0)$	0	$1/101$	$152/10201$

The two nonzero optima have the same exposure, but the full-minus-ETF-only optimal score gap is

$$V_F - V_E = \frac{1}{101} > 0. \quad (15)$$

Both trades are purchases from cash, so the shared review cost is $t/100$ and the cash constraint is $t \leq 100/101$. The return covariance has every entry $1/100$. Belief averaging gives mean factor premium $1/50$ in the first factor and mean alpha $1/100$, which yields (14). At the first parameter point the matched conditional-score difference is zero; at the second it is $a/50$. Thus the average difference uses the decision-date belief, not knowledge of the realized parameter. On E , the score $t/100 - t^2/200$ increases strictly over its allowed interval. For each fixed t in F , the score is uniquely largest at $a = t, p = 0$, and the resulting $t/50 - t^2/200$ also increases strictly. This proves the unique maximizers and the exact gap. In this instance that gap is the positive belief-mean alpha times the full optimum's active holding; matched exposures, return risk, cash and review costs contribute no difference.

This is a one-quarter score example with assumed means, belief, shocks and costs. It establishes neither a calibrated magnitude nor a realized or multiperiod wealth advantage. The second factor is unused, residual risk is zero and the covariance is singular. It is not a zero-alpha replication result, and it gives no conclusion for other initial holdings, unequal costs, binding ETF limits or another model version.

The investor can buy an ETF with the same factor exposure and risk, yet the active fund has a higher believed return in this assumed case. The positive one-quarter score gap is an alpha comparison, not evidence that the active fund will outperform in realized returns. Its equal-exposure accounting is the fixed-portfolio linearity in Proposition 1, illustrated by assumed numbers.

An ETF added to the menu changes feasible adjustments and can also change which active-fund trade is optimal.

Proposition 7 (An added ETF and incomplete feasible substitution; funded one-quarter model; machine checked). Consider two assumed menus in the funded one-quarter model that differ only by a second ETF, initially held at zero. Pre-trade wealth is one, with active holding $a^- = 1/2$, first ETF holding $p_1^- = 1/3$, and cash $k^- = 1/6$. All position limits are one, all purchase and sale rates are $1/2000$, and $\gamma = 1$. The singleton parameter support has $\lambda = (1/50, 1/200)$ and $\alpha = -1/400$; the active fund stays eligible. The fixed instrument inputs are:

Instrument	Factor loading	Residual shock size	ETF drag	Initial holding
Active fund	$(1, 1/2)$	$3/100$	none	$1/2$
First ETF	$(1, 0)$	$1/200$	$-1/10000$	$1/3$
Second ETF, when available	$(1, 1/4)$	$1/100$	0	0

The scenario law gives equal mass to all 32 combinations of five independent signs. The signs multiply factor shock sizes $3/50, 1/50$ and the three residual sizes in the table. The last sign is unused in the one-ETF menu, so the shared instruments have the same return law in both menus. Every instrument has positive gross return in every scenario. The two-ETF conditional mean and covariance are

$$\mu = \left(\frac{1}{50}, \frac{201}{10000}, \frac{17}{800} \right), \quad \Sigma = \frac{1}{40000} \begin{pmatrix} 184 & 144 & 146 \\ 144 & 145 & 144 \\ 146 & 144 & 149 \end{pmatrix}. \quad (16)$$

The one-ETF covariance is the leading two-coordinate submatrix. With w^- appropriate to each menu, both classes use $\sum_i w_i + \tau(w - w^-) \leq 1$, and the belief-average score is $\bar{Q}_0(w) = \mu^\top w - \frac{1}{2} w^\top \Sigma w - \tau(w - w^-)$. Their unique maximizers are:

Menu and class	Holding	Cash	Review cost	One-quarter score
One ETF, N	$(1/2, 1/3)$	$1/6$	0	11033/720000
One ETF, E and F	$(1/2, 3001/6003)$	0	$1/12006$	61830113/3427920000
Two ETFs, N	$(1/2, 1/3, 0)$	$1/6$	0	11033/720000
Two ETFs, E	$(1/2, 0, 2999/6003)$	0	$5/12006$	2104284079/115315228800
Two ETFs, F	$(0, 0, 11995/12006)$	0	$11/12006$	8512677811/461260915200

Thus the full-minus-ETF-only optimal score gap is zero with one ETF and

$$V_F - V_E = \frac{6369433}{30750727680} > 0 \quad \text{with two ETFs.} \quad (17)$$

Both menus' E and F optima beat no trade. With one ETF, the optimum buys 1000/6003 of that ETF and makes no active trade; it is ETF-only trading, not no trade. Adding the second ETF changes the optimal active trade to a sale of $1/2$.

The exposure facts are distinct. In the one-ETF menu, selling all holdings is a feasible full action with exposure change $(-5/6, -1/4)$, outside the ETF span, although the optimal gap is zero. In the two-ETF menu the loading rows span $\mathbb{R}^2$. Matching its full optimum while fixing $a = 1/2$ uniquely requires ETF **holdings** $(p_1, p_2) = (1/2, -11/12006)$. The second holding violates nonnegativity, so no feasible ETF-only action matches that optimum's exposure despite full linear span.

For the optimality check, independent centered signs give (16), whose quadratic form is positive definite because each instrument has a positive residual-square term. At each proposed optimum, the funded budget binds. A directional-cost subgradient and a nonnegative budget multiplier give, for any feasible comparator z in the same class, a score difference bounded above by a position-bound residual with the correct sign minus $\frac{1}{2}(z - w)^\top \Sigma(z - w)$. The residual vanishes at interior coordinates and has the sign required at zero holdings. Hence every distinct comparator scores strictly less; direct substitution gives the table and (17). The one-ETF missing direction and two-ETF prohibited short then follow by solving the exposure equations.

All means, costs and risks here are assumed. The strict gap is a one-quarter score comparison, not a calibrated effect or a multiperiod policy. The two-ETF gap is not attributed solely to the shorting obstruction: alpha, residual risks and traded amounts also differ between the optima. No global claim that adding an ETF increases active trading or the class gap follows from these two instances.

The added ETF makes a sale of the active fund optimal in this assumed case, even though the expanded ETF menu spans both factors. The ETF-only exposure match would require a short position, and the example gives no general prediction for what another ETF addition would do. A long-only bound defeating

an otherwise available linear replication is a standard constrained-replication mechanism; the result here is the exact funded example and its optimal actions.

A zero-alpha benchmark tests when an ETF can replace an active fund without losing return or increasing conditional risk.

Proposition 8 (Zero-alpha non-participation under feasible, costless ETF replacement; funded one-quarter model; machine checked). Consider an instance of the funded one-quarter model with one active fund and one ETF, a compliant initial holding (a^-, p^-, k^-) , equal loading rows $B^A = B^E$, zero ETF drag, free ETF purchases and sales, and ETF position limit $\bar{p} = 1$. The active fund may have directional purchase and sale costs and a position limit. Assume the belief-mean alpha $\bar{\alpha} = 0$. With centered conditional return shocks ξ_A and ξ_E , put $\varepsilon_s = \xi_{A,s} - \xi_{E,s}$ and $v_\varepsilon = \sum_s q_s \varepsilon_s^2$. Assume additionally $\sum_s q_s \xi_{E,s} \varepsilon_s = 0$. Finally, suppose either (A) $a^- = 0$, or (B) the active sale rate is zero. Then, for every $w = (a, p) \in F$, the holding $T(w) = (0, a + p)$ is in F , has the same factor exposure and zero review cost, and satisfies

$$\begin{aligned} k(T(w)) - k(w) &= \tau(w - w^-), \\ \bar{Q}_0(T(w)) - \bar{Q}_0(w) &= \tau(w - w^-) + \frac{\gamma}{2} a^2 v_\varepsilon \geq 0. \end{aligned} \quad (18)$$

Thus F has an optimum with no active holding. Under (A), the replacement also lies in E , so $V_F = V_E$. Every full-class optimum then has $a = 0$ if the active purchase rate is positive or $\gamma v_\varepsilon > 0$. Under (B), every full-class optimum has $a = 0$ when $\gamma v_\varepsilon > 0$. If an incumbent has $a^- > 0$, this liquidation is a full-trading action and cannot belong to E , which fixes the active holding.

The score conclusion uses belief-mean zero alpha; it does not assert equal realized returns. If alpha is zero at every parameter point and $\varepsilon_s = 0$ in every scenario, the two instruments instead have equal returns pointwise, and

$$W_1(T(w); \theta, s) - W_1(w; \theta, s) = W^- \tau(w - w^-) \quad \text{for every } \theta, s. \quad (19)$$

To see (18), the common loading and zero drag make the belief-mean return difference zero. Writing $t = a + p$, the risky return shock is $t \xi_E + a \varepsilon$; the zero cross-moment splits its conditional variance into $t^2 \sum_s q_s \xi_{E,s}^2 + a^2 v_\varepsilon$. Because w is funded, $t \leq 1$, so the uncapped, freely traded ETF can hold t . Under (A) or (B), selling the initial active holding costs zero; hence the replacement's cash rises by the original review cost. Proposition 4 supplies an attained maximum, and strictly positive terms in (18) exclude $a > 0$ in the stated cases.

The qualifications are substantive. In four assumed instances, $W^- = 1$, both loading rows are $(1, 0)$, all residual shocks vanish, ETF trading is free, the active limit is one, $\lambda = (1/50, 0)$, $\alpha = c^E = 0$, $\gamma = 1$, and equally likely factor shocks are $(\pm 1/10, 0)$. Active and ETF returns are identical, and the following outcomes are exact:

Initial (a^-, p^-, k^-)	Active buy/sell rates	ETF cap	Full optimum	Consequence
$(0, 0, 1)$	1/100 each	1	unique $(0, 1)$	No new active participation
$(1, 0, 0)$	1/100 each	1	unique $(1, 0)$	Costly incumbent stays
$(0, 0, 1)$	0 each	1	every $a + p = 1$	ETF and active optima tie
$(0, 0, 1)$	0 each	1/4	every feasible $a + p = 1$	Replacement is blocked

In the last row the ETF-only optimum is $(0, 1/4)$, and the full-minus-ETF-only score gap is 33/3200. In the incumbent row, the best zero-active holding is $(0, 99/100)$, with score 9799/2000000 $< 3/200$ for the incumbent. The common return law is positive in gross terms, and all numbers and costs are assumed.

This is a sufficient-condition result for a conditional one-quarter score, not a general zero-alpha rule or a multiperiod policy. Equal loadings give $\varepsilon = z^A - z^E$. If both residuals are uncorrelated with factor shocks and with each other, the assumed cross-moment is $-\text{Var}_q(z^E)$, so this subclass requires a noiseless ETF. The funded one-quarter model does not impose those extra uncorrelatedness conditions: a noisy ETF is compatible with the proposition when its residual shock is shared by the active fund and the active fund has an increment independent of that residual and the factor shocks. Conversely, using the preceding cases' means, loadings, risk coefficient and unit limits, with independent residual signs of size $1/100$ on both instruments, zero costs and an all-cash start, an assumed one-quarter instance has unique full optimum $(1/2, 1/2)$ and $V_F - V_E = 1/40000$; its cross-moment is $-1/10000$, outside the proposition. A zero-alpha fund then supplies diversification against ETF tracking noise. Exact return replication may leave tied optima. These instances establish no calibrated magnitude or result for another model version.

The result says that a zero-alpha active holding can be avoided when its exposure can be moved into a costless ETF without a diversification loss. It does not require liquidation of a costly incumbent, and ETF tracking noise can make the active fund useful even at zero alpha. This is a conditional analogue of zero-alpha non-participation, and its alpha is defined against the stated factor benchmark. An alpha of zero against those factors need not mean zero excess return against a tradable, noisy ETF.

The ETF-only optimum also provides a starting point for asking which active-fund beliefs leave the investor willing to make no active trade.

Proposition 9 (No-active-trade band after ETF optimization; funded one-quarter model; machine checked). Keep the initial holdings, factor-premium belief, return shocks, loadings, limits and directional costs fixed. Index the finite parameter support by h , with factor premia λ_h , active alpha α_h and belief mass π_h . Write $\bar{\lambda} = \sum_h \pi_h \lambda_h$ and $\bar{\alpha}_0 = \sum_h \pi_h \alpha_h$. For a proposed active belief mean x , replace each α_h by $\alpha_h - \bar{\alpha}_0 + x$. The translated family satisfies the model's strict gross-return condition exactly on

$$\begin{aligned} J &= (\alpha_{\min}, \infty), \\ \alpha_{\min} &= \max_{h,s} \{-1 - B^A \lambda_h - \alpha_h + \bar{\alpha}_0 - \xi_{A,s}\}, \\ \mu(x) &= (B^A \bar{\lambda} + x, B^E \bar{\lambda} - c^E). \end{aligned} \tag{20}$$

The maximum includes support points of zero belief mass. The original mean $\bar{\alpha}_0$ belongs to J . For the following formulas, extend the same one-quarter score algebraically to every real x ; outside J it is not a return-positive model instance. Let C contain the real x for which *some* full-trading optimum has $a = a^-$. The actual no-active-trade set is $C \cap J$. Choose any ETF-only optimum $w_E = (a^-, p_E)$, and write $k_E = k(w_E)$. Such an optimum exists and its set of choices does not depend on x .

The ETF smooth marginal scores and the active centering value are $g_{E,j} = B_j^E \bar{\lambda} - c_j^E - \gamma(\Sigma w_E)_{E,j}$ and $\alpha_c = \gamma(\Sigma w_E)_A - B^A \bar{\lambda}$, respectively. Thus the active smooth marginal is $x - \alpha_c$. For ETF j , let $[\ell_j, u_j]$ be the interval of marginal switching-cost slopes at $p_{E,j} - p_j^-$: it is $[\kappa_{E,j}^+, \kappa_{E,j}^+]$ after a purchase, $[-\kappa_{E,j}^-, -\kappa_{E,j}^-]$ after a sale, and $[-\kappa_{E,j}^-, \kappa_{E,j}^+]$ after no ETF trade. The compatible nonnegative cash-budget multipliers form the interval

$$\begin{aligned} I &= \{\eta \geq 0 : \eta k_E = 0, \\ g_{E,j} &\leq u_j + (1 + u_j)\eta \quad \text{if } p_{E,j} < \bar{p}_j, \\ g_{E,j} &\geq \ell_j + (1 + \ell_j)\eta \quad \text{if } p_{E,j} > 0, \quad \text{for every ETF } j\}. \end{aligned} \tag{21}$$

A zero-width ETF limit imposes neither inequality. The interval is nonempty. If $k_E > 0$, then $I = \{0\}$. Otherwise $I = [\eta_{\text{lo}}, \eta_{\text{hi}}]$, where the upper endpoint may be $+\infty$, and

$$\begin{aligned}\eta_{\text{lo}} &= \max\left(\{0\} \cup \left\{\frac{g_{E,j} - u_j}{1 + u_j} : p_{E,j} < \bar{p}_j\right\}\right), \\ \eta_{\text{hi}} &= \min\left\{\frac{g_{E,j} - \ell_j}{1 + \ell_j} : p_{E,j} > 0\right\}, \quad \min \emptyset = +\infty.\end{aligned}\tag{22}$$

For $I = \{0\}$, set both endpoints to zero. The lower endpoint is always finite and attained. If $a^- < 1$, the upper endpoint is finite. Put $b = \kappa_A^+$, $s = \kappa_A^-$, and $L = \alpha_c - s + (1 - s)\eta_{\text{lo}}$. When $\eta_{\text{hi}} < +\infty$, put $U = \alpha_c + b + (1 + b)\eta_{\text{hi}}$. The exact extended-score set is

Initial active holding	C
$0 < a^- < \bar{a}$	$[L, U]$, with finite endpoints
$a^- = 0 < \bar{a}$	$(-\infty, U]$, with finite U
$0 < a^- = \bar{a}$	$[L, \infty)$
$a^- = \bar{a} = 0$	all real x

In the interior case the full algebraic band's width is

$$U - L = b(1 + \eta_{\text{hi}}) + s(1 + \eta_{\text{lo}}) + \eta_{\text{hi}} - \eta_{\text{lo}}.\tag{23}$$

Equivalently, w_E is a full-trading optimum at a given x exactly when some $\eta \in I$ and active cost slope $t_A \in [-s, b]$ make $x - \alpha_c - \eta - (1 + \eta)t_A$ zero at an interior active holding, nonpositive at the active lower bound only, or nonnegative at its upper bound only. If $\bar{a} = 0$, no active-coordinate condition is needed. These are signs for holding bounds, not for purchase or sale. Any choice of ETF-only optimum gives the same set C .

To see necessity without assuming an interior feasible point, express each directional switching cost as the maximum of its finitely many linear pieces and lift the cost to a separate variable. The resulting concave quadratic problem has linear inequalities, whose active normals give the budget and bound multipliers at an optimum. Its coordinate conditions give (21) and the active marginal condition. Conversely, the cost-slope inequality, concavity of the score and funding complementarity make those conditions sufficient for a global optimum. Eliminating the ETF slopes gives (22); taking the union of the active marginal conditions over I gives the four cases and (23).

The set C asserts existence, since tied full-trading optima may also trade the active fund. If $\gamma > 0$ and Σ is positive definite, both action classes have unique optima, so $C \cap J$ identifies when the unique full-trading optimum leaves the active holding unchanged. Neither condition is assumed generally. For example, with one ETF, zero costs, zero loadings and shocks, and $\gamma = 0$, an initial allocation $(a^-, p^-, k^-) = (1/2, 1/2, 0)$ has tied full optima at $x = 0$, including active-trading choices. Starting instead from $(1, 0, 0)$ gives $I = [0, \infty)$ and $C = [0, \infty)$. Both examples have unit holding limits, score xa and return-positive domain $J = (-1, \infty)$.

The interval measures how active purchase and sale costs interact with the funded budget after the ETF adjustment is optimized. ETF cost kinks and holding bounds can widen it through the range of compatible budget multipliers. The intersection with J can remove an endpoint or the entire interval. This is a one-quarter belief-mean calculation, with no claim about changing the ETF menu, factor-premium belief, later trading or estimation uncertainty. The shadow-price scaling of a no-trade width is a known mechanism; this interval applies it to the funded ETF-optimized comparison.

The ETF adjustment can change when its trading cost changes, so a higher cost need not widen the active belief-mean band in Proposition 9.

Proposition 10 (A higher common switching-cost rate can narrow the no-active-trade band; funded one-quarter model; machine checked). There is an exact assumed instance with one active fund, one ETF and two factors in which raising the common purchase and sale rate for both instruments from 0 to 1/2000 strictly narrows that band. Normalize initial wealth to one. Set $\gamma = 1$, $B^A = (1, 1/2)$, $B^E = (1, 0)$, $c^E = -1/10000$, $(a^-, p^-, k^-) = (1/2, 0, 1/2)$, and $(\bar{a}, \bar{p}) = (1, 1/2)$. The belief assigns mass 1/4 to each pair $\lambda_1 \in \{3/200, 1/40\}$ and $\alpha \in \{-1/400, 1/400\}$, with $\lambda_2 = 1/200$. As in Proposition 9, replace the two alpha values by $x - 1/400$ and $x + 1/400$ to vary only their mean.

The return scenario law gives equal mass to the 32 sign vectors $(s_1, \dots, s_5) \in \{-1, 1\}^5$, with $z_s^f = (3s_1/50, s_2/50)$, $z_s^A = 3s_3/100$, and $z_s^E = s_4/200$; the fifth sign is unused. The resulting belief-mean return and conditional covariance are

$$\mu(x) = (9/400 + x, 201/10000), \quad \Sigma = \begin{pmatrix} 23/5000 & 9/2500 \\ 9/2500 & 29/8000 \end{pmatrix}, \quad J = (-183/200, \infty). \quad (24)$$

The covariance is positive definite, and J is the common return-positive domain at both rates. Let C_κ be the extended-score set of active belief means x for which the full-trading optimum leaves the active holding at $a^- = 1/2$. All four switching-cost rates equal κ ; every other input, including the complete belief at each x , stays fixed. The ETF-only optimum has zero cash in both cases:

κ	ETF-only holding p_E	Compatible interval I	Active centering α_c
0	1/2	$[0, 1319/80000]$	$-23/1250$
1/2000	1000/2001	$\{11032/690345\}$	$-61367/3335000$

κ	$C_\kappa = [L, U]$	Width $U - L$
0	$[-23/1250, -153/80000]$	$1319/80000$
1/2000	$[-808663/276138000, -38269/20010000]$	$701377/690345000$

Both bands lie wholly inside J , so restricting them to valid model instances changes neither width. Their width difference is

$$\frac{1319}{80000} - \frac{701377}{690345000} = \frac{170890979}{11045520000} > 0. \quad (25)$$

For the ETF-only class, funding gives $0 \leq p \leq 1/[2(1 + \kappa)]$. Its score increases throughout this interval at both rates, making its upper endpoint the unique ETF-only optimum. At zero cost that endpoint equals the ETF limit: the limit and cash constraint bind together, leaving a positive range of compatible budget multipliers. At rate 1/2000, the cash constraint still binds but the ETF optimum falls strictly below its limit. ETF stationarity then pins the multiplier to one value. Substitution into Proposition 9 gives the two bands above. Positive definiteness makes each full-trading optimum unique, so C_κ describes the unique action, rather than the existence of a tied no-trade action.

In expected-return units, the band narrows from 164.875 to about 10.160 basis points per quarter. This is a length comparison in the active belief mean, not an inclusion claim about the two intervals or a comparison of regions in holdings. The example relies on the zero-cost ETF purchase meeting its cap and

exhausting cash at the same point; this coincident corner is a knife-edge, so the example establishes no typical or robust effect of higher costs. It is an instance of degeneracy in parametric optimization: the ETF-only optimum has a nontrivial interval of compatible budget multipliers when both constraints bind, and the interval collapses to a single multiplier when the positive cost moves that optimum below the ETF cap. If the ETF multiplier endpoints are held fixed instead, Proposition 9's interior width increases with either active-fund directional cost. The comparison here changes ETF costs too, and therefore changes those endpoints. It says nothing about later reviews, learning or estimation uncertainty. The narrowing contradicts the common heuristic that higher costs always widen a no-trade band, not a theorem in the cited literature. It is not a general comparative static.

The two-review setting requires an attained continuation problem before an initial active-trade comparison can use future ETF adjustments consistently.

Proposition 11 (Optimal finite policies and continuation representation; funded two-review model; machine checked). Fix the finite two-review model in Section 2. Let M be the maximum of one and all specified instrument gross returns. For every pair of root and continuation classes $(D, R) \in \{F, E, N\}^2$, the policy set $\Pi_{D,R}$ is nonempty, convex and compact. Its expected terminal utility Φ is continuous and concave and attains a maximum. Every feasible policy has strictly positive wealth at the second review and at terminal marking on every specified path, and

$$W_2 \leq M^2 W_0^-, \quad -1 < V_{D,R} \leq -\exp(-\rho M^2) < 0. \quad (26)$$

Thus all nine certainty equivalents are finite. For any fixed posterior π , including one with zero masses, the conditional problem $V_1^R(x, h, \pi)$ attains its maximum at every nonnegative state (x, h) . With W_0^- , ρ , costs and return law fixed, this value is jointly concave in risky dollar holdings and cash, and nondecreasing in cash. At $x = h = 0$, zero is the only feasible trade and $V_1^R(0, 0, \pi) = -1$. The posterior and marked holdings and cash specify the review-1 conditional problem: the second scenario draw has law q conditional on θ , and no cost term depends on an earlier hidden history.

The root continuation value H_0^R is concave on the funded full root set F_0 . For every feasible root trade, conditional optima can be selected at the finitely many public observations to form one feasible observable continuation policy. In particular,

$$V_{D,R} = \max_{(u_0, u_1) \in \Pi_{D,R}} \Phi(u_0, u_1) = \max_{u_0 \in D_0} H_0^R(u_0). \quad (27)$$

For any fixed continuation class R , nesting of root classes gives $V_{N,R} \leq V_{E,R} \leq V_{F,R}$. For any fixed root class D , nesting of continuation classes gives $V_{D,N} \leq V_{D,E} \leq V_{D,F}$. The same inequalities hold for the corresponding certainty equivalents, since the certainty-equivalent transform is strictly increasing in utility value. Therefore $\Delta_R \geq 0$, and

$$\Delta_R = 0 \iff \text{some optimal policy in } \Pi_{F,R} \text{ has } u_{0,A} = 0. \quad (28)$$

The equivalence is about the existence of one full-class optimum with no active trade at the first review; other optima may trade the fund. To prove the result, funded accounting bounds every trade by incoming wealth and makes the finite policy sets compact. Strictly positive gross returns and sale rates below one preserve positive wealth. Terminal wealth is concave in policy trades because review costs are convex; increasing concave exponential utility preserves that concavity. For fixed posterior, mixing feasible trades at mixed marked states proves joint concavity of the conditional value, including the cash coordinate.

Grouping the finite expected-utility sum by public observation gives the continuation formula; conditional maximizers exist at each node, with zero trades available at zero-probability nodes. The two class orders then give the zero-gap equivalence by attained maxima.

This result makes the nine dynamic comparisons well defined. It does not calculate a particular optimal trade, give a scalar no-active-trade band, or show that future ETF adjustment changes the initial active-trade advantage in either direction. Differences such as $\Delta_E - \Delta_N$ and $\Delta_F - \Delta_E$ have no asserted sign. The concavity claim holds for fixed posterior and fixed initial-wealth normalization; it is not concavity in beliefs or in certainty equivalents. The result is finite-control groundwork, not evidence of an economic magnitude or a new dynamic trading mechanism. Its attainment and continuation identity are finite-horizon dynamic programming.

Future ETF adjustment and future active trading can change the advantage of buying the active fund today in opposite directions, even though each future opportunity weakly raises the value of every root trading class.

Proposition 12 (Opposite effects of future ETF and active trading; funded two-review model; machine checked). Consider an exact assumed instance with initial wealth one, held entirely in cash. There is one active fund and one ETF with $B^A = (1, 0)$, $B^E = (0, 1)$, zero ETF drag and $\rho = 20$. Two equally likely, persistent parameters are $\theta_+ = (1/2, -1/2, 0)$ and $\theta_- = (-1/2, 1/2, 0)$. Each quarter has one zero-shock scenario. The active and ETF gross returns are respectively $(3/2, 1/2)$ under θ_+ and $(1/2, 3/2)$ under θ_- . The first public observation reveals which parameter holds, whether or not either fund was bought. Alpha is known to be zero; the active fund is the only vehicle for the first factor, which the ETF menu does not span. Let all four active and ETF purchase and sale rates vary independently over $[0, 1/100]$, held fixed across the two reviews. Every member of this family satisfies the funded two-review model.

On the certainty-equivalent scale in units of initial wealth, the optimized root full-minus-ETF-only advantages obey, *simultaneously for every rate choice in that box*,

$$0 \leq \Delta_N \leq \frac{1}{4}, \quad \Delta_E > \frac{507}{2020}, \quad 0 \leq \Delta_F \leq \frac{3}{202}. \quad (29)$$

Consequently the matched continuation comparisons have opposite strict signs:

$$\Delta_E - \Delta_N > \frac{1}{1010} > 0, \quad \Delta_F - \Delta_E < -\frac{477}{2020} < 0. \quad (30)$$

Every full-root optimum with future ETF-only trading buys a positive active holding at the root. At zero shareholder costs, $\Delta_F = 0$, and an optimal full-root policy with future full trading keeps all wealth in cash at the root; this asserts existence, not uniqueness.

The bounds follow from feasible policies and upper bounds on whole action classes. If the second review is disabled, a post-root allocation (a, p, h) has terminal wealth $h + 9a/4 + p/4$ under θ_+ and $h + a/4 + 9p/4$ under θ_- . Their mean is at most $5/4$, which bounds $CE_{F,N}$; staying in cash gives $CE_{E,N} \geq 1$. If both reviews are ETF-only, the root active holding is zero. For its root ETF holding p , terminal wealth is at most $1 - p/2$ under θ_+ and $3/2 + 3p/4$ under θ_- . The resulting expected-utility upper bound is decreasing over $0 \leq p \leq 1$ and gives $CE_{E,E} < 21/20$.

For a funded full-root policy with future ETF-only adjustment, buy $a = (7/15)/(1 + 1/100)$ active dollars and $p = (8/15)/(1 + 1/100)$ ETF dollars today. Retain any cash left after the actual charges. At the second review sell the losing ETF under θ_+ , and otherwise make no trade. For every allowed rate the terminal wealth in each state is at least $657/505$, so $CE_{F,E} \geq 657/505$. Finally, any full policy has expected terminal wealth at most $3/2$, while an ETF-only root policy can wait in cash and buy the publicly revealed

winning fund at the second review, securing at least $150/101$ in both states. These give $CE_{F,F} \leq 3/2$ and $CE_{E,F} \geq 150/101$. Certainty equivalent lies between a policy's minimum and mean terminal wealth here; combining the bounds gives (29), then (30). The strict ETF-continuation gap and Proposition 11's attainment show that every corresponding full-root optimum buys active exposure.

The mechanism is exposure and timing. Future ETF trading lets an active purchase retain the unspanned first-factor exposure while the losing ETF can be sold after public learning. Future active trading instead lets cash wait and buy whichever fund will earn $3/2$ in the second quarter. The signs are proved for the entire stated cost box, without a cost ranking or a coincident position limit. The instance has known zero alpha, zero conditional return shocks, complete revelation and assumed quarterly returns of plus or minus 50%. Its magnitudes are not calibrated; the next result tests the signs under a small, explicitly bounded departure from those conditions. The continuation differences do not locate a general no-active-trade boundary or separate learning from the future trading opportunity; public information is the same under all controls. No sign is asserted for $\Delta_F - \Delta_N$ over the cost box. Future flexibility changing today's demand is a familiar option-value mechanism; the contribution is the two opposite signs for these explicit funded comparators throughout the stated cost box. It is not a hedge against risk: the first observation reveals the parameter and there is no conditional return risk, so future ETF adjustment is worth something here only because it redeploys wealth into returns known after the first review. Intertemporal hedging demand is a different mechanism (Section 4).

Proposition 13 (Opposite continuation effects with conditional risk and incomplete alpha learning; funded two-review model; machine checked). Fix $0 < \delta \leq 1/1000$. As in Proposition 12, initial wealth one is entirely in cash, $B^A = (1, 0)$, $B^E = (0, 1)$, ETF drag is zero, and $\rho = 20$. Now the four persistent latent parameters $\theta_{\sigma,\xi} = (\sigma/2, -\sigma/2, \xi\delta)$, for $\sigma, \xi \in \{-1, 1\}$, each have prior probability $1/4$. Each quarter independently draws a conditional shock scenario $(s_A, s_E) \in \{-1, 1\}^2$ with probability $1/4$, independent of the parameter. Factor shocks are zero, while the active and ETF residual shocks are δs_A and δs_E . Thus their gross returns are $1 + \sigma/2 + \delta(\xi + s_A)$ and $1 - \sigma/2 + \delta s_E$. The four purchase and sale rates vary independently over $[0, 1/100]$ and remain fixed across reviews. Every member of this family satisfies the funded two-review model.

The conditional return covariance is the positive-definite matrix $\delta^2 I_2$. Alpha has two nonzero, equally likely values $\pm\delta$, independent of the factor-premium regime. The first public observation reveals σ . With probability $1/2$, the active residual observation is zero and the posterior still assigns probability $1/2$ to each alpha value. Nevertheless, for *every* stated δ and rate vector, the optimized certainty-equivalent gaps, in fractions of initial wealth, obey

$$0 \leq \Delta_N \leq \frac{1}{4} + \delta^2, \quad \Delta_E > \frac{2129}{8080} - 9\delta - \delta^2, \quad 0 \leq \Delta_F \leq \frac{3}{202} + \frac{402}{101}\delta. \quad (31)$$

Hence the two matched continuation effects retain opposite strict signs throughout this family:

$$\Delta_E - \Delta_N > \frac{1}{250} > 0, \quad \Delta_F - \Delta_E < -\frac{23}{100} < 0. \quad (32)$$

Every full-root optimum with future ETF-only trading buys a positive active holding. Neither a unique optimum nor a zero root gap under future full trading is asserted.

The bounds use whole-class upper bounds and funded policies whose second-review choices depend only on the observed σ . With no future trade, mean terminal wealth under any full-root policy is at most $5/4 + \delta^2$; staying in cash gives the ETF-only root class at least one. With ETF-only trading at both reviews, pathwise return bounds for a root ETF holding p give $CE_{E,E} < 83/80 + 3\delta + \delta^2$. The same root purchases

$a = (7/15)/(1 + 1/100)$ and $p = (8/15)/(1 + 1/100)$ used in Proposition 12, followed by an ETF sale only when $\sigma = 1$, yield terminal wealth at least $657/505 - 6\delta$ on every path. This lower-bounds $CE_{F,E}$. Under full future trading, every policy has $CE_{F,F} \leq 3/2 + 2\delta$, while waiting in cash and then buying the fund favored by the revealed regime gives $CE_{E,F} \geq (3/2 - 2\delta)/(1 + 1/100)$. These estimates give (31), and their worst-case margins over $0 < \delta \leq 1/1000$ give (32). Attainment from Proposition 11 turns $\Delta_E > 0$ into the assertion about every full-root optimum.

This result shows that the signs in Proposition 12 are not confined to zero conditional risk or complete parameter revelation. It does not show a substantial-risk or alpha-driven effect: alpha and shocks are at most 0.1% per quarter, against factor premia of plus or minus 50%. The factor regime is revealed exactly, and both witness policies ignore the unresolved alpha. The active fund still uniquely carries the first factor. The law couples alpha and shock spacing so that some observations overlap; arbitrary changes to that information structure are outside the result. The values and rates are assumed, not calibrated, and no general boundary or cash-composition comparative static follows. This is a robustness example for the same continuation mechanism as Proposition 12, not a distinct principle about learning or hedging demand.

Portfolio moments can agree while public ETF returns reveal different information about a profitable active trade.

Proposition 14 (Equal portfolio geometry and covariances can hide different certification information; funded one-quarter model with public histories; machine checked). Take one active fund and one ETF, initial wealth one entirely in cash, position limits one, zero shareholder costs, $\gamma = 1$, zero ETF drag, and loadings $B^A = (0, 0)$, $B^E = (1, 0)$. The factor premia are known and $\Theta = \{(1/80, 0, \alpha) : -1/10 \leq \alpha \leq 1/10\}$. Each public record draws three independent uniform signs s, t, u . Factor shocks vanish and the active residual shock is $s/10$. Compare an *independent-residual* law $\mathcal{J}$, with ETF residual $t(3 - u)/20$, to a *revealing-residual* law $\mathcal{R}$, with ETF residual $t(3 - s)/20$. The signs are redrawn independently each quarter. Both known laws have centered shocks and positive fund gross returns.

Both laws have the same funded classes $F = \{(a, p) : a, p \geq 0, a + p \leq 1\}$ and $E = \{(0, p) : 0 \leq p \leq 1\}$, score, realized-return covariance and mean-error covariance:

$$\begin{aligned} Q((a, p); \theta) &= a\alpha + \frac{p}{80} - \frac{a^2}{200} - \frac{p^2}{80}, \\ \sup_{v \in E} Q(v; \theta) &= \frac{1}{320} \quad \text{at the unique ETF holding } (0, 1/2), \\ \Sigma &= \text{diag}(1/100, 1/40), \quad \Omega = \text{diag}(0, 0, 1/100). \end{aligned} \tag{33}$$

Their public-record covariance, each fund's marginal-return law, and the distribution of $\hat{\theta}_N$ also agree. The scores and advantages are one-quarter, initial-wealth-normalized quantities, not realized returns. At $\theta_+ = (1/80, 0, 1/10)$, the unique full optimum is $w_A = (1, 0)$, with advantage $147/1600$ against the *entire* ETF-only class. At $\theta_- = (1/80, 0, -1/10)$, every positive active holding has negative advantage.

For a full-history benchmark, allow a rule to certify any feasible active trade or fall back to $(0, 1/2)$, with an independent coin if desired. Require its probability of certifying an action with nonpositive true advantage to be at most η , uniformly for every $\theta \in \Theta$. Let $P_{\mathcal{J}}^{\max}$ and $P_{\mathcal{R}}^{\max}$ be the greatest certification probabilities at θ_+ among such rules under their respective known laws. For every $N \geq 1$ and $0 < \eta < 1$, both bounds are attained and

$$P_{\mathcal{F}}^{\max} = \min\{1, 1 - 2^{-N} + \eta\}, \quad P_{\mathcal{R}}^{\max} = 1. \quad (34)$$

Thus, for $0 \leq \beta \leq 1$, power at least $1 - \beta$ is attainable in $\mathcal{F}$ exactly when $2^{-N} \leq \eta + \beta$; $\mathcal{R}$ permits power one after one record without false certification. By contrast, the prescribed sample-mean gate with exact L_N has the *same* certification law in both models, and at $N = 1$ certifies with probability $1/2$ at θ_+ for every $\eta \in (0, 1)$. At $\eta = 1/4$, the two full-history benchmarks are $3/4$ and one.

The overlap argument makes (34) sharp. Under θ_+ and θ_- , all active returns are zero with probability 2^{-N} . In $\mathcal{F}$ the full public histories on that event have identical probabilities at both endpoints. Every active certification is false at θ_- , so at most η certification probability can be spent on the overlap; outside it, a nonzero active return identifies alpha. Certifying w_A when its identified alpha makes its advantage positive, and using a coin on the overlap, attains the bound uniformly over Θ . In $\mathcal{R}$, the ETF residual's absolute size reveals s , making alpha exactly identifiable after one record. These signals are public whether or not the ETF is held.

The sample-mean estimator discards that ETF residual in both laws. At one record its exact quantile is one. At θ_+ , its estimated alpha is zero or $1/5$ with equal probability: the first confidence set contains θ_- , and the second is $\{\theta_+\}$. The gate certifies only in the second case. With $\eta = 1/4$, its $1/4$ shortfall against $\mathcal{F}$'s benchmark comes from leaving the permitted false-certification allowance unused on all-zero histories; the gate already certifies every nonzero active-return history at θ_+ . The further $1/4$ difference between the two *full-history benchmarks* is the value of the revealing public ETF residual. The entire gap from $\mathcal{R}$'s benchmark to the gate cannot be assigned to that signal.

The obstruction is specific to a known discrete law with exact return observations, known factor premia, unknown alpha alone, zero switching costs and a fixed ETF optimum. It shows that geometry and second moments do not determine an instance's sharp full-history certification power. It does not establish a general history-length rate, an economically calibrated magnitude, or stability to rounding or unknown laws. The full-history benchmarks use rules beyond the prescribed gate; its power need not match their attainable upper values. The underlying fact that second moments do not determine a likelihood is generic, so no broad priority claim follows. The comparison uses the general two-law information-obstruction method; its specific content is the funded public-observation pair and the quantified loss of certification power.

The public-history obstruction leaves open how many records a uniformly valid decision needs when the observation law has no extra ETF signal. The next result answers that question in one bounded scalar family.

Proposition 15 (Worst-case history-length rate for a funded unspanned-factor trade; one-quarter model with known finite laws; machine checked). Start with wealth one in cash, one active fund and one ETF, zero risky incumbents, zero sale rates and ETF drag, and $\gamma = 0$. The active cap is $\bar{a} \in (0, 1]$, the ETF cap is one, and their purchase rates κ_A, κ_E lie in $[0, 1)$. Let $d > 0$, $\mu > \kappa_E$, $H > 0$, and $dH \leq 1/4$. Set $B^A = (d, 0)$, $B^E = (0, 1)$, and

$$\begin{aligned} q_E &= \frac{\mu - \kappa_E}{1 + \kappa_E}, & \lambda_0 &= \frac{\kappa_A + (1 + \kappa_A)q_E}{d}, \\ A &= \min\{\bar{a}, (1 + \kappa_A)^{-1}\}, & D &= dA > 0, \\ w_A &= \left(A, \frac{1 - (1 + \kappa_A)A}{1 + \kappa_E}\right), & v_E &= (0, (1 + \kappa_E)^{-1}). \end{aligned} \quad (35)$$

The unknown parameter is $\theta = (\lambda, \mu, 0)$, where $\lambda \in [\lambda_0 - H, \lambda_0 + H]$; alpha is known to be zero. Let $\mathcal{K}_H$ be the class of finite centered laws of a real Z supported in $[-H, H]$. For each law, the only random shock is the first factor shock Z ; all residual shocks vanish. Public observations are equivalent to independent draws of $x = \lambda + Z$: the factor record is (x, μ) , the active return is dx , and the ETF return is μ . The next holding-quarter shock is independent. The law, including its exact support and probabilities, is known before sampling.

The funded ETF-only optimum is uniquely v_E with score q_E . The full optimum is uniquely w_A when $\lambda > \lambda_0$, and every active action has negative advantage when $\lambda < \lambda_0$. For every λ , $\text{Adv}(w_A; \theta) = D(\lambda - \lambda_0)$ and $G_*(\theta) = D \max\{\lambda - \lambda_0, 0\}$. The first-factor exposure difference between w_A and every ETF-only holding is D ; funding and the cap fix that conversion from a factor-premium difference to a score difference.

Choose a score margin $0 < \delta \leq DH/2$ and error allowance $0 < \varepsilon \leq 1/16$ before sampling. Set the economic certification threshold to $\delta/4$ and the confidence error allowance to $\eta = \varepsilon$. Require, at every parameter in the interval, unconditional probability at most ε of certifying an active action with true $\text{Adv} \leq \delta/4$. Also require probability at least $1 - \varepsilon$ of certifying an active action with true $\text{Adv} > \delta/4$ whenever $G_*(\theta) \geq \delta$. A rule may depend on the known law but never on the unknown parameter; it either certifies a feasible active action or falls back to v_E . Then

$$\begin{aligned} \text{both requirements for every law in } \mathcal{K}_H &\implies N \geq \frac{1}{4\pi^2} \left(\frac{DH}{\delta} \right)^2 \log \frac{1}{\varepsilon}, \\ N \geq 32 \left(\frac{DH}{\delta} \right)^2 \log \frac{2}{\varepsilon} &\implies \text{both requirements for every law in } \mathcal{K}_H. \end{aligned} \quad (36)$$

The first implication holds even for full-public-history rules that randomize and can certify *any* funded active action. The second uses the prescribed confidence set C_N , plug-in optimizer and ETF-only fallback. With $r_B = H\sqrt{2 \log(2/\varepsilon)/N}$, it certifies the plug-in choice w_A only if $C_N \neq \emptyset$ and

$$\ell_N(w_A) = D(\hat{\lambda}_N - r_B - \lambda_0) > \delta/4, \quad \ell_N(w_A) \leq L_N(w_A) \quad \text{on these histories.} \quad (37)$$

If the set is empty or the plug-in optimum is not w_A , the rule falls back to v_E . On the calibrated coverage event at an alternative with $G_* \geq \delta$, the sufficient length in (36) gives $\ell_N(w_A) \geq \delta/2$. The certificate thus bounds the *original* optimized-ETF target L_N , rather than a comparison with one chosen ETF portfolio.

For the lower bound, put $\Delta = \delta/D$, $m = \lfloor H/\Delta \rfloor + 1$, and $\phi = \pi/(m+1)$. A centered law in $\mathcal{K}_H$ puts masses proportional to $\sin^2(i\phi)$ at $Z_i = 2\Delta[i - (m+1)/2]$, $i = 1, \dots, m$. At $\lambda_{\pm} = \lambda_0 \pm \Delta$, the full public histories have one-record affinity $\cos \phi$ and N -record affinity $\cos^N \phi$. The negative endpoint makes every active certification false, while the positive endpoint has $G_* = \delta$. A finite overlap bound for the two endpoint laws yields the first implication in (36). For the upper bound, the centered bounded shock gives $\Pr(|\hat{\lambda}_N - \lambda| \geq r_B) \leq \varepsilon$. Exact finite calibration makes the prescribed confidence radius no larger than r_B ; this proves both the lower certificate in (37) and its coverage-event power guarantee.

This is the standard scalar mean-testing rate in premium units, $(H/(\delta/D))^2 \log(1/\varepsilon)$. The funded geometry enters through D as a unit conversion. The constants in (36) differ by up to roughly 1600 after the logarithms are compared, so only the order matches. The hard law can change with the prespecified margin and needs a support size growing with DH/δ ; an individual known finite law can be much easier. The active fund uniquely carries the first factor, with known zero alpha: this is certification of a premium opportunity, not manager skill. The family has zero mean-variance penalty, deterministic ETF returns and exact public observations. It supplies neither a multivariate rate nor a guarantee under rounded or unknown laws, a moving ETF optimum, repeated live testing or calibrated market returns. The inverse-

square lower-bound mechanism is standard best-arm-style mean testing; the stated rate converts it to the funded, unspanned-factor score margin against the optimized ETF-only class.

Proposition 15 has a fixed ETF optimum. The following family makes that optimum switch with the unknown premium, so a certificate must cover two comparison directions.

Proposition 16 (Joint directional information against a moving ETF comparator; one-quarter model with known finite laws; machine checked). Take one active fund, one ETF, initial wealth one entirely in cash, caps one, zero shareholder costs and ETF drag, and $\gamma = 0$. Their loadings are $B^A = (1, 0)$ and $B^E = (0, 1)$. Write $\theta = (\lambda_1, \lambda_2, \alpha)$. Fix a known real 3×3 matrix J of Euclidean operator norm at most $1/100$, and set

$$\begin{aligned} c_0 &= (0, -1/4, 0), & c_1 &= (0, 1/4, 1/4), \\ \Theta &= \text{conv}\{c_j + Jv : j \in \{0, 1\}, v \in \{-1, 1\}^3\}, & \Omega &= JJ^\top, \\ d_0 &= (1, 0, 1), & d_1 &= (1, -1, 1), & \sigma_j^2 &= d_j^\top \Omega d_j, & \sigma &= \max\{\sigma_0, \sigma_1\}. \end{aligned} \quad (38)$$

Let $\mathcal{K}_J$ contain every *known finite* law of $U \in \mathbb{R}^3$ with $EU = 0$, $EUU^\top = I_3$ and $\|U\|_2 \leq 9$. The observation shock is $(z_1^f, z_2^f, z^A) = JU$, while the ETF residual shock is zero. Each public record is equivalent to $X = (f_1, f_2, r^A - f_1) = \theta + JU$; the ETF return is $r^E = f_2$. Histories and the next holding-quarter draw are independent conditional on fixed θ . All laws in $\mathcal{K}_J$ have the same mean-error covariance Ω . Under the nonzero-rate assumptions below, $\Omega_{33} > 0$ makes both alpha and the first premium uncertain; zero alpha-error variance is also allowed.

The funded classes are $F = \{(a, p) : a, p \geq 0, a + p \leq 1\}$ and $E = \{(0, p) : 0 \leq p \leq 1\}$. With $w_A = (1, 0)$, the whole ETF-only class and the full class give

$$\begin{aligned} \sup_{v \in E} Q(v; \theta) &= \max\{0, \lambda_2\}, \\ m_j(\theta) &= d_j^\top \theta, & \text{Adv}(w_A; \theta) &= \min_{j=0,1} m_j(\theta), \\ G_*(\theta) &= \max\{0, \min_{j=0,1} m_j(\theta)\}. \end{aligned} \quad (39)$$

Cash uniquely optimizes E when $\lambda_2 < 0$, and the ETF does so when $\lambda_2 > 0$; both regimes occur in Θ . If the minimum in (39) is positive, w_A is the unique full optimum. If it is negative, every positive active holding has negative advantage. The paired estimation errors in the two score comparisons have variances σ_j^2/N , including all alpha/premium cross-covariances. Both directions have a first-premium and an alpha component.

Assume $\Omega_{11} > 0$ and $\sigma > 0$. Fix $0 < \delta \leq \sigma/8$, $0 < \varepsilon \leq 1/16$, $\eta = \varepsilon$, and economic threshold $\delta/4$ before sampling. As in Proposition 15, a rule may depend on the known law, may randomize, and either certifies a funded active action or falls back to an ETF-only action. It must have unconditional false-certification probability at most ε at every $\theta \in \Theta$, and correct-certification probability at least $1 - \varepsilon$ whenever $G_*(\theta) \geq \delta$. Here false certification means an implemented active action with true advantage at most $\delta/4$, and correct certification means true advantage above $\delta/4$. Then

$$\begin{aligned} \text{requirements for every law in } \mathcal{K}_J &\implies N \geq \frac{\sigma^2}{16\pi^2\delta^2} \log \frac{1}{\varepsilon}, \\ N &\geq 192 \frac{\sigma^2}{\delta^2} \log \frac{6}{\varepsilon} \implies \text{requirements for every law in } \mathcal{K}_J. \end{aligned} \quad (40)$$

The first implication holds for all full-history rules, including randomized ones. For the second, let $t_{N,\varepsilon}$ be Section 2's exact finite-law critical value and set $r_N = \sqrt{t_{N,\varepsilon}/N}$. On a nonempty prescribed confidence set C_N , the rule uses the conservative certificate

$$\ell_N(w_A) = \min_{j=0,1} \{d_j^\top \hat{\theta}_N - r_N \sigma_j\} \leq L_N(w_A). \quad (41)$$

If $\ell_N(w_A) > \delta/4$, then w_A is also the actual plug-in full optimizer and the rule certifies it; otherwise it uses the prescribed ETF-only fallback. Empty C_N always forces fallback. On coverage, $\ell_N(w_A) \geq \min_j \{m_j(\theta_*) - 2r_N \sigma_j\}$. Under the sufficient length in (40), this is at least $\delta/2$ whenever $G_* \geq \delta$. Thus (41) bounds the original L_N against the *optimized* ETF class; certifying against one selected ETF action would be weaker.

For the lower bound, choose the direction j with $\sigma_j = \sigma$. A centered sine-weighted finite law is embedded along $J^\top d_j/\sigma$, with independent signs completing the three-dimensional covariance to I_3 . It preserves $\Omega = JJ^\top$, including when J is singular. Two parameters on opposite sides of that comparator face require conflicting certification decisions. Their full public histories are images of latent histories with one-record affinity $\cos(\pi/(4k))$, where $k = \lfloor \sigma/(4\delta) \rfloor \geq 2$. A finite overlap bound, applied even to rules with the latent history, gives the first implication in (40). For the upper bound, finite bounded-mean tails and a three-coordinate union bound give $t_{N,\varepsilon} \leq 12 \log(6/\varepsilon)$ under the sufficient length. A covariance-direction inequality on $\text{Im}(\Omega)$ proves (41) and the coverage-event guarantee without dropping cross terms.

The two variances expose the directional effect:

$$\begin{aligned} \sigma_0^2 &= \Omega_{11} + 2\Omega_{13} + \Omega_{33}, \\ \sigma_1^2 &= \Omega_{11} + \Omega_{22} + \Omega_{33} - 2\Omega_{12} + 2\Omega_{13} - 2\Omega_{23}, \\ \sigma^2 &\geq (\sqrt{\Omega_{11}} - \sqrt{\Omega_{33}})^2. \end{aligned} \quad (42)$$

If the alpha-error standard deviation is at most half the first-premium error standard deviation, the necessary length in (40) is at least $\Omega_{11} \log(1/\varepsilon)/(64\pi^2\delta^2)$. When $\Omega_{33} = 0$, alpha is observed exactly and $\sigma^2 \geq \Omega_{11}$. Moreover, there is a value α_0 , fixed before the history length and the law, for which the lower bound in (40) still holds when a rule is told alpha *before* sampling and its two requirements apply only to $\theta \in \Theta$ with $\alpha = \alpha_0$. This pre-disclosure conclusion is machine checked. It does not assert that the bound holds on every alpha slice. Conversely, for $s = 1/1000$, $0 < \tau \leq 1$, and

$$J = s \begin{pmatrix} 1 & 0 & 0 \\ 1 & \tau & 0 \\ 0 & \tau & 0 \end{pmatrix},$$

one gets $\sigma_1 = 0$ but $\sigma_0^2 = s^2(1 + \tau^2) > 0$. Errors cancel against the ETF, yet cash can be the better comparator and still requires data. If both $\sigma_j = 0$, outside the nonzero-rate statement, both comparisons are observed exactly after one record and a full-history rule certifies with zero false certification and power one; this does not remove the prescribed gate's empty-set fallback.

The bounds match in order only. Their constants differ by roughly 5×10^4 at the largest allowed ε . The hard law's support grows as σ/δ , so the rate is worst-case over $\mathcal{K}_J$, not a rate for each known law. The main structural result is the maximum over both comparator directions in this funded cash-versus-ETF geometry. A maximum over alternatives and a two-point lower bound are standard best-arm-identification tools; the finite testing and ellipsoidal penalties are standard mathematics. Both contrasts load on $\lambda_1 + \alpha$, so the result does not separately identify manager skill from the unspanned premium. It assumes zero costs and

risk penalty, an all-cash incumbent, exact known finite laws and a two-vertex ETF comparator. It gives no calibrated market magnitude, general multivariate menu, efficient calibration method or repeated-review guarantee.

The previous two history-length results have a fixed-size profitable active action. With a smooth active entry, the action itself becomes small as its advantage approaches zero.

Proposition 17 (Quadratic entry changes the history rate in economic advantage; funded risk-averse one-quarter model with known finite laws; machine checked). Fix $0 < s \leq 1/100$. Take one active fund and one ETF, initial wealth one in cash, caps one, zero shareholder costs and ETF drag, loadings $B^A = (1, 0)$, $B^E = (0, 1)$, and $\gamma = s^{-2}$. Set $c = (0, 1/4, 0)$ and $\Theta = c + s[-1, 1]^3$ for $\theta = (\lambda_1, \lambda_2, \alpha)$. The class $\mathcal{K}_s$ consists of known finite laws of $U \in \mathbb{R}^3$ with $\mathbb{E}U = 0$, $\mathbb{E}UU^\top = I_3$, and $\|U\|_2 \leq 9$. For each law, take $(z_1^f, z_2^f, z^A) = sU$ and zero ETF residual shock. The public record is equivalent to $X = (f_1, f_2, r^A - f_1) = \theta + sU$, and the ETF return is f_2 . Histories and the next holding-quarter draw are independent conditional on fixed θ . The two covariances have distinct roles:

$$\Omega = s^2 I_3, \quad \Sigma = \text{diag}(2s^2, s^2), \quad Q((a, p); \theta) = a(\lambda_1 + \alpha) + p\lambda_2 - a^2 - p^2/2. \quad (43)$$

Let $x = \lambda_1 + \alpha$ and $x_+ = \max\{x, 0\}$. The unique funded optima, both with strict cash slack, and the advantage over the *entire* ETF-only class are

$$\begin{aligned} v_E(\theta) &= (0, \lambda_2), & w_F(\theta) &= (x_+/2, \lambda_2), \\ \sup_{v \in E} Q(v; \theta) &= \lambda_2^2/2, & G_*(\theta) &= x_+^2/4, \\ \text{Adv}((a, p); \theta) &= ax - a^2 - (p - \lambda_2)^2/2. \end{aligned} \quad (44)$$

The ETF optimizer moves continuously with its unknown premium. At oracle advantage $\delta > 0$, the smallest positive mean signal is $x = 2\sqrt{\delta}$, and the active optimum holds $a = \sqrt{\delta}$. Thus the active trade and its paired mean-error exposure both shrink at entry. The last term in (44) is the separate score loss from using an estimated ETF holding rather than the true ETF optimum.

Fix $0 < \delta \leq s^2/128$ and $0 < \varepsilon \leq 1/16$, with confidence error $\eta = \varepsilon$ and economic threshold $\delta/4$ chosen before sampling. A rule may depend on the known law, may randomize, and must either certify a feasible active action or fall back to a funded ETF-only action. At every $\theta \in \Theta$, require probability at most ε of certifying an active action whose true advantage is at most $\delta/4$. Whenever $G_*(\theta) \geq \delta$, require probability at least $1 - \varepsilon$ of certifying an action whose true advantage exceeds $\delta/4$. Then

$$\begin{aligned} \text{requirements for every law in } \mathcal{K}_s &\implies N \geq \frac{s^2}{32\pi^2\delta} \log \frac{1}{\varepsilon}, \\ N \geq 768 \frac{s^2}{\delta} \log \frac{6}{\varepsilon} &\implies \text{requirements for every law in } \mathcal{K}_s. \end{aligned} \quad (45)$$

The first implication applies to every full-public-history rule, including randomized ones. The second is attained by a conservative version of the one-quarter model's prescribed plug-in gate. On any positive-probability history, put $\hat{x} = \hat{\lambda}_1 + \hat{\alpha}$, $\hat{a} = \hat{x}_+/2$, $\hat{p} = \hat{\lambda}_2$, and $\rho_N = s\sqrt{t_{N,\varepsilon}/N}$. Both plug-in actions $(\hat{a}, \hat{p})$ and $(0, \hat{p})$ remain funded even when the unprojected sample mean lies outside Θ . For nonempty C_N , the gate uses

$$\begin{aligned} \ell_N((\hat{a}, \hat{p})) &= \hat{a}\hat{x} - \hat{a}^2 - \sqrt{2}\hat{a}\rho_N - \rho_N^2/2 \\ &\leq L_N((\hat{a}, \hat{p})). \end{aligned} \quad (46)$$

It certifies only when $\hat{a} > 0$ and $\ell_N > \delta/4$; an empty confidence set forces the ETF fallback. The $\rho_N^2/2$ term protects against error in the ETF optimum inside the original optimized-comparator target. Comparing with only the selected ETF action would omit that loss. On coverage, if $G_* \geq \delta$ and $\rho_N \leq \sqrt{\delta}/8$, the certificate is at least $5\delta/8$, so the rule correctly certifies. A bounded-mean calculation gives $t_{N,\varepsilon} \leq 12 \log(6/\varepsilon)$ under the sufficient length in (45), which ensures that radius bound.

For the lower bound, use a centered normalized sine-weight finite law embedded along the active signal direction $(1, 0, 1)/\sqrt{2}$. Two parameters in Θ have opposite active signals of order $\sqrt{\delta}$, while both have $\lambda_2 = 1/4$. At the negative parameter every active certification is false; at the positive one $G_* \geq \delta$. The translated finite supports overlap, and their one-record affinity is $\cos(\pi/(4k))$, where $k = \lfloor s/(4\sqrt{2\delta}) \rfloor \geq 2$. Lifting any public rule to the more informative latent history and applying the finite overlap bound gives the first implication in (45). The lower pair holds the ETF premium fixed: the bound comes from active entry even though the ETF optimum varies across the full parameter domain.

The worst-case history length is therefore of order $(s^2/\delta) \log(1/\varepsilon)$ for fixed s , with constants about 4×10^5 apart. In *mean-signal* units, however, $x = 2\sqrt{\delta}$ makes this the same scalar inverse-square testing rate as the earlier examples. The exponent change in economic advantage is the conversion through the quadratic entry score, not a general effect of risk aversion; $\gamma = s^{-2}$ also ties preferences to the shock scale. The result assumes zero costs, slack funding, diagonal risky-return covariance, separable score coordinates, known finite laws and exact observations. It neither separates manager skill from the unspanned premium in $x = \lambda_1 + \alpha$ nor gives a calibrated history recommendation, a binding-budget rate or a multiperiod rule. The faster rate in the economic margin is the familiar smooth-entry conversion of a quadratic value gap, specialized here to the funded certificate and a moving ETF optimum.

The next certificate uses positive definite return risk to control a moving ETF optimum even with trading costs and binding constraints.

Proposition 18 (Curvature certificate with costs and binding constraints; funded one-quarter model with known finite laws; machine checked). Use the one-quarter statistical model of Section 2 with one active fund and one or two ETFs. Allow a compliant incumbent, position caps, directional purchase and sale costs, ETF drag, correlated risky returns and binding cash. Assume $\gamma > 0$ and a positive definite realized-return covariance Σ . The known finite law of the factor and active-residual shock is $\zeta = JU$, where J is known, $EU = 0$, $EUU^\top = I_3$, and $\|U\|_2 \leq 9$ at every scenario. ETF residual shocks may be correlated with these shocks, subject to the known joint law and positive gross returns. With $n \in \{1, 2\}$ ETFs, define the mean-exposure map and its error scale by

$$Aw = (b(w), a), \quad K = \frac{1}{\gamma} \max_{z \in \mathbb{R}^{1+n}, z \neq 0} \frac{\|J^\top Az\|_2^2}{z^\top \Sigma z}, \quad \Omega = JJ^\top. \quad (47)$$

The maximum is attained and $K \geq 0$. The covariance Ω governs mean-estimation error; Σ governs realized-return risk. Formula (47) compares their action directions without adding them.

At the unprojected estimate $\hat{\theta}_N$, let $\hat{w}_F$ and $\hat{v}_E$ be the unique funded score maximizers over F and E . Put

$$\begin{aligned} \hat{G} &= Q(\hat{w}_F; \hat{\theta}_N) - Q(\hat{v}_E; \hat{\theta}_N) \geq 0, \\ r_N &= \sqrt{t_{N,\eta}/N}, \quad u_N = Kr_N^2, \\ \ell_N(\hat{w}_F) &= \hat{G} - \sqrt{2u_N \hat{G}} - u_N/2 \leq L_N(\hat{w}_F) \quad \text{when } C_N \neq \emptyset. \end{aligned} \quad (48)$$

The lower bound compares the candidate with the *entire* ETF-only class optimized at each parameter in C_N . The gate certifies $\hat{w}_F$ only if $\ell_N > \delta_{\text{econ}} \geq 0$; otherwise, including an empty confidence set, it uses

$\hat{v}_E$. A positive certificate forces an active change $\hat{a} \neq a^-$. For every fixed true parameter, the unconditional probability of certifying an action whose true advantage is at most δ_{econ} is at most η .

Fix $\delta > 0$, $0 < \varepsilon \leq 1/16$, $\eta = \varepsilon$, and $\delta_{\text{econ}} = \delta/4$ before sampling. On coverage, $u_N \leq \delta/128$ and $G_*(\theta_*) \geq \delta$ imply $\ell_N > \delta/2$. For every law just described,

$$N \geq \max\{324, 1536K/\delta\} \log(6/\varepsilon) \implies \begin{cases} \Pr(\text{false certification}) \leq \varepsilon & \text{at every } \theta_*, \\ \Pr(\text{correct certification}) \geq 1 - \varepsilon & \text{if } G_*(\theta_*) \geq \delta. \end{cases} \quad (49)$$

Correct certification means implementing an active change whose true advantage over the optimized ETF class exceeds $\delta/4$. If $K = 0$, estimation error affects no score difference and $\ell_N = G_*(\theta_*)$ almost surely for every positive sample length; the empty-set fallback still applies.

For a matching *class-level* comparison, fix $0 < \kappa \leq 1/10000$ and $0 < \delta \leq \kappa/128$. Consider every admissible design and known law above with $K \leq \kappa$. If a common history length permits a law-specific, possibly randomized full-public-history rule meeting the same false-certification and correct-certification requirements at every parameter, then

$$N \geq \frac{\kappa}{32\pi^2\delta} \log \frac{1}{\varepsilon}, \quad N \geq 1536 \frac{\kappa}{\delta} \log \frac{6}{\varepsilon} \quad \text{is sufficient for the class.} \quad (50)$$

For (48), convex directional costs and positive definite Σ give a quadratic score gap at either optimizer, even at a cost kink or binding constraint. For a confidence error e , (47) bounds its paired score error along a holding difference z by $r_N \sqrt{\gamma K z^\top \Sigma z}$. The full-class gap bounds the difference between $\hat{w}_F$ and $\hat{v}_E$; a second quadratic gap bounds movement of the optimized ETF value by $u_N/2$. Taking the infimum over C_N yields (48). Finite-law calibration gives coverage and a bounded-mean radius for (49). The lower bound in (50) uses Proposition 17's zero-cost, slack-cash subfamily with $s = \sqrt{\kappa}$ and $K = \kappa$; its upper bound follows from (49).

Formula (48) requires exact global plug-in optima and score values. For feasible numerical candidates $\tilde{w} \in F$, $\tilde{v} \in E$, let $g = Q(\tilde{w}; \hat{\theta}_N) - Q(\tilde{v}; \hat{\theta}_N)$. Let $\varepsilon_F, \varepsilon_E \geq 0$ be *certified global* upper bounds on their respective optimizer value gaps; these imply $g + \varepsilon_F \geq 0$. A reviewed paper proof, not machine checked, gives the following lower bound when C_N is nonempty:

$$g - \varepsilon_E - \sqrt{2u_N}(\sqrt{\varepsilon_F} + \sqrt{g + \varepsilon_F}) - u_N/2 \leq L_N(\tilde{w}). \quad (51)$$

A positive bound in (51) forces an active change and retains coverage-based false-certification control, but the power rate in (49) is not proved for approximate optima. Feasibility, both value gaps and numerical rounding must be certified; a solver status or tolerance does not supply them. Simply adding the value gaps without the square-root terms can give an invalid lower bound.

The order in (50) is a worst case over designs, with loose constants; its hard design has zero costs and slack cash. It does not prove that fees or binding constraints cause that rate, or give a lower bound for each constrained instance. The upper certificate covers those constraints under strict risk curvature, but K also bounds potentially infeasible directions and can be conservative. Known finite laws, exact calibration, fixed loadings and risk inputs, and one review remain assumptions. No empirical history recommendation or separate identification of manager skill follows. Stability under strong concavity is a standard perturbation principle; the result here is the explicit whole-ETF-class certificate with directional costs and binding constraints, not a new stability theorem.

Propositions 15–18 let a rule know the exact law of the observation shocks. The next result removes that knowledge in Proposition 16’s family. It asks whether comparing with the whole optimized ETF-only class lets a rule that does not know the law do better than checking each ETF-only comparison separately.

Proposition 19 (Certification without knowing the law: separate comparison bounds attain the known-law order; one-quarter model with bounded unknown laws; machine checked). Use Proposition 16’s family: its loadings, funded classes, domain Θ , known matrix J of operator norm at most $1/100$, active action $w_A = (1, 0)$, and the score comparisons $m_j(\theta) = d_j^\top \theta$ of w_A with cash ($j = 0$) and with the ETF ($j = 1$). Call each comparison a *face* of the optimized ETF-only comparator; by (39), $\text{Adv}(w_A; \theta) = \min_j m_j(\theta)$. Replace the class $\mathcal{K}_J$ by the larger class $\mathcal{L}_J$ of finite laws of $U \in \mathbb{R}^3$ with $EU = 0$ and $\|U\|_2 \leq 9$, keeping the observation shock JU and the zero ETF residual shock. $\mathcal{L}_J$ restricts neither the covariance nor the skewness or higher moments of U . Put

$$\bar{\sigma}_j = \|J^\top d_j\|_2, \quad \bar{\sigma} = \max_{j=0,1} \bar{\sigma}_j, \quad R_j = 9\bar{\sigma}_j, \quad R = \max_{j=0,1} R_j. \quad (52)$$

Every law in $\mathcal{L}_J$ gives an admissible instance of Section 2’s statistical model with the advantage formulas (39). Each face’s per-record estimation error $d_j^\top JU$ lies in $[-R_j, R_j]$. On $\mathcal{K}_J \subset \mathcal{L}_J$, $\bar{\sigma}_j$ equals Proposition 16’s σ_j .

A *law-free* rule is a single full-history rule, possibly randomized, fixed before the law is chosen. It reads only the public history and the known J , and it either certifies a funded active action or falls back to a funded ETF-only action. Fix $\delta > 0$, $\varepsilon \in (0, 1)$, $\eta = \varepsilon$ and economic threshold $\delta/4$ before sampling. Proposition 16’s two requirements must now hold at *every* law in $\mathcal{L}_J$. The false-certification probability must be at most ε at every $\theta \in \Theta$. The probability of certifying an action with true advantage above $\delta/4$ must be at least $1 - \varepsilon$ whenever $G_*(\theta) \geq \delta$.

(i) A *joint confidence set reduces to two face bounds*. For every nonempty set $C \subset \mathbb{R}^3$,

$$\inf_{\theta \in C} \text{Adv}(w_A; \theta) = \min_{j=0,1} \inf_{\theta \in C} m_j(\theta). \quad (53)$$

Suppose a rule certifies w_A using a data-dependent set C_N that contains the true parameter with probability at least $1 - \eta$ at every parameter. Its certificate is then $\min_j \ell_j$ with $\ell_j = \inf_{C_N} m_j$, and each ℓ_j is a lower confidence bound for $m_j(\theta)$ at the full level η . Conversely, face bounds with error probabilities η_0 and η_1 give, by the union bound, a set with coverage at least $1 - \eta_0 - \eta_1$ whose certificate is their minimum. So the only difference between a joint and a face-by-face construction is the error level of each face bound, η against $\eta/2$. For face bounds whose width at level a is proportional to $\sqrt{\log(1/a)}$, this changes the required history length by the factor $1 + \log 2 / \log(1/\eta)$, which lies between 1.23 and 1.24 at $\eta = 1/20$. Bounding the ETF face through the cash face and a separate bound on the ETF premium does not help either: $d_1 = d_0 - e_2$ with $e_2 = (0, 1, 0)$, and the range of that route, $R_0 + 9\|J^\top e_2\|_2$, is never below R_1 .

(ii) A *range rule*. For $N \geq 1$ and $\eta \in (0, 1)$, put

$$r_j = R_j \sqrt{\frac{3 \log(2/\eta)}{N}}, \quad \ell_j = d_j^\top \hat{\theta}_N - r_j. \quad (54)$$

The rule certifies w_A if $\min_j \ell_j > \delta_{\text{econ}}$. Otherwise it implements the plug-in ETF-only action: the ETF if $\hat{\lambda}_2 > 0$, and cash otherwise, where $\hat{\lambda}_2$ is the second coordinate of $\hat{\theta}_N$. The rule uses J and nothing else about the law. At every law in $\mathcal{L}_J$, every θ and every $\delta_{\text{econ}} \geq 0$, the probability that it certifies w_A while $\text{Adv}(w_A; \theta) \leq \delta_{\text{econ}}$ is at most η . With $\eta = \varepsilon$ and $\delta_{\text{econ}} = \delta/4$, both requirements hold at every law in $\mathcal{L}_J$ once $N \geq \max\{3, 22R^2/\delta^2\} \log(2/\varepsilon)$. When $\bar{\sigma} \geq \delta$, this length is $1782(\bar{\sigma}/\delta)^2 \log(2/\varepsilon)$.

(iii) *The known-law order binds every law-free rule and is attained.* Let $0 < \delta \leq \bar{\sigma}/8$ and $0 < \varepsilon \leq 1/16$. Then

$$\begin{aligned} \text{a law-free rule meets both requirements on } \mathcal{L}_J &\implies N \geq \frac{\bar{\sigma}^2}{16\pi^2\delta^2} \log \frac{1}{\varepsilon}, \\ N \geq 1782 \frac{\bar{\sigma}^2}{\delta^2} \log \frac{6}{\varepsilon} &\implies \text{the range rule meets both requirements on } \mathcal{L}_J. \end{aligned} \tag{55}$$

For (53), the infimum of a minimum of two functions is the minimum of their infima. On the event that C_N contains θ , each ℓ_j is at most $m_j(\theta)$, which gives the full-level face bounds; the converse is the union bound. For (54), the face error $d_j^\top(\hat{\theta}_N - \theta)$ is the mean of N independent, identically distributed, centered variables bounded by R_j . Hoeffding's inequality for bounded independent and identically distributed variables ([39], Theorem 1), applied after an affine map of $[-R_j, R_j]$ onto $[0, 1]$, bounds each one-sided tail by $\exp(-Nr_j^2/(2R_j^2)) \leq \eta/2$. A union bound over the two faces gives $\ell_j \leq m_j(\theta)$ for both faces with probability at least $1 - \eta$, and a certification then forces $\text{Adv}(w_A; \theta) > \delta_{\text{econ}}$. If $G_*(\theta) \geq \delta$, both $m_j(\theta) \geq \delta$; with probability at least $1 - \varepsilon$ each $\ell_j \geq \delta - 2r_j$, which exceeds $\delta/4$ once $r_j < 3\delta/8$. The lower bound in (55) is Proposition 16's: a single law-free rule meeting both requirements on $\mathcal{L}_J$ meets them on $\mathcal{K}_J$, where the law-specific lower bound (40) already binds, with $\sigma = \bar{\sigma}$.

Over a bounded law class, then, the optimized ETF-only comparator in this family is exactly its two faces, and certification without knowledge of the law is separate scalar mean certification of each face. Its worst-case history length has the same order, $(\bar{\sigma}/\delta)^2 \log(1/\varepsilon)$, as with a known law. No law-free rule, whether joint, variance-adaptive or otherwise, improves that order. The funded ETF-only structure enters only through which faces exist and through their ranges. The ETF face carries the error in the ETF premium, which is why R_1 and R_0 differ, and the rate takes the larger of the two. Joint constructions and variance adaptation can change constants only, joint constructions by at most the level factor in (i).

The result does not say that the range rule is efficient at a particular law. Its constants and those of the lower bound are about 2.8×10^5 apart. At a given law, let s_j be the standard deviation of the face error $d_j^\top JU$. The range rule can then need up to $(R_j/s_j)^2$ times as many records as a rule that uses s_j ; for laws in $\mathcal{K}_J$ this factor is 81. The order match is a worst case over $\mathcal{L}_J$. The class assumes a known J and a known bound on $\|U\|_2$; laws with unknown range or unknown J are outside it. The family is Proposition 16's two-face geometry with zero costs, zero risk penalty, an all-cash incumbent and the fixed active action w_A , so this is not a result about a general plug-in gate over data-selected candidates, although (53) applies to any joint construction. The level-factor comparison in (i) covers face bounds of the stated width shape, not the best possible bound at each level. There is one review with a fixed history length: no time-uniform guarantee over repeated reviews and no sequential lower bound is proved. Both contrasts load on $\lambda_1 + \alpha$, so the result does not separate manager skill from the unspanned premium. The upper bound applies Hoeffding's inequality and a union bound, and the lower bound is Proposition 16's two-point bound; the result is an application of these tools, not a new testing principle.

We return to the funded two-review model. Propositions 12 and 13 show that future ETF adjustment can raise today's active-trading advantage, but only in specific families. The last result asks what decides the sign of that effect in every instance of the model, and whether the sign can be reversed.

Proposition 20 (Future ETF adjustment moves today's active advantage by a difference of premia at two root optima, capped by adjustable wealth, with both signs; funded two-review model; machine checked). Fix any instance of the funded two-review model of Section 2. For a feasible root trade $u_0 \in F_0$ and a continuation class $R \in \{F, E, N\}$, let $c_R(u_0) = -\rho^{-1} \log(-H_0^R(u_0))$ be the certainty

equivalent of trading u_0 today and then behaving optimally within R . Define the *premium of future ETF adjustment* and the *premium of future active trading* at u_0 by

$$\varphi(u_0) = c_E(u_0) - c_N(u_0), \quad \psi(u_0) = c_F(u_0) - c_E(u_0). \quad (56)$$

For every pair of classes, $CE_{D,R} = \max_{u_0 \in D_0} c_R(u_0)$, and the maximum is attained; φ and ψ are nonnegative on F_0 . Let A_R be any maximizer of c_R over the full root class F_0 , and B_R any maximizer over the ETF-only root class E_0 .

(i) *Sandwich*. For every choice of these maximizers,

$$\begin{aligned} \varphi(A_N) - \varphi(B_E) &\leq \Delta_E - \Delta_N \leq \varphi(A_E) - \varphi(B_N), \\ \psi(A_E) - \psi(B_F) &\leq \Delta_F - \Delta_E \leq \psi(A_F) - \psi(B_E). \end{aligned} \quad (57)$$

(ii) *Funded cap*. Let $\bar{g}$ and $\underline{g}$ be upper and lower bounds on every ETF gross return over all parameters and scenarios, and put $\text{up} = \max\{\bar{g} - 1, 0\}$ and $\text{down} = \max\{1 - \underline{g}, 0\}$. Write h_0^+ for cash after the root trade and $x_{0,j}^+$ for the dollars then held in ETF j . For every feasible root trade,

$$0 \leq \varphi(u_0) \leq \beta(u_0) = \frac{h_0^+ \text{up} + (\sum_j x_{0,j}^+) \bar{g} (\text{up} + \text{down})}{W_0^-}. \quad (58)$$

In particular $\varphi(u_0) = 0$ when u_0 leaves neither cash nor an ETF holding. Hence, if some full-root maximizer A_E leaves neither, then $\Delta_E - \Delta_N \leq -\varphi(B_N) \leq 0$.

(iii) *Movement of the no-active-trade region*. If $\Delta_N = 0$ and $\varphi(A_E) \leq \varphi(B_N)$, then $\Delta_E = 0$, and by (28) some optimal full-root policy with future ETF-only trading makes no active trade today. If $\Delta_E = 0$ and $\varphi(A_N) \geq \varphi(B_E)$, then $\Delta_N = 0$, with the corresponding no-active-trade optimum when no second trade is allowed.

(iv) *A family with the opposite sign*. Initial wealth one is held entirely in cash. There is one active fund and one ETF with $B^A = (1, 0)$, $B^E = (0, 1)$, zero ETF drag and $\rho = 20$. Each quarter has one zero-shock scenario. Two persistent parameters are $\theta_+ = (1/2, 1/2, 0)$, with prior probability $1/3$, and $\theta_- = (1/2, -1/2, 0)$, with prior probability $2/3$. The active fund's gross return is $3/2$ in every quarter under both parameters. The ETF's is $3/2$ under θ_+ and $1/2$ under θ_- , so the first public observation reveals the parameter. The four purchase and sale rates vary independently over $[0, 1/200]$ and are fixed across reviews. For every member of this family,

$$\begin{aligned} CE_{E,N} &= 1, & CE_{E,E} &\geq 1 + \frac{1}{50} - \frac{1}{320000}, & CE_{F,N} &\geq \frac{450}{201}, & CE_{F,E} &\leq \frac{9}{4}, \\ \Delta_E - \Delta_N &\leq \frac{9}{804} - \frac{1}{50} + \frac{1}{320000} < -\frac{1}{125}, \end{aligned} \quad (59)$$

and both Δ_N and Δ_E are positive. The root trade that puts all cash into the active fund is feasible and has $\varphi = 0$, while staying in cash at the root has $\varphi \geq 1/50 - 1/320000$. With Proposition 12, both signs of $\Delta_E - \Delta_N$ occur in the funded two-review model.

The attainment statements come from Proposition 11. The sandwich uses only feasibility and optimality. Since A_N is feasible for the full root under future ETF-only trading, $CE_{F,E} \geq c_E(A_N) = CE_{F,N} + \varphi(A_N)$. Since B_E maximizes c_E over E_0 and $c_N(B_E) \leq CE_{E,N}$, $CE_{E,E} \leq CE_{E,N} + \varphi(B_E)$. Subtracting gives the lower bound in (57); the other three bounds are the same argument. For the cap, fix a second-review node. An ETF-only trade there changes terminal wealth, relative to no trade, by its purchases times the ETF's excess

over cash, plus its sales times the ETF's shortfall, minus charges. Purchases are funded by cash and sale proceeds, and sales cannot exceed ETF holdings, so on every path the gain is at most $W_0^- \beta(u_0)$. Exponential utility turns a constant wealth gain into a multiplication of utility by $\exp(-\rho\beta(u_0))$, which gives $c_E \leq c_N + \beta$. Part (iii) combines (57) with $\Delta_R \geq 0$ and (28). In (iv), holding cash gives $CE_{E,N} = 1$, and no ETF-only root does better without a second trade, because the ETF's expected two-quarter gross return is $11/12$. Waiting in cash and buying the ETF only after observing its gain gives the bound on $CE_{E,E}$. Putting all cash into the active fund gives $CE_{F,N}$. No funded policy ends above $9/4$ on any path, which bounds $CE_{F,E}$.

The sign of the effect of future ETF adjustment on today's active advantage is therefore decided by comparing the premium of that future adjustment at a full-root optimum with its premium at an ETF-only-root optimum. This is the increasing-differences comparison of monotone comparative statics, written directly for these matched controls; nothing in (i) or (iii) is specific to funding. Increasing differences of c_R in the root active dollars and the future menu, ordered $N < E$, is the hypothesis of Topkis's comparative-statics theorem ([44], Theorem 5 and the definition preceding it, full text reviewed). It would make φ nondecreasing in the root active holding. If, in addition, φ did not depend on the root ETF holding and A_N held at least the incumbent active position, (57) would give a nonnegative effect. The family in (iv) shows that increasing differences fails in the funded two-review model: φ is zero at the all-active root and at least $1/50 - 1/320000$ at the all-cash root, so it is lower at the root with more active dollars.

What funding adds is the cap (58). The premium of future ETF adjustment is bounded by the wealth that can still be redeployed, cash and ETF holdings, and a root purchase of the active fund, which the ETF-only continuation cannot trade, consumes that wealth. A root that spends everything on the active fund makes future ETF adjustment a substitute for today's active trade, as in (iv). There the ETF-only root instead keeps cash, learns the parameter and then buys the ETF only if it is the winner; this is a value of waiting to learn. In Propositions 12 and 13, by contrast, the full root keeps an ETF position beside the active fund and sells it after learning, and future ETF adjustment acts as a complement to today's active trade. That contrast is a reading of the three families, not a theorem. No sign holds across the model.

Parts (i)–(iii) are exact but give no closed-form sign condition: the premia at the optimizers are defined through optimized continuation values, and this result bounds them only by the cap; Proposition 21 sharpens it. The cap is one-sided and crude. It ignores costs and uses the extreme ETF returns, so the corner condition in (ii) is sufficient for a nonpositive effect but does not characterize it. The family in (iv) is degenerate in the same way as Proposition 12's. It has no conditional return risk, full revelation, assumed returns of plus or minus 50%, $\rho = 20$ and known zero alpha: the active advantage is a known factor premium. It shows that the negative sign exists with a margin, not its magnitude or its robustness. It does not assert that the all-active root is the exact optimum when rates are positive. No movement of the no-active-trade region itself is exhibited, since the active trade is made under both continuation classes here and in Proposition 12; (iii) states what such a movement would require. Public information is the same under all controls, so the result does not separate learning from the change in the future menu, and it says nothing about alpha as distinct from factor premia. Whether the weaker single-crossing property of Milgrom and Shannon holds in this model is not examined.

Proposition 20 decides the sign by comparing premia at two optima, but it bounds each premium only from above, by a crude cap. The next result asks how sharp a cap on the premium can be. It shows that no cap built from wealth and return ranges alone can separate premia of a few basis points, and it gives sharper caps that use risk-adjusted mean returns.

We first split the premium across public observations. Fix a feasible root trade u_0 . A *node* is a first-review observation y with $P_0(y) > 0$, and its *node paths* are the pairs (θ, s_1) with $\pi_1(\theta | y) > 0$ and $q_{s_1} > 0$. In units of initial wealth W_0^- , let h be the cash left after the root trade and $m_j(y)$ the marked value of ETF j

at y . Let $c_y^R = -\rho^{-1} \log(-V_1^R(x_1^-(y; u_0), h_1^-(u_0), \pi_1(\cdot | y)))$ be the certainty equivalent at y with continuation class R , and call $G_y = c_y^E - c_y^N \geq 0$ the *node gain*: the value at y of being allowed an ETF-only trade rather than none. With weights $w_y \propto P_0(y) \exp(-\rho c_y^N)$ summing to one, write

$$\mathcal{A}_{u_0}(\ell) = -\frac{1}{\rho} \log \sum_y w_y \exp(-\rho \ell_y) \quad (60)$$

for any node-wise numbers ℓ_y . $\mathcal{A}_{u_0}$ is strictly increasing in each ℓ_y , and $\varphi(u_0) = \mathcal{A}_{u_0}(G)$. Node-wise bounds on the gains therefore give bounds on the premium. On the node paths, let g_j be ETF j 's second-quarter gross return, with largest and smallest values $\bar{g}_{j,y}$ and $\underline{g}_{j,y}$, and let W_y^N be terminal wealth without a second trade. The *tilted node measure* Q_y weights each node path by its probability times $\exp(-\rho W_y^N)$, normalized. The *risk-adjusted excess returns* of buying and selling ETF j are

$$r_{j,y}^+ = \mathbb{E}_{Q_y} \left[\frac{g_j - 1 - \kappa_j^+}{1 + \kappa_j^+} \right], \quad r_{j,y}^- = \mathbb{E}_{Q_y} [1 - \kappa_j^- - g_j].$$

Finally, let up_y and down_y be the largest $\max\{\bar{g}_{j,y} - 1, 0\}$ and $\max\{1 - \underline{g}_{j,y}, 0\}$ over ETFs, put $G_y^{\text{ub}} = h \text{up}_y + \sum_j m_j(y)(\text{up}_y + \text{down}_y)$, the node form of the cap (58), and $\beta_{\text{node}}(u_0) = \mathcal{A}_{u_0}(G^{\text{ub}})$.

Proposition 21 (Caps on the premium of future ETF adjustment: range-based caps cannot be sharp, and mean-based caps vanish exactly when no favorable trade has wealth; funded two-review model; machine checked, with a numerical comparison observed on assumed instances). Fix any instance of the funded two-review model.

(i) *Range-based caps.* Call a function C of the node data (cash h , marked ETF values m_j , purchase and sale rates, ρ , and each ETF's return range $[\underline{g}_j, \bar{g}_j]$ over the node paths, both ends attained) a *range-type cap* if $G_y \leq C$ at every node of every instance with those data. For data that some node has, and every ETF j ,

$$C \geq h \left[\frac{\bar{g}_j}{1 + \kappa_j^+} - 1 \right]^+, \quad C \geq m_j \left[1 - \kappa_j^- - \underline{g}_j \right]^+. \quad (61)$$

Each right side is the gain from putting all cash into ETF j at its best return, or selling all of it at its worst, and for every $\varepsilon > 0$ some node with the same data has a gain within ε of it. G_y^{ub} is a range-type cap. The bounds (61) do not bound G_y^{ub} from below by any factor: with one ETF, no cash and every node-path return at least one, the gain is zero while $G_y^{\text{ub}} = m(\bar{g} - 1)$.

(ii) *Tangent cap.* Let T_y be the largest first-order gain of a feasible second-review ETF-only trade u , in units of W_0^- , from the gain $\sum_j [(1 + \kappa_j^+) r_{j,y}^+ u_j^+ + r_{j,y}^- u_j^-]$, where u_j^+ and u_j^- are the purchase and sale of ETF j . Then

$$G_y \leq T_y \leq \left(h + \sum_j m_j(y) \right) \max_j ((1 + \kappa_j^+) r_{j,y}^+)^+ + \sum_j m_j(y) (r_{j,y}^-)^+ \leq G_y^{\text{ub}}. \quad (62)$$

With one ETF, $T_y = \max\{r_y^+ h, r_y^- m(y), 0\}$. $T_y = 0$ exactly when the node has *no adjustable wealth in any favorable direction*: for every ETF j , $h = 0$ or $r_{j,y}^+ \leq 0$ (no favorable purchase with cash); for every ETF k , $m_k(y) = 0$ or $r_{k,y}^- \leq 0$ (no favorable sale); and for every k with $m_k(y) > 0$ and every j , $r_{k,y}^- + (1 - \kappa_k^-) r_{j,y}^+ \leq 0$ (no favorable switch from k into j). Then $G_y = 0$. With one ETF the first two conditions suffice, since a round trip in one ETF never gains.

(iii) *Curvature cap, one ETF.* For buying, let $Z = (g - 1 - \kappa^+)/ (1 + \kappa^+)$ per dollar of cash, with $r = r_y^+$ and $\bar{\epsilon} = h$; for selling, let $Z = 1 - \kappa^- - g$ per ETF dollar, with $r = r_y^-$ and $\bar{\epsilon} = m(y)$. Let ΔW bound the range of W_y^N over the node paths, let $[g_l, g_u]$ contain the ETF's node-path returns, $s = (g_u - g_l)/2$, and let $\text{Var}_q(Z)$ be the variance of Z under the node-path probabilities. With $v_{\min} = \exp(-\rho(\Delta W + 2s\bar{\epsilon})) \text{Var}_q(Z)$, trading any amount in $[0, \bar{\epsilon}]$ in that direction gains at most

$$\bar{q}(r, v_{\min}) = \begin{cases} 0, & r \leq 0, \\ r^2 / (2\rho v_{\min}), & 0 < r \leq \rho v_{\min} \bar{\epsilon}, \\ r\bar{\epsilon} - \rho v_{\min} \bar{\epsilon}^2 / 2, & \text{otherwise,} \end{cases} \quad (63)$$

and G_y is at most the larger of the buying and selling values.

(iv) *Aggregation and sign conditions.* For any number of ETFs, let $\bar{T}_y$ be the explicit middle bound in (62); then $\varphi(u_0) \leq \mathcal{A}_{u_0}(\bar{T}) \leq \beta_{\text{node}}(u_0)$. With one ETF, let $\beta_{\text{mean}}(u_0)$ aggregate by (60) the smaller of T_y and the curvature cap at each node. Then $\varphi(u_0) \leq \beta_{\text{mean}}(u_0) \leq \beta_{\text{node}}(u_0)$. With one ETF and Proposition 20's maximizers, let ℓ_y be any node-wise lower bounds on the gains at B_N . If $\beta_{\text{mean}}(A_E) < \mathcal{A}_{B_N}(\ell)$, then $\Delta_E - \Delta_N < 0$. Symmetrically, with lower bounds on the gains at A_N , if $\beta_{\text{mean}}(B_E) < \mathcal{A}_{A_N}(\ell)$, then $\Delta_E - \Delta_N > 0$.

For (i), take one parameter, a riskless active fund and a node at which ETF j earns $\bar{g}_j$ except with a small probability η that realizes the rest of the range; buying ETF j with all cash then gains nearly the first bound in (61) as η falls to zero, with the node data unchanged, and a sale with the masses reversed gives the second. For (ii), the certainty equivalent at a node is concave in the second-review trade, so the gain is at most the first-order gain of the trade from zero. Every feasible trade is a nonnegative combination of cash purchases, sales and switches, so the first-order gain is nonpositive for all trades exactly when it is for these three kinds. For (iii), the second derivative of the certainty equivalent along one direction is $-\rho$ times the variance of Z under a tilted measure, and each tilted path weight is at least $\exp(-\rho(\Delta W + 2s\bar{\epsilon}))$ times its probability. Integrating twice bounds the gain by (63). Part (iv) follows from (60), because $\mathcal{A}$ is increasing in each node value, and from Proposition 20's sandwich.

A cap that sees only cash, holdings and return ranges cannot be smaller than the gain from staking all adjustable wealth on the best return in the range, because a return law concentrated near that return has the same data. Caps that use risk-adjusted means are sharper. They vanish exactly at nodes where no favorable purchase, sale or switch has wealth to trade. This refines the corner of Proposition 20: what matters is not idle wealth as such but wealth available in a favorable direction.

The following comparison is numerical and not part of the proposition. It uses six zero-cost, one-ETF instances of the two-review model with nondegenerate return risk, partial learning about a nonzero uncertain alpha, assumed quarterly factor premia of 1 to 4 percent, and $\rho \in \{5, 10, 20\}$, with optimizers computed by a numerical convex solver. There the premia at the root optima are 2.66 to 19.86 basis points per quarter of initial wealth. The aggregated bounds (61) at the full root's optimizers are 332 to 1023 basis points, so no range-type cap certifies any of these signs. Where the premium is positive, β_{mean} is 1.36 to 15.4 times the premium. The conditions in (iv) certify the positive sign in the two instances with $\rho = 20$. There the ETF-only root's optimizers sell the whole ETF position at the root, so at every node the only favorable direction, selling, has nothing to sell, and the premium is exactly zero. The other four signs are not certified: the premia at the two optima differ by 23 to 63 percent, while the caps at the compared optima are 1.41 to 6.5 times the premium. A cap within a factor of 1.05 to 1.96 of the premium would certify them. None is given, and no obstruction to one is shown.

The proposition applies standard tools: an attained supremum over laws with a given range, the tangent-plane bound for a concave certainty equivalent, and a lower bound on a tilted variance. The curvature cap is stated for one ETF, and its tilted-variance bound discounts by the full wealth range, which is crude. The numerical comparison uses solver optimizers of one assumed base design at zero costs, so its certifications are numerical observations, not theorems. The zero-premium corner at $\rho = 20$ is observed from those optimizers, not derived. Where the two premia are within the caps' factor, the sign has to be computed for the instance; no structural inequality here decides it. Nothing is shown about movement of the no-active-trade region.

We return to the funded one-quarter model for a different question: can the choice of factor exposure be separated from the choice of funds? A common procedure first chooses a target factor exposure and then chooses funds, charging the second stage for missing the target. The last result identifies what that procedure optimizes and bounds its loss against choosing everything jointly.

Write the factor shock covariance, and the cross and residual second moments of a holding $w = (a, p)$, as

$$\Sigma_f = \mathbb{E}_q z^f (z^f)^\top, \quad C(w) = \mathbb{E}_q [z^f (az^A + p^\top z^E)], \quad R(w) = \mathbb{E}_q [(az^A + p^\top z^E)^2], \quad (64)$$

so that $w^\top \Sigma w = b(w)^\top \Sigma_f b(w) + 2b(w)^\top C(w) + R(w)$. Let $J = \max_{w \in F} \bar{Q}_0(w)$ be the joint optimum, attained at some w_J with exposure b_J (Proposition 4).

Proposition 22 (A soft-target two-stage procedure is the joint problem with target-implied premia, and its loss is bounded by the first stage's multiplier; funded one-quarter model; machine checked). Fix any instance of the funded one-quarter model and evaluate the score at the belief mean, as in (6). In *stage 1*, choose a convex set $\mathcal{R}$ of exposures containing every funded exposure $b(F)$, and a target b^* that maximizes the factor objective $G(b) = b^\top \bar{\lambda} - (\gamma/2) b^\top \Sigma_f b$ over $\mathcal{R}$. A natural choice is $\mathcal{R}_0 = \{b(w) : w \geq 0, \sum_i w_i \leq 1\}$, which contains $b(F)$ because funded holdings are nonnegative and sum to at most one. Stage 1 sees only $\bar{\lambda}, \Sigma_f, \gamma$ and $\mathcal{R}$. Its *multiplier vector* is $v = \bar{\lambda} - \gamma \Sigma_f b^*$, the gradient of G at the target. In *stage 2*, maximize over $w \in F$

$$S(w) = a\bar{\alpha} - p^\top c^E - \tau(w - w^-) - \frac{\gamma}{2} \left[(b(w) - b^*)^\top \Sigma_f (b(w) - b^*) + 2b(w)^\top C(w) + R(w) \right], \quad (65)$$

which charges factor risk on the distance from the target rather than confining the holding to it. Let w_2 be a maximizer, with exposure b_2 , let $T_s = \bar{Q}_0(w_2)$, and write $\Lambda_s = J - T_s \geq 0$, $\Delta e = b_J - b_2$ and $\Delta w = w_J - w_2$.

(i) *Identification.* For every w ,

$$S(w) = \bar{Q}_0(w) - v^\top b(w) - \frac{\gamma}{2} b^{*\top} \Sigma_f b^*. \quad (66)$$

So stage 2 is the joint problem with the believed premia $\bar{\lambda}$ replaced by the *target-implied premia* $\gamma \Sigma_f b^*$, up to a constant. If $v = 0$, every stage-2 maximizer is a joint maximizer and $\Lambda_s = 0$.

(ii) *Multiplier bound.* For any set of holdings, convex or not, and any maximizers, $0 \leq \Lambda_s \leq v^\top \Delta e$, with equality in the second inequality exactly when w_J also maximizes S . Hence $\Lambda_s \leq \|v\| \|\Delta e\|$ in the Euclidean norm, and $\Lambda_s = 0$ when the two solutions share an exposure. Moreover $v^\top \Delta e \leq v^\top (b^* - b_2)$, so $\Lambda_s = 0$ also when stage 2 reaches the target.

(iii) *Exact form.* With F convex,

$$\Lambda_s = v^\top \Delta e - \frac{\gamma}{2} \Delta w^\top \Sigma \Delta w - s, \quad s \geq 0, \quad (67)$$

where s is the sum of the first-order optimality slack of w_2 in the direction Δw and a cost-kink term. The kink term is zero when no instrument's trade changes sign between w_2 and w_J . So the loss is at most the multiplier's value on the exposure difference, less the risk of the holding move.

(iv) *A target-confined comparison.* For any $w_T \in F$ with $b(w_T) = b^*$, $T_s \geq \bar{Q}_0(w_T) - v^\top(b^* - b_2)$. So the soft procedure gives up at most $v^\top(b^* - b_2)$ relative to any holding that meets the target exactly. The bound is vacuous when no funded holding meets it.

(v) *The multiplier.* $v = \bar{\lambda} - \gamma \Sigma_f b^*$ lies in the normal cone of $\mathcal{R}$ at b^* , so it is the multiplier of the stage-1 constraint $b \in \mathcal{R}$, and $v = 0$ when b^* is interior to $\mathcal{R}$.

Expanding the target distance in (65) gives (66). Since w_J is feasible in stage 2, $S(w_J) \leq S(w_2)$, which by (66) is the first bound in (ii). The target maximizes the concave G over the convex $\mathcal{R}$, so $v^\top(b - b^*) \leq 0$ for every $b \in \mathcal{R}$, in particular for b_J ; this gives the second bound in (ii) and part (v). For (67), move along the segment from w_2 to w_J : the score is a concave quadratic plus a convex, piecewise-linear cost, so the change in S is its one-sided directional derivative, which is nonpositive at the maximizer, minus the quadratic term and minus a nonnegative kink term. Part (iv) is (ii)'s comparison with w_T in place of w_J .

Charging the second stage for distance from the target is the same as re-running the joint problem with the returns the target implies. The loss is therefore at most the first stage's constraint multiplier applied to the exposure the joint choice takes beyond the target. The funded budget enters only through that multiplier: when the unconstrained factor optimum is admissible, $v = 0$ and the procedure recovers the joint optimum, as the classical separation results would suggest. The argument is weak duality for the relaxed stage-1 constraint and is elementary. The result says nothing about the size of the loss in particular portfolios, and the converse of (ii) fails: when $v = 0$, the loss is zero whatever the exposures. It treats only the procedure (65), in which stage 2 keeps the believed alpha and the ETF fee term; a variant that drops them is not the joint problem at any premia. It gives no ordering between this procedure and one that confines stage 2 to holdings meeting the target exactly (Proposition 23). Stage 1's maximum is attained on $\mathcal{R}_0$; for a general $\mathcal{R}$ it is assumed.

Proposition 22 lets the second stage drift from the target. The stricter procedure confines the second stage to holdings whose exposure equals the target. Its loss has an exact description in terms of how the best residual choice depends on the exposure.

Split the belief-mean score into a part that depends only on exposure and a residual part. With (64), for every w ,

$$\bar{Q}_0(w) = G(b(w)) + H(w), \quad H(w) = a\bar{\alpha} - p^\top c^E - \frac{\gamma}{2} [2b(w)^\top C(w) + R(w)] - \tau(w - w^-), \quad (68)$$

where G is Proposition 22's factor objective and H is the *residual objective*: alpha, ETF fees and drag, cross and residual risk, and trading costs. Let $B_F = b(F)$ be the set of funded exposures, and for $b \in B_F$ let

$$V(b) = \max\{H(w) : w \in F, b(w) = b\} \quad (69)$$

be the *residual value* of exposure b : the best residual outcome among funded holdings with that exposure.

Proposition 23 (Target-confined two-stage procedure: exact loss, exact separation criterion, and the funded failures of separation; funded one-quarter model; machine checked, with a numerical illustration on calibrated factor inputs). Fix any instance of the funded one-quarter model. B_F is nonempty, convex and compact, and V is attained on every exposure. $G + V$ is concave on B_F , and the joint optimum is $J = \max_{B_F}(G + V) = G(b_J) + V(b_J)$ for any joint maximizer's exposure b_J . V itself is

concave when the cross moments vanish on F ($C(w) = 0$), but not in general. In *stage 1*, choose a target b^* that maximizes G over B_F : the case $\mathcal{R} = B_F$ of Proposition 22, using only $\bar{\lambda}$, Σ_f , γ and the funded exposure set. When Σ_f is positive definite and $\gamma > 0$, the unconstrained target $\Sigma_f^{-1} \bar{\lambda} / \gamma$ is the unique such maximizer whenever it lies in B_F . In *stage 2*, choose $w \in F$ with $b(w) = b^*$ to maximize H , which attains $V(b^*)$. Write $T = G(b^*) + V(b^*)$ and $\Lambda = J - T \geq 0$.

(i) *Loss identity.*

$$\Lambda = [V(b_J) - V(b^*)] - [G(b^*) - G(b_J)], \quad (70)$$

and both brackets are nonnegative. The loss is the residual value's gain from moving the exposure from the factor optimum to the joint one, less what that move costs the factor objective.

(ii) *Exact separation.* $T = J$ if and only if b^* maximizes $G + V$ on B_F , that is, if and only if zero is a supergradient of $G + V$ at b^* relative to B_F . In particular $T = J$ when V is constant on B_F , and whenever $V(b) \leq V(b^*)$ for all $b \in B_F$. If V is concave, Σ_f is positive definite, $\gamma > 0$ and b^* lies in the relative interior of B_F , then $T = J$ exactly when $V(b) \leq V(b^*)$ on B_F : the residual value must be maximized at the factor optimum.

(iii) *Loss bound.* With $\|b\|_{\Sigma_f}^2 = b^\top \Sigma_f b$,

$$\Lambda \leq V(b_J) - V(b^*) - \frac{\gamma}{2} \|b_J - b^*\|_{\Sigma_f}^2,$$

and if V is L -Lipschitz on B_F in that norm, then $\Lambda \leq \min\{L^2/(2\gamma), L\|b_J - b^*\|_{\Sigma_f}\}$.

(iv) *When the residual value is flat, and why funding makes it not flat.* Suppose the ETFs have no residual shock, drag or trading cost, the cross moments vanish, and every exposure in B_F can be reached by a funded holding whose active position maximizes $a\bar{\alpha} - (\gamma/2)a^2 R_A - \tau_A(a - a^-)$ over the active fund's allowed range, with R_A its residual variance. Then V is constant, $T = J$, and the procedure recovers the joint optimum. With one active fund whose loading lies outside the ETF span, the exposure fixes the active position, so stage 1 chooses the manager.

For (i), w_J maximizes H among holdings with its own exposure, so $J = G(b_J) + V(b_J)$; subtracting T gives (70), and $G(b^*) \geq G(b_J)$ because b^* maximizes G on B_F . Part (ii) is (i) together with concavity of $G + V$, which holds because maximizing the concave $\bar{Q}_0$ over the holdings with a given exposure preserves concavity. The converse under concave V uses the sum rule for supergradients on the affine hull of B_F , where the gradient of G vanishes at a relatively interior optimum. Part (iii) substitutes the strong-concavity gap $G(b^*) - G(b_J) \geq (\gamma/2)\|b_J - b^*\|_{\Sigma_f}^2$ into (70) and maximizes $Lx - (\gamma/2)x^2$ over $x \geq 0$. In (iv) the residual objective depends on the active position alone, and every exposure admits the best one.

The criterion is that separation is exact when the best residual choice does not care which exposure it must deliver at the factor optimum. The flat case in (iv) is the world of the Treynor–Black benchmark discussed in Section 4. A funded, long-only menu breaks that flatness in ways a feasibility check on the target does not see, each making V depend on the exposure. First, the active fund carries factor loadings, so the exposure target bounds the active position, and fixes it when the fund's loading is outside the ETF span (manager-borne exposure). Second, when cash binds, buying ETFs to reach an exposure spends the budget the active position needs (shared budget). Third, directional ETF costs and drag put a kink in V at the incumbent's exposure. Of these, only the first is a formal conjunct, in the case stated in (iv); the shared-budget and cost channels are explanations, not theorems. Compared with Proposition 22's soft procedure, part (iv) there gives $\Lambda_s \leq \Lambda + v^\top(b^* - b_2)$ when $\mathcal{R} = B_F$. No ordering of the two losses holds in general.

The following comparison is numerical and not part of the proposition. It uses calibrated factor means and shock sizes, assumed residual risks and drag, an active cap of one, $\gamma = 5$, and three menus: an ETF that reproduces the active fund's loading, a missing factor direction, and an infeasible match. Costs are zero or 5 basis points, and the joint problem and both stages are solved numerically. The unconstrained target lies outside B_F in every case, because the long-only ETF menu cannot reach its factor-exposure ratio without shorting. The loss is at most 0.08 basis points per quarter at zero alpha. It is 0.4 to 2.9 basis points at alpha of +25 and +50 basis points, where the target caps the active holding. It is 2.4 to 22.4 basis points at alpha of -25 and -50 basis points in the missing-direction and infeasible menus, where the target forces holding a fund the joint optimum sells.

The result is an application of partial maximization, a comparison of attained maxima and a strong-concavity gap. The "only if" direction of (ii) at the residual value needs V concave; with factor-correlated residuals V can be convex at the target while $G + V$ is still maximized there. The Lipschitz constant L is not computed. The flat case in (iv) is one sufficient condition, not a characterization. The result uses belief means in one quarter and says nothing about estimated inputs or later reviews. The numerical comparison uses solver outputs on one calibration.

We return last to the repeated-sampling certification model. Proposition 19 showed that, in one family, the optimized ETF-only comparator is exactly two faces. The final result asks when the whole funded comparator reduces to a finite set of fixed ETF-only holdings in general.

In that model the conditional score is affine in the parameter: with the mean-exposure matrix A of (47), $Q(w; \theta) = \theta^\top A w + c(w) - (\gamma/2) w^\top \Sigma w$, where $c(w) = -p^\top c^E - \tau(w - w^-)$. For a sign pattern $\sigma \in \{+1, -1\}^{1+n}$, the *cost-sign cell* $F_\sigma = \{w \in F : \sigma_i(w_i - w_i^-) \geq 0 \text{ for all } i\}$ collects the funded holdings that buy or sell each instrument in the pattern's direction. The cells E_ρ of the ETF-only class are defined in the same way over the ETF coordinates, with the active holding fixed. The *face sets* $\text{Vert}(F)$ and $\text{Vert}(E)$ are the unions of the cells' extreme points. For $v \in E$, the *single-comparator contrast* is $m_v(w; \theta) = Q(w; \theta) - Q(v; \theta)$. Let $A_{N,\eta}$ be the ellipsoid of Section 2 before intersection with Θ , so that $C_N = \Theta \cap A_{N,\eta}$.

Proposition 24 (The optimized ETF-only comparator is a finite face set exactly when the score is linear or the ETF-only optimizer does not move; one-quarter model with known finite laws; machine checked, with the finiteness of the faces conditional on the linear-programming vertex theorem). Fix any admissible instance of the repeated-sampling certification model, with one or two ETFs, position caps, a compliant incumbent, directional rates and signed ETF drag.

(i) *Finite faces fixed by the funded geometry.* Each cell is a nonempty compact polyhedron containing the incumbent, cut out by $3(1+n)+1$ halfspaces for F and $3n+1$ for E . The face sets are finite, conditional on the linear-programming vertex theorem, and depend only on the caps, the incumbent, its cash and the rates, not on the parameter, the law, Σ , γ or the drag. At most $|\text{Vert}(E)| \leq 2^n \binom{3n+1}{n}$ and $|\text{Vert}(F)| \leq 2^{n+1} \binom{3n+4}{n+1}$. With one ETF,

$$\text{Vert}(E) = \left\{ (a^-, 0), (a^-, p^-), (a^-, p^- + \min\{\bar{p} - p^-, k^-/(1 + \kappa_E^+)\}) \right\} : \quad (71)$$

sell the ETF out, hold it, or buy it up to the cap or until cash runs out.

(ii) *Exact finite reduction when $\gamma = 0$.* For every parameter θ and every $w \in F$, the class optima are attained on the faces, and

$$\text{Adv}(w; \theta) = \min_{v \in \text{Vert}(E)} m_v(w; \theta), \quad \inf_{\theta \in C} \text{Adv}(w; \theta) = \min_{v \in \text{Vert}(E)} \inf_{\theta \in C} m_v(w; \theta) \quad (72)$$

for every set C of parameters, with each $m_v(w; \cdot)$ affine. In particular $L_N(w)$ is a minimum of finitely many convex programs when C_N is nonempty. Over the ellipsoid $A_{N,\eta}$, with $d_v = A(w - v)$ and $r_N = \sqrt{t_{N,\eta}/N}$,

$$\ell_N(w) = \min_{v \in \text{Vert}(E)} \left[m_v(w; \hat{\theta}_N) - r_N \sqrt{d_v^\top \Omega d_v} \right] \leq L_N(w) \quad (73)$$

when C_N is nonempty. This is a paired mean-uncertainty penalty, minimized over the finite faces. The plug-in full and ETF-only optimizers selected by the model's gate are themselves faces, so every contrast vector a history can require lies in a finite set D fixed before sampling.

(iii) *What a confidence construction needs when $\gamma = 0$.* If, on some event, each $d \in D$ has $d^\top \theta_*$ in an interval I_d , then the rule that certifies the plug-in full optimizer when $\min_v [\inf I_{A(\hat{w}_F - v)} + c(\hat{w}_F) - c(v)] > \delta_{\text{econ}}$ never falsely certifies on that event, whichever candidate the data select. It needs only $|D|$ scalar mean statements, not a vector-valued confidence set. The set of parameters with $G_* \leq \delta_{\text{econ}}$ is a finite union of polyhedra, an intersection over full-class faces of unions over ETF-only faces of halfspaces.

(iv) *Curvature deficit when $\gamma > 0$.* With positive definite Σ , let $v_E(\theta)$ be the unique ETF-only optimizer. For every finite $S \subset E$ and every θ ,

$$\max_{v \in E} Q(v; \theta) - \max_{v \in S} Q(v; \theta) \geq \frac{\gamma}{2} \min_{v \in S} \|v - v_E(\theta)\|_\Sigma^2, \quad (74)$$

and the finite contrast $\min_{v \in S} m_v(w; \theta)$ exceeds $\text{Adv}(w; \theta)$ by exactly the left side. In Proposition 17's family, every m -point comparator set has a parameter with deficit at least $s^2/(2(m+1)^2)$. There, a funded active action has advantage $-s^2/(4(m+1)^2) < 0$, yet every one of the m pairwise comparisons shows it at least $s^2/(4(m+1)^2)$ ahead. Keeping the deficit below $\delta/4$ with $\delta \leq s^2/128$ needs at least $s\sqrt{2/\delta} \geq 16$ comparators.

(v) *When a finite face set is exact under curvature.* With $\gamma > 0$, positive definite Σ and a nonempty convex parameter set T , a nonempty finite $S \subset E$ satisfies $\max_{v \in E} Q(v; \theta) = \max_{v \in S} Q(v; \theta)$ for all $\theta \in T$ if and only if some $v_0 \in S$ is the ETF-only optimizer at every $\theta \in T$. This happens in an assumed instance with $\gamma = 2$, one ETF loading on the first factor, incumbent $(3/10, 1/2, 1/5)$ in active, ETF and cash, position limits of one, equal purchase and sale rates of 20 basis points on the active fund and 5 on the ETF, shocks scaled by $1/10$, and $\Theta = [1/100, 3/100] \times [0, 1/50] \times [-1/20, 1/10]$. There the cash-exhausting purchase $(3/10, 2801/4002)$ is the ETF-only optimizer throughout Θ , because the score's derivative in the ETF holding there is at least $2600957/276000000 > 0$.

For (i), on a cell every trade has a fixed sign in each coordinate, so the trading cost and the cash constraint become linear, and the cell is a bounded polyhedron. A bounded polyhedron has finitely many extreme points, each fixed by linearly independent active constraints, and a linear objective attains its maximum at one of them ([5], Theorem 2.3, Corollary 2.1 and Theorems 2.7–2.8; [56], Section 8; both cited by theorem number, their texts not compared). The formal proof takes the finiteness part as a stated hypothesis. With $\gamma = 0$ the score is affine on each cell, which gives (ii); the infimum of a finite minimum is the minimum of the infima; and the ellipsoid bound is Cauchy–Schwarz in the Ω inner product. Part (iii) follows from (ii) on the stated event. Part (iv) is the quadratic gap of Proposition 18 at the unique optimizer, with a pigeonhole argument along Proposition 17's optimizer path. Part (v) combines that gap with continuity of the optimizer on a convex set.

With a linear score the funded whole-class comparator is exactly its finite faces, for every candidate, parameter set and law; the funded geometry only decides which faces exist. With a curved score the class

is a finite face set exactly when the ETF-only optimizer does not move on the domain. Once it moves, every finite comparator set overstates the advantage by a curvature term, and Proposition 18's whole-class certificate replaces face enumeration. That gain comes from strong concavity, a known perturbation mechanism, not from funding, costs or binding constraints. The result is deterministic: it constructs no rule for unknown laws, shows no coverage event and proves no rate. The vertex counts are crude upper bounds. Part (iv) gives a necessary condition on comparator sets, not a construction. Part (v) says nothing about how often a realistic domain has a constant ETF-only optimizer. The single-comparator instance above is one assumed design.

Proposition 21 bounded the premium of future ETF adjustment from above. The last result supplies the matching floor, using the node objects defined before Proposition 21: the node gains G_y , the weights w_y , the aggregate $\mathcal{A}_{u_0}$ of (60), the tilted node measure Q_y , the risk-adjusted excess returns $r_{j,y}^\pm$, and the node cap G_y^{ub} with its aggregate β_{node} .

Proposition 25 (The premium of future ETF adjustment decomposes across observations and is floored by adjustable wealth; funded two-review model; machine checked). Fix any instance of the funded two-review model and a feasible root trade u_0 .

(i) *Decomposition.* The premium is the aggregate of the node gains,

$$\varphi(u_0) = \mathcal{A}_{u_0}(G), \quad \min_y G_y \leq \varphi(u_0) \leq \max_y G_y, \quad (75)$$

and it is strictly increasing in each node gain. Node-wise bounds on the gains therefore aggregate to bounds on the premium by (60).

(ii) *Floors by adjustable wealth.* At a node y , with cash h and marked ETF values m_j in units of initial wealth:

- if ETF j 's second-quarter gross return is at least $1 + \delta$ on every node path, with $\delta > \kappa_j^+$, then $G_y \geq h[(1 + \delta)/(1 + \kappa_j^+) - 1]$; if it is at most $1 - \delta$ on every node path, with $\delta > \kappa_j^-$, then $G_y \geq m_j(\delta - \kappa_j^-)$;
- if $r = r_{j,y}^+ > 0$ and s is half the range of ETF j 's node-path returns, then

$$G_y \geq \begin{cases} r^2/(2\rho s^2), & r \leq \rho s^2 h, \\ rh - \rho s^2 h^2/2 \geq rh/2, & \text{otherwise,} \end{cases} \quad (76)$$

read as rh when $s = 0$; the same holds for a sale with $r = r_{j,y}^-$ and m_j in place of h .

Let G_y^{lb} be the largest applicable bound, zero if none applies, and $\alpha(u_0) = \mathcal{A}_{u_0}(G^{\text{lb}})$. Then $\alpha(u_0) \leq \varphi(u_0) \leq \beta_{\text{node}}(u_0) \leq \beta(u_0)$, with β the cap (58).

(iii) *Sign conditions.* With Proposition 20's maximizers,

$$\beta_{\text{node}}(A_E) < \alpha(B_N) \Rightarrow \Delta_E - \Delta_N < 0, \quad \alpha(A_N) > \beta_{\text{node}}(B_E) \Rightarrow \Delta_E - \Delta_N > 0. \quad (77)$$

Robust forms hold for any $u \in F_0$ and $v \in E_0$, paying each one's suboptimality under no future trade: $\Delta_E - \Delta_N \geq \alpha(u) - [CE_{F,N} - c_N(u)] - \beta_{\text{node}}(B_E)$ and $\Delta_E - \Delta_N \leq \beta_{\text{node}}(A_E) - \alpha(v) + [CE_{E,N} - c_N(v)]$.

(iv) *Two certifications at zero rates.* In Proposition 20's sure-active family, the full root under future ETF-only trading is uniquely the all-active trade, so $\beta_{\text{node}}(A_E) = 0$, and the ETF-only root's no-trade optimum is uniquely cash, so

$$\alpha(B_N) = \frac{\ln(3/2) - \ln(1 + e^{-10}/2)}{20} > \frac{1}{50}, \quad \Delta_E - \Delta_N < -\frac{1}{50}. \quad (78)$$

In Proposition 12's family, the robust lower form with the root $(7/15, 8/15)$ and the cash root gives $\Delta_E - \Delta_N > 3/100$.

For (i), group the two-review expected utility by the first public observation: the tower property gives the premium as a certainty equivalent of node certainty equivalents, which is (75). For the sure-sign floors, buying with all cash, or selling all of ETF j , gains at least the stated amount on every path. For (76), the node certainty equivalent along a purchase of size ϵ is at least $r\epsilon - \rho s^2 \epsilon^2 / 2$ by Hoeffding's lemma on the moment generating function of a variable with range at most $2s$, applied under the tilted measure ([25], inequalities (4.15)-(4.16) in the proof of Theorem 2, full text reviewed); maximizing over $\epsilon \in [0, h]$ gives the two branches. The caps are Proposition 20's argument node by node. Part (iii) is (77) substituted into (57). Part (iv) computes the two families' node gains in closed form.

Proposition 20's cap says funding limits what future ETF adjustment is worth. This result says the converse. At each observation the option to adjust is worth at least a risk-adjusted excess return times the wealth that can be redeployed, less a risk term, and at least half of it when that wealth binds. The sign of the future-ETF channel is then bounded by comparing these quantities at the two root optima. In the stylized families the bounds are tight: (78) matches the channel's numerical value, -0.020272 , to six decimals, and Proposition 12's lower form gives 0.032 against 0.0337 .

The caps are the weak side. On assumed instances with nondegenerate risk and partial learning, the floors α are within about a quarter of the premia (0.71 to 0.81 of them), but the caps β_{node} are 29 to 374 times the premia, and unbounded where the premium is zero. So these conditions certify no sign on those instances; Proposition 21 shows why no range-based cap can do much better. These ratios are numerical, from solver optimizers, and are not part of the proposition. The floors use one ETF and one trade direction at a time; joint trades across ETFs are not counted. The certifications are at zero rates, where the optimizers are identified exactly. Nothing is said about the future-active channel, or about movement of the no-active-trade region. The argument applies the tower property, Hoeffding's lemma and Proposition 20's cap and sandwich; it is not a new mechanism.

The one-quarter band of Proposition 9 describes one review. With many reviews the target moves between them, and a trade made now may be reversed later. The last result gives the exact structure of the band at quarterly reviews around a moving target, before any small-cost approximation.

Proposition 26 (Quarterly no-trade bands around a moving target: trade to the edge, a static width ceiling, and exactness for coarse innovations; multi-review tracking model with a slack budget; machine checked). Fix a slack-budget instance of the multi-review tracking model, a review $t < T$ and a state z .

(i) *One instrument: trade to the edge.* Let $n = 1$ and $c = \gamma \Sigma > 0$. There are edges $0 \leq lo_t(z) \leq hi_t(z) \leq \bar{x}$, defined by the one-sided derivatives of G_t reaching $-\kappa^+$ and κ^- , such that for every pre-trade holding $x \geq 0$, including one above the cap after marking, the optimal post-trade holding is unique and

$$x_t^+(x, z) = \min(\max(x, lo_t(z)), hi_t(z)). \quad (79)$$

The investor does not trade inside $[lo_t, hi_t]$ and otherwise trades to the nearer edge. V_t is convex with slopes in $[-\kappa^+, \kappa^-]$.

(ii) *A static width ceiling.* For every $t < T$,

$$hi_t(z) - lo_t(z) \leq \frac{\kappa^+ + \kappa^-}{c}. \quad (80)$$

At the last review the band is exactly $[x_{T-1}^* - \kappa^+/c, x_{T-1}^* + \kappa^-/c]$, clipped to $[0, \bar{x}]$; its width is the static width whenever neither bound binds. So a moving target can only narrow the band at quarterly reviews, never widen it. With the expected gross return $\bar{g}_t = \sum q_t(z', g' | z)g'$, the edges also satisfy $lo_t \geq \min(\bar{x}, x_t^* - (\kappa^+ + \beta\kappa^- \bar{g}_t)/c)$ and $hi_t \leq \max(0, x_t^* + (\kappa^- + \beta\kappa^+ \bar{g}_t)/c)$.

(iii) *Coarse innovations: the static band, tilted.* Let $t \leq T - 2$, with both edges interior. Suppose every positive-mass outcome (z', g') marks the whole current band strictly into the next review's buy region, $hi_t g' < lo_{t+1}(z')$, or strictly into its sell region, $lo_t g' > hi_{t+1}(z')$. Let U and D be the return-weighted masses of the two kinds of outcome. Then, with the tilt $\tau_t = \beta(\kappa^+ U - \kappa^- D)/c$,

$$lo_t = x_t^* - \frac{\kappa^+}{c} + \tau_t, \quad hi_t = x_t^* + \frac{\kappa^-}{c} + \tau_t, \quad hi_t - lo_t = \frac{\kappa^+ + \kappa^-}{c}. \quad (81)$$

Without marking ($g' = 1$), a sufficient condition is that every target innovation exceeds $(1+\beta)(\kappa^+ + \kappa^-)/c$ in size. With symmetric rates and $U = D$, the tilt vanishes and the band is the last review's band.

(iv) *When the ceiling is attained.* For $t \leq T - 2$ and $\kappa^+ + \kappa^- > 0$, the width equals the static width if and only if neither bound clips the band and every outcome marks the band entirely into $[0, lo_{t+1}(z')]$ or $[hi_{t+1}(z'), \infty)$. Otherwise the band is strictly narrower.

(v) *Many instruments.* For general n , the optimal post-trade holding is unique and the no-trade set is nonempty. Any two no-trade holdings satisfy

$$\gamma(x - y)^\top \Sigma(x - y) \leq \sum_i (\kappa_i^+ + \kappa_i^-) |x_i - y_i|, \quad (82)$$

so two no-trade holdings differing only in instrument i differ by at most its static width $(\kappa_i^+ + \kappa_i^-)/(\gamma \Sigma_{ii})$. At the last review, where no bound binds, the no-trade set is the parallelotope $x^* + (\gamma \Sigma)^{-1} \prod_i [-\kappa_i^+, \kappa_i^-]$. For an active fund A beside the other instruments E , its interior condition reads $d_A \in -(\Sigma_{AE} d_E)/\Sigma_{AA} + [-\kappa_A^+, \kappa_A^-]/(\gamma \Sigma_{AA})$, with $d = x - x^*$: the fund's band has its own static width but is centred at the target minus the exposure the other instruments' deviations already supply, at the rate Σ_{AE}/Σ_{AA} .

The proof is backward induction on t . The continuation V_{t+1} is convex with slopes in $[-\kappa^+, \kappa^-]$, so G_t is strictly convex, and minimizing it plus a proportional cost over an interval is a projection onto the set where its one-sided derivatives lie within the cost rates. That gives (79). The ceiling (80) holds because the continuation adds a convex term to the tracking loss, which can only steepen it. In (iii) every outcome puts the marked band on one side of the next band, where the continuation is affine with slope $-\kappa^+$ or κ^- , so it shifts the derivative by a constant, the tilt. Part (iv) is the converse: strict convexity of the continuation anywhere across the marked band strictly narrows it. For (v), comparing the objective along the segment between two no-trade holdings gives (82), and coordinate perturbations give the static shape.

At quarterly reviews, then, the band around a moving target lies between two exact anchors. One is the static width, attained exactly when the target's quarterly innovations are coarse relative to that width; there the target's motion only tilts the band. The other is strict narrowing, when innovations are fine. Instruments with small trading rates, such as ETFs, have narrow static bands and are more likely to be in the coarse regime; funds carrying loads or redemption fees have wide bands and are more likely to be in the fine regime. That assignment is an expectation, not a result.

The trade-to-the-edge band is the classical structure of discrete-time trading with proportional costs, due to Constantinides ([13], cited by title; the text has not been obtained). The proposition re-derives it only to carry the moving target, the caps and marking, which its other parts use. In continuous time, small-cost bands around a moving target have width of order the cube root of the cost and scale with the target's volatility ([57], equation (4.4), read at the level of its displayed formulas; [46], Section 4; [38], cited by title). Those are limits of continuous trading, in which a one-review width has no meaning, and they describe the fine regime this result does not cover. A binding cap reshapes a band at leading order in continuous time ([33], full text reviewed); here a cap only clips the band.

The result assumes a slack budget, so the cash coupling of Proposition 9 is absent, and it treats the target process as given. It uses nothing about how beliefs are formed, so it does not specialize to a learned, shrinking-variance target. The fine regime's leading order is not established, and a cap's effect on an interior edge is not quantified. For many instruments only the diameter inequality and the static shape are proved; the dynamic no-trade set need not be a parallelotope, and the dynamic version of the bundling shift is open. No calibration or magnitude is claimed.

The earlier dynamic results used proportional costs and a given target, and most had one active fund. With quadratic costs, many funds and learning from public returns, the optimal policy has an explicit form. The last result gives it, and shows what bundling factor exposure inside funds adds to it.

Proposition 27 (Partial adjustment toward a learning-driven aim, and exact separation of ETF exposure from fund alpha up to three bundling terms; fund-of-funds learning model, unconstrained counterfactual; machine checked, with the conditional-moment and dynamic-programming steps at paper level). Fix an instance of the fund-of-funds learning model with fixed unknown premia and alphas.

(i) *Exogenous learning.* For every policy, the posterior covariances P_t form the same deterministic sequence, and the posterior mean m_t is a martingale, $\mathbb{E}_t m_{t+1} = m_t$. Σ_t is deterministic and positive definite, and it does not increase. In the reference case the filter decouples: factor returns update the premia and fund residual returns $r^A - B^A f$ update the alphas, and

$$\Sigma_t = B(\Sigma_f + P_t^\lambda)B^\top + \text{diag}(\Sigma_A + P_t^\alpha, \Sigma_E). \quad (83)$$

So the problem is a linear-quadratic tracking problem with the martingale m_t as an exogenous state, and there is no motive to trade in order to learn.

(ii) *The policy.* Define backward from $A_T = 0$: $D_t = \Lambda + \gamma \Sigma_t + \rho A_{t+1}$ and $A_t = \Lambda - \Lambda D_t^{-1} \Lambda$, so $0 \leq A_t \leq \Lambda$. The optimal policy exists, is unique, and is partial adjustment toward an aim,

$$x_t = x_{t-1} + \Gamma_t(\text{aim}_t - x_{t-1}), \quad \Gamma_t = D_t^{-1}(\gamma \Sigma_t + \rho A_{t+1}), \quad (84)$$

$$\text{aim}_t = (\gamma \Sigma_t + \rho A_{t+1})^{-1} [\gamma \Sigma_t M k_t + \rho A_{t+1} \mathbb{E}_t \text{aim}_{t+1}], \quad \text{aim}_{T-1} = M k_{T-1}. \quad (85)$$

Γ_t has eigenvalues in $(0, 1)$, so every trade is a genuine partial adjustment. Unrolled, the aim is a matrix-weighted average of $(\gamma \Sigma_s)^{-1} \mu_t$ over the remaining reviews $s \geq t$, with weights summing to the identity. It is the current Markowitz portfolio re-weighted toward the smaller predictive covariances that learning will bring.

(iii) *Exposure-and-fund coordinates.* Suppose $M = K$ and B^E is invertible, so ETFs span the factors exactly, and write $R = (B^E)^{-1}$. Use the total factor exposure $y = B^\top x$ and the fund positions x^A as coordinates;

then $x^E = R^\top(y - B^{A\top}x^A)$. If ETFs cost nothing to trade, carry no residual risk and charge no fee ($\lambda_E = 0$, $\Sigma_E = 0$, $c^E = 0$), the problem separates exactly:

$$y_t = (\gamma(\Sigma_f + P_t^\lambda))^{-1} \hat{\lambda}_t, \quad x_t^A = x_{t-1}^A + \Gamma_t^A(\text{aim}_t^A - x_{t-1}^A), \quad x_t^E = R^\top(y_t - B^{A\top}x_t^A), \quad (86)$$

where the fund rule is (84)-(85) with cost $\lambda_A I_N$, risk $\gamma(\Sigma_A + P_t^\alpha)$ and the alpha means $\hat{\alpha}_t$. Exposure follows premium beliefs myopically, funds follow alpha beliefs with partial adjustment, and ETFs net the exposure the funds carry as a by-product, with either sign.

(iv) *The three bundling terms.* In the reference case, the only couplings between the two problems are: an ETF trading cost, which adds a cost cross term, because a fund trade must be netted by a costly ETF trade and exposure is then no longer myopic; ETF residual risk, which adds a risk cross term; and ETF fees $\phi = Rc^E$, which shift the exposure mean by $-\phi$ and the fund mean by $+B^A\phi$, since a fund's by-product exposure saves the fee ETF exposure would pay. With all three set to zero the policy is (86); in general it is (84)-(85) in the new coordinates.

(v) *Trading speeds.* In (86) with homogeneous funds ($\Sigma_A = \sigma_A^2 I$, prior alpha variance $s^2 I$, no pooled component), the fund rate is scalar, $\Gamma_t^A = g_t I$ with $g_t \in (0, 1)$, decreasing in λ_A and increasing in γ , and the ETF exposure adjusts in full. As the horizon and then t grow, g_t converges to a_∞/λ_A , where a_∞ is the unique root in $[0, \lambda_A]$ of

$$a = \lambda_A - \frac{\lambda_A^2}{\lambda_A + \gamma\sigma_A^2 + \rho a}. \quad (87)$$

For (i), the Kalman covariance recursion involves only the prior, the observation matrix and the noise covariance, none of which depends on positions. Transforming the observation to $(f, r^A - B^A f, r^E - B^E f + c^E)$ makes its matrix and noise block diagonal in the reference case, which decouples the filter. For (ii), backward induction verifies a value function quadratic in the position, with the stated recursion, and the maximizer of each Bellman objective is affine in the state. Part (iii) is the same problem in new coordinates: the transformed risk, cost and mean have zero cross blocks exactly when the three ETF frictions vanish. Part (v) reduces the fund block to a scalar recursion.

The machine check covers each numbered part. Two steps are kept at the level of a paper proof. One is the Gaussian conditional-moment facts that give the predictive moments and the martingale property, which discharge a hypothesis of the formal Bellman verification. The other is the dynamic-programming step from Bellman optimality to optimality among all admissible policies.

Partial adjustment toward an aim under quadratic costs is Gârleanu and Pedersen's rule ([1], Propositions 1-3, full text reviewed), and filter/control separation with a Riccati control law is standard linear-quadratic-Gaussian control ([2], Assumption 1 and Theorems 2.1-2.2, full text reviewed). Those results are stationary. The time-varying recursion that learning produces is proved directly, so neither is used as a lemma. In (v), a_∞/λ_A coincides with Gârleanu and Pedersen's trading rate after matching their discounting convention; that identification is checked. What the fund-of-funds structure adds is (iii)-(iv): a dynamic, costly version of Treynor and Black's netting ([60], equation (16), full text reviewed), in which ETFs carry exposure and funds carry alpha, with three explicit terms that couple them.

The result is for an unconstrained position space. Its policy may short funds and use leverage, and it is not implementable by a long-only fund of funds without the proportional-cost and cap corrections that Proposition 26 begins. Part (iii) needs exactly as many ETFs as factors, with full rank. The case where ETFs miss a factor direction, redundant ETFs, a pooled alpha prior with two speeds, and unequal persistence

of premia and alphas are not treated. The infinite horizon is reached only through the limit in (v). No magnitude is claimed.

Proposition 27's exact split needs the ETFs to span every factor. When some factor direction is reachable only through the funds, premium beliefs reach fund choice. The last result says exactly how.

Proposition 28 (Premium beliefs leak into fund choice through the factor directions ETFs cannot reach; fund-of-funds learning model, unconstrained counterfactual; machine checked, with the conditional-moment and dynamic-programming steps at paper level). Fix an instance of the fund-of-funds learning model in the reference case with fixed unknown means. Switch ETF frictions off ($\lambda_E = 0$, $\Sigma_E = 0$, $c^E = 0$), so only funds cost Λ_A to trade, and let B^E have full row rank $M \leq K$. Its row space L_E is the *reachable* subspace of factor directions, and its orthogonal complement $L_E^\perp$ the *unreachable* one, with orthogonal projections Π_R and Π_U . Let $\tilde{\Sigma}_t = \Sigma_f + P_t^\lambda$ be the predictive factor covariance, return risk plus premium-estimation risk, with blocks $\tilde{\Sigma}_{RR}, \tilde{\Sigma}_{RU}, \tilde{\Sigma}_{UU}$ on the two subspaces, and let $\tilde{\Sigma}_{U \cdot R} = \tilde{\Sigma}_{UU} - \tilde{\Sigma}_{UR} \tilde{\Sigma}_{RR}^{-1} \tilde{\Sigma}_{RU}$ be the Schur complement: the unreachable directions' variance net of their best reachable hedge. The *hedge map* $J_t = \Pi_U - \Pi_R \tilde{\Sigma}_{RR}^{-1} \tilde{\Sigma}_{RU}$ takes a factor vector to its unreachable part minus the reachable position that hedges it at minimum variance. Define the *reduced fund moments*

$$\alpha_t^{\text{red}} = \hat{\alpha}_t + B^A J_t^\top \hat{\lambda}_t, \quad \Sigma_t^{\text{red}} = \Sigma_A + P_t^\alpha + B^A \tilde{\Sigma}_{U \cdot R} B^{A\top}. \quad (88)$$

(i) *The ETF exposure is myopic plus a hedge.* The total exposure is the reachable exposure y_R plus the funds' unreachable exposure $\Pi_U B^{A\top} x_t^A$, which the funds alone carry. At every review the optimal reachable exposure is

$$y_{R,t} = \tilde{\Sigma}_{RR}^{-1} \left[\frac{1}{\gamma} \Pi_R \hat{\lambda}_t - \tilde{\Sigma}_{RU} B^{A\top} x_t^A \right], \quad (89)$$

the Markowitz exposure on the reachable directions minus the minimum-variance hedge, through ETFs, of the funds' unreachable exposure. The ETFs then net the funds' reachable by-product. When $\tilde{\Sigma}_{RU} = 0$ the hedge vanishes.

(ii) *The fund problem.* The fund positions follow Proposition 27's partial adjustment, with the time-varying mean map that gives the mean α_t^{red} , the risk $\gamma \Sigma_t^{\text{red}}$ and the cost Λ_A . Their one-quarter Markowitz portfolio is $(\gamma \Sigma_t^{\text{red}})^{-1} (\hat{\alpha}_t + B^A J_t^\top \hat{\lambda}_t)$.

(iii) *The leak.* Along an unreachable direction $v \in L_E^\perp$, the fund policy's sensitivity to premium beliefs at review t is

$$D_t^{-1} M_t B^A v, \quad M_t = I + \rho \Lambda_A D_{t+1}^{-1} M_{t+1}, \quad M_{T-1} = I, \quad (90)$$

with D_t the fund problem's matrix from Proposition 27. If every fund loading lies in L_E , the funds' factor exposure is replicable, $B^A J_t^\top = 0$, the risk term in (88) vanishes and there is no leak: Proposition 27's separation. If some fund loading has an unreachable component ($B^A v \neq 0$ for some $v \in L_E^\perp$), the sensitivity is nonzero whenever M_t is nonsingular. That holds at the last review, at every review when $T \leq 3$, in the stationary limit, and whenever a symmetric-part condition holds along the recursion. In that case the risk term $B^A \tilde{\Sigma}_{U \cdot R} B^{A\top}$ is also nonzero, and it changes the policy's own-position coefficient at every review.

(iv) *Learning and the leak.* Along the learning path, P_t^λ shrinks, so $\tilde{\Sigma}_{U \cdot R}$ decreases to its return-only value and the risk leak shrinks. The mean leak $B^A J_t^\top \hat{\lambda}_t$ persists. An unreachable premium is a permanent reason for fund positions, and its estimation error a transient one.

Maximizing the joint problem over the costless reachable exposure first gives (89). Substituting it back leaves a fund problem whose mean and risk gain the unreachable premium and variance net of the reachable hedge, the Schur complement in (88). The joint Riccati matrix's Schur complement over the ETF block is the reduced one, so the fund positions follow the reduced problem's policy. The leak (90) follows by differentiating the backward recursion along v , where $J_t^\top v = v$. The risk term raises the Riccati matrix in the positive semidefinite order, so it changes the own-position coefficient.

When ETFs cannot reach a factor, then, premium beliefs and their estimation error reach fund choice exactly through the unhedgeable part of the funds' loadings: as a mean term that persists and a risk term that learning removes. This is the dynamic, quantified form of the one-quarter geometry of Proposition 5, which separates a missing direction from a match blocked by funding. Treynor and Black's explicit market position nets a fund's by-product exactly because the market asset spans their single factor ([60], equation (16), full text reviewed). The unreachable direction is the case it does not cover. The Schur-complement step is elementary partial maximization of a quadratic.

The status is as for Proposition 27: the same two steps, the Gaussian conditional moments and the passage to all admissible policies, are reviewed paper proofs. That M_t is nonsingular at every review for every horizon was observed in 100,000 random cases and is not proved. So the leak's nonvanishing beyond the cases listed in (iii) is a numerical observation. ETF frictions together with a missing direction, redundant ETFs, long-only and funding constraints, and the leak's magnitude at calibrated scale are not treated. The policy is the unconstrained counterfactual of Proposition 27.

Proposition 27's aim leans toward the smaller risk charges that learning will bring. The last result makes that anticipation explicit in the separated case, and shows how a prior shared across funds changes the trading speeds.

Proposition 29 (Learning is anticipated only where trading costs, and a pooled alpha prior gives two trading speeds; fund-of-funds learning model, unconstrained counterfactual; machine checked, with the conditional-moment and dynamic-programming steps at paper level). Take the separated case of Proposition 27 (iii), with homogeneous funds, $\Sigma_A = \sigma_A^2 I_N$ and $\Lambda_A = \lambda_A I_N$ with $\lambda_A > 0$, and a *pooled* alpha prior with covariance $\bar{s}^2 \mathbf{1}\mathbf{1}^\top + s^2 I_N$, $s > 0$, $\bar{s} \geq 0$: the funds' alphas share a common component. Let $\Pi_c = \mathbf{1}\mathbf{1}^\top / N$ project onto the *common direction*, the average fund position, and $\Pi_r = I - \Pi_c$ onto the **relative directions**.

(i) *Two posterior variances, one learning rate.* For every t ,

$$P_t^\alpha = p_t^c \Pi_c + p_t^r \Pi_r, \quad \frac{1}{p_t^c} = \frac{1}{s^2 + N\bar{s}^2} + \frac{t}{\sigma_A^2}, \quad \frac{1}{p_t^r} = \frac{1}{s^2} + \frac{t}{\sigma_A^2}. \quad (91)$$

The common direction starts more uncertain, $p_t^c \geq p_t^r$ with equality exactly when $\bar{s} = 0$, and both precisions grow by $1/\sigma_A^2$ a quarter. The gap $p_t^c - p_t^r$ falls like $1/t^2$.

(ii) *Two speeds.* The fund trading rate is $\Gamma_t^A = g_t^c \Pi_c + g_t^r \Pi_r$. Each scalar rate comes from the scalar recursion of Proposition 27 with risk $\gamma(\sigma_A^2 + p_t^{\text{dir}})$, and $g_t^c \geq g_t^r$, with equality exactly when $\bar{s} = 0$. The average fund position moves toward its aim faster than relative positions, and both rates have the same stationary limit.

(iii) *The learning factor.* In each direction the fund aim is the shrunk Markowitz position scaled up,

$$\text{aim}_t^{A,\text{dir}} = \ell_t^{\text{dir}} \frac{\sigma_A^2 + p_t^{\text{dir}}}{\sigma_A^2} \frac{\hat{\alpha}_t^{\text{dir}}}{\gamma(\sigma_A^2 + p_t^{\text{dir}})}, \quad \frac{\sigma_A^2}{\sigma_A^2 + p_t^{\text{dir}}} \leq \ell_t^{\text{dir}} \leq 1, \quad (92)$$

where ℓ_t^{dir} is a weighted average of $\sigma_A^2/(\sigma_A^2 + p_s^{\text{dir}})$ over the remaining reviews, with positive weights from the Riccati recursion that sum to one. The factor $\ell_t^{\text{dir}}(\sigma_A^2 + p_t^{\text{dir}})/\sigma_A^2$, the *learning factor*, is at least one, and equals one only at the last review. The aim therefore lies above today's shrunk position, which reflects today's estimation variance, and never above the position with alpha known, $\hat{\alpha}_t^{\text{dir}}/(\gamma\sigma_A^2)$.

(iv) *Only where trading costs.* In the same case the ETF exposure is $y_t = (\gamma(\Sigma_f + P_t^\lambda))^{-1}\hat{\lambda}_t$, shrunk by today's premium uncertainty with no forward-looking factor, because it is costless to adjust and is re-set every review.

For (i), the pooled prior covariance and the posterior covariance share the eigenvectors Π_c and Π_r , and in each eigenspace the precision grows by one observation's precision each quarter. For (ii), the recursion of Proposition 27 splits along the same eigenspaces, and its scalar rate is nondecreasing in the whole future path of the risk charge, which is larger in the common direction. For (iii), unrolling the aim recursion gives the weighted average, and each term lies between today's shrinkage factor and one because the posterior variance only falls. Part (iv) is Proposition 27 (iii).

Where a position is costly to change, then, its target anticipates the precision it will have while held; where it is costless, only today's precision matters. The aim formula is Gârleanu and Pedersen's average of current and expected future Markowitz portfolios ([1], Proposition 3, full text reviewed), with a martingale signal and a time-varying risk charge: what changes over time here is estimation variance, not the signal's decay. Their result that trading is faster with more risk aversion (Proposition 2(ii)) is the monotonicity that (ii) extends to a whole path of risk charges. Estimation risk about a constant premium produces a hedging demand in Brennan's continuous-time setting ([8], cited for the mechanism; not compared at theorem level). With a per-quarter mean-variance objective, the forward-looking effect appears here only through trading costs. The pooled prior is Pástor and Stambaugh's learning across funds ([64], full text reviewed), made dynamic.

The status is Proposition 27's: the conditional-moment and dynamic-programming steps are reviewed paper proofs. The result needs homogeneous funds, and it does not treat ETF trading costs, under which the exposure would acquire its own learning factor. Two regularities observed numerically are not proved: that the learning factor is smaller in the common direction, and that it falls with fund costs. At the instances checked the learning factor is at most 3%. Magnitudes are not established.

Propositions 27-29 assume the return covariance is known. A manager estimates it. The last result measures what running the policy of Proposition 27 with an estimated covariance costs.

Proposition 30 (Value loss from an estimated covariance: an exact identity, a non-local Riccati bound, and a guarantee for an estimate made before the run; fund-of-funds learning model, unconstrained counterfactual; machine checked in its finite-dimensional core, with the summation to the loss identity, the second-moment step and the state-moment recursion at paper level). Fix an instance of the fund-of-funds learning model with fixed unknown means, and let the filter use the true inputs. The manager runs the policy of Proposition 27 computed from an estimated covariance path $\tilde{\Sigma}_t$, which gives coefficients $(\tilde{K}_t, \tilde{L}_t, \tilde{l}_t)$ and the plug-in policy $x_t = \tilde{K}_t x_{t-1} + \tilde{L}_t m_t + \tilde{l}_t$. Write (K_t, L_t, l_t) for the true coefficients, $\delta K_t = \tilde{K}_t - K_t$ and so on, and D_t for the true curvature matrix of Proposition 27. Use the cost-weighted norms $\|X\|_\Lambda = \|\Lambda^{1/2} X \Lambda^{-1/2}\|$ for a matrix and $\|Y\|_* = \|\Lambda^{-1/2} Y \Lambda^{-1/2}\|$ for a symmetric one, with $\|\cdot\|$ the spectral norm.

(i) *Loss identity.* For every admissible policy π with finite second moments,

$$V_0^* - V_0^\pi = \mathbb{E}_0 \sum_{t=0}^{T-1} \rho^t \frac{1}{2} e_t^\top D_t e_t, \quad e_t = x_t^\pi - (K_t x_{t-1}^\pi + L_t m_t + l_t). \quad (93)$$

The loss is the discounted, curvature-weighted sum of the squared deviations of the policy's trades from the optimal trade at the policy's own state.

(ii) *The plug-in loss.* For the plug-in policy, $e_t = \delta K_t x_{t-1} + \delta L_t m_t + \delta l_t$, so the loss is an explicit quadratic in the coefficient errors and the state's first two moments. Those moments follow an exact linear recursion, so the loss is computable without simulation. A norm bound follows: $V_0^* - V_0^\pi \leq \frac{1}{2} \sum_t \rho^t \|D_t\|_* (\|\delta K_t\|_\Lambda r_t^x + \|\Lambda^{1/2} \delta L_t\| r_t^m + \|\Lambda^{1/2} \delta l_t\|)^2$, with r_t^x and r_t^m the root mean squares of $\Lambda^{1/2} x_{t-1}$ and m_t .

(iii) *Non-local stability of the Riccati recursion.* If each $\tilde{\Sigma}_s$ is positive semidefinite, then for every $t < T$ and every size of error,

$$\|\delta M_t\| \leq \sum_{s=t}^{T-1} \rho^{s-t} \gamma \|\tilde{\Sigma}_s - \Sigma_s\|_*, \quad \|\delta K_t\|_\Lambda \leq \gamma \|\tilde{\Sigma}_t - \Sigma_t\|_* + \rho \|\delta M_{t+1}\|, \quad (94)$$

where $M_t = \Lambda^{-1/2} A_t \Lambda^{-1/2}$. The errors in L_t and l_t are bounded by similar discounted sums. An indefinite estimate can break every bound.

(iv) *A guarantee for an estimate made before the run.* Suppose the return covariance is estimated once, from a history separate from the run, and the estimate is positive semidefinite, with an event of probability at least $1 - \alpha$ over that history on which its error is at most r in the $\|\cdot\|_*$ norm. Then on that event the bounds of (ii)-(iii) hold with r in place of every covariance error:

$$\Pr(V_0^* - V_0^\pi \leq B(r)) \geq 1 - \alpha, \quad (95)$$

with $B(r)$ the explicit number that (ii)-(iii) give.

For (i), completing the square in each Bellman step of Proposition 27 writes the objective at any trade as the value minus $\frac{1}{2} e^\top D_t e$, and summing under the policy's conditional expectations gives (93). For (iii), the recursion's one-step map $R \mapsto I - (I + R)^{-1}$ is 1-Lipschitz on positive semidefinite matrices, so errors propagate backward at most additively with discounting. Part (iv) is (ii)-(iii) on the confidence event.

The machine check covers each one-step identity, the pathwise error bounds, all four Riccati bounds, and the bounds on the confidence event. Three steps are reviewed paper proofs, not machine checked: summing the one-step identity into (93) under conditional expectations, the passage from pathwise bounds to second moments, and the recursion for the state moments.

The shape of the result is known. For stationary linear-quadratic-Gaussian control, the value loss of the certainty-equivalent controller is of second order in the parameter error ([35], Theorems 3-4 and Proposition 2, cited for context as audited in the ledger). The stationary Riccati equation has non-local perturbation bounds with a condition number ([32], Theorems 3.2-3.3, cited for context as audited in the ledger). Both are stationary and are cited for context; the finite-horizon, time-varying statements here are proved directly, with the cost-weighted norm in the role of the condition number.

A numerical comparison, not part of the proposition, shows which part is usable at realistic scale. On a calibrated instance with 30 funds, 8 ETFs, five factors and 40 reviews, with the covariance estimated from 160 quarters, the exact loss (ii) is 5.9 to 16.7 bp per quarter on five estimation draws, or 1.0 to 2.8% of the optimal value. The norm bound is 330 to 1,150 times larger. The Riccati bound (iii) exceeds 57 where the true quantity is at most 1, because the cost-weighted norm divides by the small ETF trading cost and the sum runs over 40 reviews. So the identity is the usable object, and a guarantee at calibrated scale has to be computed from it over the confidence set, not read off the norm bounds. These figures belong to the

unconstrained, levered policy. With funded long-only caps, a learning-aware policy gains only about 0.03 bp per quarter over a myopic one, so the covariance's estimation error matters on that order.

The estimated input is the covariance only: estimated loadings and costs, and a misestimated filter, are not treated. The estimate is made once. A manager who re-estimates at every review is not a fixed plug-in policy, and a time-uniform guarantee for that case is not given. Whether an a-priori bound can be non-vacuous at realistic scale is open.

Proposition 26 took the target process as given. The last result puts the band on the learning path: the target is the learning model's Markowitz portfolio, and the risk charge falls as premia and alphas are learned.

Proposition 31 (Quarterly bands on the learning path: the static width rises, the target drifts outward, and pure learning ends the coarse regime; learning-driven tracking model, finite-law variant, slack budget; machine checked, with the innovation's unconditional moments at paper level). Fix a slack-budget instance of the learning-driven tracking model in its finite-law variant, and write $c_t = \gamma \Sigma_{t,ii}$ for the curvature of the instrument considered.

(i) *Transfer.* The instance is a multi-review tracking model with a time-varying covariance. Every part of Proposition 26 holds with c replaced by c_t at every review and state: the band and trade to the edge, the ceiling $(\kappa^+ + \kappa^-)/c_t$, the edge brackets, the coarse regime with tilt $\beta(\kappa^+ U - \kappa^- D)/c_t$, the equality characterization, and, for many instruments, the diameter inequality and the static shape with Σ_{T-1} . In the coarse-regime condition without marking, each target innovation must exceed $(\kappa^- + \beta\kappa^+)/c_t + (\kappa^+ + \beta\kappa^-)/c_{t+1}$ in size. With a diagonal Σ_t the instruments separate.

(ii) *The static width rises.* $P_t^{-1} = P_0^{-1} + tH^\top R^{-1}H$, so P_t falls to zero, Σ_t falls to Σ_r , c_t is nonincreasing, and the static width

$$w_t = \frac{\kappa^+ + \kappa^-}{c_t} \uparrow \frac{\kappa^+ + \kappa^-}{\gamma \Sigma_{r,ii}}. \quad (96)$$

As estimation risk is learned away, the same deviation costs less and the ceiling loosens quarter by quarter.

(iii) *Drift and tilt.* The learning drift $[(\gamma \Sigma_{t+1})^{-1} - (\gamma \Sigma_t)^{-1}] \mu_t$ is known at t . For one instrument it has the sign of μ_t when the instrument's risk charge learns, and is zero otherwise, so it pushes the target away from zero. Both it and the innovation covariance V_t are of order $1/t^2$. In the coarse regime without marking, with symmetric trading rates κ and a conditional innovation law symmetric about zero, the tilt is zero or has the drift's sign, with

$$|\tau_t| = \frac{\beta\kappa}{c_t} \Pr(|(\gamma \Sigma_{t+1})^{-1} G \epsilon_{t+1}| \leq |\delta_t| \mid z), \quad (97)$$

where δ_t is the drift. A martingale target with symmetric innovations has no tilt; the learning drift is what tilts the band.

(iv) *Pure learning ends the coarse regime.* Suppose the instrument's gross return is one on every path, so that only learning moves the deviation from the target (*pure-learning marking*), and $\kappa^+ + \kappa^- > 0$. If the band at $t \leq T - 2$ has the static width, every outcome moves the target by at least

$$\frac{\kappa^-(1 - \beta) + \beta(\kappa^+ + \kappa^-)U}{c_t} \text{ up} \quad \text{or} \quad \frac{\kappa^+(1 - \beta) + \beta(\kappa^+ + \kappa^-)D}{c_t} \text{ down}, \quad (98)$$

where U and D are the probabilities of landing wholly above or below the next band, with $U + D = 1$. Since learning shrinks the target's moves while the static width rises, there is a review t_0 , determined by the model's inputs and not by the horizon, after which the band is strictly narrower than the static width at every review and history, for every horizon.

For (i), every argument of Proposition 26 works at a fixed review and state, and uses the covariance only through its value there. Part (ii) is the information form of the filter with fixed means. Part (iii) differentiates the target along the filter and computes the coarse-regime tilt from the up and down masses. For (iv), a band of static width requires every outcome to land it wholly on one side of the next band, which forces a target move at least the size in (98). The information form bounds the target's moves uniformly over histories by a quantity that tends to zero.

Along the learning path the two anchors of Proposition 26 move apart. The static width rises, and the target's learning-driven moves shrink, so a pure-learning band passes from the coarse to the fine regime and is strictly narrower thereafter. The continuous-time small-cost band around a target moved by a mean-reverting factor with constant coefficients ([46], Section 4, full text reviewed) therefore has to be read with time-varying curvature and a drift term before it can describe a learned target.

The machine check covers each numbered part in the finite-law variant, including the exact $1/t^2$ limits and the existence of t_0 independent of the horizon. The unconditional mean and covariance of the belief-mean innovation over the finite law, used only to describe it, are a reviewed paper step and enter no formal conclusion. The results are for the finite-law variant. Under the Gaussian law the second-moment facts in (ii)-(iii) still hold, but the band statements are not proved. Part (iv) needs pure-learning marking: with real returns the deviation also moves by the instrument's return, which does not learn away, so an instrument can stay in the coarse regime, and which ones do is a calibration question. The symmetry in (iii) is a hypothesis. The fine regime's leading order, per-direction bands under the pooled prior, a binding budget and non-diagonal many-instrument learning statements are not treated. No calibration is claimed.

Proposition 30 wrote the plug-in loss as a quadratic in the coefficient errors. The last result writes it as an explicit function of the model's inputs: starting holdings, prior means, precision paths, costs, fees, spanning, risk aversion and horizon.

Proposition 32 (The plug-in value loss as a function of the inputs; fund-of-funds learning model, unconstrained counterfactual; machine checked in its deterministic and pathwise content, with the expectation identities at paper level). Use Proposition 30's setting, allowing Λ positive semidefinite provided each D_t is positive definite. Write $V_s = P_s - P_{s+1}$ for the *precision decrement* at review s , the covariance of the belief update $\eta_s = m_{s+1} - m_s$, and $\Phi_{t,s} = \tilde{K}_t \cdots \tilde{K}_{s+1}$ for the plug-in transition products.

(i) *General case.* The trade error is $e_t = \mu_t + \sum_{u < t} W_{t,u} \eta_u$, where μ_t is an explicit affine function of the starting holding x_0 and the prior mean m_0 , and $W_{t,u} = \delta K_t \sum_{s=u+1}^{t-1} \Phi_{t-1,s} \tilde{L}_s + \delta L_t$. The value loss is

$$V_0^* - V_0^\pi = \frac{1}{2} \sum_{t=0}^{T-1} \rho^t \left[\mu_t^\top D_t \mu_t + \sum_{u < t} \text{tr}(D_t W_{t,u} V_u W_{t,u}^\top) \right] : \quad (99)$$

a quadratic polynomial in (x_0, m_0) plus a term linear in the precision decrements, with coefficients that are explicit products of the policy coefficients. It is zero exactly when every μ_t , and every $W_{t,u}$ with $V_u \neq 0$, vanishes.

(ii) *Closed form in the separated homogeneous case.* In Proposition 29's setting without the pooled component, with the fund residual variance σ_A^2 and the factor covariance Σ_f estimated, the loss splits into an

exposure part and a fund part. The exposure part is the myopic Markowitz loss of using the wrong factor covariance, averaged over the premium belief's second moment,

$$\text{Loss}_y = \frac{1}{2\gamma} \sum_t \rho^t \text{tr}(\Delta_t(\hat{\lambda}_0 \hat{\lambda}_0^\top + P_0^\lambda - P_t^\lambda)), \quad (100)$$

with $\Delta_t = (\tilde{\Sigma}_t^{\lambda-1} - \Sigma_t^{\lambda-1})\Sigma_t^\lambda(\tilde{\Sigma}_t^{\lambda-1} - \Sigma_t^{\lambda-1})$ and $\Sigma_t^\lambda = \Sigma_f + P_t^\lambda$. The fund part, for each fund with starting holding x_0 and prior mean $\hat{\alpha}_0$, is $\text{Loss}_A = \frac{1}{2} \sum_t \rho^t d_t[\mu_t^2 + \sum_{u < t} w_{t,u}^2(p_u - p_{u+1})]$, with the scalar recursion of Proposition 29 and explicit μ_t and $w_{t,u}$. With a pooled prior the same holds in the common and relative directions.

(iii) *How the fund loss depends on the inputs.* As a function of the residual-variance error $\epsilon = \tilde{\sigma}_A^2 - \sigma_A^2$, $\text{Loss}_A = C\epsilon^2 + O(\epsilon^3)$, with an explicit constant C that is a quadratic form in $(x_0, \hat{\alpha}_0)$ plus a term linear in the precision decrements. As the fund cost $\lambda_A \rightarrow 0$, the loss tends to the myopic loss $\frac{1}{2} \sum_t \rho^t r_t(1/\tilde{r}_t - 1/r_t)^2(\hat{\alpha}_0^2 + p_0 - p_t)$, with $r_t = \gamma(\sigma_A^2 + p_t)$, and the starting holding drops out. As $\lambda_A \rightarrow \infty$, the loss is of order $1/\lambda_A$, exactly when $x_0 \neq 0$, and of order $1/\lambda_A^3$ when $x_0 = 0$: both policies stay near the starting holding. The loss need not be monotone in the cost. With one review,

$$\text{Loss}_A = \frac{1}{2}(\lambda_A + r_0) \left[(\lambda_A x_0 + \hat{\alpha}_0) \left(\frac{1}{\lambda_A + \tilde{r}_0} - \frac{1}{\lambda_A + r_0} \right) \right]^2, \quad (101)$$

which vanishes where $\lambda_A x_0 + \hat{\alpha}_0 = 0$, since the trade is then zero under either variance.

(iv) *Spanning decides whether the errors compound.* In the separated case the factor-covariance error enters only the exposure part and the residual-variance error only the fund part. With spanning, the fund coefficients do not depend on the factor covariance at all. Without spanning (Proposition 28's setting), the fund coefficients can depend on it, through the reduced risk and the hedge map: in every such setting, doubling the factor covariance changes the last review's fund coefficient.

Part (i) unrolls the plug-in path in Proposition 30's identity. Part (ii) specializes it to the separated case, where Proposition 27 (iii) splits the problem. Part (iii) differentiates the scalar recursion in the residual variance and takes limits in the cost. Part (iv) reads the dependence off Proposition 28's reduced fund problem.

The loss of trading on a misestimated risk matrix is thus explicit in where the investor starts and what it believes, plus a term for how much its beliefs will still move, with coefficients set by costs, risk aversion and horizon. The second-order shape in the parameter error is known for stationary linear-quadratic-Gaussian control ([35], Theorems 3-4, cited for context as audited in the ledger). What is added here is the constant in front of it, and the whole loss, as an explicit expression in the inputs.

The machine check covers the deterministic and pathwise content: the product form, the trade-error decomposition and when the loss vanishes, (100) as an identity, the scalar recursion, the expansion in ϵ , both cost limits with the exact orders, the one-review formula and its positivity at intermediate cost, and the spanning statements. The expectation identities over the Gaussian filtering model, which turn the pathwise forms into (99)-(100) and the expectation form of C , are reviewed paper proofs. The estimated inputs are the covariance blocks only. Monotonicity in the precision inputs or the horizon is not established, and heterogeneous funds and ETF frictions are covered only by (99), not in closed form. The re-estimating guarantee is a separate result.

Propositions 30 and 32 assume the covariance is estimated once, before the run. A manager who re-estimates at every review is not following a fixed plug-in policy. The last result bounds that manager's expected loss.

Proposition 33 (An expected-loss guarantee for a manager who re-estimates at every review; fund-of-funds learning model, unconstrained counterfactual; machine checked in its pathwise core, with the expectation, split and confidence-sequence steps at paper level). Use Proposition 30's setting. At each review t the manager holds a positive semidefinite covariance estimate $\tilde{\Sigma}_r^{(t)}$, formed from data available at t , recomputes the coefficients of Proposition 27 from it, and trades with the review- t coefficients. Write $g = \|\Lambda^{-1/2}G\|$ and $e = \|\Lambda^{-1/2}(0, c^E)\|$, and $M^* = \sup_{t \leq T} \|m_t\|$ for the largest belief mean along the path.

(i) *A priori bounds.* For every positive semidefinite estimate, whatever its error,

$$\|K_s^{(t)}\|_{\Lambda} \leq 1, \quad \|\Lambda^{1/2}L_s^{(t)}\| \leq g(T-s), \quad \|\Lambda^{1/2}l_s^{(t)}\| \leq e(T-s), \quad (102)$$

and the same holds for the true coefficients. The realized position is therefore bounded along every path, $\|\Lambda^{1/2}x_t\| \leq X_t = \|\Lambda^{1/2}x_0\| + \sum_{s \leq t} (T-s)(gM^* + e)$.

(ii) *On the good event.* Let E_T be any event on which the estimation error is at most r_t in the cost-weighted norm at every review. On E_T , each review's curvature-weighted squared trade error is at most an explicit function $L_t^{\text{on}}(r_t; x_0, M^*)$, built from Proposition 30's bounds at the constant error r_t .

(iii) *The guarantee.* Off E_T , (102) bounds the same quantity by an explicit $L_t^{\text{off}}(x_0, M^*)$ with no dependence on the error. If $\Pr(E_T^c) \leq \alpha$, then

$$V_0^* - V_0^\pi \leq \mathbb{E} \left[\sum_t \rho^t L_t^{\text{on}}(r_t; x_0, M^*) \right] + \sqrt{\alpha} \left(\mathbb{E} \left[\left(\sum_t \rho^t L_t^{\text{off}}(x_0, M^*) \right)^2 \right] \right)^{1/2}. \quad (103)$$

Both expectations are polynomials in $\|\Lambda^{1/2}x_0\|$, the radii and M^* , and M^* 's moments are bounded by Doob's inequality in terms of m_0 and $P_0 - P_T$. So the expected loss is an explicit function of the inputs and the radii: of order $\sum_t \rho^t r_t^2$ on the good event, plus $\sqrt{\alpha}$ times an α -free constant. Both terms vanish only as the data before the run grow, with α falling slowly enough that the radii still shrink.

A time-uniform confidence sequence for the covariance supplies such an event ([26], Theorem 1, full text reviewed), applied entrywise with a union bound. It is applied to differences of consecutive observations, which have mean zero whatever the unknown means, since the plain sample covariance does not qualify when the means are unknown. The sequence's constants are not evaluated: the result takes the radii as inputs.

The structural fact is (i). With a positive definite trading cost, a rule built from any positive semidefinite risk estimate never moves more than a cost-set fraction of the way, and never responds to beliefs by more than a horizon-set multiple. So the loss is bounded on every path, without any smallness condition on the estimation error and without truncating the policy. The known fixed-error bound for stationary linear-quadratic-Gaussian control needs such a smallness condition, with constants from stability margins ([35], Theorems 3-4, cited for context as audited in the ledger).

The machine check covers the pathwise core: the bounds (102) for any positive semidefinite estimate and for the true coefficients, the position bound along the re-estimating path, and the pathwise on-event and off-event bounds. The following are reviewed paper proofs, not machine checked:

- the loss identity in expectation for the re-estimating policy;
- the split of the expectation over E_T and its complement, with Cauchy-Schwarz and Doob's inequality for M^* ;
- the existence of E_T from the confidence sequence.

The on-event constants can be loose by large factors at realistic inputs, and whether the $\sqrt{\alpha}$ term can be improved to α is not pursued. Estimated loadings, costs or fees, and a misestimated filter, are not treated.

Propositions 27 and 29 show that the aim anticipates learning. The last result asks when that anticipation, or anticipation of mean reversion in alpha, changes today's trade by a material amount, and answers with explicit conditions on the inputs.

Proposition 34 (When anticipating learning or mean reversion changes today's trade; fund-of-funds learning model, separated homogeneous case; machine checked, with the persistence aim formula and two readings at paper level). Take Proposition 29's separated homogeneous case without the pooled component, with fund cost $\lambda_A > 0$. Let $\kappa_t = p_t/(\sigma_A^2 + p_t)$ be the *Kalman gain* on alpha, the weight of one quarter's residual return in the belief update. A *static-belief* manager freezes the risk path at today's value and re-solves each review. Its aim is $\text{aim}_t^S = \hat{\alpha}_t/(\gamma(\sigma_A^2 + p_t))$, and its rate g_t^S comes from the recursion with constant risk $r_t = \gamma(\sigma_A^2 + p_t)$. The *learning-aware* manager of Proposition 29 has aim aim_t and rate g_t . Today's trades are $u_t = g_t(\text{aim}_t - x_{t-1})$ and $u_t^S = g_t^S(\text{aim}_t^S - x_{t-1})$.

(i) *Learning moves the aim by at most the gain's odds.* $\text{aim}_t = L_t \text{aim}_t^S$, with Proposition 29's learning factor

$$1 \leq L_t < \frac{1}{1 - \kappa_t} = 1 + \frac{\kappa_t}{1 - \kappa_t}, \quad (104)$$

and $L_t = 1$ exactly at the last review. So anticipating learning raises today's target by a relative amount above θ only if the Kalman gain exceeds $\theta/(1 + \theta)$, whatever the cost, horizon and starting holding.

(ii) **Learning slows trading, and the trade changes to first order in the gain.** $g_t \leq g_t^S$: a manager who expects a falling risk charge trades more slowly. With $g(r; n)$ the constant-risk rate with n reviews to go,

$$|u_t - u_t^S| \leq [g(r_t; T - t) - g(\gamma\sigma_A^2; T - t)] |\text{aim}_t^S - x_{t-1}| + g_t \frac{\kappa_t}{1 - \kappa_t} |\text{aim}_t^S|, \quad (105)$$

where the first bracket is at most $(T - t)\lambda_A\gamma p_t/(\lambda_A + \gamma\sigma_A^2)^2$. Both terms are of first order in the gain, with explicit constants in the cost, the risk aversion, the horizon and the position. A checkable condition for a material change is that the right side exceed a stated fraction of $|u_t^S|$.

(iii) *Mean reversion.* If alphas revert to a long-run mean $\bar{\alpha}$ at persistence $\phi \in [0, 1]$, the aim splits into a long-run part and a deviation part, and the deviation part is scaled by a weight $M_t(\phi)$ that is nondecreasing in ϕ , with $M_t(1) = 1$ and

$$\tilde{w}_{t,t} + (1 - \tilde{w}_{t,t})\phi^{T-1-t} \leq M_t(\phi) \leq 1, \quad (106)$$

where $\tilde{w}_{t,t}$ is the recursion's weight on today, renormalized by the precision ratios. Mean reversion changes the deviation part by more than a fraction θ only if $(1 - \tilde{w}_{t,t})(1 - \phi^{T-1-t}) > \theta$. That requires a cost high enough to put weight on the future, and a persistence low enough over the remaining horizon.

(iv) *Only where trading costs.* As $\lambda_A \rightarrow 0$, $L_t \rightarrow 1$ and $M_t(\phi) \rightarrow 1$ for every ϕ : both anticipation effects are cost effects. The costless ETF exposure is re-set to its myopic position every review, so premium learning and premium mean reversion never change today's exposure trade beyond today's estimate.

Part (i) is Proposition 29's weighted average, whose terms lie between today's precision ratio and one. For (ii), the constant-risk rate is increasing in the risk and Lipschitz in it with an explicit constant. For (iii), the recursion's weights, renormalized, form a probability distribution over the remaining reviews, and M_t is its average of ϕ^{s-t} . Part (iv) takes the limit of the weights.

The benchmark splits the optimal allocation under a mean-reverting premium into a myopic term and a hedging term, which disappears when the investment opportunity set is constant ([62], equation (35) and p. 73, full text reviewed). Here the objective is a per-review certainty equivalent, so there is no hedging demand at all. What remains are the cost effects above, which vanish for costless instruments. Under proportional costs and caps, the same anticipation appears as the outward drift and tilt of the band in Proposition 31.

The machine check covers each numbered part on a general nonincreasing positive precision path, including the exact form of $L_t - 1$, the threshold condition, the rate bounds and Lipschitz constant, the trade identity and bound, the properties of M_t , and the zero-cost limits. Three steps are reviewed paper proofs:

- the aim formula under mean reversion, which extends Proposition 27's recursion to persistent alphas;
- the reading that the objective carries no hedging demand, which is a matter of its definition;
- the reading against Proposition 31's band.

Heterogeneous funds, ETF trading costs, and the pooled prior beyond the direction-by-direction reading are not treated. That the learning factor rises with cost and horizon was observed in every case tested but is not proved.

Proposition 28 showed, under quadratic costs, that premium beliefs reach fund choice only through the factor directions ETFs cannot reach. The last result states what that means for a single review with proportional costs: when a premium error changes a fund's hold, buy or sell decision, by how much, at what cost, and what a rule that picks funds on total estimated return loses.

Proposition 35 (How premium error enters fund choice: not through spanning ETFs, through the hedged unreachable part of a fund's loading, and in full under naive selection; learning-driven tracking model at one review, ETFs frictionless; machine checked, with the expectations over the premium error at paper level). Fix one review of the learning-driven tracking model in the reference case, with N funds, M ETFs with B^E of full row rank $M \leq K$, fund purchase and sale rates $\kappa_i^\pm$, caps $\bar{x}_i$ and incumbents x_i^- . ETFs are frictionless, and the budget and every ETF bound are slack at the optima compared. Let $v_i = \Sigma_{A,ii} + P_{ii}^\alpha$, let $e = \hat{\lambda} - \lambda$ be the premium error, and use Proposition 28's hedge map J and Schur complement $\tilde{\Sigma}_{U,R}$. For each fund define the reduced moments $\alpha_i^{\text{red}} = \hat{\alpha}_i + B_i^A J^\top \hat{\lambda}$ and $s_i^{\text{red}} = v_i + B_i^A \tilde{\Sigma}_{U,R} B_i^{A\top}$, and the *alpha-band holding*

$$a_i = \text{clip}\left(x_i^-, \frac{\alpha_i^{\text{red}} - \kappa_i^+}{\gamma s_i^{\text{red}}}, \frac{\alpha_i^{\text{red}} + \kappa_i^-}{\gamma s_i^{\text{red}}}\right), \text{ clipped to } [0, \bar{x}_i] : \quad (107)$$

hold inside the band, otherwise trade to its nearer edge.

(i) *Spanning ETFs keep premium error out.* If $M = K$ with B^E invertible, every fund is held at its alpha-band holding with $\alpha_i^{\text{red}} = \hat{\alpha}_i$ and $s_i^{\text{red}} = v_i$. Fund holdings do not depend on the premium belief, its uncertainty or the loadings. Premium error reaches only the ETF exposure, at the Markowitz cost $e^\top \tilde{\Sigma}_f^{-1} e / (2\gamma)$.

(ii) **One unreachable fund: a persistent shift and a transient narrowing.** If exactly one fund i has a loading with a component outside the ETFs' reach, the joint optimum holds it at (107) with its reduced moments and the others at their spanning holdings. The premium reaches fund i 's decision only through the scalar $\delta = B_i^A J^\top e$, the hedged unreachable part of its loading times the error, which shifts the band's centre. Premium uncertainty narrows the band by the factor v_i / s_i^{red} , and that narrowing learns away while the shift does not. The holding moves by at most $|\delta| / (\gamma s_i^{\text{red}})$, exactly that on a common trading piece, and

not at all on a common hold piece. A held fund is moved by no error with $|\delta|$ at most the band's slack. A bought fund is turned into a hold or a sale only by a negative δ at least as large as its margin over the purchase threshold. The loss in the fund's reduced objective satisfies

$$0 \leq \text{loss} \leq \frac{\gamma s_i^{\text{red}}}{2} \Delta^2 + (\kappa_i^+ + \kappa_i^- + \mu) |\Delta|, \quad |\Delta| \leq \frac{|\delta|}{\gamma s_i^{\text{red}}}, \quad (108)$$

with μ the cap's multiplier, zero at an interior holding. With $\ell_i^2 = B_i^A J^\top P^\lambda J B_i^{A\top}$, the leak variance, and $\mu = 0$, the expected loss is at most $\ell_i^2 / (2\gamma s_i^{\text{red}}) + (\kappa_i^+ + \kappa_i^-) \ell_i / (\gamma s_i^{\text{red}})$: second order in the premium error, and first order only through the trading cost of the misplaced trade.

(iii) *Naive selection lets the whole premium in.* With spanning, suppose each fund is chosen from its total estimated mean $\hat{\alpha}_i + B_i^A \hat{\lambda}$ and total variance $v_i + B_i^A \hat{\Sigma}_f B_i^{A\top}$ by the same band rule, with ETFs then re-optimized. Its loss against the joint optimum is exactly

$$\Lambda^{\text{naive}} = \sum_i [\psi_i(a_i) - \psi_i(a_i^{\text{naive}})] \geq 0, \quad \psi_i(a) = \hat{\alpha}_i a - \frac{\gamma}{2} v_i a^2 - \kappa_i^+ (a - x_i^-)^+ - \kappa_i^- (x_i^- - a)^+, \quad (109)$$

zero exactly when every naive holding is the alpha-band holding. The premium error enters the naive holdings in full. From a zero incumbent, the naive rule buys fund i when $\hat{\alpha}_i + B_i^A \hat{\lambda} > \kappa_i^+$, and the hedged rule when $\hat{\alpha}_i > \kappa_i^+$. So a fund with negative net alpha but a positive loaded premium is bought naively, at an explicit loss.

For (i), with spanning ETFs the exposure can be set freely, so the fund problem separates from it and involves only alpha. For (ii), the fund's problem is Proposition 28's reduced problem at one review, and its solution is the band (107), which is Lipschitz in the reduced alpha with constant $1/(\gamma s_i^{\text{red}})$. The loss bound (108) compares the reduced objective at two points on either side of a kink. Part (iii) sums the reduced-objective gaps, since the ETFs re-optimize around any fund holdings.

In the inputs, then, premium error enters fund choice through one scalar per fund, $B_i^A J^\top e$. It is zero for every fund a spanning menu of frictionless ETFs can hedge, and nonzero exactly for loadings the menu cannot reach. A rule that picks funds on total estimated return forgoes the hedge and admits the whole premium error, at the explicit loss (109); that is the gain from hedging a fund's factor part with ETFs. The static analogue of the unhedged part is the "nonbenchmark" component of a fund's return in Pástor and Stambaugh's Bayesian fund choice ([64], full text reviewed).

The machine check covers each numbered statement pointwise: (i), the joint optimum and each part of (ii) including the move and loss bounds, and the exact identity and zero condition of (iii). The expectations over the premium error, the expected squared move and expected loss in (ii) and the expected squared error in (iii), are reviewed paper proofs, each applying the formal pointwise bound. The result is for one review with one unreachable fund. Several unreachable funds couple through the Schur complement. ETF frictions, and ETF bounds that bind (an ETF at zero cannot be sold to hedge), are not treated. The premium error is a fixed perturbation, with covariance P^λ only for the expectations. No coverage statement or magnitude is claimed.

Propositions 15-19 measured how much history certifies an active trade against the optimized ETF-only class, in fixed families. The last result restates that question for a fund decision, as conditions in the inputs: when can a history of returns support buying (or selling) a fund?

Proposition 36 (When data can support an active change, in the inputs; learning-driven tracking model at one review, spanning frictionless ETFs; machine checked in its pointwise and de-

terministic content, with the probability steps at paper level). Fix one review with spanning, frictionless ETFs and slack budget and caps, and one fund i with residual variance $\sigma_{A,i}^2$, posterior variance p_i , $v_i = \sigma_{A,i}^2 + p_i$, incumbent x_i^- and trading rates $\kappa_i^\pm$. Define the *buy threshold* and the *gap*

$$b_i = \kappa_i^+ + \gamma v_i x_i^-, \quad \Delta_i = \alpha_i - b_i. \quad (110)$$

A rule maps an n -quarter history of the fund's residual returns $\alpha_i + z_{ij}^A$ to "buy" or "no change". It *certifies* at (δ, ε) if it buys with probability at most ε whenever $\Delta_i \leq 0$ and at least $1 - \varepsilon$ whenever $\Delta_i \geq \delta$, for every law in the stated class.

(i) *The decision is the gap's sign.* The review's optimum for fund i is unique, and lies above x_i^- exactly when $x_i^- < \bar{x}_i$ and $\Delta_i > 0$; a fund at its cap cannot be bought. Selling is symmetric, with sale threshold $s_i = -\kappa_i^- + \gamma v_i x_i^-$, and a fund at zero cannot be sold. Premium beliefs, their precision and the loadings do not enter. Supporting an active change therefore means certifying the sign of Δ_i , a hypothesis about the fund's own alpha, with costs and the incumbent inside the threshold.

(ii) *Known Gaussian law.* The rule "buy if $\hat{\alpha}_i - \sigma_{A,i} z_\varepsilon / \sqrt{n} > b_i$ ", with $\hat{\alpha}_i$ the sample mean and z_ε the normal quantile, certifies once $n \geq 4\sigma_{A,i}^2 z_\varepsilon^2 / \delta^2$. For $\varepsilon < 1/4$, no rule certifies unless

$$n \geq \frac{4\sigma_{A,i}^2}{\delta^2} \log \frac{1}{4\varepsilon}, \quad (111)$$

so the needed history is of order $(\sigma_{A,i}^2 / \delta^2) \log(1/\varepsilon)$. A normal prior with variance s^2 is worth $\sigma_{A,i}^2 / s^2$ quarters of data.

(iii) *Unknown bounded law.* With residuals bounded by R and otherwise unknown, the Hoeffding rule certifies once $n \geq (8R^2 / \delta^2) \log(1/\varepsilon)$, and a two-point law shows that no rule certifies unless $n \geq (R^2 / \delta^2) \log(1/\varepsilon)$, for $0 < \delta \leq R/2$ and $\varepsilon \leq 1/16$. Not knowing the law costs no order, only the range in place of the standard deviation, as in Proposition 19.

(iv) *Persistent residuals.* If the residual process is stationary and satisfies a generalized Bernstein inequality with an effective number of observations $n_{\text{eff}} \leq n$, with constants c_σ, c_B, C , variance bound σ^2 and range bound B , the certificate with margin $\delta/2$ works once $n_{\text{eff}} \geq (4c_\sigma \sigma^2 + 2c_B \delta B) \log(C/\varepsilon) / \delta^2$. For independent data this is of order (iii)'s length; persistence of the residuals only discounts the history.

(v) *Persistent alpha: a floor no history removes.* If alpha itself follows $\alpha_{t+1} = \phi \alpha_t + (1 - \phi) \bar{\alpha} + \eta_t$ with innovation variance $q > 0$ and $|\phi| < 1$, the prediction variance of the current alpha falls monotonically to

$$p_\infty = \frac{1}{2} \left[(q - \sigma_{A,i}^2 (1 - \phi^2)) + \sqrt{(q - \sigma_{A,i}^2 (1 - \phi^2))^2 + 4q\sigma_{A,i}^2} \right] > 0 \quad (112)$$

from any prior at least as diffuse, and never below it. So for $\varepsilon \leq 1/2$ no history certifies a buy at posterior confidence $1 - \varepsilon$ unless the estimated gap exceeds $z_\varepsilon \sqrt{p_\infty}$. That is a ceiling on precision, not a discount on n ; an effective sample size cannot express it. It vanishes as $q \rightarrow 0$, and rises to the alpha process's own stationary variance as the residuals become uninformative.

(vi) *Spanning keeps premium error out.* Under spanning, premium precision plays no role. Without it, with premia estimated from the same history, the needed history in (ii) is multiplied by

$$1 + \frac{B_i^A J^\top \Sigma_f J B_i^{A^\top}}{\sigma_{A,i}^2}, \quad (113)$$

the unhedgeable part of the loading's factor variance over the residual variance, with J Proposition 28's hedge map. Part (iii) is not covered, since Gaussian premium error is unbounded.

(vii) *A pooled prior makes a common tilt N times cheaper.* With the pooled prior of Proposition 29 and homogeneous residuals, the average alpha across N funds has posterior variance $\frac{1}{N}(1/(s^2 + N\bar{s}^2) + n/\sigma_A^2)^{-1}$. Certifying a common tilt, the average net alpha above a common threshold, needs a history of order $(\sigma_A^2/(N\delta^2))\log(1/\varepsilon)$, N times shorter than one fund's own alpha, while a relative direction keeps the single-fund rate.

The upper bounds are concentration: the Gaussian tail in (ii) and Hoeffding's inequality in (iii) ([25], Theorem 1, full text reviewed). The lower bounds are Le Cam's two-point method ([9], Lemma 1, full text reviewed), as in Propositions 15-16. Part (iv) is Hang and coauthors' generalized Bernstein inequality with an effective number of observations ([23], Assumption 2, inequality (7) and Section 3.3, full text reviewed). Part (v) is the steady state of the scalar Kalman filter. What the fund-of-funds structure adds is the threshold that the costs and the incumbent set, the role of spanning, the direction split under the pooled prior, and the precision floor from persistent alpha.

The machine check covers the one-review decision (i), the certifying rule and the arithmetic of the lower bound in (ii), the Hoeffding rule and the two-point lower bound in (iii), the certificate under the effective-sample-size inequality in (iv), the fixed point, monotone convergence and limits in (v), and the Gaussian computations in (vi)-(vii). The following are reviewed paper proofs, not machine checked:

- the probability step of (ii)'s lower bound, which takes Le Cam's inequality as a premise, since no formal structure for non-finite laws exists;
- the mixing instantiations in (iv), which are citations;
- the Gaussian and Kalman posterior forms;
- the reduction of the full review to the one-fund problem.

The threshold's general form with ETF frictions and several funds traded at once are not treated. Lower bounds for persistent residuals and a frequentist version of the floor in (v) are not given.

Propositions 26-36 were stated in two cost models, quadratic and proportional. The last result says what the two share, what they do not, and where the one-quarter theory of Propositions 9 and 22-23 sits in both.

Proposition 37 (One tracking problem under two cost models: a common target and loss, one trade map per cost, and an anticipation order set by the cost's smoothness; fund-of-funds learning model and learning-driven tracking model; machine checked in its new steps, with the component results through their formal statements, and the proportional-cost anticipation at paper level, cited at leading order). A *quarterly tracking instance* has the fund-of-funds learning model's instruments, factor model, pooled Gaussian prior and Kalman filter with fixed means. Its predictive moments μ_t and Σ_t are deterministic, with Σ_t positive definite and nonincreasing, and its *target* is $x_t^* = (\gamma\Sigma_t)^{-1}\mu_t$. Given a discount $\rho \in (0, 1]$, a horizon T , a convex trading cost C_t and a closed convex set X_t of post-trade holdings, the objective is

$$\mathbb{E} \sum_{t=0}^{T-1} \rho^t \left[\mu_t^\top x_t - \frac{\gamma}{2} x_t^\top \Sigma_t x_t - C_t(x_t - x_t^*) \right], \quad x_t \in X_t, \quad (114)$$

with x_t^- the pre-trade holding. The fund-of-funds learning model is the instance with $C_t(u) = \frac{1}{2}u^\top \Lambda u$ and no constraint; the learning-driven tracking model is the instance with directional proportional rates $\kappa_i^\pm$, caps, a cash constraint and marking by gross returns.

(i) *The bridge.* For every positive definite Σ ,

$$\mu^\top x - \frac{\gamma}{2} x^\top \Sigma x = -\frac{\gamma}{2} (x - x^*)^\top \Sigma (x - x^*) + \frac{1}{2\gamma} \mu^\top \Sigma^{-1} \mu, \quad (115)$$

so both models minimize expected discounted tracking loss against the same target and in the same metric. Only the cost and the constraints differ, and nothing below transfers a policy from one model to the other.

(ii) *The target is common, and its learning drift is second order in the gain.* The target's split into ETF exposure and fund alpha, the leak of premium beliefs through directions ETFs cannot reach, and its motion as belief innovation plus a known learning drift are properties of the target (Propositions 27, 28 and 31), so they hold in both models. In Proposition 29's scalar fund block with Kalman gain $\kappa_t = p_t/(\sigma_A^2 + p_t)$, the target's cumulative relative drift from review t to review $s \geq t$ satisfies

$$\frac{\sigma_A^2 + p_t}{\sigma_A^2 + p_s} - 1 = \frac{p_t - p_s}{\sigma_A^2 + p_s} \leq (s - t) \left(\frac{\kappa_t}{1 - \kappa_t} \right)^2, \quad (116)$$

while the belief innovation has variance $p_t \kappa_t$, which moves the target by a relative standard deviation of order $\sqrt{p_t \kappa_t}/|\hat{\alpha}_t|$.

(iii) *The trade is one map of the gap per cost, and anticipation has the order of the cost's smoothness.* Under quadratic costs the trade is partial adjustment toward an aim (Propositions 27, 29 and 34); under proportional costs it is the projection onto a band around the target (Propositions 26 and 31). With $w_{t,s}$ the positive weights, summing to one over the remaining reviews $s \geq t$, of Proposition 29's weighted average, the learning factor L_t satisfies

$$L_t - 1 = \sum_{s \geq t} w_{t,s} \left[\frac{\sigma_A^2 + p_t}{\sigma_A^2 + p_s} - 1 \right] \leq \left(\frac{\kappa_t}{1 - \kappa_t} \right)^2 D_t, \quad D_t = \sum_{s \geq t} w_{t,s} (s - t), \quad (117)$$

strictly positive before the last review, with the aim's look-ahead duration $0 \leq D_t \leq T - 1 - t$ tending to zero with the fund cost. So under quadratic costs anticipating learning changes the aim by a fraction θ only if $(\kappa_t/(1 - \kappa_t))^2 D_t \geq \theta$: order κ_t^2 times a bounded duration, which sharpens (104). Under proportional costs in the coarse regime the band's shift is Proposition 31's tilt, one step of the drift, which vanishes with it. In the fine regime no bound is proved; the leading-order displacement from continuous trading is about two thirds of the band's mixing time times the per-step drift, which along the fixed-means path is of order $\kappa_t^{4/3}$ if the half-width falls like $t^{-2/3}$. That order is a reading, not a result. No condition on the relative change of today's trade is claimed, since the static trade can be zero.

(iv) *The value loss of any policy is one identity.* On a finite-law problem, for every feasible policy π and initial law, $V^* - V^\pi = \sum_t \rho^t \mathbb{E} \Delta_t$, where $\Delta_t \geq 0$ is the policy's Bellman residual at its own state; with stage rewards bounded by c and H_t the remaining discounted horizon, $|J_t| \leq cH_t$ and $\Delta_t \leq 2cH_t$. Under quadratic costs the residual is Proposition 30's completed square; under proportional costs it is the excess tracking loss and cost of the chosen holding over the band projection. The a priori device for re-estimation is Proposition 33's contraction under quadratic costs, and the caps under proportional costs: every admissible stage reward is at most $|\mu_t|^\top \bar{x} + \frac{\gamma}{2} \|\Sigma_t\| \|\bar{x}\|^2 + \sum_i (\kappa_i^+ + \kappa_i^-) \max(\bar{x}_i, x_{t,i}^-)$ in size, with no

condition on the estimate. At the last review, with a slack budget and one instrument, the band built from estimates $\tilde{c}$ of $c = \gamma\Sigma$ and $\tilde{x}^*$ of x^* loses at most

$$(|\mu| + c\bar{x} + \max(\kappa^+, \kappa^-)) \max(|\Delta_{lo}|, |\Delta_{hi}|), \quad |\Delta_{lo}|, |\Delta_{hi}| \leq |\tilde{x}^* - x^*| + \kappa^\pm \left| \frac{1}{\tilde{c}} - \frac{1}{c} \right|, \quad (118)$$

with κ^+ for the lower edge and κ^- for the upper.

(v) *The horizon-one case is the one-quarter theory.* At $T = 1$, or at the last review, the proportional-cost instance is Proposition 26's last-review band, which is Proposition 9's multiplier criterion with n instruments and the predictive covariance, and Propositions 22-23 apply to it. The quadratic-cost instance is $x_0 = (\Lambda + \gamma\Sigma_0)^{-1}(\Lambda x^- + \mu_0)$, the unique maximizer, which trades unless $\mu_0 = \gamma\Sigma_0 x^-$. At the same inputs the contrast with Proposition 9 is the cost model, not the horizon.

What composes is the target and its structure, the loss metric and the residual identity, the shape of the re-estimation guarantee, the horizon-one reduction, and the anticipation orders in (iii), which neither model gives alone. What does not compose is the trade rule: the aim has no proportional-cost counterpart and the band has no quadratic-cost counterpart, so one trade rule for both would need to drop one cost model.

Part (i) is the expansion of the quadratic. For (116), the scalar filter with a constant state has $1/p_s = 1/p_t + (s - t)/\sigma_A^2$. Part (iii)'s bound applies (116) termwise to Proposition 29's exact form of $L_t - 1$. Part (iv) telescopes the value function along the policy, and the residual is nonnegative because the Bellman equation maximizes over a set containing the policy's choice. The last-review bound in (118) uses that clipping is 1-Lipschitz in its edges and that the one-review objective is Lipschitz on $[0, \bar{x}]$. Part (v) is the first-order condition of a concave quadratic.

The fine-regime displacement is Martin's leading-order law for bands around a moving target ([38], equations (9)-(10), read at the level of its displayed formulas), in the discrete form of Proposition 31's model. It agreed with exact dynamic programs to about one percent in the stationary case and to 10-30 percent on the learning path; that agreement illustrates the law and does not prove it. The residual identity is elementary dynamic programming. Matched asymptotic expansions, an outer solution away from a boundary joined to an inner one near it, would give one composite rule, with the quadratic-cost policy outside and the band as the inner correction ([27] and, for one asset with a small quadratic cost, [11], both read at the level of their abstracts). That rule is not built here.

A multi-review estimation bound under proportional costs, a proved bound on the fine-regime band centre, the tilt's order in the gain, and the return-driven motion of the target through marking are not treated.

Proposition 26 gave one instrument's band across reviews and, in (v), the static shape of the no-trade set when a fund sits beside other instruments. The last result follows a fund's band across reviews when the ETFs beside it are traded optimally.

Proposition 38 (A fund's band beside optimally traded ETFs: a residual-variance ceiling, a bracket through the ETFs' cost bands, and the one-instrument band when ETFs are frictionless; learning-driven tracking model with a slack budget, one fund and $M \geq 1$ ETFs; machine checked). Fix a slack-budget instance of the multi-review tracking model with covariance Σ_t depending on the review and the state, as Proposition 31 reads the learning-driven model, one fund A and M ETFs E , and the stay objective G_t of Proposition 26. Partition Σ_t into Σ_{AA} , Σ_{AE} and Σ_{EE} , and define the *hedge ratios*, the fund's *residual variance given the ETFs* and its *residual curvature*,

$$\rho_t = \Sigma_{AE} \Sigma_{EE}^{-1}, \quad \sigma_{A,E,t}^2 = \Sigma_{AA} - \Sigma_{AE} \Sigma_{EE}^{-1} \Sigma_{EA}, \quad c_t^{\text{res}} = \gamma \sigma_{A,E,t}^2. \quad (119)$$

With $\bar{g}_j$ the expected gross returns, the *leak bounds*

$$\begin{aligned} H_t^+ &= \sum_j [\rho_j^+ (\kappa_{E,j}^+ + \beta \kappa_{E,j}^- \bar{g}_j) + \rho_j^- (\kappa_{E,j}^- + \beta \kappa_{E,j}^+ \bar{g}_j)], \\ H_t^- &= \sum_j [\rho_j^+ (\kappa_{E,j}^- + \beta \kappa_{E,j}^+ \bar{g}_j) + \rho_j^- (\kappa_{E,j}^+ + \beta \kappa_{E,j}^- \bar{g}_j)] \end{aligned} \quad (120)$$

are the most the ETFs' cost bands and continuation slopes, weighted by the hedge ratios, can raise or lower the fund's marginal. The fund's *ETF-optimized value* $U_t(a; p^-)$ is the least of $G_t((a, p), z) + C_E(p - p^-)$ over ETF holdings p in the ETF box.

(i) *One-instrument structure.* U_t is continuous and $U_t - \frac{1}{2} c_t^{\text{res}} a^2$ is convex. The full optimum is the fund problem $\min_a [U_t(a; p^-) + C_A(a - a^-)]$ followed by the ETF problem at that a . So the fund's post-trade holding is the clip (79) of a^- to an *effective band* $[lo_t, hi_t] \subseteq [0, \bar{x}_A]$, defined by U_t 's one-sided derivatives reaching $-\kappa_A^+$ and κ_A^- , and the fund is held exactly when a^- lies in it. The ETF incumbents enter only through U_t .

(ii) *A residual-variance ceiling.* At every review and state, and for every ETF incumbent,

$$hi_t - lo_t \leq \frac{\kappa_A^+ + \kappa_A^-}{\gamma \sigma_{A,E,t}^2}. \quad (121)$$

This is the fund's static width with its residual variance in place of its total variance, wider than a fund alone by $\Sigma_{AA}/\sigma_{A,E,t}^2$, which is $1/(1 - \text{corr}^2)$ for one ETF: the ETFs absorb the fund's factor risk.

(iii) *A bracket through the ETFs' bands.* With a_t^* the fund's target coordinate and the ETF optimizer interior to the ETF box,

$$hi_t \leq a_t^* + \frac{\kappa_A^- + \beta \kappa_A^+ \bar{g}_A + H_t^+}{c_t^{\text{res}}} \quad (hi_t > 0), \quad lo_t \geq a_t^* - \frac{\kappa_A^+ + \beta \kappa_A^- \bar{g}_A + H_t^-}{c_t^{\text{res}}} \quad (lo_t < \bar{x}_A). \quad (122)$$

The ETFs' costs enter the fund's band at the hedge ratios, and each instrument's continuation at β times its rates times its expected gross return; nothing else about the future enters.

(iv) *The last review: three regimes.* At $T - 1$, with the edges and the ETF optimizer interior,

$$hi_{T-1} = a^* + \frac{\kappa_A^- + \rho^\top c_E(hi)}{c^{\text{res}}}, \quad lo_{T-1} = a^* - \frac{\kappa_A^+ - \rho^\top c_E(lo)}{c^{\text{res}}}, \quad (123)$$

with c_E the ETFs' cost slopes at the ETF-optimized point. If the ETFs re-hedge the same way at both edges, the width is the ceiling (121). If every ETF re-hedges in opposite directions at the two edges, it is the ceiling minus $\sum_j |\rho_j| (\kappa_{E,j}^+ + \kappa_{E,j}^-) / c^{\text{res}}$. If the ETFs are untraded at both edges, it is exactly the fund-alone width $(\kappa_A^+ + \kappa_A^-) / (\gamma \Sigma_{AA})$, for any M and without the interiority of the ETF optimizer. Every band interior to the fund's box is at least that wide, so the last-review width lies between the fund-alone width and the residual ceiling.

(v) *An outer parallelotope.* For any number of instruments, every no-trade holding strictly inside the box lies in

$$x_t^* + (\gamma \Sigma_t)^{-1} \prod_i [-(\kappa_i^+ + \beta \kappa_i^- \bar{g}_i), \kappa_i^- + \beta \kappa_i^+ \bar{g}_i], \quad (124)$$

the static parallelotope of Proposition 26 (v) with each rate widened by β times the opposite rate times the expected gross return. With (82), the dynamic no-trade region never exceeds the static one's Σ_t -diameter, whatever the targets' motion.

(vi) *Frictionless ETFs*. If the ETFs are frictionless, residual-free, fee-free and spanning, and the hedge $x_E^* - \rho_t(a - a_t^*)$ lies in the ETF box for every $a \in [0, \bar{x}_A]$ at every review and state, then, for every ETF incumbent, U_t is up to a constant the stay objective of a one-instrument instance with curvature c_t^{res} , target a_t^* and the fund's own rates, cap and gross returns. The fund's band is then that instance's band, so Proposition 26 applies to it, and so, on paper, do Proposition 31's learning-path statements; the ETFs carry no band of their own.

The proof minimizes over the ETFs first. The Schur identity $d^\top \Sigma d - \sigma_{AE}^2 d_A^2 = (d_E + \Sigma_{EE}^{-1} \Sigma_{EA} d_A)^\top \Sigma_{EE} (d_E + \Sigma_{EE}^{-1} \Sigma_{EA} d_A)$ makes $G_t - \frac{1}{2} c_t^{\text{res}} a^2$ jointly convex, and a partial minimum of a jointly convex function over a convex set is convex. That gives (i), and Proposition 26's argument applied to U_t gives (121). For (122) the fund moves along the hedge direction $(1, -\rho_t)$, and the continuation's slopes are bounded by β times the rates times the expected gross returns. At the last review there is no continuation, so (123) is exact; with idle ETFs their slopes move with the fund's holding, and $c^{\text{res}} + \gamma \Sigma_{AE} \Sigma_{EE}^{-1} \Sigma_{EA} = \gamma \Sigma_{AA}$ gives the fund-alone width. Part (v) compares one-sided differences coordinate by coordinate, and (vi) is backward induction, since frictionless ETFs can be moved to any exposure at no cost.

In continuous time with small costs, the multi-asset no-trade region solves a corrector equation that has no explicit solution beyond one dimension ([52], read at the level of its abstract and introduction). The result above is exact at quarterly reviews and places the fund's band between explicit bounds in the inputs instead. A binding cap reshapes a band at leading order in continuous time ([33], full text reviewed); here the budget is slack and the caps only clip.

The shape of the dynamic region inside (124), its dependence on the correlation of the targets' innovations, and the fine-regime leading order for two instruments are not treated. Parts (iii)-(iv) assume the ETF optimizer interior: an ETF at zero can carry an edge beyond (122), but not beyond (121). A binding budget and several funds, whose bands then couple through the Schur complement of the fund block, are not treated either.

Propositions 4-8 compared ETF-only actions with active ones in the one-quarter model, one fund at a time. The last result prices the ETF-only restriction at one review with N funds, as a function of the inputs, and separates it from the frozen-funds baseline.

Proposition 39 (The cost of the ETF-only restriction in the inputs: the value of the forgone fund purchases, with the frozen-funds baseline adding the forgone sales; learning-driven tracking model at one review; machine checked, with the multiplier test in (i) conditional on the polyhedral Karush-Kuhn-Tucker theorem). Fix one review as in Proposition 35, with N funds, $v_i = \Sigma_{A,ii} + P_{ii}^\alpha$, rates $\kappa_i^\pm$, caps $\bar{x}_i$ and incumbents $x_i^- \in [0, \bar{x}_i]$. Compare three action classes: full trading F with value J ; ETF-only with funds that may be sold but not bought, E^- , with value J^- ; and ETF-only with funds frozen, E^0 , with value J^0 . The *cost of the ETF-only restriction* is $C^- = J - J^-$, and the *frozen-funds cost* is $C^0 = J - J^0$. For each fund let $\psi_i(a) = \hat{\alpha}_i a - \frac{\gamma}{2} v_i a^2 - \kappa_i^+ (a - x_i^-)^+ - \kappa_i^- (x_i^- - a)^+$, and define the *purchase excess* and *sale excess*

$$p_i = \hat{\alpha}_i - \kappa_i^+ - \gamma v_i x_i^-, \quad s_i = \gamma v_i x_i^- - \hat{\alpha}_i - \kappa_i^- : \quad (125)$$

net alpha over the purchase rate plus the risk charge on the incumbent, and the reverse over the sale rate. At most one of them is positive.

(i) *Structure, for any inputs.* $J^0 \leq J^- \leq J$ and $C^0 = C^- + (J^- - J^0)$. $C^- = 0$ exactly when no fund is bought at the joint optimum, and $C^0 = 0$ exactly when no fund is traded there. Conditional on the polyhedral Karush-Kuhn-Tucker theorem ([53], Theorems 27.4 and 28.2-28.3, cited by theorem number as a textbook result), the latter holds exactly when a multiplier test on the funds' marginals passes at the ETF-only optimum. $J^- - J^0$, the value of the permitted fund sales, is zero exactly when no fund is sold at the E^- optimum. The restriction costs exactly the value of the purchases it forbids.

(ii) *Spanning frictionless ETFs.* With $M = K$, B^E invertible, frictionless residual-free fee-free ETFs, and the budget and every ETF bound slack at the three optima,

$$C^- = \sum_{i: p_i > 0} \frac{p_i^2}{2\gamma v_i}, \quad J^- - J^0 = \sum_{i: s_i > 0} \frac{s_i^2}{2\gamma v_i}, \quad (126)$$

where a fund whose purchase edge $(\hat{\alpha}_i - \kappa_i^+)/(\gamma v_i)$ lies above its cap contributes $\psi_i(\bar{x}_i) - \psi_i(x_i^-)$ instead, and a fund whose sale edge lies below zero contributes $\psi_i(0) - \psi_i(x_i^-)$. So the restriction costs, for each fund it would have bought, the squared purchase excess over twice the fund's curvature γv_i . Premium beliefs, their precision, the loadings and the ETF sleeve do not enter, and alpha precision enters only through v_i . From zero incumbents, with no cap binding, $C^- = \sum_i [(\hat{\alpha}_i - \kappa_i^+)^+]/(2\gamma v_i)$ and $J^- = J^0$: the frozen baseline then agrees with the restriction proper.

(iii) *One unreachable fund.* If, as in Proposition 35 (ii), exactly one fund has a loading outside the ETFs' reach and everything else is frictionless, (126) holds with that fund's reduced moments α_i^{red} and s_i^{red} in place of $\hat{\alpha}_i$ and v_i . A missing exposure raises the cost by the unreachable premium net of its hedge, credited to the fund's alpha, and lowers it through the unhedgeable variance.

(iv) *ETF frictions: a bracket through the re-hedge costs.* With spanning ETFs that carry rates and fees but no residual risk, a slack budget, and room at the three optima for the netting trades and their cash, let $r_i = R^\top B_i^{A^\top}$ be fund i 's netting vector, h_i^+ and h_i^- the ETF cost of netting a unit purchase and a unit sale, and $A_i^0 = \hat{\alpha}_i + r_i^\top c^E - \gamma v_i x_i^-$ the fund's reduced marginal at the incumbent. With

$$g_i(m, d) = \max_{0 \leq \delta \leq d} \left[m\delta - \frac{\gamma v_i}{2} \delta^2 \right], \quad (127)$$

$$\begin{aligned} \sum_i g_i(A_i^0 - \kappa_i^+ - h_i^+, \bar{x}_i - x_i^-) &\leq C^- \leq \sum_i g_i(A_i^0 - \kappa_i^+ + h_i^-, \bar{x}_i - x_i^-), \\ \sum_i g_i(-A_i^0 - \kappa_i^- - h_i^-, x_i^-) &\leq J^- - J^0 \leq \sum_i g_i(-A_i^0 - \kappa_i^- + h_i^+, x_i^-). \end{aligned} \quad (128)$$

ETF frictions enter the restriction's cost only through the fee credit $r_i^\top c^E$ and an effective purchase rate between $\kappa_i^+ - h_i^-$ and $\kappa_i^+ + h_i^+$.

Part (i) follows from the nesting $E^0 \subseteq E^- \subseteq F$ and the uniqueness of each class's optimum. For (ii), in exposure-and-fund coordinates the three classes differ only in each fund's interval, $[0, \bar{x}_i]$, $[0, x_i^-]$ or $\{x_i^-\}$, and the exposure part is maximized at the same point in all three. Each ψ_i is concave with its kink at x_i^- ; for a bought fund the gain from the incumbent to the purchase edge is $\frac{\gamma v_i}{2}(\text{edge} - x_i^-)^2 = p_i^2/(2\gamma v_i)$. Part (iii) is the same computation after Proposition 35's reduction to one variable, which constrains only fund coordinates. For (iv), from each class's optimum a fund move with its netting trade leaves the exposure unchanged, and the ETF cost of a combined netting is at most the sum of the unit costs by subadditivity.

Part (ii) is the Treynor-Black logic of valuing an active position by its own alpha and residual variance ([60], p. 67 and equations (13) and (16), full text reviewed), net of the purchase rate and the incumbent's risk charge, and summed over the funds a restriction forbids. Earlier numerical comparisons reported the

ETF-only gap at fixed inputs; (i) shows that a baseline with frozen funds measures C^0 , which adds the value of selling the incumbents it assumes, and the restriction proper does not forbid those sales.

The result is for one review. The multi-review cost, ETF residual risk in (iv), an ETF at zero that cannot be sold to net a purchase, several unreachable funds, and the pooled prior's coupling of the v_i are not treated. No magnitude is claimed.

Propositions 35 and 39 kept every ETF strictly inside its bounds. The last result drops that assumption for the two-stage procedures of Propositions 22 and 23: when some ETFs sit at zero, when is each procedure exact, and what does the target-confined one lose?

Proposition 40 (Two-stage exactness with ETFs at zero: stage 1's one-sided slacks are the joint multipliers, the one-fund fibre interval, and the soft procedure; learning-driven tracking model at one review, spanning frictionless ETFs without fees; machine checked, with the reduction to ETF coordinates at paper level). Fix one review as in Proposition 39 (ii), with fees off, no ETF caps, a slack budget and a fund box $0 \leq x^A \leq \bar{x}^A$. In ETF coordinates, let $Q = R^\top B^{A\top}$ be the netting matrix with columns r_i , $w = x^E + Qx^A$ the exposure, $\mu_E = B^E \hat{\lambda}$, $\Sigma_{EE} = B^E \tilde{\Sigma}_f B^{E\top}$ and $G_E(w) = \mu_E^\top w - \frac{\gamma}{2} w^\top \Sigma_{EE} w$; nonnegative ETFs mean $w \geq Qx^A$. Let $H(x^A) = \hat{\alpha}^\top x^A - \frac{\gamma}{2} x^{A\top} V x^A - C_A(x^A - x^{A-})$ be the funds' part, $w_{TB} = (\gamma \Sigma_{EE})^{-1} \mu_E$ the factor Markowitz exposure, $V_E(q) = \max\{G_E(w) : w \geq q\}$, and $W_F = \{w : w \geq Qx^A \text{ for some } x^A \text{ in the box}\}$. The joint problem maximizes $G_E(w) + H(x^A)$ over $w \geq Qx^A$, with value J and holding x_J . The *target-confined* procedure (Proposition 23) takes stage 1's exposure $w^* = \arg \max_{W_F} G_E$, then maximizes H over the fibre $\{x^A : Qx^A \leq w^*\}$ at x_2 , with value T and loss $\Lambda = J - T$. *Stage 1's slacks* are $\zeta^* = \gamma \Sigma_{EE} w^* - \mu_E$.

(i) *Stage 1's slacks are the joint multipliers.* $\zeta^* \geq 0$, and $\zeta_j^* = 0$ wherever $w_j^* > (Qx_2)_j$. The procedure is exact, $T = J$, exactly when every fund's marginal at stage 1's prices,

$$g_i = \hat{\alpha}_i - \gamma(Vx_2)_i - r_i^\top \zeta^*, \quad (129)$$

equals κ_i^+ where x_2 buys, $-\kappa_i^-$ where it sells, and lies in $[-\kappa_i^-, \kappa_i^+]$ where it holds, with the one-sided forms at the box bounds. Stage 1 never sees the funds, yet its one-sided marginals at the ETFs at zero are the prices the joint optimum puts on the funds' by-product there. For any number of funds, $V_E(Qx_2) = G_E(w^*)$: stage 1's exposure is already optimal for stage 2's by-product, so the loss has no exposure-misfit part.

(ii) *One fund: the fibre interval.* With one fund and netting vector r , stage 2 sees the interval

$$[x_{lo}, x_{hi}] = \left[\max\left(0, \max_{j: r_j < 0} \frac{w_j^*}{r_j}\right), \min\left(\bar{x}, \min_{j: r_j > 0} \frac{w_j^*}{r_j}\right) \right], \quad (130)$$

and holds the fund at $x_2 = \text{clip}(x_f, x_{lo}, x_{hi})$, with x_f the frictionless band holding of Proposition 35 (107) at spanning moments. An ETF at zero that a purchase would sell caps the fund from above, and one that a sale would sell floors it from below. With $\Phi(x) = H(x) + V_E(rx)$,

$$T = J \iff x_J \in [x_{lo}, x_{hi}], \quad \Lambda = \Phi(x_J) - \Phi(x_2) \geq 0 : \quad (131)$$

the loss is the fund's misplacement in the joint objective, zero exactly when the joint holding lies in the room stage 1 leaves.

(iii) *The soft procedure.* If stage 1's set contains w_{TB} , in particular if stage 1 is unconstrained, Proposition 22's multiplier v_E is zero and the soft procedure is exact, with ETFs at zero or not. If stage 1 is confined to W_F , the procedure is exact when $w_{TB} \in W_F$, that is, when some fund holding's by-product lies below w_{TB}

in every ETF coordinate; for one fund, when (130) evaluated at w_{TB} is nonempty and $w_{TB,j} \geq 0$ wherever $r_j = 0$. Otherwise $v_E \neq 0$ lies in the normal cone of W_F at w^* , the soft loss satisfies $0 \leq \Lambda_s \leq v_E^\top (w_J - w_2)$ with w_2 stage 2's exposure, and $\Lambda_s = 0$ exactly when the joint optimum also solves the v_E -tilted stage 2. A nonzero multiplier need not cost anything: when the fund sits at a box bound in both problems and every ETF direction with a nonzero multiplier sits at the funds' by-product, at zero, in both, the tilted stage 2 can return the joint optimum.

For (i), W_F is closed upward, so moving w^* up in any coordinate cannot help stage 1; that gives $\zeta^* \geq 0$. The set $\{w \geq Qx_2\}$ lies inside W_F and contains w^* , which gives complementarity and $V_E(Qx_2) = G_E(w^*)$. The joint problem's optimum and multiplier are unique, so exactness is the fund lines at ζ^* . In (ii), $V_E(rx) = G_E(w^*)$ on the whole fibre interval, so the joint holding, if it lies there, maximizes H on it. Part (iii) applies Propositions 22 and 23 with the ETF bound as part of the feasible set. The first-order facts are proved by difference quotients, without a Karush-Kuhn-Tucker citation.

Dropping the ETFs at zero from the menu would turn the fund's by-product in their directions into an unreachable exposure, as in Proposition 35 (ii). Stage 1 here keeps those ETFs available upward, so its exposure is not the reduced-menu one, and the exactness condition reads on stage 1's own slacks. In Jagannathan and Ma's terms (Section 4) the slacks are the binding bounds' multipliers; here the exposure stage computes them without seeing the funds.

Fees, ETF frictions, ETF caps and a binding budget are not treated, nor are an explicit form of stage 1's implied fund holdings for several funds, bounds on $\Phi(x_J) - \Phi(x_2)$ in the inputs, and the soft loss beyond the multiplier bound.

Proposition 40 kept the other ETFs frictionless, fees off and the budget slack. The last result puts all of these back: some ETFs at zero, the others costly, fees on and a budget that may bind.

Proposition 41 (The fund decision and two-stage exactness with ETFs at zero, costly ETFs, fees and a binding budget; learning-driven tracking model at one review, spanning ETFs; machine checked, with the joint criterion in (i) and the exactness criterion in (iv) conditional on the polyhedral Karush-Kuhn-Tucker theorem). Fix one review with spanning ETFs ($M = K$, B^E invertible, netting matrix Q with columns r_i), fees c^E , ETF rates $\kappa_{E,j}^\pm$, ETF residual covariance Σ_E , the zero bound $x^E \geq 0$ and no ETF caps, a fund box, and the funded budget $k(x) \geq 0$ with multiplier $\eta \geq 0$. Let $\mu_E = B^E \hat{\lambda} - c^E$, $\Sigma_{EE} = B^E \tilde{\Sigma}_f B^{E\top}$, $\tilde{\alpha} = \hat{\alpha} + Q^\top c^E$, $w = x^E + Qx^A$, g the smooth marginal of every instrument, and $A_i = \tilde{\alpha}_i - \gamma(Vx^A)_i + \gamma r_i^\top \Sigma_E x^E$ the fund's reduced marginal, so that $g_i = A_i + r_i^\top g_E$. Trading slopes t lie in the trade-sign sets.

(i) *The joint criterion.* A feasible x is the optimum exactly when there are η , slopes t and slacks $\zeta_j \geq 0$, zero unless ETF j is at zero, such that

$$g_j = \eta + (1 + \eta)t_j - \zeta_j, \quad A_i + r_i^\top (\eta \mathbf{1} + (1 + \eta)t_E) - \sum_{j \text{ at zero}} r_{ij} \zeta_j = \eta + (1 + \eta)t_i, \quad (132)$$

for every ETF j and every fund i , with the box signs on the funds and $\eta k(x) = 0$. This composes the earlier one-sided and friction criteria and adds nothing to them: the idle ETFs' slopes and the at-zero slacks are tied to the state, not given.

(ii) *Explicit forms given the ETFs' statuses* ($\Sigma_E = 0$). Split the ETFs at the optimum into *traded* ones T (bought or sold, slope pinned), with pinned marginals $\pi_T = \eta \mathbf{1} + (1 + \eta)t_T$, and *fixed* ones F (at zero, or idle at a positive incumbent, holding known). With the hedged blocks of Σ_{EE} and the *effective netting weights*

$$\mu_{F:T} = \mu_F - \Sigma_{FT} \Sigma_{TT}^{-1} \mu_T, \quad \Sigma_{FF:T} = \Sigma_{FF} - \Sigma_{FT} \Sigma_{TT}^{-1} \Sigma_{TF}, \quad \rho_{iT} = r_{iT} + \Sigma_{TT}^{-1} \Sigma_{TF} r_{iF}, \quad (133)$$

and $w_F = Q_F x^A + x_F$ the exposure held in the fixed directions, the traded exposure is $w_T = \Sigma_{TT}^{-1}[(\mu_T - \pi_T)/\gamma - \Sigma_{TF} w_F]$, the fixed ETFs' marginals are $g_F = \mu_{F,T} + \Sigma_{FT} \Sigma_{TT}^{-1} \pi_T - \gamma \Sigma_{FF,T} w_F$, which gives each at-zero slack and each idle band, and every fund's marginal is

$$g_i = \alpha_i^{F,T} - \gamma(V^{F,T} x^A)_i, \quad \alpha_i^{F,T} = \tilde{\alpha}_i + r_{iF}^\top \mu_{F,T} - \gamma r_{iF}^\top \Sigma_{FF,T} x_F + \rho_{iT}^\top \pi_T, \quad V^{F,T} = V + Q_F^\top \Sigma_{FF,T} Q_F. \quad (134)$$

The fund line then reads

$$\alpha_i^{F,T} - \rho_{iT}^\top \pi_T - \gamma(V^{F,T} x^A)_i + (1 + \eta) \rho_{iT}^\top t_T - \eta(1 - \rho_{iT}^\top \mathbf{1}) = (1 + \eta) t_i. \quad (135)$$

So a fund's by-product in every fixed direction, at zero or idle alike, earns those directions' hedged premium and pays their hedged risk charge, and the funds interact through $V^{F,T}$. The traded ETFs' pinned slopes enter at ρ_{iT} , which adds to the fund's own netting weight the traded ETFs' hedge of its fixed-direction by-product, and the shadow price η is charged on $1 - \rho_{iT}^\top \mathbf{1}$, the net cash of a unit netted through the traded ETFs only. Selling a correlated traded ETF raises an at-zero ETF's slack and buying one lowers it. With no ETF fixed, this is the pinned-slope threshold with the cash shift $\eta(1 - \sum_j r_{ij})$; with none traded, it is the fund-alone form with the idle holdings added to the by-product. For an ETF starting at zero, with threshold $\eta + (1 + \eta) \kappa_{E,j}^+$, being at zero implies g_j at most the threshold, being bought implies equality, and g_j below it implies being at zero; the ETF stays at zero exactly when $g_j + \gamma \Sigma_{jj} x_j^E$ is at most the threshold. Over the traded ETFs' sign sets, $(1 + \eta) \rho_{iT}^\top t_T$ ranges over an interval given by the re-hedge costs computed with ρ_{iT} .

(iii) *One fund and one ETF.* With netting weight r , ETF variance σ_{EE} , fund holding a and ETF holding p , each line with the fund's box signs:

$$\begin{aligned} \text{ETF at zero from } p^- = 0 &\iff \mu_E - \gamma \sigma_{EE} r a \leq \eta + (1 + \eta) \kappa_E^+, \\ \text{ETF traded: } \tilde{\alpha} + r(\eta + (1 + \eta) t_E) - \gamma v a &= \eta + (1 + \eta) t_A, \\ \text{ETF fixed: } \tilde{\alpha} + r \mu_E - \gamma(v + r^2 \sigma_{EE}) a - \gamma r \sigma_{EE} p &= \eta + (1 + \eta) t_A. \end{aligned} \quad (136)$$

A traded ETF leaves the fund its own curvature and a net cash of $1 - r$ per netted unit; a fixed one gives it the fund-alone curvature, the ETF's full premium at weight r and the full cash of its unit.

(iv) *Two-stage exactness with everything on.* Take Proposition 40's target-confined stage 1 (fees in μ_E ; no ETF rates, residuals or budget) with slacks ζ^* , and a stage 2 that keeps every friction and the budget on the fibre, at x_2 . The procedure is exact exactly when the criterion (132) holds at x_2 ; there the fund marginals are $\tilde{\alpha}_i - \gamma(V x_2^A)_i - r_i^\top \zeta^*$, the residual terms cancelling. With $\Sigma_E = 0$, an ETF ending interior at x_2 has $\zeta_j^* = 0$, so its line reads $0 = \eta + (1 + \eta) t_j$. Exactness therefore needs every ETF bought and ending interior to be free with a slack budget, every ETF sold to an interior holding to satisfy $(1 + \eta) \kappa_{E,j}^- = \eta$, a coincidence of the inputs, and every untraded interior ETF to have $\kappa_{E,j}^- \geq \eta/(1 + \eta)$. A sale to zero needs only $-\zeta_j^* \leq \eta - (1 + \eta) \kappa_{E,j}^-$, which holds on a set of inputs of positive measure. So the procedure is generically inexact whenever it trades an ETF that ends interior. The soft procedure with an unconstrained stage 1 stays exact with every friction and the budget on. With stage 1 confined to the reachable exposures, $0 \leq \Lambda_s \leq \zeta^{*\top} (w_{s2} - w_j)$, with w_{s2} stage 2's exposure, and it is exact exactly when the joint optimum solves the ζ^* -tilted stage 2.

The proof of (i) is the Karush-Kuhn-Tucker condition of the lifted polyhedral problem with the identity $g_i = A_i + r_i^\top g_E$ substituted. For (ii), the traded ETFs' lines $g_T = \pi_T$ are solved for w_T and substituted into the fixed rows and the fund marginals: a block elimination given the statuses. Since $\Sigma_{FF,T}$ is a Schur complement of a positive definite matrix, the fund problem given the statuses stays strictly concave. For

(iv), x_2 is feasible for the joint problem, so it is exact exactly when it is the unique joint optimum; stage 1's complementarity gives $\zeta_j^* = 0$ at every ETF interior at x_2 . The soft stage 2 differs from the joint problem by $v_E^\top w$, as in Proposition 22.

In the terms of Jagannathan and Ma (Section 4), $V^{F,T}$ is the status-wise fold-in of the fixed ETFs into the funds' covariance, here over idle ETFs as well as those at zero, and with the traded ETFs' costs pinned. Without cross covariance between traded and fixed ETFs, $\rho_{iT} = r_{iT}$ and nothing interacts.

The explicit forms take $\Sigma_E = 0$. ETF caps, a stage 1 that sees costs or the budget, bounds on the target-confined loss in the inputs, and the coincidence events in (iv) are not treated. With several ETFs, the statuses are decided by a finite complementarity problem, not in closed form. No magnitude is claimed.

Propositions 26 and 38 bound quarterly bands exactly but leave the fine regime, where the target's quarterly step is small against the band, to leading-order asymptotics. The last result gives that leading order for a fund beside one costly ETF, at its two ends and in the weight between them.

Proposition 42 (A fund's fine-regime band beside one costly ETF: an error coordinate, three exact reductions, the cited cube-root law at each end, and one dimensionless weight between; learning-driven tracking model with a slack budget, one fund and one ETF; machine checked in its algebra and reductions, with the cube-root law resting on the cited corrector solution and the limits between the ends as readings). Fix Proposition 38's setting with one fund a and one ETF b , rates $\kappa_A^\pm$ and $\kappa_E^\pm$, risk correlation r_c , and targets whose per-review innovations have variances v_A , v_B and correlation r . Let $y_a = a - a^*$, $y_b = b - b^*$, $\rho_h = \Sigma_{AE}/\Sigma_{EE}$ the hedge ratio, $\rho' = \Sigma_{AE}/\Sigma_{AA}$, $c^{\text{res}} = \gamma \Sigma_{AA}(1 - r_c^2)$, and

$$v^{\text{idle}} = v_A + \rho'^2 v_B + 2\rho' r \sqrt{v_A v_B}, \quad v_B^{\text{eff}} = v_B + \rho_h^2 v_A + 2\rho_h r \sqrt{v_A v_B}, \quad (137)$$

the innovation variances of $a^* + \rho' b^*$, the fund's effective target when the ETF is frozen, and of $b^* + \rho_h a^*$, the ETF's effective target when it hedges the fund.

(i) *The error coordinate.* With $e = y_b + \rho_h y_a$, the ETF's tracking error against the position that hedges the fund,

$$\frac{\gamma}{2} (y_a, y_b) \Sigma (y_a, y_b)^\top = \frac{c^{\text{res}}}{2} y_a^2 + \frac{\gamma \Sigma_{EE}}{2} e^2. \quad (138)$$

A fund trade moves (y_a, e) along $(1, \rho_h)$, an ETF trade moves e alone, and the innovations of (y_a, e) have variances v_A and v_B^{eff} . The ETF's cost reaches the fund only through e .

(ii) *Three exact reductions.* (a) If $\Sigma_{AE} = 0$ at every review and state, the value is the sum of two one-instrument values, and the fund's band is its own one-instrument band with curvature $\gamma \Sigma_{AA}$, whatever the targets' innovation correlation or the ETF's rates. (b) With a frictionless ETF, the fund's problem is the one-instrument problem with curvature c^{res} and innovation variance v_A (Proposition 38 (vi)). (c) With the ETF frozen at b , by prohibitive rates in both directions or a pinned box, the fund's problem is the one-instrument problem with curvature $\gamma \Sigma_{AA}$, target $a^* + \rho'(b^* - b)$ and innovation variance v^{idle} .

(iii) *The cube-root law at each end.* For tracking a Brownian target of variance v per unit time with holding cost $\frac{c}{2} y^2$ and rates $\kappa^\pm$, the first corrector problem's no-trade band is $[-\Delta, \Delta]$, centred on the target whatever the split of the rates, with

$$\Delta^3 = \frac{3(\kappa^+ + \kappa^-)v}{4c}, \quad \text{average cost } \frac{c\Delta^2}{2} \quad (139)$$

([57], equations (4.3)-(4.5), read at the level of its displayed formulas; [46], Section 4, full text reviewed). Each reduction in (ii) meets its hypotheses, so the fund's fine-regime half-width is Δ_0 with $(c, v) = (\gamma \Sigma_{AA}, v_A)$ for uncorrelated risks, Δ_{free} with (c^{res}, v_A) for a frictionless ETF, and Δ_{frozen} with $(\gamma \Sigma_{AA}, v^{\text{idle}})$ for a frozen one. Their ratio, $\Delta_{\text{frozen}}^3 / \Delta_{\text{free}}^3 = (1 - r_c^2) v^{\text{idle}} / v_A$, can lie on either side of one.

(iv) *One weight between the ends.* With both instruments costly, the fine-regime problem is a two-dimensional ergodic problem in (y_a, e) . Scaled by Δ_{free} and by the ETF's own scale $\Delta_E^3 = 3(\kappa_E^+ + \kappa_E^-) v_B^{\text{eff}} / (4\gamma \Sigma_{EE})$, it depends on the rates only through

$$\xi^3 = \frac{\Delta_E^3}{\Delta_{\text{free}}^3} = \frac{\kappa_E^+ + \kappa_E^-}{\kappa_A^+ + \kappa_A^-} \cdot \frac{v_B^{\text{eff}}}{v_A} \cdot \frac{c^{\text{res}}}{\gamma \Sigma_{EE}}, \quad (140)$$

together with the correlations, the variance and curvature ratios, and the cost asymmetries. The fund's half-width is $\Delta_{\text{free}} F(\xi, \dots)$, with $F = 1$ exactly at $r_c = 0$. As a reading, if the problem's cross-section is continuous at the ends of ξ and its pushing set has measure zero, then $F \rightarrow 1$ as $\xi \rightarrow 0$, $F \rightarrow \Delta_{\text{frozen}} / \Delta_{\text{free}}$ as $\xi \rightarrow \infty$, and the ETF's idle probability, which tends to one at every ξ , is not the weight; ξ is. At basis-point ETF rates and quarterly target moves of about a percent, the ETF is not in its fine regime at all, so the relevant pairing is a fine fund beside a coarse ETF, whose leading order is (b) perturbed by an error bounded by the ETF's static half-width (also a reading).

Part (i) completes the square in y_a with ρ_h . Part (ii) (a) holds because the objective is a sum with separate controls and no shared constraint, (b) is Proposition 38 (vi), and (c) completes the square with ρ' at fixed b . In (iii), the quartic corrector $w(y) = -\frac{c}{12v} y^4 + \frac{\lambda}{v} y^2 + \frac{\kappa^- - \kappa^+}{2} y$ with $\lambda = c\Delta^2/2$ satisfies $\frac{v}{2} w'' + \frac{c}{2} y^2 = \lambda$ on the band, has $w'' = 0$, $w'(-\Delta) = -\kappa^+$ and $w'(\Delta) = \kappa^-$ at its edges exactly when (139) holds, and keeps $-\kappa^+ \leq w' \leq \kappa^-$ on the band and $\frac{c}{2} y^2 \geq \lambda$ off it: the rate asymmetry enters only the odd term, so the band does not shift. Part (iv)'s ξ is dimensional analysis.

The one-instrument law is the sources'; what is added is the error coordinate that makes the pair's loss diagonal, the three exact ends in one frame, and ξ as the weight. The two-dimensional corrector problem is the multidimensional object of [52], whose no-trade region has no explicit solution beyond one dimension (read at the level of its abstract and introduction). The identification of the quarterly fine-regime band with the corrector's half-width is cited, not shown, as for Proposition 26. F between the ends, its end limits as theorems, several ETFs, and the fine-fund coarse-ETF pairing as a theorem are not treated.

Proposition 38 bounded a fund's band beside one ETF at each ETF incumbent. The last result gives how that band, and the ETF's, move with the other instrument's holding: exactly at the last review, and monotonically at every review.

Proposition 43 (The shape of the bundling band: median edges at the last review, monotone edges at every review, and the frozen-ETF band; learning-driven tracking model with a slack budget, one fund and one ETF; machine checked, with the monotonicity in (ii) conditional on Topkis's lattice theorems and its nonpositive-covariance case at paper level). Fix Proposition 38's setting with one fund and one ETF, finite caps and gross returns $g' > 0$. Let $\rho = \Sigma_{AE} / \Sigma_{EE}$, $\rho_A = \Sigma_{AE} / \Sigma_{AA}$, and let $c^{\text{res}} = \gamma \Sigma_{AA} (1 - \text{corr}^2)$ and $c_E^{\text{res}} = \gamma \Sigma_{EE} (1 - \text{corr}^2)$ be the two residual curvatures. The fund's band $[lo_t, hi_t]$ at ETF incumbent p^- is Proposition 38's effective band; the ETF's band given a fund holding a is Proposition 26's band for $p \mapsto G_t((a, p), z)$; and NT_t is the set of holdings from which nothing trades.

(i) *The last review: median edges.* At $T - 1$, for every ETF incumbent at which the ETF optimizer at the fund's edge is strictly inside the ETF box, each fund edge is the median of its *bought level*, its **idle line** and its *sold level*, clipped to $[0, \bar{x}_A]$:

$$\begin{aligned}
lo_{T-1}(p^-) &= \text{med}\left(a^* - \frac{\kappa_A^+ - \rho\kappa_E^+}{c^{\text{res}}}, a^* - \rho_A(p^- - p^*) - \frac{\kappa_A^+}{\gamma\Sigma_{AA}}, a^* - \frac{\kappa_A^+ + \rho\kappa_E^-}{c^{\text{res}}}\right), \\
hi_{T-1}(p^-) &= \text{med}\left(a^* + \frac{\kappa_A^- + \rho\kappa_E^+}{c^{\text{res}}}, a^* - \rho_A(p^- - p^*) + \frac{\kappa_A^-}{\gamma\Sigma_{AA}}, a^* + \frac{\kappa_A^- - \rho\kappa_E^-}{c^{\text{res}}}\right).
\end{aligned} \tag{141}$$

For $\Sigma_{AE} > 0$, each edge is continuous and nonincreasing in p^- : flat at its bought level while the ETF is bought at that edge, falling along the idle line at slope $-\rho_A$, and flat at its sold level beyond. The idle segment has length

$$\frac{\kappa_E^+ + \kappa_E^-}{c_E^{\text{res}}} \tag{142}$$

in the ETF incumbent, the ETF's own residual static width. So the fund's edge moves with the ETF's holding only while the ETF is idle at that edge. For $\Sigma_{AE} < 0$ the picture mirrors, and for $\Sigma_{AE} = 0$ the three levels coincide. The width is the residual ceiling (121) on two bought or two sold levels, the fund-alone width on the idle lines, and the ceiling minus the leak with the lower edge bought and the upper sold, as in Proposition 38 (iv), now read off p^- . The last-review region is Proposition 26 (v)'s static parallelotope.

(ii) *Every review: monotone edges and a two-band region.* If $\Sigma_{AE} \geq 0$ at every review and state, then V_t has increasing differences in the two holdings, and so do G_t and the ETF-optimized value in (a, p^-) . Hence both fund edges are nonincreasing in the ETF incumbent and both ETF edges are nonincreasing in the fund holding, at every review, whatever the rates, the horizon or the innovation law, and

$$NT_t = \{(a, p) : lo_t(p) \leq a \leq hi_t(p), lo_t^E(a) \leq p \leq hi_t^E(a)\}, \tag{143}$$

the intersection of two bands with monotone edges, inside Proposition 38's parallelotope (124). With $\Sigma_{AE} \leq 0$ everywhere the directions reverse, and with $\Sigma_{AE} = 0$ no edge depends on the other holding.

(iii) *A frozen ETF.* If the ETF cannot trade and sits at p_0 , under pure-learning marking the fund's problem at every review is the one-instrument problem with curvature $\gamma\Sigma_{AA}$, the fund's rates and cap, and the idle target $a_t^* - \rho_{A,t}(p_0 - p_t^*)$, so Proposition 26 applies to it. With constant covariance, the idle target's innovation has variance v^{idle} of (137), increasing in the targets' innovation correlation when $\rho_A > 0$. The correlation enters the frozen-ETF band only through the law of the idle target's innovation, whose variance is v^{idle} ; it enters through v^{idle} alone when that law is fixed by its variance, as for Gaussian innovations or in the fine-regime reading. With a frictionless ETF it does not enter at all (Proposition 38 (vi)).

(iv) *The last review is free of the innovation law.* Two instances that agree at the last review on the premia, the covariance, γ , the rates and the caps have the same last-review bands and region, whatever their outcome laws. At earlier reviews the innovation law enters only through the continuation.

For (i), the ETF's optimizer at fund holding a is a median of its own band edges and p^- , the ETF-optimized value is differentiable with derivative $\partial G/\partial a$ at that optimizer, and a monotone affine map sends a median to a median. So the value's derivative is a median of three strictly increasing affine functions, and its root is the median of their roots. For (ii), in coordinates with the fund's holding negated, the review's minimand is a sum of terms each with decreasing differences in a pair of coordinates, hence submodular, and submodularity survives partial minimization over a product of intervals ([59], Theorems 3.1, 3.2 and 4.3, read at the level of the theorems); backward induction carries it through the tree, and increasing differences move the one-sided derivatives, and hence the edges, monotonically. Part (iii) completes the square at fixed p_0 .

The small-cost literature gives the leading-order multi-asset region from a corrector equation (Section 4). With independent assets every boundary is flat in the other holdings ([33], full text reviewed), which is the $\Sigma_{AE} = 0$ case of (ii). Here the bend in (i) is exact at a finite cost, and its length is the other instrument's residual static width.

How the targets' innovation correlation moves the band at reviews before the last is not claimed. Experiments suggest a mixture of the idle and re-hedging variances weighted by the future idle share, which is a conjecture. A bound on the edges' slope in the ETF incumbent, the ETF at a bound in (i), several ETFs in (ii), general marking in (iii), and a binding budget are not treated.

Propositions 35-43 took the one-review fund decision as known. The last result states it: when a fund is held, bought or sold, against adjusting exposure through ETFs, at one review of the learning-driven tracking model.

Proposition 44 (The fund's hold, buy and sell criterion at one review: exact in general, explicit under frictionless spanning ETFs, bracketed by the re-hedge cost otherwise, with the budget, the bounds and several reviews; learning-driven tracking model; machine checked, with the multiplier criterion and the parts built on it conditional on the polyhedral Karush-Kuhn-Tucker theorem). Fix one review with N funds and M ETFs, predictive moments μ and Σ as in Proposition 35, rates $\kappa_i^\pm \in [0, 1]$, caps $0 \leq x \leq \bar{x}$, pre-trade holdings x^- and cash h^- , and funded cash $k(x) = h^- - \mathbf{1}^\top(x - x^-) - C(x - x^-) \geq 0$. Write $g(x) = \mu - \gamma \Sigma x$ for the smooth marginal.

(i) *The exact criterion.* The optimum exists and is unique. A feasible x is optimal exactly when there is $\eta \geq 0$ with $\eta k(x) = 0$ such that each instrument bought in the interior has $g_i = \eta + (1 + \eta)\kappa_i^+$, each sold in the interior has $g_i = \eta - (1 + \eta)\kappa_i^-$, and each held has g_i between the two, with the one-sided forms at zero and at the cap. The criterion is exact but implicit, since g is evaluated at the optimum.

(ii) *Is any fund traded?* With x_E the optimum over ETFs alone and I the shadow prices compatible with it, no fund is traded at the full optimum exactly when some $\eta \in I$ puts every fund's marginal at x_E in its held band.

(iii) *Frictionless spanning ETFs.* If $M = K$ with B^E invertible, ETFs are frictionless, V is diagonal with entries v_i , and at the optimum the budget is slack and every ETF strictly inside its box, then

$$y^* = (\gamma \tilde{\Sigma}_f)^{-1} \hat{\lambda}, \quad x_i = \text{clip}\left(x_i^-, \frac{\hat{\alpha}_i - \kappa_i^+}{\gamma v_i}, \frac{\hat{\alpha}_i + \kappa_i^-}{\gamma v_i}\right) \text{ in } [0, \bar{x}_i], \quad x^E = R^\top(y^* - B^{A^\top} x^A). \quad (144)$$

Fund i is bought exactly when $\hat{\alpha}_i - \gamma v_i x_i^- > \kappa_i^+$ and $x_i^- < \bar{x}_i$, sold exactly when $\hat{\alpha}_i - \gamma v_i x_i^- < -\kappa_i^-$ and $x_i^- > 0$, and held otherwise. Premium beliefs, their precision and the loadings do not enter the decision.

(iv) *ETF frictions: a bracket.* With spanning, diagonal V , a slack budget and every ETF strictly inside its box, let the re-hedge costs $h_i^\pm$ be as in Proposition 39 (iv) and the reduced marginal be $A_i = \hat{\alpha}_i + r_i^\top c^E - \gamma v_i x_i + \gamma r_i^\top \Sigma_E x^E$. Then

$$\text{bought} \Rightarrow A_i \geq \kappa_i^+ - h_i^-, \quad \text{sold} \Rightarrow A_i \leq -\kappa_i^- + h_i^+, \quad \text{held} \Rightarrow -\kappa_i^- - h_i^- \leq A_i \leq \kappa_i^+ + h_i^+. \quad (145)$$

Under the purchase re-hedge an interior purchase has $A_i = \kappa_i^+ + h_i^+$ (at least that at the cap), and the sale is the mirror; a held fund can satisfy the same equality, so it is necessary, not a test. If every ETF the fund's netting uses is traded, the slopes are pinned and an interior purchase has $A_i = \kappa_i^+ - \sum_j r_{ij} t_j$. With no ETF residual risk, $A_i(x^-) > \kappa_i^+ + h_i^+$ forces a purchase and $A_i(x^-) < -\kappa_i^- - h_i^-$ a sale.

(v) *A binding budget.* The thresholds on g_i become $\eta \pm (1 + \eta)\kappa_i^\pm$. On the reduced marginal the shift is $\eta(1 - \sum_j r_{ij})$, the shadow price on the net cash of a netted unit, and in (iii) fund i is bought exactly when

$$\hat{\alpha}_i - \gamma v_i x_i^- > \eta \left(1 - \sum_j r_{ij}\right) + (1 + \eta)\kappa_i^+, \quad (146)$$

with exposure $y^*(\eta) = (\gamma \tilde{\Sigma}_f)^{-1}(\hat{\lambda} - \eta R \mathbf{1})$.

(vi) *Several reviews.* In (iii)'s separated setting with discount β and pure-learning marking, each fund is a one-instrument problem with curvature $c_t = \gamma v_{i,t}$, and

$$\hat{\alpha}_{i,t} \leq (1 - \beta)\kappa_i^+ \Rightarrow \text{no purchase from zero}, \quad \hat{\alpha}_{i,t} > \kappa_i^+ + \beta\kappa_i^- \Rightarrow \text{a purchase from zero}, \quad (147)$$

with the mirror at the cap, carrying the cap's risk charge $c_t \bar{x}_i$, for sales. Looking ahead lowers the alpha a purchase needs, because the position persists.

Part (i) is the polyhedral Karush-Kuhn-Tucker theorem applied to the lifted problem ([53], Theorems 27.4 and 28.2-28.3, cited by theorem number), with the lifted multipliers eliminated. Part (iii) solves out the frictionless ETF block. For (iv) and (v), the identity $g_i = A_i + r_i^\top g_E$ moves the ETF lines into the fund's, each ETF's slope lying in its trade-sign set. Part (vi) reads Proposition 26's edge brackets for each fund's reduced problem.

The fund decision is thus a band in net alpha, of width the round-trip rate widened by the re-hedge cost, centred at the risk charge on the current holding. Exposure is adjusted through the ETFs and enters the fund decision only through the re-hedge cost, the fee credit and, with a binding budget, the net cash of a netted unit. This is Proposition 9's multiplier condition for n instruments; Treynor and Black's separation ([60], p. 67, full text reviewed) is (iii).

ETFs at zero or at their caps, the common long-only case, are Propositions 40-41's; they are not covered by (iii)-(iv). A binding budget with ETF frictions is covered by (iv)-(v) only as stated. No magnitude is claimed.

Propositions 22 and 23 priced two-stage implementation in the one-quarter model. The last result carries them to one review of the learning-driven tracking model with several funds and ETFs, and states in the inputs when each procedure is exact and what it loses.

Proposition 45 (When two-stage implementation is exact and what it loses, at one review; learning-driven tracking model in the reference case; machine checked, with the joint criterion, the exactness criterion under frictions and the binding-budget case conditional on the polyhedral Karush-Kuhn-Tucker theorem). Fix one review as in Proposition 44, in the reference case, and split the objective as $Q(x) = G(b(x)) + H(x)$, with exposure $b = B^\top x$, factor part $G(b) = \hat{\lambda}^\top b - \frac{\gamma}{2} b^\top \tilde{\Sigma}_f b$ and residual part H (alpha, fees, residual risk and costs). Let B_F be the funded feasible exposures, J the joint optimum's value, at exposure b_J . The *target-confined* procedure (Proposition 23) chooses b^* maximizing G over B_F , then maximizes H on its fibre at x_2 ; its loss is $\Lambda = J - T$. The *soft* procedure (Proposition 22) chooses b^* over a convex $\mathcal{R} \supseteq B_F$, with multiplier $\nu = \hat{\lambda} - \gamma \tilde{\Sigma}_f b^*$, and loses Λ_s .

(i) *Transfer.* Propositions 22 and 23 hold for every such instance: the loss identity, the exactness criterion, the loss bounds, and the soft procedure as the joint problem with the premia replaced by $\gamma \tilde{\Sigma}_f b^*$.

(ii) *Frictionless spanning ETFs.* With spanning, frictionless ETFs, diagonal V , and a slack budget with every ETF interior at the joint optimum, both procedures are exact for every value of the fund and premium inputs: stage 1 chooses $(\gamma \tilde{\Sigma}_f)^{-1} \hat{\lambda}$, and stage 2 on its fibre returns (144)'s fund holdings.

(iii) *Spanning ETFs with frictions.* With a slack budget and every ETF interior at x_2 , the target-confined procedure is exact exactly when x_2 is jointly optimal. In the inputs that is two conditions together:

$$-c_j^E - \gamma(\Sigma_E x_2^E)_j \in T_j(x_2) \text{ for every ETF,} \quad \hat{\alpha}_i - \gamma(Vx_2^A)_i \in T_i(x_2) \text{ for every fund,} \quad (148)$$

each ETF's fee and residual marginal in its own cost band with the sign of its trade, and each fund in its frictionless alpha band. The first alone is necessary, not sufficient. With fees and traded ETFs, it holds only on a set of inputs of measure zero. The soft procedure is exact whenever $(\gamma\tilde{\Sigma}_f)^{-1}\hat{\lambda} \in \mathcal{R}$, whatever the frictions.

(iv) *A fund with an unreachable loading.* With one fund whose loading has a component outside the ETFs' reach, Proposition 35's reduced moments and ψ_i , stage 1 fixes the fund at the premium-implied holding a_i^* and the joint optimum at the band holding a_J , and

$$\Lambda = \psi_i(a_J) - \psi_i(a_i^*) \geq 0, \quad \frac{\gamma s_i^{\text{red}}}{2}(a_J - a_i^*)^2 \leq \Lambda \leq \frac{\gamma s_i^{\text{red}}}{2}(a_J - a_i^*)^2 + (\kappa_i^+ + \kappa_i^- + \mu_J)|a_J - a_i^*|, \quad (149)$$

with μ_J the box multiplier at a_J , zero when a_J is interior. Stage 1 chooses the manager, and the procedure is exact exactly when the premium-implied holding lies in the fund's alpha band.

(v) *What the frictions cost.* In (iii)'s setting,

$$\Lambda \leq \min\left(\frac{L_E^2}{2\gamma}, L_E\|b_J - b^*\|_{\tilde{\Sigma}_f}\right), \quad L_E = \|\tilde{\Sigma}_f^{-1/2}R\|(\|c^E\| + \gamma\|\Sigma_E\|\|\bar{x}^E\| + \|\kappa_E^{\max}\|). \quad (150)$$

With a binding budget, the procedure is exact exactly when x_2 meets the joint criterion for some shadow price, and the bound holds with $L_E(\eta_2)$ computed at the fibre problem's own cash multiplier; there it holds for the supergradient that multiplier supplies, not for every one.

The criterion is the polyhedral Karush-Kuhn-Tucker theorem applied to a lifted problem with one cost bound per instrument ([53], Theorems 27.4 and 28.2-28.3, cited by theorem number). Part (ii) solves the frictionless block as in Proposition 44 (iii). For (iii), at x_2 the exposure error vanishes because stage 1 chose the Markowitz exposure, so joint optimality reduces to (148). Part (iv) reduces the joint and fibre problems to the one variable a by Proposition 28's elimination. Part (v) bounds the residual value's supergradients at b^* by the ETF frictions.

So choosing exposure first is harmless exactly in Treynor and Black's world ([60], p. 67, full text reviewed) and costs, in the inputs, the fund's misplacement for a missing direction and at most $L_E^2/(2\gamma)$ for ETF frictions, while the soft procedure with an unconstrained first stage stays exact. ETFs at zero are Propositions 40-41's.

The bound in (v) is stated at b^* only: at a boundary point of B_F the relative supergradients are unbounded. No magnitude is claimed.

Propositions 44 and 45 keep every ETF strictly inside its box. The last result is the fund decision when some ETFs sit at their zero bound, the common long-only case, on which Propositions 40-41 build.

Proposition 46 (The fund decision with ETFs at zero: one-sided slacks price the funds' by-product, the at-zero set follows from the inputs, and the response is piecewise with direction-dependent curvature; learning-driven tracking model at one review; machine checked, with the general criterion in (i) conditional on the polyhedral Karush-Kuhn-Tucker theorem). Fix one review as in Proposition 44. Parts (ii)-(v) assume spanning frictionless ETFs (fees allowed), ETF caps absent

or slack and the budget slack, and use Proposition 40's ETF coordinates: the netting matrix Q with columns r_i , the exposure $w = x^E + Qx^A$, $\mu_E = B^E \hat{\lambda} - c^E$, $\Sigma_{EE} = B^E \tilde{\Sigma}_f B^{E\top}$, and $\tilde{\alpha} = \hat{\alpha} + Q^\top c^E$. The bound $x^E \geq 0$ reads $w \geq Qx^A$.

(i) *The criterion with one-sided ETF marginals.* With every ETF below its cap, at the optimum each ETF j at zero satisfies $g_j = \eta + (1 + \eta)t_j - \zeta_j$ with a *slack* $\zeta_j \geq 0$, and with spanning fund i 's line is Proposition 44's with the extra term $-\sum_{j \text{ at zero}} r_{ij}\zeta_j$: the fund's by-product in the at-zero directions is priced at minus their slacks.

(ii) *Which ETFs are at zero.* For a by-product $q = Qx^A$, the exposure problem $\max\{\mu_E^\top w - \frac{\gamma}{2} w^\top \Sigma_{EE} w : w \geq q\}$ has a unique solution $w(q)$ with a unique multiplier $\zeta(q) \geq 0$. For a candidate set Z with complement c ,

$$w_c = \Sigma_{cc}^{-1} \left(\frac{\mu_c}{\gamma} - \Sigma_{cZ} q_Z \right), \quad \zeta_Z = \gamma \Sigma_{ZZ \cdot c} q_Z - \mu_{Z \cdot c}, \quad (151)$$

and Z is an at-zero set exactly when $\zeta_Z \geq 0$ and $w_c \geq q_c$; every such set gives the same w and ζ . The value $V_E(q) = \max$ is concave and differentiable with gradient $-\zeta(q)$.

(iii) *The fund marginal and the criterion.* With the exposure optimized out, the funds' smooth marginal is

$$G_i(x^A) = \alpha_i^Z - \gamma(V^Z x^A)_i, \quad \alpha^Z = \tilde{\alpha} + Q_Z^\top \mu_{Z \cdot c}, \quad V^Z = V + Q_Z^\top \Sigma_{ZZ \cdot c} Q_Z, \quad (152)$$

at $Z = Z(Qx^A)$. The incumbent is optimal exactly when $-\kappa_i^- \leq G_i(x^{A-}) \leq \kappa_i^+$ for every fund, one-sided at the bounds, for any N . For one fund, it is bought exactly when $G(x^-) > \kappa^+$ and $x^- < \bar{x}$, sold exactly when $G(x^-) < -\kappa^-$ and $x^- > 0$, and held otherwise. A premium error e shifts G_i by $r_{iZ}^\top B_{Z \cdot c}^E e$, with $B_{Z \cdot c}^E$ the at-zero ETFs' loadings hedged by the free ones, which is zero for every e exactly when $r_{iZ} = 0$.

(iv) *Along a fund's trade* ($N = 1$). The marginal $m(x) = G(x)$ is continuous, strictly decreasing and piecewise linear, with slope $-\gamma s^Z$, $s^Z = v + r_Z^\top \Sigma_{ZZ \cdot c} r_Z$, on each piece where $Z(rx)$ is constant; the pieces end where a free ETF's holding or an at-zero slack reaches zero, both affine in x . The optimum is the incumbent or the clipped root of $m = \kappa^+$ or $m = -\kappa^-$. The *drop-Z rule*, Proposition 44 (iii) on the menu without the incumbent's at-zero set Z^- ,

$$\tilde{x} = \text{clip}\left(x^-, \frac{\alpha^{Z^-} - \kappa^+}{\gamma s^{Z^-}}, \frac{\alpha^{Z^-} + \kappa^-}{\gamma s^{Z^-}}\right) \text{ in } [0, \bar{x}], \quad (153)$$

agrees with the optimum exactly when their unclipped roots agree or both clip to the same bound, and the unclipped roots agree exactly when either lies in the incumbent's piece. If the at-zero set contains Z^- all along the way to the drop-Z root, the true trade is no longer than the drop-Z trade; if it is contained in Z^- , no shorter. A purchase and a sale of the same fund face different pieces, so different curvatures.

(v) *One ETF.* With $M = K = 1$ and $q = \sum_i r_i x_i$, the ETF is at zero exactly when $\mu_E \leq \gamma \sigma_{EE} q$, and then every fund's marginal is $\tilde{\alpha}_i + r_i \mu_E - \gamma(v_i x_i + r_i \sigma_{EE} q)$: its total expected return less the full risk charge.

Part (i) is Proposition 45's criterion with the ETF bound's multiplier kept. Part (ii) is the Karush-Kuhn-Tucker system of a strictly concave quadratic over an orthant translate, solved by a Schur complement on the free block; its uniqueness uses the complementarity estimate $\|w(q') - w(q)\|_\Sigma \leq \|q' - q\|_\Sigma$. Part (iii) differentiates the reduced objective, using (ii)'s gradient. Part (iv) follows the pieces of (ii) along the line rx .

With Z given, (152) is Proposition 28's reduced moments with the at-zero directions as the unreachable ones, produced here by the bound rather than by the menu, and premium beliefs reach the fund decision

through them although the ETFs span. In the language of Jagannathan and Ma ([28], Proposition 1, read at the level of the proposition), the at-zero multipliers fold into the moments; here the fold-in is an adjusted premium and an adjusted covariance V^Z , positive semidefinite of rank at most $|Z|$, with the multipliers fixed by complementarity from the inputs rather than read off the solution, and it holds piecewise along a trade.

ETF frictions and a binding budget with ETFs at zero are Proposition 41's. An ETF at its cap, the per-fund direction for several funds, and magnitudes are not treated.

Proposition 41 made every fund's marginal explicit once the ETFs' statuses are known, but with several ETFs those statuses come from a finite complementarity problem. With one ETF the last result removes that step: two scalar prices decide everything.

Proposition 47 (One ETF and any number of funds: the optimum through two monotone scalars, the exposure price and the cash price; learning-driven tracking model at one review; machine checked, with the existence of a binding budget's cash price at paper level through the polyhedral Karush-Kuhn-Tucker theorem). Take Proposition 41's setting with one ETF and one factor and a diagonal V with entries v_i : fund holdings $x_i \in [0, \bar{x}_i]$, an ETF holding $p \geq 0$ with no cap, loading $b_E > 0$, fee c^E , rates $\kappa_E^\pm$ and residual variance σ_E , and the funded budget with multiplier η . Write $r_i = b_{A,i}/b_E$, $\mu_E = b_E \hat{\lambda} - c^E$, $\sigma_{EE} = b_E^2 \tilde{\sigma}_f$, $\tilde{\alpha}_i = \hat{\alpha}_i + r_i c^E$, the exposure price $m(w) = \mu_E - \gamma \sigma_{EE} w$ at total exposure $w = p + \sum_i r_i x_i$, the scaled ETF thresholds $\theta_b = \eta + (1 + \eta) \kappa_E^+$ and $\theta_s = \eta - (1 + \eta) \kappa_E^-$, and the one-fund clip at prices (m, η)

$$x_i(m, \eta) = \text{clip}\left(x_i^-, \frac{\tilde{\alpha}_i + r_i m - \eta - (1 + \eta) \kappa_i^+}{\gamma v_i}, \frac{\tilde{\alpha}_i + r_i m - \eta + (1 + \eta) \kappa_i^-}{\gamma v_i}\right) \text{ in } [0, \bar{x}_i], \quad (154)$$

with $q(m, \eta) = \sum_i r_i x_i(m, \eta)$ and $p(m, \eta) = (\mu_E - m)/(\gamma \sigma_{EE}) - q(m, \eta)$.

(i) *Every fund sees the ETF through one price.* Fund i 's marginal is $\tilde{\alpha}_i + r_i m(w) - \gamma v_i x_i$, so at the optimum every fund holds its clip (154) at (m^*, η^*) . $q(\cdot, \eta)$ is continuous, nondecreasing and bounded, $p(\cdot, \eta)$ is continuous, strictly decreasing and onto, and each x_i is nonincreasing in η at fixed m .

(ii) *The ETF's status.* For each $\eta \geq 0$, let m_0 be the root of $p(m, \eta) = 0$ and, when $p^- > 0$, m_I the root of $p(m, \eta) = p^-$; both are unique. With the ETF's own marginal $e(m) = m - \gamma \sigma_E p(m, \eta)$,

$$\begin{aligned} p^- > 0 : & \text{bought iff } e(m_I) > \theta_b, \quad \text{idle iff } \theta_s \leq e(m_I) \leq \theta_b, \quad \text{sold iff } e(m_I) < \theta_s, \\ p^- = 0 : & \text{at zero iff } m_0 \leq \theta_b, \text{ else bought,} \end{aligned} \quad (155)$$

a sale ending at a positive holding when $m_0 > \theta_s$ and at zero otherwise. In each regime m^* is the stated price: m_I when idle, m_0 at zero, and in a traded regime the unique root of $e(m) = \theta$, which is θ itself when $\sigma_E = 0$.

(iii) *The cash price.* The cash left by (ii)'s solution, $k(x(\eta))$, is nondecreasing in η . If $k(x(0)) \geq 0$ the budget is slack and $\eta^* = 0$; otherwise any $\eta > 0$ with $k(x(\eta)) = 0$ gives the optimum, and all such η give the same holdings.

(iv) *Readings.* At the optimum's prices, fund i is bought exactly when

$$\tilde{\alpha}_i + r_i m^* - \gamma v_i x_i^- > \eta^* + (1 + \eta^*) \kappa_i^+ \quad \text{and} \quad x_i^- < \bar{x}_i, \quad (156)$$

sold under the mirror condition with $x_i^- > 0$, and held otherwise: Proposition 44 (iii)'s rule with the exposure price added, exact for every N . A fund's holding moves by at most $|r_i|/(\gamma v_i)$ per unit of exposure price, and at a fixed exposure price it does not rise with the cash price.

For each η , the Lagrangian with the budget priced at η has exactly one maximizer over the funds' boxes and $p \geq 0$; its fund coordinates are the clips at the exposure price it induces, and (155) reads the ETF's line against its scaled cost band. The monotonicity of q and p in m makes each root unique. The existence of a positive η that exhausts the budget, when $\eta = 0$ leaves it negative, is the multiplier of the polyhedral Karush-Kuhn-Tucker theorem ([53], Theorems 27.4 and 28.2-28.3, cited by theorem number) through Proposition 44 (i).

A manager with one broad ETF therefore needs no solver. Two monotone scalar searches, each exact on finitely many linear pieces, give the cash price and the exposure price; every fund is then its own band decision, and the ETF's status is read off (155). Frictionless ETFs with a slack budget are the case $\theta_b = \theta_s = 0$ of Proposition 46 (v).

Several ETFs, and a formal link from these one-ETF coordinates to the general review model, are not treated. No magnitude is claimed.

Proposition 41 (iv) showed that a first stage fixing the exposure by the frictionless one-sided problem is generically inexact once a costly ETF ends interior. The last result gives a first stage that prices the ETF trades, and a loss bound for any first stage.

Proposition 48 (An incumbent-aware first stage, and the two-stage loss through the ETF-line residual; learning-driven tracking model at one review; machine checked, with the existence of the second stage's multipliers at paper level through the polyhedral Karush-Kuhn-Tucker theorem, and the converse of (ii) conditional on it). Take Proposition 41's setting with no ETF residual risk and the joint optimum x_J , value J and exposure w_J . The *fibre value* $V(w)$ is the best objective the funds can reach when the ETFs must deliver exposure w under the zero bound and the budget. A *fibre-confined* procedure picks w_1 , then maximizes on its fibre at x_2 , losing $\Lambda = J - V(w_1)$. The *incumbent-aware* first stage solves the review with the funds frozen at their incumbents, ETF costs, fees, zero bound and budget on, and takes its exposure as w_1 . With multipliers of the fibre problem at x_2 (its cash price η_2 , cost slopes $t^{(2)}$ and zero-bound multipliers $\zeta^{(2)}$), the *ETF-line residual* is

$$s = g_E(x_2) - \eta_2 \mathbf{1} - (1 + \eta_2)t^{(2)} + \zeta^{(2)}, \quad g_E(x_2) = \mu_E - \gamma \Sigma_{EE} w_1, \quad (157)$$

the amount by which each ETF's joint line fails at x_2 .

(i) *A loss bound.* For any fibre-confined procedure and any valid multipliers, with $d = w_J - w_1$,

$$\frac{\gamma}{2} [(x_J^A - x_2^A)^\top V(x_J^A - x_2^A) + \|d\|_{\Sigma_{EE}}^2] \leq \Lambda \leq \min\left(s^\top d, \frac{s^\top \Sigma_{EE}^{-1} s}{2\gamma}\right), \quad \gamma \|d\|_{\Sigma_{EE}} \leq \|s\|_{\Sigma_{EE}^{-1}}, \quad (158)$$

and $\Lambda = 0$ exactly when $x_2 = x_J$.

(ii) *Exactness.* Multipliers with $s = 0$ exist exactly when Proposition 41's joint criterion holds at x_2 , and then $x_2 = x_J$. Conversely, an exact split admits such multipliers, and at an ETF traded to an interior holding under a slack budget $s_j = 0$ for every valid choice.

(iii) *When the incumbent-aware split is exact.* It is exact if every fund's marginal at the frozen-funds optimum lies in its scaled held band (then that optimum is the joint one); if every ETF trades to an interior holding at the first stage and keeps its direction to an interior holding at the second, with a slack budget at both; and, when some ETFs are fixed and keep their statuses, if every fund's line holds at x_2 with the first stage's ETF marginals, the residual then living only in the fixed directions.

Part (i) bounds the joint objective at any feasible point by its value at x_2 plus the residual's linear term less the strong-concavity terms in the exposure and the funds, and evaluates it at x_J ; the lower bound is

strong concavity along the segment from x_J . Part (ii) matches the residual's zero with the joint criterion, moving an at-zero ETF's slope to Proposition 46's convention. Part (iii) reads the residual at the first stage's statuses.

Pricing the ETF trades at the first stage removes the failure that ETF costs alone cause: the incumbent-aware split can fail only when the funds' trades move the by-product in fixed directions, the cash, or an ETF's status. Its fibre is always fundable. It is the better first stage on exactness and fundability, not on loss (experiment 045's illustration, in Appendix A). The bound (158) sharpens Proposition 45 (v)'s norm bound with the exact multiplier vector.

ETF residual risk, the repeated split from the second stage's holdings, and an input-level condition under a binding budget beyond (iii)'s first case are not treated. No magnitude is claimed.

Proposition 42 left the fine-regime problem between its frictionless and frozen ends as a two-dimensional corrector problem. The last result traps that problem's leading-order cost between explicit bounds in the inputs.

Proposition 49 (The fine-regime cost with one costly ETF, sandwiched in the inputs; learning-driven tracking model with one fund and one ETF, at the ergodic level; machine checked for every eigenvalue the comparison principle brackets, with its application to the pair's corrector cited). Take Proposition 42's setting with symmetric per-side rates $\kappa_A, \kappa_E > 0$, $v_A, v_B > 0$, $|r| < 1$, $c_E = \gamma \Sigma_{EE}$, and v_B^{eff} from (137). The *corrector problem of the pair* is the first corrector equation for the state (y_a, e) with running cost $\frac{c^{\text{res}}}{2} y_a^2 + \frac{c_E}{2} e^2$, innovation covariance W with entries $v_A, r\sqrt{v_A v_B} + \rho_h v_A$ and v_B^{eff} , and admissible gradients $C = \{p : |p_a + \rho_h p_e| \leq \kappa_A, |p_e| \leq \kappa_E\}$; write a for its eigenvalue, the pair's leading-order average cost. With the one-instrument eigenvalues at per-side rate k ,

$$a_A(k) = \frac{c^{\text{res}}}{2} \left(\frac{3(2k)v_A}{4c^{\text{res}}} \right)^{2/3}, \quad a_E(k) = \frac{c_E}{2} \left(\frac{3(2k)v_B^{\text{eff}}}{4c_E} \right)^{2/3}, \quad (159)$$

and $x = |\rho_h| \kappa_E / \kappa_A$:

(i) *The sandwich.* If $x < 1$,

$$a_A(\kappa_A - |\rho_h| \kappa_E) + a_E(\kappa_E) \leq a \leq a_A(\kappa_A + |\rho_h| \kappa_E) + a_E(\kappa_E), \quad (160)$$

and in general a is at least the largest $a_A(k'_A) + a_E(k'_E)$ over $k'_A + |\rho_h| k'_E \leq \kappa_A$, $k'_E \leq \kappa_E$. A third lower bound works through the ETF's own gap, which only the ETF moves: $a \geq a_b(\kappa_E)$, the one-instrument eigenvalue with curvature $c^{\text{res}} c_E / (c^{\text{res}} + \rho_h^2 c_E)$ and variance v_B .

(ii) *Uncorrelated risks.* If $\Sigma_{AE} = 0$, $a = a_A(\kappa_A) + a_E(\kappa_E)$, whatever r .

(iii) *The gap.* Relative to $a_A(\kappa_A)$, the bounds differ by $(1+x)^{2/3} - (1-x)^{2/3} \leq 2[1 - (1-x)^{2/3}]$, of first order in x .

(iv) *An effective rate and the ends.* When $x \leq 1$, $a = a_A(k^*) + a_E(\kappa_E)$ for some k^* within $|\rho_h| \kappa_E$ of κ_A . As $\kappa_E \rightarrow 0$, $a \rightarrow a_A(\kappa_A)$, the frictionless end; as $\kappa_E \rightarrow \infty$, $a \rightarrow \infty$ through the third bound, so the frozen end is not an ergodic object.

The bounds come from separable test functions built from the one-instrument corrector solutions of Proposition 42 (iii), which are classical sub- and supersolutions of the pair's problem with matching growth. The comparison principle for the multidimensional corrector ([52], Theorems 3.1-3.2 and Corollary 6.1, audited for ledger entry AX-16) places the eigenvalue between any such sub- and supersolution's values; the one-instrument solutions are ledger entry AX-15 ([57], equations (4.3)-(4.5); [46], Section 4). Part (iii) is the concavity of $t \mapsto t^{2/3}$.

At leading order, then, the ETF's cost reaches the fund as a shift of the fund's rate by at most the re-hedge cost $|\rho_h|\kappa_E$, with the fund's residual curvature and its own innovation variance; the idle target's variance of Proposition 43 enters no bound. This is Proposition 38's static re-hedge term transported to the fine regime. The result bounds the pair's cost, not the fund's band: where k^* lies, and the non-contact set between the ends, are open. The identification of the discrete quarterly problem with the corrector problem is cited, not shown, as for Proposition 42.

Every result so far with a funded budget is for one review. The last result puts a budget that may bind into a two-review problem with one fund, one ETF and cash, and says exactly what the budget adds to today's decision.

Proposition 50 (Two reviews with a binding budget: the dynamic cash price and the incumbent values; learning-driven tracking model, finite-law variant, one fund and one ETF; machine checked, with necessity of the lines and the myopic-optimality test conditional on the polyhedral Karush-Kuhn-Tucker theorem). Take the finite-law variant of the learning-driven tracking model with one fund and one ETF, holdings $x = (a, p)$, reviews $t = 0, 1$ and marking at 2, a finite state law $q(z')$ for tomorrow with gross returns $g(z') > 0$, predictive moments (μ_t, Σ_t) , discount β , rates $\kappa_i^\pm$, caps, and the funded budget at both reviews, $h_0^+ \geq 0$ and $h_1^+(z') \geq 0$, cash earning nothing. Write $g_{t,i} = (\mu_t - \gamma \Sigma_t x_t)_i$ for the smooth marginals and T for Proposition 44's trade-sign slope sets. The *repeated one-review* (myopic) policy solves Proposition 47's one-review problem today, and again in each state tomorrow from the marked holdings and the cash left.

(i) *Structure.* An optimal policy exists; tomorrow's value is concave in the carried holdings and cash and nondecreasing in cash; and the root problem is a concave maximization over a polyhedron.

(ii) *Today's lines.* A policy is optimal exactly when, for multipliers $\eta_0, \eta_1(z')$ complementary to the two budgets and slopes $t_0, t_1(z')$ in their trade-sign sets, tomorrow's one-review lines hold in every state and today's lines hold with the *dynamic cash price* and the *incumbent values*

$$\hat{\eta}_0 = \eta_0 + \beta \sum_{z'} q(z') \eta_1(z'), \quad S_i = \beta \sum_{z'} q(z') g_i(z') s_i(z'), \quad s_i(z') = \eta_1(z') + (1 + \eta_1(z')) t_{1,i}(z'), \quad (161)$$

$$g_{0,i}(x_0) + S_i - \hat{\eta}_0 - (1 + \hat{\eta}_0) t_{0,i} \begin{cases} = 0 & 0 < x_{0,i} < \bar{x}_i, \\ \leq 0 & x_{0,i} = 0, \\ \geq 0 & x_{0,i} = \bar{x}_i. \end{cases} \quad (162)$$

The incumbent value $s_i(z')$ of a unit carried into state z' is tomorrow's marginal score if the unit is then held inside its box, its scaled purchase threshold if bought, and its scaled sale threshold if sold. So today's decision is Proposition 47's one-review rule with each instrument's net return raised by S_i and the cash price replaced by $\hat{\eta}_0$.

(iii) *What the budget adds.* If tomorrow's budget is slack in every state, $\eta_1 = 0$, $\hat{\eta}_0 = \eta_0$ and $s_i = t_{1,i}$: today's lines are Proposition 26's edge brackets with today's scaling. A budget that may bind tomorrow therefore adds exactly two things: an upward shift of every threshold today, the purchase threshold by $\beta E \eta_1 (1 + \kappa_i^+)$ and the sale threshold by $\beta E \eta_1 (1 - \kappa_i^-)$, the option value of cash (so the manager buys less and sells more readily today; the claim's own wording, a fall of the sale threshold, is wrong, as red's read of the manuscript found); and the factor $1 + \eta_1(z')$ on tomorrow's slopes inside the incumbent values. In the inputs,

$$\beta E [g_i(\eta_1 - (1 + \eta_1) \kappa_i^-)] \leq S_i \leq \beta E [g_i(\eta_1 + (1 + \eta_1) \kappa_i^+)]. \quad (163)$$

(iv) *When repeating the one-review rule is optimal.* The myopic policy is dynamically optimal exactly when its root holdings satisfy (162) for some multipliers and slopes of its own tomorrow problems and some admissible η_0, t_0 . Tomorrow's multiplier is unique in any state that trades some instrument strictly inside its box. If tomorrow's budget is slack everywhere, the test reads $g_{0,i}(x_0^{\text{my}}) + \beta E[g_i t_{1,i}] - \eta_0 - (1 + \eta_0)t_{0,i}$ with (162)'s signs; with today's budget slack as well, an instrument traded strictly inside its box today needs $\beta E[g_i t_{1,i}] = 0$. For a myopic purchase strictly inside the box with today's budget slack, the root objective along that instrument, with the other holding fixed, lies below the line of slope $r = S_i - \beta E[\eta_1](1 + \kappa_i^+)$ through the myopic root, in both directions.

(v) *What survives.* The two-price rule at both reviews; the band at every review, in the marginal $g_{0,i} + S_i$, of width $(1 + \hat{\eta}_0)(\kappa_i^+ + \kappa_i^-)$ at fixed S_i and $\hat{\eta}_0$; and a costless instrument still anticipates the budget, through the cash price alone: $S_i = \beta E[g_i \eta_1]$.

The problem is one concave program over a polyhedron in today's and tomorrow's holdings. Sufficiency of the lines is a direct Lagrangian bound. Necessity is the polyhedral Karush-Kuhn-Tucker theorem ([53], Theorems 27.4 and 28.2-28.3, cited by theorem number) on the lifted two-review problem, whose tomorrow multipliers become $\hat{\eta}_0$ and S_i when today's lines are read.

The result is the dynamic cash price and the incumbent values: a binding budget tomorrow moves today's thresholds by the expected cash price and credits each position with what a unit of it is worth to tomorrow's problem. The slope r in (iv) says only how the root objective moves along one instrument with the other held fixed: negative reads as *hoarding* cash, positive as *front-loading* a purchase. It is a one-instrument reading, not a rule for what the dynamic policy buys. With both holdings free the joint move can reverse by substitution through the covariance and the shared budget; experiment 047 found this in 12 of 82 cases, one with no binding budget at either review. The two signs are Propositions 12-13's opposite continuation effects, here attributed to the budget.

More than two reviews, several funds or ETFs, the joint comparative static, and the width of today's band in holdings with a binding budget today are not treated. No magnitude is claimed.

Proposition 50's lines use two numbers read from tomorrow's solutions: the cash price and the incumbent values. The last result bounds both in the inputs without solving tomorrow, and turns the myopic test into a statement about when repeating the one-review rule can be right at all.

Proposition 51 (The two-review criterion bounded in the inputs; learning-driven tracking model, finite-law variant, one fund and one ETF; machine checked given tomorrow's multipliers, with the sign result's link to the two-review problem at paper level). Take Proposition 50's setting. Write

$$\bar{\eta}(z') = \max_i \frac{(\mu_{1,i}(z') - \kappa_i^+)^+}{1 + \kappa_i^+}, \quad \hat{x}_i(z') = \frac{(\mu_{1,i}(z') - \kappa_i^+)^+}{\gamma \Sigma_{1,ii}(z')}, \quad \text{need}(z') = \sum_i (1 + \kappa_i^+) (\hat{x}_i(z') - g_i(z') x_{0,i})^+, \quad (164)$$

the best expected return tomorrow net of its purchase rate, instrument i 's solo target, and the cash those targets would need.

(i) *Tomorrow's cash price.* With nonnegative covariances tomorrow, some admissible cash price in each state is at most $\bar{\eta}(z')$, and 0 is admissible when the cash carried covers $\text{need}(z')$. For any family of tomorrow multipliers, $\eta_1 \leq \bar{\eta}$ in every state that trades something or has cash left, so $\eta_0 \leq \hat{\eta}_0 \leq \eta_0 + \beta E[\bar{\eta}]$ unless some state trades nothing with all cash spent.

(ii) *The incumbent values and the residual.* With any admissible η_1 ,

$$-\beta\mathbb{E}[g_i]\kappa_i^- \leq S_i \leq \beta\mathbb{E}[g_i(\eta_1 + (1 + \eta_1)\kappa_i^+)], \quad (165)$$

and the residual $R_i = S_i - \beta\mathbb{E}[\eta_1](1 + \kappa_i^+)$ lies in an explicit bracket. It is negative when tomorrow's cash price is high in the states where the instrument is marked down, by more than its purchase rate covers, and positive when the cash price is high where its marked gain, net of the round trip, covers the purchase.

(iii) *When repeating the one-review rule can be right.* With today's budget slack at the myopic root, an instrument the myopic policy trades strictly inside its box passes the test of Proposition 50 (iv) exactly when $R_i = 0$ for a purchase, or $S_i = \beta\mathbb{E}[\eta_1](1 - \kappa_i^-)$ for a sale, for some admissible tomorrow family. That is an identity for an instrument costless on both sides with a slack budget tomorrow, and otherwise one equation on the inputs. With today's budget binding at the myopic root, the condition is the inequality

$$S_i \geq (\beta\mathbb{E}[\eta_1] - \eta_0^{\text{my}})(1 + \kappa_i^+), \quad (166)$$

which holds on open sets of inputs.

(iv) *A restricted comparative static.* With a frictionless, uncapped ETF held strictly inside its box at the dynamic root and at the reduced myopic point (the joint optimum with the fund held at its myopic holding), today's budget slack at both, and a myopic purchase of the fund strictly inside its box,

$$\text{sign}(a^{\text{dyn}} - a^{\text{my}}) = \text{sign}(R_A^{\text{red}} - \rho_0 R_E^{\text{red}}) \quad (167)$$

whenever the right side is nonzero, with the residuals evaluated at the reduced point and ρ_0 today's hedge ratio.

The bounds in (i) come from tomorrow's one-review lines: a cash price above $\bar{\eta}$ would make every purchase unattractive, and a state whose cash covers the solo targets needs none. (ii) bounds each incumbent value by its regime's thresholds. (iii) evaluates the root line at an interior trade. For (iv), the fund's root line at the reduced point is $R_A^{\text{red}} - \rho_0 R_E^{\text{red}}$, and a concave function of one variable with a positive right derivative at the myopic holding is maximized to its right.

So, with cash left today, repeating the one-review rule is dynamically optimal only where it trades nothing, trades to a bound, or trades an instrument costless both ways with a slack budget tomorrow, apart from inputs on which one equation happens to hold; with cash spent today it can be optimal on open sets. The sign in (iv) is the corrected, restricted form of the hoarding and front-loading reading of Proposition 50. With a costly ETF no comparative static is claimed: experiment 047's instance 106 has the dynamic policy holding more of the fund than the myopic one although the fund's residual is negative, because it sells the ETF instead.

More than two reviews, several funds or ETFs, and the joint admissibility of a per-state selection of tomorrow's multipliers are not treated. No magnitude is claimed.

Propositions 50-51 use the finite-law variant, whose filter is the best linear predictor, not the posterior. The last result puts learning, predictive risk, marking and the budget into one model under either shock law, says what the filter's output means under each, and works an example.

Proposition 52 (One worked model for learning, predictive risk, marking and the budget, under two laws; the learning-driven tracking model with one fund, one ETF and cash over two reviews; machine checked in its finite-law consistency, moment content and the example's first-review numbers, with the filter's two meanings cited and the transfer and later tables at paper level). Take the learning-driven tracking model with one fund (loading b_A) and one ETF (loading $b_E > 0$,

fee c^E), cash, reviews $t = 0, 1$, the funded budget allowed to bind, and a diagonal prior. Under the *Gaussian law* premia, alphas and shocks are Gaussian; under the *finite law* each takes finitely many values with the same first two moments (two values $m_0 \pm \sqrt{P_0}$ per parameter block), so the public histories form a finite tree.

(i) *Consistency*. Under either law the posterior variance path is deterministic and independent of trades, with $P_1^{-1} = P_0^{-1} + \text{diag}(1/\sigma_f^2, 1/\sigma_A^2)$; the predictive covariance is positive definite and the target well defined at every history. Under the finite law marking keeps holdings nonnegative, every review's feasible set is nonempty, its optimum unique, and the two-review problem has an optimal policy.

(ii) *The filter's two meanings*. Under the Gaussian law the filter's mean is the exact posterior mean ([47], the conjugate Gaussian update, ledger entry AX-18). Under any law with the same first two moments it is the minimum-mean-squared-error affine predictor ([61], on Kalman's Corollary 1, ledger entry AX-17, which states the Gaussian case as sufficient only). Under the finite law the innovation has mean zero and covariance $P_t - P_{t+1}$, the exact posterior mean averages to m_0 , and the two means differ at a history reached from exactly two hidden states $m_0 \pm \sqrt{P_0}$ in a block whose filter innovation is nonzero: the posterior returns to m_0 while the filter moves.

(iii) *What transfers*. Every one-review result holds at each review under the finite law, and under the Gaussian law wherever the marked holdings are nonnegative, since the stage score uses only the first two moments. The multi-review band results hold under the finite law with a slack budget, which holds at every pair of in-box holdings when initial cash is at least $2(1 + \kappa) \sum_i \bar{x}_i$ with every purchase rate at most κ . From the quadratic-cost model only the target's structure transfers. Unproved: the dynamic optimum under the Gaussian law, the exact-posterior manager's policy under the finite law, and any dependence of the dynamic policy on the finite law beyond its moments.

(iv) *The worked example* (all inputs assumed; an illustration). With $b_A = 0.9$, $b_E = 1$, $c^E = 5$ basis points, prior premium 1.0% (sd 0.4%) and alpha 0.4% (sd 2.0%) a quarter, $\gamma = 2.5$, $\beta = 1$, rates of 50 basis points for the fund and 10 for the ETF, caps of 1, and incumbents fund 0.40, ETF 0, cash 0.06: the gains are 1/402 and 1/11, today's target is (0.406, 0.225), the budget binds with a cash price of 0.174%, and both the one-review rule and the two-review optimum hold the fund and buy the ETF to 0.060. After an ambiguous observation the filter raises the alpha estimate by 0.55 percentage points while the exact posterior stays at the prior; both managers hold. A manager using the exact posterior gains 8.6 basis points over the two quarters, all at the second review, with decisions differing at 42 of the 72 public nodes.

Parts (i)-(ii) combine the scalar filter recursion with the finite tree's Bayes computation; the two meanings are the cited ledger entries, whose hypotheses the model's laws meet. Part (iii) reads each result's hypotheses against the model. Part (iv) evaluates the formulas and solves the finite two-review program.

In this model, current learning, today's updated means and predictive risk, enters every decision under either law; the law changes what the filter's mean is, not the rule that uses it. The cheap ETF, not cash, is the reserve for tomorrow's adjustment at these inputs.

Propositions 50-52 say how tomorrow enters today's decision in the two-review model. The last result reads them as a rule for the reserve: which part of today's ETF and cash beyond the one-review optimum answers tomorrow's revision, when none is needed, where it is held, how it grows and how large it can be.

Proposition 53 (The reserve rule with one fund and one ETF; the two-review worked model of Proposition 52, finite law; machine checked in its bounds, certificate and shift, with the direction reading, the computation's fixed point and the Gaussian tail bound at paper level). Take Proposition 52's model. Let $x_0^{\text{my}} = (a^{\text{my}}, p^{\text{my}})$ with leftover cash h^{my} be the one-review optimum today,

x_0^{dyn} the dynamic optimum, and, for holdings x_0 and a state z' , the solo targets $\hat{x}_i(z')$ and $\text{need}(z'; x_0)$ of (164), with $N(x_0) = \max_{z'} \text{need}(z'; x_0)$ the largest funding need tomorrow. The alpha revision has variance $v^\alpha = k_0^\alpha p_0^\alpha$, the gain times the prior variance, and extreme $\epsilon_{\max}^\alpha$; $g_i^{\min}$ is instrument i 's smallest gross return.

(i) *Two channels, two reserves.* Today's ETF and cash differ from the one-review ones through Proposition 50's two channels. The *cost-channel reserve*, present with slack budgets, is carried by $S_E = \beta \mathbb{E}[g_E t_{1,E}]$, the ETF's expected marked cost slope tomorrow: the ETF is held below its one-review level when that slope is expected negative and above when positive, with cash taking the difference. The *budget-channel reserve* comes from states that price cash.

(ii) *When no reserve is needed.* The cost channel vanishes exactly when $\mathbb{E}[g_E t_{1,E}] = 0$. The budget channel vanishes at any root x_0 with cash $h > 0$ and $h \geq N(x_0)$, and in the inputs

$$N(x_0^{\text{my}}) \leq \sum_{i \in \{A,E\}} (1 + \kappa_i^+) \left[\frac{(\mu_{1,i}^{\max} - \kappa_i^+)^+}{\gamma \Sigma_{1,ii}} - g_i^{\min} x_{0,i}^{\text{my}} \right]^+, \quad (168)$$

with $\mu_{1,A}^{\max} = \mu_{0,A} + b_A \epsilon_{\max}^\lambda + \epsilon_{\max}^\alpha$ and $\mu_{1,E}^{\max} = \mu_{0,E} + b_E \epsilon_{\max}^\lambda$: leftover cash covering the largest solo-target purchases at the revision's extreme suffices. As a certificate, a root of Proposition 50's lines with $\eta_1 = 0$ in every state whose cash is positive and covers its own need is the dynamic optimum. When cash does not cover every state, the budget channel's terms are bounded by the uncovered states' moments, $\beta \mathbb{E}[\eta_1] \leq \beta \mathbb{E}[\bar{\eta} \mathbf{1}_U]$, and ignoring the reserve errs in each purchase threshold by at most $(1 + \kappa_i^+)$ times that.

(iii) *ETF against cash, and the size.* With today's budget slack, the ETF's root line differs from its one-review line by the *reserve premium* $\Pi_E = S_E - \beta \mathbb{E}[\eta_1](1 + t_{0,E})$, which by state reads as the ETF's marked proceeds in the states where cash is dear against its cost today. With both budgets slack and the fund fixed at a^{my} ,

$$p^{\text{fix}} - p^{\text{my}} = \frac{S_E - (t_{0,E}^{\text{fix}} - t_{0,E}^{\text{my}})}{\gamma \Sigma_{0,EE}}, \quad |p^{\text{fix}} - p^{\text{my}}| \leq \frac{\beta \max(\kappa_E^+, \kappa_E^-) \mathbb{E}[g_E] + \kappa_E^+ + \kappa_E^-}{\gamma \Sigma_{0,EE}}, \quad (169)$$

so the shift is exactly $S_E/(\gamma \Sigma_{0,EE})$ when the ETF trades the same way today under both, and zero when the ETF is held untraded inside today's band and the band absorbs S_E . Here S_E is read from the tomorrow of the point p^{fix} itself: (169) is a fixed-point condition, not a size read from the one-review tomorrow.

(iv) *How the need grows.* On its active set the fund's need is affine in the alpha revision,

$$\frac{\partial \text{need}_A}{\partial \epsilon^\alpha} = \frac{1 + \kappa_A^+}{\gamma \Sigma_{1,AA}}, \quad (170)$$

and an ETF reserve funding a need n must be $n/((1 - \kappa_E^-)g_E^{\min})$ units. The reserve itself is not monotone in the inputs.

(v) *The maximum.* If some state prices cash for every admissible family of tomorrow multipliers, the cash carried at the dynamic optimum satisfies

$$h^{\text{dyn}} \leq N(x_0^{\text{dyn}}), \text{ strictly when } h^{\text{dyn}} > 0. \quad (171)$$

The results are Propositions 50-51's objects sized in the model's inputs, the revision's law included; no new mechanism is claimed. The bounds read tomorrow's one-review lines at the revision's extremes; (169)

solves the ETF's root line with the fund fixed; (171) holds because a positive cash price tomorrow in some state requires that state's need to exceed the cash carried.

The rule's computation. Fix the fund at a^{my} . If the one-review root holds the ETF untraded and $t^{\text{my}} + S_E$ stays in its band, hold it; otherwise solve $g_{0,E}(a^{\text{my}}, p) + S_E(p) - t_{0,E}(p) = 0$ for p , with $S_E(p)$ read from the tomorrow of p itself. A one-shot reading from the one-review tomorrow can mislead: experiment 053 found it worse than repeating the one-review rule in 16 of 84 cells. Applied as a fixed point, it tied the dynamic optimum in all 93 of experiment 054's cells where its hypotheses hold, and where the dynamic policy also sells the fund it leaves 0.10 to 0.26 basis points (illustrations of the tested cells).

Several funds, an ETF reserve under the Gaussian law beyond the tail bound, and the reserve's joint move with the fund free are not treated. No magnitude is claimed.

Proposition 53 is for one fund. The last result extends the reserve to several funds sharing one ETF and cash, and says exactly what a fund-by-fund rule misses.

Proposition 54 (Several funds sharing one ETF reserve; the two-review model with N funds and one ETF, finite law; machine checked, with the cash price's common root, the pooling remark and the carried-over cost channel at paper level). Take Proposition 50's model with N funds of loadings $b_i > 0$ and one ETF of loading $b_E > 0$, with Proposition 47's netting weights $r_i = b_i/b_E$, exposure price m_i and net alphas $\tilde{\alpha}_{t,i}$. Add the ETF's solo sale threshold $\hat{x}_E^s(z') = (\mu_{1,E} + \kappa_E^-)/(\gamma \Sigma_{1,EE})$, the aggregate need over every instrument, and the liquidity $\text{liq}(z') = h_0^+ + (1 - \kappa_E^-)(g_E(z')p_0 - \hat{x}_E^s(z')^+)^+$, today's cash plus the ETF's sale proceeds down to that threshold.

(i) *Two shared scalars.* Every fund's root holding is Proposition 47's clip at the root's exposure price m_0 and the dynamic cash price $\hat{\eta}_0$, with its incumbent value S_i added to its net alpha. Given the vector of incumbent values, the funds couple only through m_0 and $\hat{\eta}_0$.

(ii) *The aggregate no-reserve check.* If $\text{liq}(z') \geq \text{need}(z')$ at a state, $\eta_1(z') = 0$ is admissible there; if it holds in every state, some admissible family has $\eta_1 = 0$ everywhere, so $\hat{\eta}_0 = \eta_0$ and every incumbent value lies in Proposition 26's bracket, and with positive cash today every family does. This strengthens Propositions 51 (i) and 53 (ii) by counting the ETF's sale proceeds and summing the need over the funds. The largest need is the reserve's ceiling, and $\max_{z'} \text{need}(z') \leq \sum_i \max_{z'} \text{need}_i(z')$, with equality when one state attains every instrument's maximum.

(iii) *What fails fund by fund.* If fund i trades strictly inside its box in the same direction at the dynamic and the myopic roots, with slope κ ,

$$a_{0,i}^{\text{dyn}} - a_{0,i}^{\text{my}} = \frac{S_i + r_i(m_0^{\text{dyn}} - m_0^{\text{my}}) - (\hat{\eta}_0 - \eta_0^{\text{my}})(1 + \kappa)}{\gamma v_i} : \quad (172)$$

the fund's own residual, corrected by the changes in the two shared prices, which are the same for every fund and can flip its sign. At a state whose budget binds, the cash price is one number for all funds, so a rule that prices each fund's reserve with its own cash price is wrong whenever two funds would buy there with liquidity short.

(iv) *The reserve.* The reserve at par, $R = (p_0 + h_0^+)^{\text{dyn}} - (p_0 + h_0^+)^{\text{my}}$, equals the funds' holdings forgone plus the cost difference,

$$R = \sum_i (a_{0,i}^{\text{my}} - a_{0,i}^{\text{dyn}}) + [C(u_0^{\text{my}}) - C(u_0^{\text{dyn}})], \quad (173)$$

and is bounded by (172)'s numerators over γv_i plus the cost difference; it can be negative when the dynamic policy front-loads funds. With every state covered the bound has no cash-price terms.

Part (i) substitutes the incumbent values into Proposition 47's one-review reduction. Part (ii) builds a feasible tomorrow without the budget from the ETF's sale down to its solo threshold. Part (iii) subtracts the two roots' fund lines on a common trading piece. Part (iv) is the wealth identity with the clip's Lipschitz bound.

A shared reserve therefore pools across funds only through the states: the ceiling is the worst state's aggregate need, and what the cash price weighs is that state's probability. Several ETFs, and a bound on the exposure price's change in the inputs, are not treated.

Propositions 50-54 have one ETF. The last result takes many funds and many ETFs and asks whether spanning separates the flexibility problem as it separates the one-review problem (Propositions 27 and 45).

Proposition 55 (Many funds and many ETFs over two reviews: the two-number structure, and spanning separates the root's lines given the tomorrow numbers; the two-review model with N funds and M ETFs, finite law; machine checked, with necessity of the lines conditional on the polyhedral Karush-Kuhn-Tucker theorem and the failure with costly ETFs a reading). Take Proposition 50's model with N funds, M ETFs and K factors.

(i) *Two numbers per instrument, one cash price.* A policy is optimal exactly when tomorrow's lines hold in every state and every instrument's root line holds with the dynamic cash price $\hat{\eta}_0$ and its own incumbent value S_i , as in (162). Given $(S, \hat{\eta}_0)$, the root lines are the one-review lines with μ_0 replaced by $\mu_0 + S$ and the cash price by $\hat{\eta}_0$; nothing couples the instruments beyond the one-review coupling.

(ii) *Spanning separates the lines.* With $M = K$, B^E invertible ($R = (B^E)^{-1}$, netting vectors r_i), frictionless residual-free ETFs without fees, a diagonal fund covariance, and today's budget slack with the ETFs strictly inside their boxes at the dynamic root, write

$$\hat{\lambda}^{\text{res}} = \hat{\lambda} + R(S_E - \hat{\eta}_0 \mathbf{1}), \quad \alpha_i^{\text{res}} = \hat{\alpha}_i + S_{A,i} - r_i^\top S_E - \hat{\eta}_0(1 - r_i^\top \mathbf{1}). \quad (174)$$

Then the dynamic root is Proposition 45 (ii)'s optimum at these tilted inputs:

$$\begin{aligned} \gamma \tilde{\Sigma}_{f,0} b_0 &= \hat{\lambda}^{\text{res}}, \quad x_0^E = R^\top(b_0 - B^{A\top} x_0^A), \\ x_{0,i} &= \text{clip}\left(x_{0,i}^-, \frac{\alpha_i^{\text{res}} - (1 + \hat{\eta}_0)\kappa_i^+}{\gamma v_i}, \frac{\alpha_i^{\text{res}} + (1 + \hat{\eta}_0)\kappa_i^-}{\gamma v_i}\right) \text{ in } [0, \bar{x}_i]. \end{aligned} \quad (175)$$

So, given the tomorrow numbers at the optimum, the root's optimality conditions separate into an *ETF reserve problem*, the frictionless exposure at a premium tilted by the ETFs' incumbent values net of the cash price, and a *fund alpha problem*, each fund's clip at a residual alpha that carries its own incumbent value, less the netting-weighted ETF incumbent values, less the cash price on its net cash. The tomorrow numbers are set by the whole root, so the two problems share them and are a fixed point, not two independent problems as at one review.

(iii) *The ETFs' interaction.* The exposure departs from the untilted target by

$$\gamma \tilde{\Sigma}_{f,0}(b_0 - b^{\text{TB}}) = R(S_E - \hat{\eta}_0 \mathbf{1}), \quad S_{E,j} - \hat{\eta}_0 = \beta \mathbb{E}[(g_{E,j} - 1)\eta_1], \quad (176)$$

since frictionless ETFs have zero slope tomorrow: the ETFs share the reserve only through the factor covariance's inverse, and those marked up in the states where cash is dear are held as the reserve. With one ETF this is Proposition 53's shift.

(iv) *Where separation fails.* With costly ETFs, or an ETF at a bound today, the root is Proposition 41's one-review problem at the tilted inputs, whose fund lines carry the ETFs' statuses: the lines fail to separate exactly as at one review. With one ETF, Proposition 47's two-scalar reduction holds at the root.

Part (i) is Proposition 50's argument, which never used the number of instruments. Part (ii) substitutes the tilted inputs into Proposition 45 (ii); sufficiency is direct, and necessity reads the lines at an optimum with today's budget slack. Part (iii) solves (175)'s exposure equation and uses $t_{1,E} = 0$.

Spanning thus still splits the manager's question into exposure and alpha, but over two reviews the split is of the optimality conditions, not of the optimization: both halves read the same tomorrow numbers, which depend on the whole of today's holdings. Separation in the optimization itself, and many ETFs with costs, are not treated.

Propositions 51, 53 and 54 test tomorrow's funding state by state and ask that every state be covered. The last result states the test for the full menu, lets a stated fraction of states fail, and bounds what repeated one-review optimization then loses, in the inputs.

Proposition 56 (A flexibility test for the full menu, and the loss of repeated one-review optimization; the two-review model with N funds and M ETFs, finite law; machine checked, with the multipliers' existence conditional on the polyhedral Karush-Kuhn-Tucker theorem and the one-ETF form of the band term at paper level). Take Proposition 55's model, with tomorrow's predictive covariance entrywise nonnegative in every state. At today's holdings x_0 and cash h , take (164)'s solo targets, need and cash-price bound $\bar{\eta}$ over every instrument, each ETF's solo sale threshold $\hat{x}_E^s(z')$ as in Proposition 54, and write

$$\begin{aligned} \text{liq}(z') &= h + \sum_E (1 - \kappa_E^-) (g_E(z') x_{0,E} - \hat{x}_E^s(z')^+)^+, & D(z') &= (\text{need}(z') - \text{liq}(z'))^+, \\ T_\varepsilon(Y) &= \min_c \left[c + \frac{\mathbb{E}(Y - c)^+}{\varepsilon} \right], & \varepsilon &\in (0, 1], \end{aligned} \quad (177)$$

the *liquid reserve*, today's cash plus every ETF's sale proceeds down to its solo sale threshold net of its rate; the *shortfall*; and the tail measure of Rockafellar and Uryasev at level $1 - \varepsilon$ ([54], audited for the ledger).

(i) *Coverage.* If $\text{liq}(z') \geq \text{need}(z')$, then $\eta_1(z') = 0$ is admissible at z' : tomorrow's budget is slack there.

(ii) *The quantile test.* For $\varepsilon \in [0, 1]$ the test *passes at level ε* when the shortfall's $(1 - \varepsilon)$ -quantile is zero, equivalently $\Pr(D(z') > 0) \leq \varepsilon$: at most a fraction ε of tomorrow's states are uncovered. When the liquid reserve does not depend on the state, this reads $\text{liq} \geq \text{VaR}_{1-\varepsilon}(\text{need})$. At $\varepsilon = 0$ it is Proposition 54 (ii)'s all-states check. Whenever the test passes at a level $\varepsilon > 0$, $\mathbb{E}[\bar{\eta}D] = \varepsilon T_\varepsilon(\bar{\eta}D) \leq \varepsilon \max_{z'} \bar{\eta}D$.

(iii) *The loss bound.* Let $S_i = \beta \mathbb{E}[g_i t_{1,i}]$ be the incumbent values at the myopic root's *relaxed* tomorrow, with tomorrow's budget dropped, and take D and the uncovered fraction ε at the myopic root $(x_0^{\text{my}}, h^{\text{my}})$. With V^{dyn} the dynamic optimum's value and $J(x_0^{\text{my}})$ the repeated one-review policy's,

$$\begin{aligned} V^{\text{dyn}} - J(x_0^{\text{my}}) &\leq \frac{1}{2} S^\top (\gamma \Sigma_0)^{-1} S + \beta \mathbb{E}[\bar{\eta}(z') D(z')] \\ &\leq \frac{\beta^2}{2} \|(\gamma \Sigma_0)^{-1}\| \sum_i (\mathbb{E}[g_i] \max(\kappa_i^+, \kappa_i^-))^2 + \beta \varepsilon T_\varepsilon(\bar{\eta}D), \end{aligned} \quad (178)$$

a *band term* plus a *tail term*, with the operator norm, the tail term read as zero at $\varepsilon = 0$. The second line is in the inputs and needs no solution of tomorrow's problems. At $\varepsilon = 0$ only the band term remains, second order in the rates. With one ETF and the funds' incumbent values zero, the band term is $S_E^2 / (2\gamma \Sigma_{EEF})$,

with Σ_{EEF} the ETF's variance net of its regression on the funds; it equals Proposition 53's cost-channel shift times $S_E/2$ only with the funds fixed or uncorrelated with the ETF.

Part (i) builds a fundable tomorrow without the budget, as in Proposition 54 (ii), selling each ETF down to its threshold. Part (ii) reads the quantile on the finite law; when $\Pr(Y > 0) \leq \varepsilon$ the tail objective's least value is $\mathbb{E}[Y]/\varepsilon$, attained at $c = 0$. Part (iii) bounds the dynamic value by the relaxed problem, whose gain over the myopic root is at most the completed square of Proposition 48 by today's strong concavity and the relaxed tomorrow's supergradient S ; and it bounds the myopic policy's loss from tomorrow's budget by the Lagrangian at the cash price $\min(\eta_1, \bar{\eta})$, which costs nothing in covered states.

A manager can therefore sort tomorrow's states by how far the purchases they call for exceed the cash plus what the ETFs would fetch. If at most a fraction ε are short, re-optimizing each quarter with bands loses at most a term of second order in the rates plus ε times the short states' average cash-price bound times the shortfall. The bound takes premia and alphas as fixed; the test with time-varying premia is not treated here.

Propositions 50-56 take premia and alphas as fixed and learned, so the target moves only because beliefs are revised. The last result lets them move on their own.

Proposition 57 (The predictable and unpredictable target move under time-varying premia and alphas; Proposition 52's model with a state equation, one fund and one ETF over two reviews, finite law; machine checked in the decomposition, the width's direction, the tilt's probabilities and the one-coordinate front-loading with its threshold, with the case of a free ETF, the planned purchase, the transfer and the steady-state filter's form at paper level). Take Proposition 52's model and let premia and alphas follow $\theta_{t+1} = \Phi\theta_t + (I - \Phi)\bar{\theta} + w_{t+1}$, with persistence $\Phi = \text{diag}(\phi_\lambda, \phi_\alpha)$, long-run mean $\bar{\theta}$ and state noise of covariance $Q = \text{diag}(q_\lambda, q_\alpha)$, not both $\Phi = I$ and $Q = 0$. The filter carries the mean m_t , the variance P_t entering review t and P_t^u after its update, the gain K_t and the innovation v_{t+1} ; G maps the state to the instruments' expected returns, μ_t and Σ_t are the predictive moments, $x_t^* = (\gamma\Sigma_t)^{-1}\mu_t$ is the target, and $c_t = \gamma\Sigma_{t,ii}$ an instrument's curvature.

(i) *Two parts.* $x_1^* - x_0^* = \Delta_0^P + \Delta_1^u$, with

$$\begin{aligned} \Delta_0^P &= (\gamma\Sigma_1)^{-1}G(\Phi - I)(m_0 - \bar{\theta}) + [(\gamma\Sigma_1)^{-1} - (\gamma\Sigma_0)^{-1}]\mu_0, & \Delta_1^u &= (\gamma\Sigma_1)^{-1}G\Phi K_0 v_1, \\ \mathbb{E}\Delta_1^u &= 0, & \text{Cov } \Delta_1^u &= (\gamma\Sigma_1)^{-1}GV_0G^\top(\gamma\Sigma_1)^{-1}, & V_0 &= \Phi(P_0 - P_0^u)\Phi^\top. \end{aligned} \quad (179)$$

The *predictable* part Δ_0^P is known today, since Σ_1 is deterministic; it has two sources, the pull toward $\bar{\theta}$ at rate $I - \Phi$ and the change in the risk charge, which now has either sign. The *unpredictable* part Δ_1^u is centred, with the filter's propagated innovation covariance.

(ii) *The static width's direction.* $\Sigma_1 - \Sigma_0 = G(P_1 - P_0)G^\top$ with $P_1 - P_0 = Q - (P_0 - \Phi P_0^u \Phi^\top)$. An instrument's static width $(\kappa_i^+ + \kappa_i^-)/c_t$ of Proposition 26 rises from t to $t+1$ exactly when $(G(Q - (P_t - \Phi P_t^u \Phi^\top))G^\top)_{ii} < 0$. If learning removes more variance than the state noise adds in both blocks, every width rises; if less, every width falls; when the blocks move oppositely, the fund's and the ETF's widths can move in opposite directions. Each block's variance recursion converges monotonically to a fixed point, unique when $q > 0$ or $\phi^2 < 1$, at which the widths are constant.

(iii) *The band leans toward the predictable move.* In Proposition 26's coarse regime the band is the static band shifted by $\tau_t = \beta(\kappa^+ U - \kappa^- D)/c_t$, with $U = \Pr(\Delta_{t+1,i}^u > -\Delta_{t,i}^P)$ and $D = \Pr(\Delta_{t+1,i}^u < -\Delta_{t,i}^P)$. With a symmetric innovation law and $\kappa^+ = \kappa^- = \kappa$, τ_t has the sign of $\Delta_{t,i}^P$ (or is zero) and, without an atom at $-\Delta_{t,i}^P$, $|\tau_t| = \beta\kappa \Pr(|\Delta_{t+1,i}^u| \leq |\Delta_{t,i}^P|)/c_t$: the band leans toward the predictable move by the probability that the

innovation is smaller than the predictable move, the full $\beta\kappa/c_t$ when the move exceeds the innovation's support. With $\Phi = I$ and $Q = 0$ this is Proposition 31's tilt.

(iv) *Front- and back-loading.* Suppose the budget is slack today at the dynamic and the myopic roots and tomorrow in every state, the ETF's holding is fixed, $\kappa_A^+ > 0$, and tomorrow's purchase edge exceeds the myopic root's marked fund holding $g_A(z')a_0^{\text{my}}$ in every state. Then the myopic tomorrow pins the fund's incumbent value at $S_A = \beta E[g_A]\kappa_A^+$, and $a_0^{\text{dyn}} \geq a_0^{\text{my}}$. At the dynamic root its own incumbent value satisfies $S_A^{\text{dyn}} \leq \beta E[g_A]\kappa_A^+$, so on a purchase

$$\kappa_A^+ - S_A^{\text{dyn}} \geq (1 - \beta E[g_A])\kappa_A^+, \quad (180)$$

with equality exactly when every state's slope tomorrow is κ_A^+ , the fund bought, or held at its purchase edge, from the dynamic root's marked holding. The predictable rise is bought today at a threshold cut by at most $\beta E[g_A]\kappa_A^+$, because tomorrow would pay the rate anyway; since a larger holding makes tomorrow's purchase less likely, the full cut is the exception. A predictable fall is the mirror, with $\kappa_A^- > 0$, $a_0^{\text{dyn}} \leq a_0^{\text{my}}$ and the sale threshold. With the ETF free, the sign of the fund's move is (172)'s: S_A corrected by the changes in the exposure and cash prices, so the ETF's own predictable move can offset the fund's.

(v) *The planned purchase.* Tomorrow's solo target for the fund is $\hat{x}_A(z') = (\mu_{1,A}^p + (G\Phi K_0 v_1(z'))_A - \kappa_A^+)^+ / (\gamma \Sigma_{1,AA})$, with $\mu_1^p = G(\Phi m_0 + (I - \Phi)\bar{\theta}) - (0, c^E)$ tomorrow's *predictable mean*. The need (164) at $v_1 = 0$ with the worst marking, the *planned purchase* $(1 + \kappa_A^+)(\hat{x}_A^p - g_A^{\min} a_0)^+$ with $\hat{x}_A^p = (\mu_{1,A}^p - \kappa_A^+)^+ / (\gamma \Sigma_{1,AA})$, is a base set by the predictable mean; the innovation contributes the rest of the need (the positive parts do not split in general). Propositions 53-56 read with this need: the reserve prices the unpredictable part, through the innovation covariance in (179), while the predictable part is front-loaded when today's budget is slack or, when it binds, is a planned purchase that tomorrow's liquidity must cover, best provided by the ETF's sale since its size is known today. A predictable fall beyond the innovation's support needs no reserve.

(vi) *What transfers.* The one-review results hold at each review. Propositions 50, 51, 53 and 54 hold verbatim on the finite tree, which the state equation changes only through the law of tomorrow's moments. Proposition 26's band holds at every review with c_t from the variance path, its tilt given by (iii). Proposition 31's rising width and outward drift hold for an instrument only while (ii)'s condition does, that is, while learning outweighs the state noise; in the steady state the width is constant and learning does not end the coarse regime. Under quadratic costs the aim's structure is Proposition 34 (iii)'s, as a structure, not a policy.

Part (i) is the filter's predict-and-update algebra. Part (ii) writes the variance change and uses that each block's recursion $p \mapsto \phi^2 ps / (p + s) + q$ is increasing, concave and bounded. Part (iii) is Proposition 26's tilt with the learning drift replaced by Δ^p . Part (iv) places the root objective below the myopic root's tangent plane with slopes S , pins S_A from the myopic tomorrow's purchases, shows every smaller holding lowers the root objective strictly, and bounds S_A^{dyn} by the bracket $s_A \leq \kappa_A^+$. Part (v) substitutes μ_1 into (164). Part (vi) checks each result's hypotheses against the state equation.

So when the target moves on its own, its next move splits into the part known today and the surprise. The known part tilts the band toward it and, when tomorrow would trade anyway, is traded today instead, or else is a purchase to plan cash for; the surprise is what a reserve is for, sized by its dispersion as before. The exact posterior with a state transition and several ETFs under the state equation are not treated here; the flexibility test with time-varying premia is the next result.

Proposition 56's test takes premia and alphas as fixed. The last result combines it with Proposition 57's split of tomorrow's need.

Proposition 58 (The quantile flexibility test with time-varying premia, and the timing of the planned purchase; Proposition 57's model with one fund and one ETF over two reviews, finite law; machine checked in the split test, the bound at any holding and the pinned incumbent value, with the front-loaded point's zero residual conditional on the polyhedral Karush-Kuhn-Tucker theorem at paper level, and the assembly of (183), the mirror, the front-loading reading and the Gaussian quantile at paper level). Take Proposition 57's model and Proposition 56's objects at today's holdings x_0 and cash h : the need (164), the liquid reserve, the shortfall D , the bound $\bar{\eta}$ and the tail measure T_ε of (177). With tomorrow's predictable mean μ_1^p of Proposition 57 (v) and $g^{\min}$ the smallest gross return on the support, write

$$\hat{x}_i^p = \frac{(\mu_{1,i}^p - \kappa_i^+)^+}{\gamma \Sigma_{1,ii}}, \quad \text{PP}(x_0) = \sum_i (1 + \kappa_i^+) (\hat{x}_i^p - g_i^{\min} x_{0,i})^+, \quad \Delta(z') = \text{need}(z') - \text{PP}(x_0), \quad (181)$$

the *planned purchase*, the need at zero innovation with the worst marking, known today, and the *innovation-driven part*, of either sign. For a holding x_0 in today's feasible set C with cash $h(x_0)$, the *root residual* $\rho(x_0)$ is the shortest vector, in the norm of $(\gamma \Sigma_0)^{-1}$, of the form $g_0(x_0) + S - l$. Here S ranges over the admissible incumbent values at the relaxed tomorrow from x_0 , tomorrow's budget dropped, where an instrument held at a bound tomorrow has any slope in the interval its one-sided line cuts from $[-\kappa_i^-, \kappa_i^+]$; and l ranges over today's line values $\eta \mathbf{1} + (1 + \eta)t + n$, with $\eta \geq 0$ (zero when $h(x_0) > 0$), t in today's trade-sign slope sets and n in the box's normal cone.

(i) *The test in split form.* Proposition 56 holds on this model's tree verbatim. The test at level ε passes exactly when $\Pr(\Delta(z') > \text{liq}(z') - \text{PP}(x_0)) \leq \varepsilon$, and, with a reserve that does not depend on the state, exactly when $\text{liq} - \text{PP}(x_0) \geq \text{VaR}_{1-\varepsilon}(\Delta)$: the liquid reserve covers the planned purchase in full and the $(1 - \varepsilon)$ -quantile of the innovation-driven part.

(ii) *The loss bound at any holding.* For every x_0 in C , with D , $\bar{\eta}$ and ε at $(x_0, h(x_0))$ and $J(x_0)$ the value of holding x_0 today and optimizing tomorrow,

$$V^{\text{dyn}} - J(x_0) \leq \frac{1}{2} \rho(x_0)^\top (\gamma \Sigma_0)^{-1} \rho(x_0) + \beta \mathbb{E}[\bar{\eta}(z') D(z')]. \quad (182)$$

At the myopic root the band term is at most Proposition 56's, so (182) contains (178). At the *front-loaded point* x_0^s , the relaxed two-review optimum, $\rho(x_0^s) = 0$; under the test at a level $\varepsilon > 0$ there, $V^{\text{dyn}} - J(x_0^s) \leq \beta \varepsilon T_\varepsilon(\bar{\eta} D)$. Front-loading the planned purchase removes the band term and leaves the uncovered states' loss.

(iii) *The pinned band term.* If today's budget is slack at the myopic root and tomorrow's in every state, $\kappa_A^+ > 0$, and the fund is bought in every state tomorrow from the myopic root's marked holding, then $S_A = \beta \mathbb{E}[g_A] \kappa_A^+$ exactly, and the band term at the myopic root is

$$\frac{1}{2} S^\top (\gamma \Sigma_0)^{-1} S = \frac{(\beta \mathbb{E}[g_A] \kappa_A^+)^2}{2 \gamma \Sigma_{0,AA-E}} + S_A S_E [(\gamma \Sigma_0)^{-1}]_{AE} + \frac{1}{2} S_E^2 [(\gamma \Sigma_0)^{-1}]_{EE}, \quad \Sigma_{0,AA-E} = \Sigma_{0,AA} - \frac{\Sigma_{0,AE}^2}{\Sigma_{0,EE}}. \quad (183)$$

The fund's part is the discounted marked purchase rate squared over twice its curvature net of the ETF: the price of not front-loading a rise that tomorrow buys anyway. A fall sold in every state is the mirror, with κ_A^- .

Part (i) rewrites Proposition 56’s test with the need split. Part (ii) runs Proposition 56’s proof at any holding: every admissible slope family gives a supergradient of the relaxed continuation, and the completed square is taken about the nearest line value, so the bound holds for every admissible choice and therefore at the minimum; at the front-loaded point the relaxed program’s optimality conditions select a choice with zero residual. Part (iii) is Proposition 57 (iv)’s pinned value read in the block inverse.

Which point to hold is decided by the two bounds, both in the inputs once the relaxed tomorrow is solved. When the test passes at a small level at the front-loaded point, front-loading is adequate. When today’s cash is short, the front-loaded point spends cash the uncovered states would want, and the myopic root can do better: in the check’s four tight instances the front-loaded point’s realized loss exceeded the myopic policy’s, while both bounds held. Front-loading the planned purchase as far as the test still passes is a reading, not a result. The Gaussian quantile, several ETFs under the state equation and more reviews are not treated.

4. Related work

Propositions 1–3 unfold self-financing accounting, bilinearity and finite averaging. Proposition 4 applies compact-set attainment; Proposition 5 applies bounded linear-image feasibility; and Proposition 6 is a numerical use of Proposition 1. Proposition 11 applies finite-horizon dynamic programming and compact attainment. These are supporting facts, not new general theorems. The portfolio specificity lies in the common funded action classes and in what their comparison does and does not establish.

Gârleanu and Pedersen solve a dynamic trading problem with predictable security returns and quadratic transaction costs. Their policy trades partway toward a position that incorporates future expected returns. The solved control problem chooses positions in a directly tradable security universe without the funded, long-only ETF-only action class considered here. Accordingly, that policy provides a comparison for dynamic trading, while the comparison between ETF-only and full-trading feasible exposure sets remains a separate question ([1], full text reviewed).

Gallien, Kassibrakis and Malamud study an illiquid, alpha-generating fund hedged with a liquid futures contract. With a lower bound on terminal wealth and small proportional costs for trading the fund, they derive an approximate region in which the fund position is left unchanged. Their continuously traded futures hedge differs from a funded, long-only ETF adjustment made at a quarterly review. We have not established that their result is a limiting case of the model here ([19], full text reviewed).

Bichuch and Guasoni solve a continuous-time, infinite-horizon CRR problem with a risk-free asset, a liquid index, and one illiquid asset with expected excess return $\mu_2 = \beta_2\mu_1 + \alpha$ and small proportional trading costs. With positive idiosyncratic risk, their zero-alpha result says that the illiquid asset is never worth holding. It is a benchmark for a zero-alpha limiting case, but does not establish a liquidation rule in Section 2’s model: their assets are not exact return replicas, whereas that model permits exact replication, an incumbent active holding with sale costs, and ETF position limits that can obstruct replacement. Their model also has no funded long-only restriction or belief uncertainty in (α, β) , and uses one illiquid instrument rather than an ETF-versus-active-fund menu with several factors ([6], full text reviewed). Proposition 8 supplies a limited one-quarter analogue: when ETF replacement is feasible and costless, belief-mean alpha is zero, and the ETF shock has zero cross-moment with the fund-minus-ETF shock, a zero-active full optimum exists. With no incumbent, its score equals the ETF-only optimum; with a free active sale and positive penalized residual-difference variance, every full optimum liquidates the fund. Costly incumbent sales can instead preserve a zero-alpha holding, costless exact replication can leave ties, and an ETF limit can block replacement. Independent ETF tracking noise can also make a zero-alpha active holding optimal

through diversification: the cross-moment assumption fails in the exact example accompanying Proposition 8. Zero alpha against the specified economic factors does not mean zero expected return advantage against a tradable, noisy ETF; the latter benchmark changes both the mean and risk comparison. These are obstructions in the funded one-quarter model to an unqualified transfer, not counterexamples to the source's long-run result.

Dai, Jin and Liu study a finite-horizon fund manager with a risk-free asset, a liquid stock, and an illiquid stock subject to exogenous position limits and proportional costs, with fixed, known return parameters and no alpha/premium decomposition. Their trading boundaries are monotone in the position limits, and a limit not currently binding can still affect current trading through its anticipated future binding. Neither the ETF span restriction nor belief uncertainty appears in their model ([14], full text reviewed). Proposition 12 has no extra position limit: it compares initial active-trade advantages under matched future trading menus after a public signal reveals factor premia, including a factor direction unavailable through the ETF. The fund's alpha is known to be zero in that example. Its opposite continuation effects come from the future trading opportunities, not an anticipated binding position limit or learning about manager skill.

Liu and Muhle-Karbe study one risky asset with small proportional trading costs and a binding upper portfolio constraint. Their no-trade boundaries depend on that constraint, and the region's leading-order width changes from the unconstrained cubic-root scale to a square-root scale in the spread ([33], Theorem 2.3, full text reviewed). This is a precedent for a constraint changing a no-trade band beyond the traded asset's own cost. Its continuous-time, infinite-horizon, single-risky-asset problem does not characterize a quarterly funded investor's active-trade boundary after optimizing an ETF menu. Proposition 9 gives that one-quarter boundary as an exact interval in the active belief mean, conditional on fixed other inputs. For an interior active position, its algebraic width depends on active buy and sell costs and the endpoints of the compatible ETF budget-multiplier interval. That interval reflects ETF trade kinks, position bounds and cash funding. Its position-bound cases differ from the cited dynamic boundary; the qualitative effect of a binding constraint on a no-trade region is already known from this precedent. Proposition 10 illustrates degeneracy of parametric optimization: at the coincident ETF cap and cash-exhausting purchase, compatible budget multipliers form an interval; a positive cost moves the optimum off the cap and pins the multiplier. Its knife-edge narrowing contradicts a common heuristic that higher costs widen a no-trade band, not any cited theorem. It does not imply a robust effect or a narrowing of the dynamic holding bands studied in these sources.

Pástor and Stambaugh combine fund and passive-index return histories with prior beliefs about pricing and manager skill. Their portfolio may include active funds even when the investor believes managers cannot outperform passive indexes. They distinguish pricing-model error from skill, and report that their sampled fund universe has no close substitutes for the pricing-model benchmarks. Their portfolio is selected jointly; they do not analyze a factor-allocation stage followed by manager selection or compare an ETF-only adjustment to full trading under Section 2's common funded budget ([64], full text reviewed). Platanakis, Sutcliffe and Ye study asset allocation followed by stock selection within asset classes. Under a common idiosyncratic variance, their Proposition 1 gives formulas for the expected-loss difference between mean-variance and equal weighting, separating an alpha-driven component and specifying when idiosyncratic variance changes each component in a given direction. This is a formal comparison of methods used within an already-fixed two-stage procedure. They explicitly do not compare one-stage with two-stage portfolio formation, so the result gives neither conditions for the staged procedure to recover a joint optimum nor a bound on the loss from staging ([51], Proposition 1, full text reviewed).

Proposition 22's two-stage procedure has classical precedents in which separation is exact. Treynor and Black combine an active portfolio with a passive index in "an idealized world in which there are no

restrictions on borrowing, or on selling securities short” ([60], p. 67, full text reviewed). Positions in the analyzed securities are taken “purely on the basis of expected independent return and variance”, and an explicit market position complements the market exposure those positions accumulate (equations (13) and (16)). Merton’s mutual-fund theorems derive two-fund separation for mean-variance investors, with borrowing and short sales allowed ([41], Theorems I and II, full text reviewed). Cass and Stiglitz treat separation as a property under which a few mutual funds span every investor’s relevant opportunities ([10], full text reviewed). Tobin’s analysis of several non-cash assets keeps holdings nonnegative with total at most one, so it has no borrowing ([58], Section 3.6, full text reviewed). None of these gives a loss bound for a first stage restricted to funded, long-only exposures. In Proposition 22, an unrestricted first stage has a zero multiplier, and the soft procedure then coincides with the joint problem. The funded, long-only budget is what makes the multiplier nonzero, and the loss bound is the value of that multiplier, an application of weak duality. Proposition 23’s flat case is the Treynor–Black world read in these terms: residuals uncorrelated with the market (their equation (2)) and an explicit market position free in sign, which “may, of course, be negative, requiring an explicit position in the market that is short” (equation (16), p. 72), net any exposure the residual choice carries. Its three funded failures remove those assumptions one at a time: no short netting position, no unrestricted borrowing, and no frictions. Tobin’s no-borrowing budget suggests that the shared-budget failure bites where cash binds, not from the absence of borrowing as such.

Jones and Shanken model fund alphas through a shared population distribution with fund-specific deviations, so information about one fund changes beliefs about others ([29], full text reviewed). That shared component concerns manager skill, rather than a common factor-premium error. Barras, Scaillet and Wermers use false-discovery control to estimate the fractions of unskilled, zero-alpha and skilled funds in the presence of dependence across estimated alphas ([4], full text reviewed). These are distinct ways to handle cross-fund estimation uncertainty; neither gives the funded ETF-versus-active decision in Section 2 or shows that a one-quarter belief-average score responds to belief dispersion at fixed means. Proposition 12 instead learns factor premia with known zero manager alpha. Proposition 13 allows uncertain alpha, but its witness policies use only the revealed factor regime. Neither result separates the effects of learning about manager alpha and factor premia on the active-trading decision.

Estimation uncertainty and trading costs are also studied jointly elsewhere: DeMiguel, Martín-Utrera and Nogales analyze multiperiod losses and shrinkage policies ([15]); Olivares-Nadal and DeMiguel express a turnover norm as the worst mean-error loss in the trade direction over a dual-norm uncertainty set, including an ellipsoidal case, and relate this to regularized and Bayesian portfolio problems ([48], Proposition 1, full text reviewed); and Hautsch and Voigt connect turnover penalties with covariance shrinkage under model uncertainty ([24]; all full texts reviewed). Garlappi, Uppal and Wang use multiple priors to represent ambiguity aversion about expected returns ([21], full text reviewed). These cost and uncertainty results optimize a single action class; they do not supply a comparison with the best feasible ETF-only action. Section 2’s one-quarter belief-average score depends on the belief mean alone, as Proposition 3 states; it does not implement their robust or multiperiod objectives.

Palczewski and coauthors give a numerical method for a dynamic portfolio problem with state-dependent drift and proportional trading costs, rather than a closed-form no-active-trade boundary ([49], full text reviewed). Petrik, Chow and Ghavamzadeh compare a candidate policy with a fixed baseline under the same uncertain transition model, before minimizing their performance difference. They bound loss relative to the true optimum, exhibit tightness of that bound, and prove their robust comparison is computationally hard ([50], Definition 2 and Theorems 5 and 6, full text reviewed). Shared uncertainty and paired subtraction therefore have a direct precedent. Their fixed-baseline decision process does not

compare a candidate portfolio with the best feasible ETF adjustment under one funded budget or give a sampling-information lower bound for that decision. Howard and coauthors give time-uniform confidence boundaries for sequential estimation; their theorem does not by itself cover a changing latent conditional mean or establish a portfolio decision rule. A specified process scale is required to apply the stitched boundary in that theorem ([26], Theorem 1, full text reviewed).

For bounded independent and identically distributed observations, Maurer and Pontil bound a mean using the sample variance, without specifying the observation law beyond a known range. Their finite-class extension controls all members of a fixed finite function class by a union bound ([39], Theorem 4 and Corollary 5, full text reviewed). These are single-sample bounds for bounded variables. They offer a general ingredient for an unknown-law comparison across finitely many portfolio contrasts; they do not themselves certify an action against a continuum of feasible ETF adjustments or control a rule used at repeated reviews. The certificates of Propositions 15–18 instead calibrate to a known finite observation law; Proposition 19 uses only a known range.

Manski gives finite-sample bounds on the expected welfare of empirical treatment rules in a binary-choice problem with bounded outcomes. The lower welfare bound controls loss relative to the choice that knows the true mean outcomes, but does not give a sample-size lower bound applying to every treatment rule ([36], Proposition, full text reviewed). Those bounds concern expected welfare across samples, rather than the probability of a false decision after a particular sample; both treatment outcome means are unknown to the empirical rule. His adaptive minimax-regret rule uses accumulated evidence to choose treatment allocations for successive cohorts. Its per-cohort criterion does not establish a finite-sample loss rate for a funded portfolio decision ([37], Section 2.2, full text reviewed).

Mohajerin Esfahani and Kuhn size a Wasserstein ambiguity ball from independent samples. Under their tail condition, the robust solution’s worst-case cost bounds its true expected cost with the stated confidence ([18], Theorem 3.5, full text reviewed). This one-sided coverage statement applies to a data-selected decision. The ball’s radius also depends on concentration constants determined by tail conditions and dimension; the displayed concentration theorem treats dimension two separately ([18], Assumption 3.3, Theorem 3.4 and equation (8), full text reviewed). The guarantee does not compare the optimized full and ETF-only classes under a common funded budget, nor give a matching lower bound on the sample size needed for that comparison. The paper’s one-quarter belief-average score does not impose a Wasserstein ambiguity set.

Proposition 14 uses a two-law information comparison: equal funded geometry and second moments leave different public-history likelihoods, so they cannot determine the sharp certification power. Its exact finite-law pair and the quantified gap for the prescribed gate are the application here. The general fact that moments do not determine a likelihood, and the use of two laws to bound testing power, are not new statistical mechanisms.

Garivier and Kaufmann’s best-arm-identification lower bound optimizes information against alternative arm models and bounds the expected stopping time of a fixed-confidence procedure ([20], Theorem 1, full text reviewed). Kaufmann, Cappé and Garivier give a divergence lower bound for identifying the best arms in an identifiable bandit family ([31], Theorem 4, full text reviewed). The hardest-alternative logic, and inverse-square gap behavior in regular mean models, behind Propositions 15–16 are therefore established testing mechanisms. Those theorems concern adaptive allocation and stopping among a finite set of arms; the propositions here use fixed public histories, known bounded finite laws and a funded ETF-only action class. Neither cited theorem directly yields the stated portfolio bounds or their constants. In particular, Proposition 15’s fixed-sample lower bound uses a two-law overlap inequality, while its upper bound uses bounded-mean concentration. The adaptive best-arm results are a benchmark for the mechanism, not

the source of either stated bound. The overlap step in Propositions 15–17 is Le Cam’s two-point method: the affinity of N -fold product laws is the N -th power of the one-observation affinity, which bounds their distance ([9], Lemma 1 and its proof, full text reviewed). The propositions prove that step directly on finite supports rather than citing it.

Proposition 15 converts a scalar premium-testing rate to an economic score margin through funded exposure D , with a fixed ETF optimum and known zero alpha. Proposition 16 applies the hardest-alternative idea when the ETF comparator switches between cash and an ETF. Its specific two-direction variance calculation uses both joint alpha/premium error directions, rather than only the direction chosen at one estimate. The finite two-point constructions and ellipsoidal upper penalties are standard statistical tools. What is established here is their exact application to these stated funded comparison classes, not a new general lower-bound method or an optimal sampling policy. The difficulty these rates express in premium units is the classical one of estimating an expected return. In Merton’s setting of prices with a constant mean and variance in continuous time, the precision of the estimated mean depends only on the calendar length of the record, not on how often returns are sampled within it ([43], Appendix A, equation (A.3), full text reviewed). The propositions here use quarterly records and define no within-quarter returns, so that result is context, not an input.

The inverse-square dependence on the economic score margin in Propositions 15 and 16 belongs to their stated fixed-action, linear-gap families. Proposition 17 gives a distinct smooth-entry family with a continuously moving, interior ETF optimum. Its inverse-first-power economic-margin rate follows from the quadratic map from active mean signal to optimized score advantage; the underlying mean-testing rate remains inverse-square in the signal. This conversion from a smooth mean signal to a quadratic optimized-value gap is a fast-rate pattern, rather than a new testing method. A full-text comparison has not determined whether the specific quadratic-entry certificate is already a corollary of an existing statistical-decision theorem.

Proposition 18 extends that certificate to directional costs and binding constraints under positive definite return risk. Its strong-concavity and optimized-value stability arguments are standard mathematical tools; the class-level lower bound still comes from Proposition 17’s zero-cost family. The constraint extension supplies neither a new statistical method nor a lower bound for each costly, binding instance. Whether the whole-ETF-class certificate follows from a general optimization-sensitivity theorem has not been verified by a full-text comparison.

Proposition 19 is an application of known tools. Its upper bound is Hoeffding’s inequality for bounded independent and identically distributed variables ([25], Theorem 1 and the remark following it on variables in an interval $[a, b]$, full text reviewed; restated as [39], Theorem 1), with a union bound over the two comparator faces. Its lower bound is Proposition 16’s two-point bound. The empirical Bernstein bounds of Maurer and Pontil and the time-uniform confidence sequences of Howard and coauthors are variance-adaptive or sequential alternatives to the range bound ([39], Theorem 4 and Corollary 5; [26], Theorems 1 and 4; full texts reviewed). Proposition 19 implies that in its family they can improve the worst-case history length by constants only; it does not analyze their constants or give a time-uniform guarantee. Adaptive best-arm identification over the two faces ([20], [31]) is the sequential analogue of the maximum over faces; neither is used. Proposition 24 generalizes the two-face reduction to any funded menu with a linear score, using the linear-programming vertex theorem ([5], Theorem 2.3, Corollary 2.1 and Theorems 2.7–2.8; [56], Section 8; cited by theorem number, texts not compared). Its curvature deficit is the strong-concavity gap behind Proposition 18, and its reading, that curvature rather than funding removes the finite reduction, is a known perturbation mechanism.

The earlier feasibility and continuation examples have the same boundary: Proposition 7 applies constrained replication geometry, Proposition 8 depends on the chosen definition and tradability of alpha's benchmark, and Propositions 12–13 illustrate continuation value from future flexibility and learning. Their proved portfolios, signs and limits are the paper's results; these general mechanisms are not claimed new. That measured performance depends on the benchmark is itself a known point. Roll shows that betas against an index that is not mean-variance efficient need not line up with mean returns. For every performance ranking obtained with such an index, another non-efficient index reverses it ([55], statements S4 and S6, full text reviewed). Proposition 8's alpha is defined against a stated factor benchmark and inherits that dependence.

Proposition 20's sandwich is the increasing-differences comparison of monotone comparative statics. Topkis proves that optimizer sets move monotonically with a parameter when the objective has monotone differences in the choice and the parameter and the constraint sets are ascending in the parameter; his theorem is stated for minimizing submodular functions ([59], Theorem 6.1, full text reviewed). Milgrom and Shannon restate that theorem for maximization and characterize monotone maximizers by quasisupermodularity and the weaker, ordinal single-crossing property ([44], Theorems 4 and 5, full text reviewed). Amir's survey gives a version with box constraints whose endpoints increase in the parameter, and shows that maximizing a supermodular function over some coordinates of a fixed rectangle preserves supermodularity in the rest ([3], Theorems 8 and 11, full text reviewed). Proposition 20 does not apply these theorems. It writes the comparison as an inequality between four attained maxima and shows that increasing differences fails in the funded two-review model; it does not examine single crossing.

Their constraint hypotheses show where funding enters. The continuation value under future ETF-only trading is a maximum over the second-review ETF trade. That trade can buy at most the cash left after the root trade, net of the purchase charge, and a root purchase of the active fund reduces that cash one for one plus costs. So the later choice's feasible range shrinks as the root active holding rises, rather than being a fixed rectangle or ascending. The general route from a supermodular objective to increasing differences of the continuation value is therefore unavailable here. The funded cap (58) is the value-side form of that constraint, and it is elementary accounting rather than a new mechanism.

The negative sign in Proposition 20's example comes from the value of waiting to learn before committing. Merton's changing-expectations examples show a related effect with a fixed menu: when future returns are expected to become more favorable, an investor with exponential utility holds less of the risky asset now and keeps riskless wealth "as a reserve" for later investment ([40], Section 9, equation (129) and the discussion following it, full text reviewed). There the opportunity set moves while the menu is fixed. In Proposition 20 public information is the same under every control and only the future menu changes. The positive sign of Proposition 12 is not hedging demand either: its family has no conditional return risk and reveals the parameter at the first review, so future ETF adjustment gains only by redeploying wealth into returns already known. Gârleanu and Pedersen's signal persistence, which tilts today's trade toward where a persistent predictor is heading, does not arise, because the two-review model has fixed conditional means ([1], full text reviewed).

Proposition 21's caps use standard tools and no cited theorem. Its impossibility for range-based caps is the elementary fact that a bound depending only on a payoff's range is attained by a law concentrated near the best payoff. Its tangent cap is the tangent-plane bound for a concave function, here the exponential certainty equivalent, whose concavity in the trade follows from convexity of the log-sum-exp function. Its curvature cap bounds the second derivative of that function by a variance under a tilted measure.

Propositions 26 and 27 extend the dynamic results to many reviews. With proportional costs, the continuous-time small-cost literature gives a no-trade band of width of order the cube root of the cost

around a target that moves with a mean-reverting state variable ([46], Section 4, full text reviewed; its homogenization technique is [57], equation (4.4), read at the level of its displayed formulas, and the multi-dimensional [52]). That target does not learn: its motion comes from a stationary factor, not a posterior mean whose variance shrinks. Proposition 26 is exact at a fixed quarterly interval, where a one-review width is meaningful. It shows that such asymptotics describe only the regime of fine target innovations. With quadratic costs, Gârleanu and Pedersen’s continuous-time extension to many securities and predictors imposes conditions that make the predictors stationary ([22], Section 1, full text reviewed). That excludes a learned posterior mean, which is a martingale. Proposition 27 handles that case with a time-varying recursion. Its filter/control separation is the standard linear-quadratic-Gaussian one ([2], full text reviewed), and its new content is the exposure/alpha split for bundled instruments.

Propositions 30 and 32-34 concern trading on estimated inputs. In one period, Kan and Zhou derive the expected loss of the plug-in mean-variance rule exactly, as a function of the number of assets and the length of the estimation window: $N/(2\gamma T)$ when only the means are estimated. They then treat the case of an estimated covariance with known means ([30], equation (17) and the discussion following it, full text reviewed). Propositions 30 and 32 are the dynamic counterpart for the covariance, with trading costs and learning. There the loss is a discounted sum over reviews, explicit in the starting holdings, the prior means and the precision still to be gained. Proposition 33 adds re-estimation at every review. Proposition 34’s benchmark is Wachter’s split of a dynamic allocation into a myopic term and a hedging term ([62], p. 73, full text reviewed). Its point is that under a per-review objective the anticipation that remains is a cost effect.

Two further sources fix how these results read. Without transaction costs, “a greedy strategy that only considers one period at a time is optimal, since performance for the current period does not depend on previous holdings”, while with costs “current holdings strongly affect whether a return prediction can be profitably acted on” ([7], Chapter 5, p. 45, full text reviewed). Propositions 29, 31 and 34 are that distinction made exact for a learning-driven target: anticipation of future learning enters today’s trade only through trading costs. And the Kalman recursion is the minimum-mean-squared-error filter among affine rules for any noise with finite second moments, and the exact posterior when the noise is Gaussian ([61], Section 2, on Kalman’s Corollary 1, full text reviewed). The source states only that sufficiency: under other laws the posterior mean can differ from the filter’s, but need not, as the ledger’s audit of its entry records. That is the difference between the fund-of-funds learning model, whose filter is the exact posterior, and the finite-law variant of the learning-driven tracking model, whose same linear filter is the best linear predictor.

Propositions 26 (v) and 38 concern several instruments under proportional costs. For two correlated risky assets, Lynch and Tan find numerically that the no-trade region is a rectangle at zero correlation and a parallelogram otherwise, distorted to exploit the substitutability that positive correlation induces, and that with predictable returns the conditional correlation sets the distortion ([34], read at the level of its introduction). Proposition 26 (v) gives that parallelogram exactly at the last review. Proposition 38 bounds a fund’s band at every review between explicit anchors in the inputs, the fund-alone width and a ceiling set by the residual variance given the ETFs, but it does not give the region’s dynamic shape. When some instruments sit at a bound, Jagannathan and Ma show that the multipliers of the binding bounds fold into an adjusted covariance, so the constrained minimum-variance portfolio is an unconstrained one for that matrix ([28], Proposition 1, read at the level of the proposition). Propositions 35, 38 and 39 exclude ETFs at their bounds, and that fold-in is the benchmark for extending them there. Proposition 40 takes the first such step for the two-stage procedures.

Propositions 42 and 43 concern the fine regime and the shape of a fund’s band beside a costly ETF. For a position hedged through an imperfectly correlated asset with fixed and proportional costs, Whalley

derives the leading-order no-transaction band by asymptotic analysis; lowering the absolute correlation narrows the band and moves its centre toward zero ([63], read at the level of its abstract). Proposition 42 is the proportional-cost counterpart at quarterly reviews. It is exact at the frictionless and frozen ends, where the correlation enters through the residual curvature and the idle target's variance, and it leaves the band between the ends open. With a binding constraint, the effect of a small spread on the policy changes order from $\epsilon^{1/3}$ to $\epsilon^{1/2}$, because the unconstrained optimum lies outside the feasible set ([33], Section 3.1, full text reviewed). Propositions 40 and 41 are exact at one review with ETFs at zero and a binding budget, and any leading-order band for that case must allow for this change of order.

Propositions 50-57 concern flexibility over two reviews. Moallemi and Saglam evaluate a class of linear rebalancing rules against computed upper bounds on any policy's value: in an execution example the best linear rule comes within 5% of the value any policy can achieve, and in a mean-variance trading example its improvement over the best alternative policy reaches 18% as the horizon and costs vary ([45], pp. 5, 30 and 32, read at those passages). The lab's registered comparisons are against the exact dynamic optimum on a finite tree instead, and Proposition 56 (iii) bounds the loss of re-optimizing each quarter without solving the dynamic problem at all. That bound's tail term is the Rockafellar-Uryasev tail measure, the conditional expectation of a loss beyond its quantile, obtained as the minimum of a convex function of the quantile level ([54], Theorem 1 and the definitions before it, read at that level); Proposition 56 uses only its elementary identity on a finite law. Collin-Dufresne, Daniel and Saglam show that with costly trading and a stochastic opportunity set a long-horizon investor trades toward an aim portfolio that depends on costs and on each asset's correlation with changes in the opportunity set, at a speed set by the shock's persistence and how well it can be hedged ([12], read at the level of its abstract). Proposition 57's predictable move is the persistent part under a per-review score, where no hedging demand arises: it enters through the band's lean and front-loading, and the unpredictable part through the reserve. The standard references on pooling one buffer across several sources of need ([17]), on intertemporal separation with several hedging funds ([42]) and on investment under uncertainty ([16]) are registered but not read; how Propositions 54, 55 and 57 relate to them is not checked.

These source comparisons cover registered texts, not every general result that could contain these mechanisms as special cases. The search for a two-stage separation result beyond the sources above is also incomplete. Candidate sources on constrained replication, learning and flexibility, smooth decision rules and optimization sensitivity have only been identified from titles or abstracts; their precise coverage has not been checked against full texts. No priority claim or transfer theorem for the funded ETF-only comparison follows from this review.

5. Outlook

This draft establishes one-quarter accounting, feasibility geometry, an exact no-active-trade band after ETF optimization for fixed other inputs, exact assumed examples and finite two-review policy attainment. An exact two-review example gives opposite effects of future ETF and active trading on the root active advantage. It does not estimate how often these mechanisms occur in observed portfolios or give a general rule with statistical coverage for choosing an active trade. The one-quarter belief-average score cannot express a preference against uncertainty about conditional means when their belief averages are unchanged.

The two-review model supplies an explicit law for public learning and an attained continuation value. The opposite-sign example shows one structural effect of future adjustment. A bounded extension retains the signs with positive conditional return risk and alpha that remains uncertain after some observations. Its alpha and shocks are small relative to the assumed factor premia, and the mechanism still uses the revealed

factor regime. Proposition 20 shows that no sign of the future-ETF effect holds across the model. In every instance the effect lies between differences of the premium of future ETF adjustment at the two root optima. That premium is capped by the cash and ETF holdings left after today's trade, and an explicit family has the opposite sign. Proposition 25 supplies a floor on the premium by adjustable wealth, and its floors come within about a quarter of the premia on assumed instances. The caps are the weak side, 29 to 374 times too large there, and no instance in which the no-active-trade region itself moves is exhibited. Proposition 21 shows that no cap built from wealth and return ranges alone can be sharp enough. Caps using risk-adjusted mean returns are sharper and vanish exactly where no favorable trade has wealth, but in the assumed numerical instances they decide only two of six signs. Elsewhere the sign has to be computed instance by instance. A statistical account would need a stated estimation procedure and an evidential comparison with the best feasible ETF-only alternative. Proposition 14 supplies a finite-law benchmark against the optimized ETF-only class and shows that the prescribed sample-mean gate can miss information in public ETF returns. Its exact support calculations require a known law and exact observations. Proposition 15 gives a matching-order worst-case history-length envelope over a class of bounded known finite laws, with a valid conservative certificate against the whole funded ETF-only class. Its active trade earns an uncertain factor premium with known zero alpha; the envelope is scalar mean testing after converting the score margin by the funded exposure D . Proposition 16 lets alpha and premia vary together and makes the optimized ETF comparator switch between cash and the ETF. Its worst-case rate depends on the larger of two covariance directions. Both the covariance floor when alpha is observed exactly and the lower bound on a fixed known-alpha slice are machine checked; the latter asserts the existence of one alpha value, not a bound for every alpha value. An error-free comparison against the ETF can still leave a noisy comparison against cash. The result uses a bounded known-law class, two comparator vertices, and loose constants. It does not give a general empirical certificate or dynamic boundary. Proposition 17 instead makes the active holding shrink smoothly at entry while the optimal ETF holding moves continuously. Its known-law worst-case rate is inverse-first-power in the economic advantage margin and inverse-square in the active mean signal. The result ties risk aversion to the shock scale, assumes zero costs and slack funding, and does not identify a general rate under binding constraints or a calibrated live-trading rule. Proposition 18 gives a valid whole-class certificate with costs, caps and binding funding when return risk is strictly curved. Its worst-case lower bound uses Proposition 17's slack, zero-cost subfamily. Numerical use requires certified global optimizer values or the reviewed paper-proof gap bound; the exact-optimizer power rate does not transfer to arbitrary solver outputs. Neither result gives a calibrated history recommendation, unknown-law coverage or repeated-review rule. Proposition 19 removes knowledge of the law in the two-face family, keeping only a known bound on the shocks. There the optimized ETF-only comparator is exactly its two faces: a rule that certifies each face separately from its range attains the known-law worst-case order, and no law-free rule does better in order. What remains open in that family is constants, which decide whether any law-free rule has useful power at realistic history lengths, and a time-uniform version for repeated reviews. Proposition 22 shows that choosing a factor target first and funds second, with a soft penalty for missing the target, loses only through the multiplier of the funded first-stage constraint. It gives a bound, not a magnitude, for how large that loss is in realistic portfolios. Proposition 23 identifies the loss exactly when the second stage must meet the target, and names three ways funded long-only trading breaks separation. Only the first of these, manager-borne exposure, is proved; the shared budget and the cost kink are explanations. The loss in its numerical illustration is below 3 basis points per quarter except where the target forces holding a fund the joint optimum would sell. Neither procedure dominates the other in general. Proposition 24 shows that with a linear score the whole funded ETF-only comparator is its finite faces, so any certification rule reduces to finitely many scalar mean statements. With a curved score this holds only when the ETF-only optimizer is constant on the domain. It says nothing about

which law classes supply those statements or at what rate. Proposition 26 fixes the exact structure of quarterly no-trade bands around a moving target, with a slack budget. The band is never wider than the one-review band, and it equals it, shifted, when target innovations are coarse. What remains open is the leading order of the band when innovations are fine, the effect of a binding budget and caps on its edges, the dynamic shape with many instruments, and the specialization to a target driven by learning. Proposition 27 gives the exact policy when positions are unconstrained and costs are quadratic. It shows partial adjustment toward an aim that anticipates learning, and an exact split of ETF exposure from fund alpha that only ETF costs, residual risk and fees couple. The policy is a counterfactual: it may short funds, and its implementable, long-only version is open. Proposition 28 shows that when ETFs miss a factor direction, premium beliefs leak into fund choice through that direction, as a persistent mean term and a risk term that learning removes. Open there are whether the leak is nonzero at every review for every horizon, beyond the proved cases, and its size at calibrated scale. Proposition 29 shows that learning is anticipated only in positions that are costly to trade, through a learning factor at most a few percent in the instances checked, and that a pooled alpha prior makes the average fund position trade faster than relative positions. Proposition 30 gives the exact value loss from an estimated covariance and a guarantee for an estimate made before the run. Its a-priori norm bounds are vacuous at realistic scale, and a time-uniform guarantee for a manager who re-estimates is open. Proposition 31 puts Proposition 26's band on the learning path: the static width rises while the learned target's moves shrink, so pure learning ends the coarse regime. Which instruments stay coarse once real returns move the deviation, and the fine regime's leading order, are open. Proposition 32 writes the plug-in loss as an explicit function of the inputs, with limits in the fund cost and a statement of when spanning keeps the two estimation errors apart. Its monotonicity in precision and horizon is open. Proposition 33 bounds the expected loss of a manager who re-estimates at every review, with no smallness condition, as an explicit function of the inputs and any time-uniform confidence radii. The confidence sequence's constants, and so the bound's size at given data, are not evaluated. Proposition 34 gives explicit conditions for when anticipating learning or mean reversion materially changes today's trade: the Kalman gain for learning, and the future's weight times the persistence shortfall for mean reversion. Both effects vanish for costless instruments. Heterogeneous funds and ETF costs are open. Proposition 35 says in the inputs how premium error enters fund choice: not at all through spanning frictionless ETFs, through one hedged scalar per fund otherwise, and in full under naive selection. Several unreachable funds, ETF frictions and binding ETF bounds are open; the last is the common case at long-only instances. Proposition 36 states when data can support buying a fund, as a margin over a cost-and-incumbent threshold of order the noise scale times $\sqrt{\log(1/\varepsilon)/n_{\text{eff}}}$, with a precision floor from persistent alpha that no history removes. Several funds traded at once, ETF frictions, and lower bounds under persistence are open. Proposition 37 places the two cost models in one tracking problem with a common target and loss, and shows that anticipating learning has order κ^2 times a bounded duration under quadratic costs. The proportional-cost order, about $\kappa^{4/3}$ in the fine regime, rests on a cited leading-order law; a proved bound on the band's centre, and a composite rule matched between the two costs, are open. Proposition 38 bounds a fund's band beside optimally traded ETFs at every review, between the fund-alone width and a ceiling set by its residual variance given the ETFs. The band's dependence on the correlation of the targets' motion, and several funds whose bands couple, are open. Proposition 39 prices the ETF-only restriction at one review as the value of the fund purchases it forbids, and shows that a frozen-funds baseline adds the value of the forgone sales. The multi-review cost and ETFs at their bounds are open. Proposition 40 gives the two-stage procedures' exactness with ETFs at zero, through stage 1's own slacks. With fees, ETF frictions and a binding budget it is open. Proposition 41 adds them: given the ETFs' statuses, every fund's marginal is explicit in the inputs, and the target-confined procedure is generically inexact once it trades a costly ETF that ends interior, while the soft procedure with an unconstrained first

stage stays exact. ETF caps and a first stage that sees costs or the budget are open. Proposition 42 gives a fund's fine-regime band beside one costly ETF at its frictionless and frozen ends, through the cited cube-root law, and names the weight between them. The band between the ends, and its end limits as theorems, are open. Proposition 43 gives the shape of the two-instrument band: exact median edges at the last review, bending over the ETF's residual width, and monotone edges at every review. How the targets' correlation moves the band before the last review is a conjecture. Proposition 44 states the one-review fund criterion on which the bundling results rest: exact, explicit under frictionless spanning ETFs, and bracketed by the re-hedge cost otherwise. Proposition 45 gives two-stage exactness and loss at one review in the inputs; a first stage that sees costs or the budget is open. Proposition 46 gives the fund decision with ETFs at zero, on which the later results with bounds rest; an ETF at its cap is open. Proposition 47 reduces the one-ETF case to two monotone scalar prices, the exposure price and the cash price. Proposition 48 gives a first stage that prices the ETF trades and a loss bound for any split, through the ETF-line residual. Proposition 49 sandwiches the fine-regime cost of a fund and one costly ETF in the inputs; where the effective rate lies in the sandwich, and the fund's band between the ends, are open. Proposition 50 extends the one-review rule to two reviews with a budget that may bind, through the dynamic cash price and the incumbent values; more reviews and the joint comparative static are open. Proposition 51 bounds tomorrow's cash price and the incumbent values in the inputs, and restricts the sign reading to a frictionless interior ETF. Proposition 52 puts learning, predictive risk, marking and the budget into one worked model under either shock law; the dynamic optimum under the Gaussian law is open. Proposition 53 turns Propositions 50-51 into a reserve rule sized in the inputs, and Proposition 54 extends it to several funds sharing one ETF. Proposition 55 shows that with many ETFs spanning splits the two-review optimality conditions, not the optimization; many costly ETFs are open. Proposition 56 gives the full menu a quantile flexibility test and bounds the loss of re-optimizing each quarter under it; the test with time-varying premia is open. Proposition 57 lets premia and alphas move on their own: the target's move splits into a predictable part, which tilts the band and is front-loaded or planned for, and an unpredictable part, which the reserve prices; the rising width holds only while learning outweighs the state noise. Proposition 58 extends the test to that case: cover the planned purchase in full and a quantile of the rest, and front-loading the planned purchase removes the loss bound's band term.

Appendix A. Machine verification

The repository version codes appear here only to trace paper results to their formal statements. The general review notation is a notation-level setting; the funded one-quarter and two-review models are the working specializations of Section 2. The multi-review tracking model of Section 2 is the version labelled M6, the fund-of-funds learning model is M5, and the learning-driven tracking model is M7. The two-review worked model is M8, and M8 with M5's state equation is M9.

Paper result	Claim file	Model version
Proposition 1	'claims/002-paired-mean-error.md'	M0
Proposition 2	'claims/001-self-financing-identity.md'	M0
Proposition 3	'claims/003-m2-score-accounting.md'	M2
Proposition 4	'claims/004-m2-action-classes.md'	M2
Proposition 5	'claims/005-m2-etf-exposure-geometry.md'	M2
Proposition 6	'claims/006-m2-complete-substitution.md'	M2
Proposition 7	'claims/007-m2-incomplete-substitution.md'	M2
Proposition 8	'claims/008-m2-zero-alpha-limiting-case.md'	M2

Paper result	Claim file	Model version
Proposition 9	'claims/009-m2-no-active-trade-band.md'	M2
Proposition 10	'claims/010-m2-higher-cost-narrower-band.md'	M2
Proposition 11	'claims/011-m3-finite-continuation.md'	M3
Proposition 12	'claims/012-opposite-continuation-effects.md'	M3
Proposition 13	'claims/013-continuation-with-risk-and-partial-learning.md'	M3
Proposition 14	'claims/014-m4-public-information-obstruction.md'	M4
Proposition 15	'claims/015-m4-bounded-law-history-rate.md'	M4
Proposition 16	'claims/016-m4-joint-directional-rate.md'	M4
Proposition 17	'claims/017-m4-curved-entry-rate.md'	M4
Proposition 18	'claims/018-m4-curvature-certificate.md'	M4
Proposition 19	'claims/021-m4-law-free-faces-match-order.md'	M4
Proposition 20	'claims/022-m3-etf-channel-sandwich.md'	M3
Proposition 21	'claims/026-m3-mean-type-caps.md'	M3
Proposition 22	'claims/028-m2-soft-target-two-stage.md'	M2
Proposition 23	'claims/027-m2-two-stage-separation.md'	M2
Proposition 24	'claims/020-m4-finite-comparator-faces-iff.md'	M4
Proposition 25	'claims/023-m3-premium-node-bounds.md'	M3
Proposition 26	'claims/029-m6-quarterly-band-static-ceiling.md'	M6
Proposition 27	'claims/030-m5-partial-adjustment-split.md'	M5
Proposition 28	'claims/031-m5-missing-direction-leak.md'	M5
Proposition 29	'claims/032-m5-learning-aim-two-speeds.md'	M5
Proposition 30	'claims/033-m5-plug-in-value-loss.md'	M5
Proposition 31	'claims/100-m7-learning-band-transfer.md'	M7
Proposition 32	'claims/034-m5-plug-in-loss-inputs.md'	M5
Proposition 33	'claims/035-m5-reestimation-guarantee.md'	M5
Proposition 34	'claims/036-m5-when-anticipation-matters.md'	M5
Proposition 35	'claims/105-m7-premium-error-in-fund-choice.md'	M7
Proposition 36	'claims/039-when-data-support-an-active-change.md'	M7
Proposition 37	'claims/038-integration-theorem.md'	M5, M7
Proposition 38	'claims/107-m7-dynamic-bundling-band.md'	M7
Proposition 39	'claims/106-m7-etf-only-restriction-cost.md'	M7
Proposition 40	'claims/041-m7-two-stage-with-etfs-at-zero.md'	M7
Proposition 41	'claims/109-m7-fund-decision-etfs-at-zero-costs-budget.md'	M7
Proposition 42	'claims/042-m7-fine-band-with-one-costly-etf.md'	M7
Proposition 43	'claims/108-m7-dynamic-bundling-shape.md'	M7
Proposition 44	'claims/102-m7-fund-hold-buy-sell-criterion.md'	M7
Proposition 45	'claims/104-m7-two-stage-exactness-loss.md'	M7
Proposition 46	'claims/040-m7-fund-decision-with-etfs-at-zero.md'	M7
Proposition 47	'claims/110-m7-one-etf-two-scalars.md'	M7
Proposition 48	'claims/111-m7-incumbent-aware-first-stage.md'	M7
Proposition 49	'claims/043-m7-fine-cost-sandwich-with-one-costly-etf.md'	M7
Proposition 50	'claims/044-m7-two-reviews-binding-budget.md'	M7
Proposition 51	'claims/046-m7-two-reviews-bounds-in-the-inputs.md'	M7
Proposition 52	'claims/112-m8-worked-learning-example.md'	M8
Proposition 53	'claims/047-m8-reserve-rule-one-fund-one-etf.md'	M8
Proposition 54	'claims/113-m7-several-funds-one-etf-reserve.md'	M7, M8
Proposition 55	'claims/048-m7-many-funds-many-etfs-two-reviews.md'	M7
Proposition 56	'claims/049-m7-quantile-flexibility-test-full-menu.md'	M7

Paper result	Claim file	Model version
Proposition 57	'claims/114-m9-predictable-and-unpredictable-target-move.md'	M9
Proposition 58	'claims/115-m9-quantile-flexibility-test-time-varying.md'	M9

Proposition 1 corresponds to `claims/002-paired-mean-error.md`. Its identity, universal and particular cancellation statements, exposure-matching consequences, and a nonzero-exposure example of particular cancellation are machine checked in `lean/Standalone/PairedMeanError.lean` and `lean/Novel/PairedMeanErrorProof.lean`. No cited result or project hypothesis structure is used. The formal statement fixes the same loadings, covariance, ETF drag and switching-cost function across score evaluations. It does not verify feasibility, an optimized comparator, an error distribution, or a claim for either one-quarter specialization; no simulation is presented as evidence for Proposition 1.

Proposition 2 corresponds to `claims/001-self-financing-identity.md`. The identity, funded long-only equivalence, feasible-trade cost bound and zero-trade conclusion are stated in `lean/Standalone/SelfFinancingIdentity.lean` and machine checked by `lean/Novel/SelfFinancingIdentityProof.lean`. The identity and equivalence need no sign assumption on the trade or cost; the bound uses a nonnegative cost and funded long-only feasibility; the zero-trade conclusion additionally requires nonnegative initial holdings and cash and $C_t(0) = 0$. No cited result or project hypothesis structure is used. The formal statement verifies no score, terminal-wealth or normalized one-quarter funding claim.

Proposition 3 corresponds to `claims/003-m2-score-accounting.md`. The cost normalization, funding transfer, gain and conditional-moment identities, score formula, belief-mean evaluation, positivity at that mean and same-mean ranking and maximizer-set conclusions are machine checked in `lean/Standalone/M2ScoreAccounting.lean` and `lean/Novel/M2ScoreAccountingProof.lean`. The formal statement covers every holding and arbitrary finite instrument and factor counts, so it includes the funded one-quarter model's restricted counts and feasible holdings. Its funding algebra is re-derived in that model's index type rather than imported from Proposition 2's Lean proof. No cited result or project hypothesis structure is used. The machine check does not establish attainment, an effect of belief dispersion, a predictive-variance objective, dynamic learning, or empirical relevance.

Proposition 4 corresponds to `claims/004-m2-action-classes.md`. Nesting, nonemptiness, closedness, boundedness and convexity of the classes; continuity and concavity of the belief-average score; attained maxima; and the weak value chain are machine checked in `lean/Standalone/M2ActionClasses.lean` and `lean/Novel/M2ActionClassesProof.lean`. The formal statement reuses Proposition 3's one-quarter definitions, permits arbitrary finite instrument counts, and drops hypotheses where a particular part does not need them. The Lean proof uses compactness of a closed finite box and an extreme-value theorem for attainment; the reviewed paper proof uses an explicit subsequence construction. No cited result or project hypothesis structure is used. The machine check gives no uniqueness, strict value gap or empirical materiality.

Proposition 5 corresponds to `claims/005-m2-etf-exposure-geometry.md`. The finite directional-cash representation, ETF-only action and exposure-image identities, compactness and convexity, exact matching and obstruction test, one-ETF interval, invertible two-ETF case, and active-loading span criterion are machine checked in `lean/Standalone/M2EtfExposureGeometry.lean` and `lean/Novel/M2EtfExposureGeometryProof.lean`. The formal statement reuses Proposition 3's one-quarter objects and Proposition 4's initial-compliance and nonnegative-rate predicates. It proves the set properties under the hypotheses needed for each part; these cover Section 2's stated restrictions. Lean

proves the closedness of the exposure images using compactness, while the reviewed paper proof uses a bounded-sequence construction. No cited result or project hypothesis structure is used. The machine check establishes feasibility geometry only: it proves no score comparison, strict value gap, economic magnitude or transfer to another model version.

Proposition 6 corresponds to `claims/006-m2-complete-substitution.md`. The literal one-quarter data, positive gross returns, exact funded action classes and exposure images, feasible replacement for every full action, matched risk, cost and cash, belief-average and conditional score differences, unique class maximizers, their scores and the strict 1/101 gap are machine checked in `lean/Standalone/M2CompleteSubstitution.lean` and `lean/Novel/M2CompleteSubstitutionProof.lean`. The formal instance uses Proposition 3’s one-quarter definitions, Proposition 4’s compliance and rate predicates, and Proposition 5’s exposure-set definitions. The score identities are proved for all nonnegative holdings, not only feasible ones. The formal proof computes the one-ETF image directly; it does not import Proposition 5’s image lemma. No cited result or project hypothesis structure is used. The machine check proves this assumed instance, not its empirical relevance, another instance or a multiperiod return advantage.

Proposition 7 corresponds to `claims/007-m2-incomplete-substitution.md`. The two literal one-quarter menus, all 32 scenario returns and their positivity, the shared return law of common instruments, all five unique class optima with cash, costs and scores, the zero and positive optimal gaps, the active-trade comparison, the one-ETF missing direction and the two-ETF full-span but infeasible match are machine checked in `lean/Standalone/M2IncompleteSubstitution.lean` and `lean/Novel/M2IncompleteSubstitutionProof.lean`. The proof reuses Proposition 3’s one-quarter definitions, Proposition 4’s feasibility hypotheses and Proposition 5’s ETF-span definition. It certifies strict optimality through the funded score inequality and a positive quadratic form. The paper proof displays particular multiplier and subgradient vectors in the claim file; those vectors are not separate formal conjuncts, although the optimizer statements are checked. No cited result or project hypothesis structure is used. The machine check covers these assumed instances only; it does not attribute the gap solely to the shorting obstruction or establish economic materiality.

Proposition 8 corresponds to `claims/008-m2-zero-alpha-limiting-case.md`. The general replacement map, its feasibility, exposure and cash identities, its belief-average score identity, attained zero-active optimum, $V_F = V_E$ under no incumbent, and the stated strict non-participation conclusions are machine checked in `lean/Standalone/M2ZeroAlphaLimitingCase.lean` and `lean/Novel/M2ZeroAlphaLimitingCaseProof.lean`. The stronger pointwise equal-return and terminal-wealth identities, and all four literal examples with their one-quarter model validity, optimizer descriptions and scores, are checked there too. The Lean statement uses Proposition 3’s one-quarter definitions; it proves attainment directly rather than importing Proposition 4’s proof. It permits arbitrary finite factor, scenario and belief sets in its general part, and omits unused one-quarter hypotheses such as positive gross returns there. No cited theorem or project hypothesis structure is used. The registered source’s continuous-time conclusion is compared in prose, not imported or machine checked. The zero cross-moment remains an explicit formal hypothesis. It forces zero ETF residual variance when both residuals are uncorrelated with factor shocks and with each other; shared ETF and active residual noise can instead satisfy it with positive ETF residual variance. The independent-noise counterexample above lies outside the formal hypothesis and is checked by exact finite calculation, not a formal conjunct. The machine check establishes no general zero-alpha rule. The four formal examples and the counterexample use assumed inputs, not empirical evidence.

Proposition 9 corresponds to `claims/009-m2-no-active-trade-band.md`. The translated belief-mean family, return-positive domain, existence and alpha-independence of ETF-only optima, multiplier interval and its endpoints, all four active-bound cases, interior width, global marginal equivalence, conditional uniqueness and both degenerate counterexamples are machine checked in `lean/Standalone/M2NoActiveTradeBand.lean` and `lean/Novel/M2NoActiveTradeBandProof.lean`. The formal theorem permits any finite ETF menu and factor dimension; the paper uses the one- or two-ETF specialization. Some unused model hypotheses, including centered shocks, are omitted in the general formal result. The Lean proof uses feasible-move necessity instead of the paper proof's lifted-cost finite-cone argument, so that particular proof device is not separately machine checked. The four-row table is formalized for the algebraically extended score, and the return-positive domain is characterized separately; the paper's $C \cap J$ applies both results. No cited theorem or project hypothesis structure is imported. The assumed examples are exact counterexamples, not empirical estimates. The independent paper-to-formal fidelity review matches the claim, with no added or strengthened hypotheses. The review also confirms that the lifted-cost finite-cone argument is a paper proof, while its optimality conclusion is machine checked by a different route.

Proposition 10 corresponds to `claims/010-m2-higher-cost-narrower-band.md`. The two literal cost cases, their return-positive domains, positive definite covariance, unique ETF-only and full-trading optima, ETF multiplier intervals, complete bands inside the domain, exact widths and strict width difference are machine checked in `lean/Standalone/M2HigherCostNarrowerBand.lean` and `lean/Novel/M2HigherCostNarrowerBandProof.lean`. The formal proof imports Proposition 9's proof to use its band characterization; no cited theorem or hypothesis structure is assumed. The two widths and their positive difference are exact arithmetic identities linked to the formally characterized bands. The formal scope statement checks the algebraic slopes of the interior width at fixed ETF multiplier endpoints; their positivity follows from Proposition 9's nonnegative multiplier interval, and is not a separate formal conjunct. The cap-perturbation scope analysis and the comparison to dynamic no-trade literature are not formal conjuncts. This is one assumed instance, not evidence about calibrated rates or a general cost comparative static. The independent paper-to-formal fidelity review matches the claim with no added or strengthened hypotheses.

Proposition 11 corresponds to `claims/011-m3-finite-continuation.md`. The formal statement defines the two-review observation, posterior, dollar funding classes, nine policy sets, terminal-utility sum, conditional and root continuation values, and certainty-equivalent comparison in `lean/Standalone/M3FiniteContinuation.lean`. `lean/Novel/M3FiniteContinuationProof.lean` machine checks policy compactness and convexity, objective concavity and attainment, pathwise positive wealth and value bounds, conditional attainment and state concavity, finite Bayes regrouping, the attained Bellman representation, both class orders and the existence-only zero-gap equivalence. The root cash constraint is explicit in the formal feasibility definition. The proof chooses conditional optimizers at every observation, including zero-mass ones; the paper may choose zero trades there. The formal theorem permits arbitrary finite instrument and factor counts and omits centered-shock hypotheses unused by this finite-control argument. Conditional independence is encoded in the two scenario weights of the explicit utility sum. No cited theorem or project hypothesis structure is imported. The machine check establishes neither a dynamic trade boundary nor a sign or magnitude for the effect of future ETF adjustment. The independent paper-to-formal fidelity review matches the claim, with no added or strengthened hypotheses. Its state-sufficiency content is the proved finite conditional regrouping; the conditional objective depends on the marked state and posterior by definition. The value bounds make the logarithmic certainty equivalent meaningful. No nondegenerate example is a formal conjunct.

Proposition 12 corresponds to `claims/012-opposite-continuation-effects.md`. The literal two-parameter, one-scenario family with four independently variable rates over $[0, 1/100]$, its valid two-review instances and public revelation, all three certainty-equivalent gap bounds, the two strict continuation signs, the positive active root trade under ETF-only continuation, and the zero-cost waiting optimum are machine checked in `lean/Standalone/OppositeContinuationEffects.lean` and `lean/Novel/OppositeContinuationEffectsProof.lean`. The proof imports Proposition 11’s machine-checked attainment and class orders. It proves the strict ETF-only upper bound directly with exponential inequalities; the paper’s intermediate logarithmic expression and derivative argument are not separate formal conjuncts. The exposure and timing mechanism is an interpretation of the proved policies, not a formal statement. No cited theorem or project hypothesis structure is imported. Alpha is known zero and the active fund alone carries the first factor. The law has no conditional shock risk and reveals the parameter completely; the assumed returns and preference coefficient are uncalibrated. The uniform result concerns only the stated cost box. The independent paper-to-formal fidelity review matches the claim, with no added or strengthened hypotheses; it also confirms that the mechanism narrative and intermediate derivative calculation are paper arguments rather than separate formal conjuncts.

Proposition 13 corresponds to `claims/013-continuation-with-risk-and-partial-learning.md`. The literal four-parameter, four-scenario family, for every $0 < \delta \leq 1/1000$ and all four independent rates in $[0, 1/100]$, is machine checked in `lean/Standalone/ContinuationRobustness.lean` and `lean/Novel/ContinuationRobustnessProof.lean`. The formal statement proves model admissibility, positive-definite conditional covariance $\delta^2 I_2$, independent nonzero alpha uncertainty, and the observation event of probability $1/2$ on which the alpha posterior remains split equally. It also proves every gap bound in (31), both strict signs in (32), and a positive root active purchase at every full-root optimum with ETF-only continuation. The proof imports Proposition 11’s attainment and class orders and Proposition 12’s elementary exponential facts; it proves its utility bounds from the specified returns and funded policies, without a project hypothesis structure or cited theorem. Its ETF-only upper bound uses an exponential inequality rather than the paper proof’s intermediate derivative argument, so that intermediate calculation is not a separate formal conjunct. The machine-checked limits are the stated small δ and rate boxes. Partial learning concerns alpha only; neither the proof nor the witness policies attribute either sign to unresolved alpha. Numerical probes at larger shocks are not part of this result. No calibrated magnitude, alpha-driven mechanism or general dynamic boundary is machine checked.

Proposition 14 corresponds to `claims/014-m4-public-information-obstruction.md`. The two literal public-return laws, their common funded action sets and score, the covariance and ETF-optimum identities in (33), and the endpoint advantages are machine checked. The statement is in `lean/Standalone/M4InformationObstruction.lean`. Its proof is in `lean/Novel/M4InformationObstructionProof.lean`. The formal statement defines the fixed-parameter, finite public sampling law, sample-mean estimate, mean-error covariance, discretely calibrated confidence set, optimized ETF comparator, and exact selected-action gate. For every positive history length and error allowance in $(0, 1)$, it proves both attained full-history benchmark powers in (34) against rules uniformly valid over the entire parameter interval, allowing randomized active actions. It proves that the prescribed gate has the same decision law in both models at every parameter and sample length, and certifies with probability $1/2$ after one record at the favorable endpoint. The two benchmark values at $\eta = 1/4$ are formal too. No cited result, project hypothesis structure or earlier claim is imported into the proof. The paper’s alternative confidence-set construction for the full-history rules is not a formal conjunct; neither are red’s finite calculations of gate power at two and three records. Uniform coverage of the prescribed confidence set for general instances of the sampling model is not a result here. In particular

the gate is not asserted to have monotone power. The one-record attribution of its shortfall follows the stated overlap and gate calculations; no separate general attribution theorem is claimed. Known discrete shocks, exact observations, zero costs, known premia and the fixed ETF optimum delimit the machine-checked example. No realistic magnitude, general sample-size rate or robustness to rounding is proved.

Proposition 15 corresponds to `claims/015-m4-bounded-law-history-rate.md`. The machine-checked statement and proof modules are listed in the proof map. The statement reuses the one-quarter sampling definitions from Proposition 14’s formal statement file, but imports no result from its proof. The machine check covers the funded class and unique ETF optimum, the advantage and exposure formulas in (35), both quantified implications in (36), the conservative certificate against the original L_N in (37), its coverage-event power, and the logarithm comparison. The lower-bound quantifiers allow a different full-history randomized rule for each known finite law but require validity over every parameter in the interval. The formal proof constructs the sine-weighted hard law and a finite overlap bound; it uses an affinity-squared bound rather than the paper proof’s total-variation inequality. Both yield the stated constant. The bounded-mean tail and discrete confidence radius are proved from finite sums, including zero-variance laws. No rate, tail inequality, cited result or project hypothesis structure is assumed; the axiom audit reports only standard axioms.

The formal result is a worst case over laws whose finite support may grow with the prespecified margin, and a sufficient length for a specific conservative gate. It proves neither matching constants nor monotone power for a particular law. Alpha is known zero, and the ETF comparator is fixed in this family. Unknown or rounded shock laws, a moving ETF optimum, multivariate uncertainty and realized economic magnitudes are outside its machine-checked scope.

Proposition 16 corresponds to `claims/016-m4-joint-directional-rate.md`. The machine-checked statement and proof modules are listed in the proof map. Red independently checked the claim and PM approved it before formalization. The formal statement reuses M4’s definitions and Proposition 15’s rule requirements; its proof imports Proposition 15’s proof module and re-derives the paired score errors used here. It proves the admissibility and covariance of the literal family, the moving ETF optimum and two-direction advantage in (38)–(39), both quantified length implications in (40), the original optimized-comparator certificate and its coverage-event power in (41), and the variance formulas and examples in (42). The lower bound allows a separate randomized full-history rule for each known law and remains valid for singular J : the proof lifts a public rule to a more informative finite latent history. Its adverse law retains covariance $JJ^\top$. Finite concentration and calibration prove the upper bound, including the empty-set fallback. The formal statement also covers a one-record full-history benchmark when both comparison variances vanish. No rate, tail or covariance inequality is assumed as a project hypothesis; the axiom audit reports only standard axioms.

When $\Omega_{33} = 0$, the formal statement proves that alpha is read exactly from each record and that the unspanned-premium variance floor remains. Its separate known-alpha conjunct machine checks the lower bound when both rule requirements are imposed only on the domain slice $\alpha = \alpha_0$, for an α_0 fixed before the history length and law. The hard pair shares that alpha. The formal statement asserts existence of this slice, not the bound for every alpha value in the domain.

The formal scope is a worst case over a bounded class of known finite laws, with support size allowed to grow as σ/δ . The lower and upper constants differ by roughly 5×10^4 at $\varepsilon = 1/16$, so only their order matches. The two contrasts combine the unspanned premium and alpha; the result does not isolate manager skill. A continuously moving risk-averse ETF optimum, general menus, estimated loadings, rounded data, efficient calibration and repeated live decisions remain outside the machine-checked result.

Proposition 17 corresponds to `claims/017-m4-curved-entry-rate.md`. Red independently checked it and PM approved it before formalization. The statement and proof modules listed in the proof map machine check the funded optima and optimized-class advantage in (43)–(44), both quantified history-length implications in (45), and the conservative original-target certificate and its coverage-event power in (46). The formal proof imports Proposition 16’s proof module for the finite-law tools and re-establishes the required specialization; no cited result or project hypothesis structure supplies the rate. The lower implication covers law-specific randomized rules using the full public history. The upper implication covers the stated plug-in gate, including feasibility of its unprojected estimates, the moving ETF comparator’s quadratic penalty and fallback on an empty confidence set. The axiom audit reports only standard axioms.

The machine check concerns the one-quarter family with $\gamma = s^{-2}$, known exact finite laws, zero costs and strict cash slack. The hard law’s support grows as $s/\sqrt{\delta}$. It does not establish an optimal rate for each individual law, unknown shock laws, binding funding, general preferences, rounded observations or repeated reviews. Its mean signal combines the first factor premium and manager alpha; it does not identify their separate contributions.

Proposition 18 corresponds to `claims/018-m4-curvature-certificate.md`. Red independently checked the numbered claim and PM approved it before formalization. The statement and proof modules in the proof map machine check the admissible funded setting, attainment and uniqueness of the exact plug-in optima, the curvature-normalized scale in (47), and the original-target certificate, active-change condition and uniform false- certification bound in (48). They also check the coverage-event power and sufficient length in (49), the $K = 0$ identity, and both class-level implications in (50), including a member with $K = \kappa$. The formal proof imports Proposition 4’s action-class results and Proposition 17’s lower-bound family; it uses the finite-law machinery from Proposition 16. No cited result, project hypothesis structure, optimizer-stability assumption or statistical rate is imported as an axiom. The axiom audit reports only standard axioms. The formal admissibility condition keeps the paper’s one- or two-ETF scope. The independent fidelity review matches the numbered statement without added hypotheses; it confirms that the later approximate-optimizer remark is outside the formal statement.

The approximate-optimizer formula (51) is a ***reviewed paper proof, not machine checked***. Red checked its derivation, exact counterexamples to simpler value-gap shortcuts, and finite numerical attacks; those checks do not establish a floating-point certification algorithm. It requires certified feasibility, global value gaps, the radius and directed rounding. It preserves a pointwise lower bound and hence coverage-based false-certification control, but no power rate for approximate outputs is proved. The machine-checked class lower bound uses Proposition 17’s zero-cost subfamily; it is not a rate for each binding face or cost schedule.

Proposition 19 corresponds to `claims/021-m4-law-free-faces-match-order.md`. Red independently checked it, including a recheck after its tail bound was replaced by a citation, and PM approved it before formalization. The statement and proof modules in the proof map machine check the setting (every law in $\mathcal{L}_J$ gives an admissible instance, the face formulas, the face ranges, $\mathcal{K}_J \subset \mathcal{L}_J$ and $\bar{\sigma}_j = \sigma_j$ there), all of part (i) including the identity (53), the full-level face bounds, the union-bound converse, the level-factor identities with its value at $\eta = 1/20$ and the decomposition inequality, the range rule’s membership in the rule class, its false-certification bound and both sufficient lengths in (ii), and both implications in (55). The formal statement reuses Proposition 16’s family, rule class and law class $\mathcal{K}_J$, Proposition 15’s requirement predicates and Proposition 14’s sampling objects. A law-free rule is one formal rule fixed before the law is quantified, and the range rule’s estimate is shown to be the sample mean under every law.

The paper proof cites Hoeffding’s inequality ([39], Theorem 1). The formal proof instead uses Hoeffding’s lemma as already proved in the pinned Mathlib library, followed by a finite Chernoff step, so no tail

inequality is assumed and nothing is re-proved; the axiom audit reports only standard axioms. Several formal parts are stronger than the prose: the range rule’s validity holds for every history length, whereas the claim restricts it to $N \geq 3 \log(2/\eta)$; part (i)’s joint and union-bound statements hold for any data-dependent sets and bounds under any finite law; the sufficient lengths in (ii) need only $\varepsilon < 1$; and the lower bound in (55) needs neither $(JJ^\top)_{11} > 0$ nor a minimum history length. Infima are taken in the extended reals. The formal lower bound applies to rules in Proposition 16’s rule class, which is the class the prose uses. The independent paper-to-formal fidelity review matches the claim with no added or strengthened hypotheses, and confirms the more general formal parts listed here. It also notes that the width and history-length reading of the level factor in (i) holds only for face bounds of the stated width shape, as the proposition says.

The machine check establishes worst-case order only, over a bounded class with known J and known range, in the two-face, zero-cost, all-cash family with the fixed active action w_A . Its constants are about 2.8×10^5 apart. It gives no per-law efficiency, no analysis of empirical Bernstein constants, no time-uniform or sequential guarantee over repeated reviews and no result for laws of unknown range; it does not separate manager skill from the unspanned premium.

Proposition 20 corresponds to `claims/022-m3-etf-channel-sandwich.md`. Red independently checked it, including a recheck after the monotone-comparative-statics reading was added, and PM approved it before formalization. The statement and proof modules in the proof map machine check, for every instance of the funded two-review model: attainment of $CE_{D,R}$ as the maximum of c_R over the root class with $\varphi, \psi \geq 0$; both sandwiches in (57) for every choice of maximizers; the cap (58), the zero premium at a root with neither cash nor ETF holdings, and the corner sign condition; and both region-movement statements in (iii), each with an optimal policy that makes no active root trade. For the family in (iv) they check that every member is an instance whose first observation reveals the parameter, all four certainty-equivalent bounds and the channel bound in (59), positivity of both gaps, feasibility and zero premium of the all-active root, and the lower bound on the all-cash root’s premium. Both signs are formal: this family gives a negative channel and Proposition 12’s family a positive one. The formal cap holds for any upper and lower bounds on ETF gross returns, which includes the tightest choice in the prose. The proof imports the proof modules of Propositions 11 and 12. No cited result or project hypothesis structure is used, and the axiom audit reports only standard axioms.

The reading of increasing differences, single crossing, hedging demand and signal persistence against the model is prose; only its numerical fact about the two premia is a formal conjunct. The machine check gives no closed-form sign condition, no lower bound on the premium, no movement of the region itself and no calibrated magnitude; its example has no conditional risk, full revelation and known zero alpha. The independent paper-to-formal fidelity review matches parts (i)–(iv) and the numerical fact about the two premia, with no added or strengthened hypotheses. It notes that the fall of the premium in the root active holding is formal only as the comparison between the all-active and all-cash roots, which is how the text states it.

Proposition 21 corresponds to `claims/026-m3-mean-type-caps.md`, which refiles two earlier versions that red refuted for wording errors: a misdescribed certified case, a false comparison, a buy-side cap written with the wrong wealth amount, and a zero-gain condition without switches. Red checked the refile, including 600 random two-ETF nodes solved exactly, and PM approved it before formalization. The statement and proof modules in the proof map machine check parts (i)–(iv) for every instance of the funded two-review model: the near-attaining node constructions and the lower bounds (61) on every range-type cap, the realizability constraint on node data, that G_y^{ub} is a range-type cap, and the zero-gain example; the tangent cap (62), the equivalence of $T_y = 0$ with the three no-favorable-direction conditions, zero gain

there, and the one-ETF formula; the one-ETF curvature cap (63) in each direction; and part (iv) as stated, including both sign conditions with β_{mean} built from T_y . The node objects, the weights, the aggregate (60) and the identity $\varphi = \mathcal{A}(G)$ come from the formal modules of a companion result on node-wise lower bounds, which enter with this result and are machine checked. That companion result is Proposition 25; Proposition 21 uses only its definitions and the identity. Several formal parts are stronger than the prose: parts (iii) and (iv) hold for every valid wealth-range bound and return bracket, and the sign conditions hold for any node-wise lower bounds. The formal proof uses Jensen’s inequality under the tilted node measure for (ii) and a monotonicity argument on the second derivative for (iii), rather than the paper’s one-sided derivative and double integration. The proof imports the proof modules of Propositions 11 and 20. No cited result or project hypothesis structure is used, and the axiom audit reports only standard axioms.

The numerical comparison after the proposition has no formal counterpart. It is computed from solver optimizers, not certificates, on one assumed base design. The zero-premium corner at $\rho = 20$ is read from those optimizers.

The independent paper-to-formal fidelity review matches parts (i)–(iv) with no added hypotheses. Its first version found that the formal cap in part (iv) was built from the looser middle bound in (62) rather than T_y . The formal statement was then restated with T_y , and the re-audit confirmed that part (iv) is machine checked as stated. The numerical comparison computes the same β_{mean} that the formal statement defines.

Proposition 22 corresponds to `claims/028-m2-soft-target-two-stage.md`. Red independently checked it, including 200 random instances with factor-correlated residual shocks, and PM approved it before formalization. The statement and proof modules in the proof map machine check that $b(F) \subset \mathcal{R}_0$ with $\mathcal{R}_0$ convex and compact, and that stage 1 on $\mathcal{R}_0$, stage 2 and the joint problem attain their maxima. They also check the identification (66) and the $\nu = 0$ case; all of (ii), including the equality condition, the Euclidean bound and the target-reached case; the exact form (67) with nonnegative slack and kink terms and the sign condition for a zero kink; the target-confined comparison (iv); and the normal-cone statement (v). The formal statement is more general than the prose: it allows any numbers of active funds, ETFs and factors and a score at any parameter, which by Proposition 3 includes the belief-mean score. Part (ii) is proved for any holding set, and the directional derivatives are proved to be right derivatives along the segment. The files are self-contained. They import only Proposition 4’s proof module, and nothing rests on the target-confined procedure’s separate analysis. No cited result or project hypothesis structure is used, and the axiom audit reports only standard axioms. Not formal: the reading of the slack as binding-constraint multipliers times the displacement, and the face-normal reading of ν beyond the normal-cone statement. An independent paper-to-formal fidelity review of this result has not yet been recorded.

Proposition 23 corresponds to `claims/027-m2-two-stage-separation.md`. Red refuted its first filing, which asserted that the residual value V is always concave; red and lean gave a counterexample with factor-correlated residuals. The amended claim states concavity for $G + V$ in general and for V only under zero cross moments. Red checked the amendment and PM approved it before formalization. The statement and proof modules in the proof map machine check: the split (68) at the belief mean; B_F nonempty, convex and compact; attainment of V ; concavity of $G + V$ and, under zero cross moments, of V ; $J = \max_{B_F}(G + V)$; stage 1’s maximizer and the unconstrained target’s uniqueness when it lies in B_F ; the identity (70) with both brackets nonnegative; the separation criterion of (ii), including the relative-interior converse for concave V (with the relative interior taken as the intrinsic interior); both bounds of (iii); and the flat case of (iv) with the missing-direction statement for one active fund. The formal statement allows any numbers of active funds, ETFs and factors, and its loss bounds need no positive definiteness. It re-derives the belief-average identity locally and imports only Proposition 4’s proof module. No cited result or project hypothesis structure is used, and the axiom audit reports only standard axioms.

The shared-budget and cost-kink failures of separation are qualitative explanations and have no formal counterpart. The numerical comparison after the proposition is computed from floating solver outputs on one calibration, not certificates. The comparison with Proposition 22’s soft procedure uses that result’s part (iv); no ordering between the two is proved. The independent paper-to-formal fidelity review matches parts (i)–(iv) with no added or strengthened hypotheses. It confirms that, of the funded failures of separation, only the missing-direction fact is formal, as the text says.

Proposition 24 corresponds to `claims/020-m4-finite-comparator-faces-iff.md`, which refiles a claim red refuted: that filing asserted that a finite face set is never exact under a curved score, and red’s single-comparator instance in (v) shows otherwise. Red checked the refile, including the vertex counts, the closed form (73) with singular Ω , the covering bound in (iv) and the instance in (v) exactly. PM approved it before formalization. The statement and proof modules in the proof map machine check all five parts. That covers (i) with the halfspace descriptions, finiteness, the counts, independence from the parameter and law, and (71). It covers (ii)’s attainment on faces, (72) for every parameter set, the faces as plug-in selections, and (73) for nonempty C_N . It also covers (iii)(a) and the polyhedral structure of (iii)(b); (74), the exact excess and the $m + 1 \geq 16$ count in Proposition 17’s family; and (v) with its instance. The finiteness of the faces and their counts in (i), and the finite-union-of-polyhedra statement in (iii), are conditional on the linear-programming vertex theorem. It enters as the audited hypothesis structure LPVertex (ledger entry AX-05), whose instance, the empty constraint system in dimension zero, is degenerate and disclosed, because a faithful instance would re-derive the cited theorem. An earlier version of the formal proof proved the theorem in general for these cells. The re-audit replaced that with the cited hypothesis, as the project’s rules require for a known general theorem. Every other part of (i)–(v) is unconditional. The face maxima use the model’s lexicographic selection. The proof imports Proposition 18’s proof module, and the axiom audit reports only standard axioms. Formal choices: the ETF-only cells use the compliant incumbent, (73) needs a nonempty C_N , infima over parameter sets are in the extended reals, and (v) needs T and S nonempty. The reading after the proposition has no formal counterpart. The independent paper-to-formal fidelity review matches the claim, with several formal parts more general than the prose. Its re-audit confirms that the vertex theorem is a hypothesis of the finiteness conclusions only.

Proposition 16’s existential known-alpha pre-disclosure conclusion is machine checked. Proposition 25 corresponds to `claims/023-m3-premium-node-bounds.md`. Red checked it by hand, recomputed the two certifications independently, tested the node lemmas on random nodes and computed the true premia at the numerical comparison’s optimizers. After Red’s recheck of the revision that replaced a self-proved curvature step with the citation of Hoeffding’s lemma, PM approved it before formalization. The statement and proof modules in the proof map machine check all five parts. They cover the decomposition (75) with its bounds and monotonicity; the sure-sign and risk-adjusted floors (76), including the $rh/2$ floor; the caps and the chain $\alpha \leq \varphi \leq \beta_{\text{node}} \leq \beta$; the sign conditions (77) and their robust forms; and both certifications, with the uniqueness claims, closed-form aggregates and margins in (78). These files entered main with Proposition 21, which builds on them, and are unchanged. Hoeffding’s lemma, ledger entry AX-08, enters through the pinned Mathlib library’s machine-checked lemma, applied under the tilted node measure to a variable of range at most $2s$. It is not an assumed hypothesis, and nothing is re-proved. Several formal parts are more general than the prose. The floors and caps hold for every valid bracket of the node-path returns, the sure-sign bounds for every δ , and the caps for every nonnegative bound on gains and losses. In (iv), uniqueness of the cash root is proved with exponential inequalities instead of the paper’s concavity argument. The proof imports the proof modules of Propositions 11, 12 and 20, and the axiom audit reports only standard axioms. The numerical ratios after the proposition, and the reading that the caps are the

weak side, have no formal counterpart. The independent paper-to-formal fidelity review matches the claim.

Proposition 26 corresponds to `claims/029-m6-quarterly-band-static-ceiling.md`. Red checked parts (i)–(v) by hand and ran an independent backward induction on 40 random instances (345 bands) with no failure. PM approved the claim before formalization. The statement and proof modules in the proof map machine check every numbered part.

- (i): strict convexity and continuity of G_t , convexity of V_t with slopes in $[-\kappa^+, \kappa^-]$, the ordering of the edges, and uniqueness and optimality of the clip (79) for every real pre-trade holding.
- (ii): the ceiling (80), the last-review band and the edge brackets.
- (iii): the tilted band (81) with its sufficient conditions.
- (iv): the equality characterization and strict narrowing otherwise.
- (v): uniqueness, the nonempty no-trade set, (82) with its single-coordinate and Euclidean corollaries, the static shape and the bundling equivalence.

The formal model is the slack-budget instance, so cash does not appear, with a finite state set and finite outcome laws. Part (i) uses Mathlib’s one-sided derivatives of convex functions, and (v) uses the comparison along a segment, with no subdifferential sum rule. No cited result, project hypothesis structure or other claim’s proof module is used, and the axiom audit reports only standard axioms. The trade-to-the-edge structure re-derives Constantinides’ classical band. When that source’s text is obtained, the plan is to cite it through a ledger entry and keep only the additions. Not formal: the coarse/fine reading for ETFs and funds, the remark relating (80) to Proposition 9, and the regression reading of the bundling shift. The independent paper-to-formal fidelity review matches the claim, with no added or strengthened hypothesis and several conjuncts stronger than the prose.

Proposition 27 corresponds to `claims/030-m5-partial-adjustment-split.md`. Red checked parts (i)–(v) by hand and confirmed the separated case (iii) with an independent recursion to 2×10^{-15} , together with each coupling term alone. PM approved the claim, with two corrections that changed no result: the eigenvalue range of Γ_t , and an induction step in (v). The statement and proof modules in the proof map machine check, over arbitrary finite index sets:

- the filter facts of (i), including positive definiteness, monotonicity, the innovation covariance, the decoupling and (83);
- the Bellman verification of (ii) with the recursion, uniqueness, the eigenvalue and similarity facts for Γ_t , the partial-adjustment form and the aim’s weights summing to the identity;
- the coordinate change, reference-case blocks and exact separation (86) of (iii);
- the scalar speeds, monotonicity, the stationary root (87) and the two limits of (v), together with the identification with Gârleanu and Pedersen’s rate.

At PM’s instruction, under the formalization boundary for measure-theoretic infrastructure, two steps are reviewed paper proofs, not machine checked. The first is the Gaussian conditional moments, which give the predictive moments and discharge the martingale hypothesis of the formal Bellman verification. The second is the dynamic-programming step from Bellman optimality to optimality over all admissible measurable policies. Ledger entry AX-10 (the stationary linear-quadratic-Gaussian separation) is cited as context and not used; no hypothesis structure or other claim’s proof module is used, and the axiom

audit reports only standard axioms. Not formalized: that the value function’s intercept is quadratic in the belief mean, and the readings of (iii)-(iv) beyond the block formulas. The independent paper-to-formal fidelity review matches the claim within the disclosed formal scope, with no added hypothesis; the formal statement is more general in allowing $\rho \geq 0$ and arbitrary finite index sets.

Proposition 28 corresponds to `claims/031-m5-missing-direction-leak.md`. Red found that the first version’s converse in (iii), nonvanishing at every review, was unproved beyond $T \leq 3$ and the stationary case, and required a correction. PM withdrew the claim for revision, and red passed the revision, which asserts nonvanishing only where M_t is nonsingular. PM approved it before formalization. The statement and proof modules in the proof map machine check:

- (i): the projection, the coordinate bijection, the reachable exposure (89) and the vanishing hedge;
- (ii): the fund positions as the reduced problem’s policy, through the Schur complement of the joint Riccati matrix, with the reduced Markowitz portfolio;
- (iii): the sensitivity (90) with its recursion, the vanishing case, the proved nonsingular cases (last review, $T \leq 3$, the symmetric-part condition, and the stationary limit), the positive semidefinite, nonzero risk term, and its effect on the own-position coefficient at every review;
- (iv): the monotone convergence of the Schur complement and the hedge map along the learning path.

Proposition 27’s formal problem was generalized to a deterministic, time-varying mean map for this purpose. As for Proposition 27, the Gaussian conditional moments and the passage to all measurable policies are reviewed paper proofs, not machine checked. Not formalized: the numerical observation that M_t is nonsingular at every review, and the reading of (iv) that the premium estimate has a nonzero limit. No cited result or project hypothesis structure is used, and the axiom audit reports only standard axioms. The independent paper-to-formal fidelity review matches the claim within Proposition 27’s disclosed formal scope, with no added hypothesis.

Proposition 29 corresponds to `claims/032-m5-learning-aim-two-speeds.md`. Red’s verdict required two corrections, which were met before PM’s approval: the $1/t^2$ rate in (i), and the dependence on Proposition 27. No result changed. The statement and proof modules in the proof map machine check:

- (i): (91) with its equality condition, the monotone gap and its $1/t^2$ limit;
- (ii): the two-direction decomposition of the rate and the Riccati matrix, $g_t^c \geq g_t^r$ with its equality condition, and the common stationary limit;
- (iii): the positive weights summing to one, the bounds in (92), the learning factor at least one with equality exactly at the last review, and the aim and Markowitz formulas in each direction;
- (iv): as Proposition 27 (iii).

The proof imports Proposition 27’s proof module. Its paper-level steps are Proposition 27’s. Not formalized: the effect of ETF costs on the exposure, and the two numerically observed orderings. No cited result or project hypothesis structure is used, and the axiom audit reports only standard axioms. The independent paper-to-formal fidelity review matches the claim within Proposition 27’s disclosed formal scope, with no added hypothesis and several conjuncts stronger than the prose.

Proposition 30 corresponds to `claims/033-m5-plug-in-value-loss.md`. Red’s first review led to a revision that PM approved after red’s recheck. The revision assumes a positive semidefinite estimate in

(iii), states the indefinite failure mode, and restricts (iv) to a covariance estimated once before the run. The statement and proof modules in the proof map machine check:

- (i): the one-step identity at every review and state, the Bellman objective at any trade equal to the value minus $\frac{1}{2}e^\top D_t e$;
- (ii): the pathwise error decomposition and bounds, including $\|D_t\|_* \leq 1 + \|R_t\|$;
- (iii): all four Riccati bounds, for positive semidefinite true and estimated risk matrices, positive definite Λ and $\rho \in [0, 1]$;
- (iv): the bounds on the confidence event, whose probability carries over by inclusion.

Three steps are reviewed paper proofs, within the scope PM confirmed under the formalization boundary: summing the one-step identity into (93) under conditional expectations, the passage to second moments, and the recursion for the state moments. Ledger entries AX-11 and AX-12 (the stationary perturbation and certainty-equivalence results) are cited as context and not used. The proof imports Proposition 27's proof module, and the axiom audit reports only standard axioms. The calibrated numbers come from a check script, not a formal statement. The independent paper-to-formal fidelity review matches the claim on the finite-dimensional core the formal file declares, with the expectation-level steps at paper level, as disclosed.

Proposition 31 corresponds to `claims/100-m7-learning-band-transfer.md`. Red passed it and PM approved it. Prose corrections followed at Lean's and red's request and changed no result: the drift's sign requires a learning risk charge, the landing probabilities in (iv) are closed, and t_0 does not depend on the horizon. The statement and proof modules in the proof map machine check, in the finite-law variant:

- (i): the instance is a multi-review tracking model with covariance $\Sigma(t, z) = \Sigma_t$, and Proposition 26's statement transfers, proved by its own proofs at a fixed review and state, including the two-date coarse condition and the separation for diagonal Σ_t ;
- (ii): the information form and the monotone convergence of P_t, Σ_t, c_t and the static width;
- (iii): the target decomposition, the positive semidefinite bracket, the drift's sign, the exact $1/t^2$ limits, and the tilt formula (97), whose zero case needs $\kappa > 0$, a condition the formal statement adds;
- (iv): the landing dichotomy with $U + D = 1$, the move bounds (98), and eventual strict narrowing from a t_0 fixed before the horizon.

The proof imports Proposition 26's proof module. The unconditional moments of the belief-mean innovation over the finite law are a reviewed paper step that enters no formal conclusion. No cited result or project hypothesis structure is used, and the axiom audit reports only standard axioms. The independent paper-to-formal fidelity review matches the claim, with no added hypothesis and a formal model more general than the claim's.

Proposition 32 corresponds to `claims/034-m5-plugin-in-loss-inputs.md`. Red passed it and PM approved it, with the formal scope confirmed as for Propositions 27 and

1. The statement and proof modules in the proof map machine check:

- (i): the plug-in path's product form, the decomposition $e_t = \mu_t + \sum_u W_{t,u} \eta_u$ for every innovation sequence, and the loss's vanishing condition, for arbitrary plug-in coefficients;

- (ii): (100) as an identity for symmetric covariances, and the scalar recursion as Proposition 27's in one dimension;
- (iii): the derivatives of the coefficients and the $C\epsilon^2 + O(\epsilon^3)$ expansion, the costless limit, the $1/\lambda_A$ order (exact for $x_0 \neq 0$ with a nonzero error) and $1/\lambda_A^3$ for $x_0 = 0$, the one-review formula (101) with its positivity at intermediate cost, and the monotonicity of the posterior variance in the prior;
- (iv): separation of the two risk blocks, invariance of the fund positions to the factor covariance under spanning, and a family of instances in every non-spanning setting where the fund coefficient changes.

The expectation identities over the Gaussian filtering model, and the identification of C 's expectation form with its derivative form, are reviewed paper proofs. The proof imports the proof modules of Propositions 27 and 28. Ledger entry AX-12 is context only, no cited result or project hypothesis structure is used, and the axiom audit reports only standard axioms. The independent paper-to-formal fidelity review matches the claim on the deterministic core the file declares, with several conjuncts stronger than the claim. It notes three points where the formal statement is narrower than the prose. The vanishing condition in (i) is formal only when every precision decrement is positive definite. Of (iii)'s precision statement, only the monotonicity of the posterior variance in the prior is formal. And the split of the total loss into the exposure and fund parts in (ii) is not itself a formal conjunct: it follows from Proposition 27's separation and was confirmed numerically.

Proposition 33 corresponds to `claims/035-m5-reestimation-guarantee.md`. Red passed it and PM approved it, with the formal scope split as for Propositions 30 and

1. The statement and proof modules in the proof map machine check (102) for any positive semidefinite estimate and for the true coefficients, the position bound along the re-estimating path, and the pathwise on-event and off-event bounds. The good event is a hypothesis of the formal statement. The loss identity in expectation, the split with Cauchy-Schwarz and Doob's inequality, and the event's existence from ledger entry AX-04 (the confidence sequence) are reviewed paper proofs. The proof imports Proposition 30's proof module; no project hypothesis structure is used, and the axiom audit reports only standard axioms. The independent paper-to-formal fidelity review matches the claim on the pathwise core the file declares, with the expectation-level combination and the confidence event at paper level, as disclosed.

Proposition 34 corresponds to `claims/036-m5-when-anticipation-matters.md`. Red passed it and PM approved it. The statement and proof modules in the proof map machine check, for one fund or direction with a general nonincreasing positive precision path:

- (i): $\text{aim}_t = L_t \text{aim}_t^S$, the bounds (104) with the strict upper bound at every review, the exact form of $L_t - 1$, $L_t = 1$ exactly at the last review for a strictly decreasing path, and the threshold;
- (ii): the rate ordering, the monotonicity and Lipschitz constant of the constant-risk rate, and the trade identity and bound (105);
- (iii): given the aim formula, the split, the renormalized weights as a probability, the properties of M_t and (106);
- (iv): the zero-cost limits and Proposition 27's separation, together with the identity linking Proposition 31's drift to the learning factor's one-step increment.

The aim formula under mean reversion, the no-hedging reading, and the band reading are reviewed paper proofs. The proof imports Proposition 29's proof module. No cited result or project hypothesis

structure is used, and the axiom audit reports only standard axioms. The independent paper-to-formal fidelity review matches the claim, with several conjuncts stronger than the prose and two steps at paper level, as disclosed.

Proposition 35 corresponds to `claims/105-m7-premium-error-in-fund-choice.md`. Red passed it and PM approved it. The claim uses three lemmas from an approved companion result on two-stage implementation, since written up as Proposition 45. The formal modules of that companion result entered main with this one, are imported by its proof and are machine checked. The alpha-band holding and the reduced objective are defined in the proposition itself. The statement and proof modules in the proof map machine check:

- (i): fund holdings at slack joint optima are the band holdings, independent of premia, factor covariance and loadings, and the exposure loss;
- (ii): the joint optimum's form, the invariance of the holding to the premium belief beyond $B_i^A J^\top \hat{\lambda}$, the narrowing factor and its limit along the learning path, the Lipschitz move bound, the exact move on a common trading piece and none on a common hold piece, the decision conditions, and the loss bracket (108);
- (iii): the exact loss identity (109) with its zero condition, the naive move bound, and the zero-incumbent readings.

In (ii), Lean corrected the decision conditions at two boundaries: the purchase and sale conditions include the cap and the zero bound, and an equality case at the threshold. The text states the corrected form. The expectations over the premium error are reviewed paper proofs. No cited result or project hypothesis structure is used, and the axiom audit reports only standard axioms. The independent paper-to-formal fidelity review finds a match except at those two boundaries in (ii), where the formal statement is the correct one and the claim's prose, not yet revised, is not; the text uses the formal form. The expectations are at paper level.

Proposition 36 corresponds to `claims/039-when-data-support-an-active-change.md`. Red's review required two corrections, the proved form of (ii)'s necessary length and (v)'s hypothesis that the prior is at least as diffuse as the floor. Math made them, red's recheck passed them, and PM approved the claim. The formal forms follow math's later revision, which Lean B's report prompted. The claim proves the one-fund, one-review threshold it uses in (i) and names the general threshold of claim 102 without relying on it. The statement and proof modules in the proof map machine check:

- (i): existence and uniqueness of the one-fund optimum and the buy and sell conditions;
- (ii): the certifying rule's two errors under independent Gaussian residuals, the lower bound's arithmetic from the Gaussian affinity, $z_\epsilon^2 \geq \log(1/(4\epsilon))$, and the Bayesian condition;
- (iii): the Hoeffding rule for independent centred residuals bounded by R , not necessarily identically distributed, and, in a second stage, the two-point lower bound for any randomized rule, through Proposition 15's finite Le Cam lemma;
- (iv): the certificate under inequality (7) as a hypothesis;
- (v): the fixed point (112), its uniqueness and position below the stationary prior, the monotone convergence, the floor, and both limits;
- (vi): the Gaussian variance of the decision quantity and the length ratio (113);

- (vii): the direction variances under the pooled posterior and the averaged rule's length.

Hoeffding's inequality enters through the pinned Mathlib library, as for Proposition 19. Part (ii)'s lower bound takes ledger entry AX-09's inequality as a premise, and together with the mixing instantiations, the Gaussian and Kalman posterior forms, the reduction to the one-fund problem (claims 030 and 104) and the reading of (113)'s loading as $J(B_t^A)^\top$, they are reviewed paper proofs. The proof imports the proof modules of Propositions 15 and 29, and the axiom audit reports only standard axioms. The independent paper-to-formal fidelity review matches the claim for the first stage, with the lower bounds' probability steps at paper level as then declared; the second stage's formal two-point bound in (iii) has not yet been reviewed.

Proposition 37 corresponds to `claims/038-integration-theorem.md`, which refiles the refuted claim

1. Red refuted 037 on one point: under proportional costs in the fine regime the band's centre anticipates about two thirds of a mixing time of the learning drift, not one step. Red's review of the refile then required the orders in the gain to be stated separately for the two costs; math made that correction, red's recheck passed it, and PM approved. The component results enter through the formal statements of claims 009, 027-036 and 100, and the proof imports their proof modules. The statement and proof modules in the proof map machine check the new steps:

- (i): the tracking identity (115) for any positive definite covariance;
- (ii): the scalar Kalman path, the drift identity and bound (116), the innovation variance $p_t \kappa_t$, and the one-step form;
- (iii): the quadratic-cost bound (117), its strict positivity, the bounds and zero-cost limit of D_t , and the condition for a change of the aim by a fraction θ ;
- (iv): the residual identity and its bounds on a finite-law decision problem, the stage-reward bound from the caps, and the last-review bound (118);
- (v): the quadratic-cost rule as the unique maximizer, and when it trades. Reviewed paper proofs, not machine checked, are the proportional-cost side of (iii), whose coarse-regime part is claim 100's statement and whose fine-regime part is Martin's cited leading-order law in the corrected form of refuted conjecture 101, checked numerically by red; the off-event expectation step of the re-estimation guarantee under proportional costs; the reading of the innovation variance as a standard deviation of the target; the attachment of claim 104 at the last review; and the composition reading. No cited result or project hypothesis structure enters the formal proof, and the axiom audit reports only standard axioms. An independent paper-to-formal fidelity review of this result has not yet been recorded.

Proposition 38 corresponds to `claims/107-m7-dynamic-bundling-band.md`. Red's review found the theorems sound but required a correction to the reading of (iv), which had the idle-ETF regime widening the band beyond (121); mathb restated it, red's recheck passed it, and PM approved. Experiment 036 compared the result with an exact two-instrument dynamic program and agrees within grid resolution; that is an illustration, not part of the proof. The statement and proof modules in the proof map machine check every numbered part:

- (i): the positive residual curvature, continuity and strong convexity of U_t , the split of the full optimum into the fund and ETF problems, and the clip;
- (ii): the ceiling (121);

- (iii): each edge's bracket in (122), under its stated edge condition and an interior ETF optimizer;
- (iv): the exact edges (123), the width formula, the ceiling for equal slopes, the opposite-direction width, the fund-alone width for untraded ETFs, and the lower bound on any interior band's width;
- (v): the outer parallelotope (124) for any number of instruments, with the diameter bound taken from Proposition 26's formal (82);
- (vi): the reduced one-instrument instance, the value and stay objective up to constants, and the equality of the edges, so that Proposition 26's formal statement applies; carrying Proposition 31's learning-path statements to the reduced instance is prose. The claim lists the per-review optimality condition AX-13, but the formal proof does not use it: every first-order fact is proved by one-sided difference quotients. The proof imports claim 029's proof module, and the axiom audit reports only standard axioms. Reading a_t^* in (vi) as claim 104's reduced target is prose. The hypothesis in (vi) that the hedge lies in the ETF box at every fund holding, not only at optimal ones, was Lean B's formal reading; mathb has since written it into the claim's statement as what the paper proof uses. The independent paper-to-formal fidelity review matches the claim and finds several formal conjuncts more general than the prose: (vi) needs only zero ETF rates among its four conditions on the ETFs, and the narrowing under Proposition 26's condition in (ii) is prose, not a conjunct.

Proposition 39 corresponds to `claims/106-m7-etf-only-restriction-cost.md`. Red found every part sound with four nits: the room restriction in (iv)'s display, the cap form of the zero-incumbent reading, the netting trades' cash room, and one inequality's direction in the proof's prose. PM approved, and mathb made the nits without changing a result. The statement and proof modules in the proof map machine check:

- (i): the ordering of the values, both zero-cost conditions, the value-of-sales condition, the decomposition of C^0 , and, given AX-13, claim 104's multiplier criterion at the ETF-only optimum;
- (ii): (126) with the cap and floor forms and the zero-incumbent readings, at the three optima with a slack budget and interior ETFs;
- (iii): the same forms with the unreachable fund's reduced moments;
- (iv): the four brackets (128) with the room restriction in (127), the room for the netting trades and their cash entering as the feasibility of the four netted holdings. Ledger entry AX-13, the polyhedral Karush-Kuhn-Tucker theorem, is needed only by the multiplier criterion in (i). The formal statement makes all of (i) conditional on it, so (i) is machine checked conditional on the citation and (ii)-(iv) unconditionally. The proof imports the proof modules of claims 104 and 105, the companions of Proposition 35, and the axiom audit reports only standard axioms. The reading of the earlier experiments' ETF-only baseline is paper level. The independent paper-to-formal fidelity review matches the claim, with two bookkeeping points: all of (i) is formally conditional on AX-13 although one conjunct needs it, and the claim's list of axioms used omitted AX-13.

Proposition 40 corresponds to `claims/041-m7-two-stage-with-etfs-at-zero.md`. Lean B's prose check found (iii)'s "iff" false for the confined soft procedure, with a one-fund counterexample, and three narrower points; the analyst's experiment 040 found the exposure-misfit term identically zero. Math revised, red passed the revision with one wording correction, and PM approved. The claim builds on claim 040 (the fund decision with ETFs at zero), which is approved but neither formalized nor written up here; the objects the result uses from it, V_E , its multiplier and its at-zero set, are defined in the statement above.

The statement and proof modules in the proof map machine check, for any maximizers, whose existence and stage 1's uniqueness are proved:

- (i): the sign and complementarity of ζ^* , exactness iff the fund lines (129) hold at x_2 , and $V_E(Qx_2) = G_E(w^*)$ for any number of funds;
- (ii): stage 1 as a choice of fund holding, the interval of such holdings, the fibre interval (130) with stage 1's holding at the end an ETF at zero selects, the clip, the equivalence and loss (131), and the one-fund reachability condition;
- (iii): the soft procedure's exactness when $v_E = 0$, the normal-cone property and bound otherwise, and the tilted-stage criterion. The fund covariance V is only assumed positive semidefinite, more generally than in the claim. AX-13 is listed by the claim but not used formally. A second formal stage replaced the first stage's temporary copy of claim 040's ETF-coordinate objects with claim 040's own module, which is now formalized. The reduction of the model to ETF coordinates and the comparison with the dropped-ETF benchmark are paper level. Experiment 029, which prompted the claim, is an illustration; after approval, experiment 040 led math to sharpen the gloss on exactness under a nonzero multiplier in (iii), which the text follows. The independent paper-to-formal fidelity review matches the revised statement, rechecked after the second stage.

Proposition 41 corresponds to `claims/109-m7-fund-decision-etfs-at-zero-costs-budget.md`. Lean B's prose check found the status tests in (ii) wrong as two equivalences and questioned the untraded-ETF condition in (iv); mathb corrected both before red's review, and red's nit then derived the condition's simple form from stage 1's complementarity. Red passed the claim with one required correction: the target-confined procedure can be exact while trading costly ETFs, when every ETF is sold to zero, so its inexactness reading is restricted to ETFs that end interior. Red's recheck passed the correction, and PM approved. The criterion in (i) composes claim 040's one-sided line (Proposition 46) with claim 102's friction and budget criterion (Proposition 44) and claim 104's exactness forms (Proposition 45); the claim says plainly that nothing in (i) goes beyond that composition. The statement and proof modules in the proof map machine check:

- (i), conditional on AX-13: a feasible holding is the optimum iff (132) holds, stated in claim 104's reference case with spanning ETFs; necessity is claim 040's formalized criterion and the converse claim 104's;
- (ii), with $\Sigma_E = 0$ and given the traded ETFs' lines: the traded exposure, the fixed marginals, (134) and (135), the three special cases, the status implications and the one-quantity test, and the range of $(1 + \eta)\rho_{iT}^\top t_T$;
- (iii): (136);
- (iv): the exactness criterion at x_2 , conditional on AX-13; the fund marginals at x_2 ; $\zeta_j^* = 0$ at interior ETFs and the three conditions; and the soft procedure's exactness, bound and tilted-stage criterion, for any convex feasible set and concave remainder. Paper level are the restatement of (i) as consistency of a status partition with the uniqueness of the optimum, the display in (iv) with ETF residual risk, the reading of the range in (ii) as the re-hedge bracket, the one-sided line for a sale to zero, and the comparison with the composition. The proof imports the proof modules of claims 040, 041 and 104, and the axiom audit reports only standard axioms. The independent paper-to-formal fidelity review matches the claim as revised, with (i) and (iv)'s exactness criterion conditional on AX-13 and the rest unconditional. Several parts are proved more generally than stated, and the headline of (iv), "exact only if", is a composition of formal pieces rather than one theorem.

Proposition 42 corresponds to `claims/042-m7-fine-band-with-one-costly-etf.md`. Red’s first review required two corrections: the corrector verification had the rates on the wrong sides and wrongly let the band’s centre shift with the rate asymmetry, and (iv)’s end limits and the idle-probability negative were stated as results. PM also asked that the one-instrument law be cited through a ledger entry (AX-15), not re-proved. Math revised; red’s recheck passed it, and PM approved. The statement and proof modules in the proof map machine check:

- (i): the identity (138), and the innovation moments (137) as second moments of a finite law;
- (ii): (a) in the multi-review tracking model with zero cross covariance at every review and state, as a sum of two one-instrument instances whose optima separate; (b) as Proposition 38’s formal (vi); (c) the completed square at a fixed ETF holding;
- (iii): the corrector lemma behind (139), its twice-differentiable linear continuation off the band, the symmetric band, the attainment identity, and the end ratio;
- (iv): the identity (140). AX-15 is cited and enters no formal conjunct beyond the formula it displays; its corrector framework and the identification of the discrete band with the corrector’s are paper level, as are (ii) (c)’s dynamic reading, (iv)’s dimensional analysis, end readings and idle fractions, and the checks, which illustrate. The proof imports claim 107’s proof module, and the axiom audit reports only standard axioms. An independent paper-to-formal fidelity review of this result has not yet been recorded.

Proposition 43 corresponds to `claims/108-m7-dynamic-bundling-shape.md`. Red passed it on its own exact grid dynamic program. PM approved it and asked, under rule 21, that the proof cite Topkis’s theorems through a ledger entry (AX-14) instead of re-proving their finite case; mathb made that change without changing a result, and red checked it. The statement and proof modules in the proof map machine check:

- (i): the median edges (141) for an interior ETF optimizer, their continuity, direction and bend length (142) for either sign of the covariance, the coinciding levels at zero covariance, the regime widths, and the static last-review region;
- (ii), for $\Sigma_{AE} \geq 0$ along the tree and conditional on AX-14: increasing differences of V_t , G_t and the ETF-optimized value, the monotone edges of both bands, and (143);
- (iii): the frozen value and stay objective as the idle instance’s up to constants, and v^{idle} as a second moment;
- (iv): equality of the last-review bands and region across outcome laws. Topkis’s Theorems 3.1, 3.2 and 4.3 enter as a hypothesis structure, audited as nontrivial, so (ii) is conditional on that citation; the convex-difference lemma, the coordinate flip and the induction are proved. The cases $\Sigma_{AE} \leq 0$ and $\Sigma_{AE} = 0$ in (ii) are paper level, as PM allowed, as are the conjecture and the checks. The proof imports the proof modules of claims 029 and 107, and the axiom audit reports only standard axioms. The independent paper-to-formal fidelity review matches the claim, with (ii) conditional on AX-14 and formal for nonnegative covariance. Its one precision point is on (iii): formally the frozen band depends on the whole law of the idle target’s innovation, and two finite laws with the same variances and correlation can give different bands, so “through v^{idle} alone” holds when that law is fixed by its variance. The text above states that form.

Proposition 44 corresponds to `claims/102-m7-fund-hold-buy-sell-criterion.md`, written up after the propositions that build on it. Red’s review required corrections that restricted (iii)-(iv) to ETFs strictly

inside their boxes, sending the at-zero case to Propositions 40-41; mathb made them and red's recheck passed them before PM's approval. Later revisions, none changing an approved result, cited AX-13 in place of re-proving the multiplier criterion, corrected (v)'s shift on the reduced marginal to $\eta(1 - \sum_j r_{ij})$ after experiment 034 and red found the literal form wrong, and adopted Lean's forms at the box edges in (iii), (iv) and (vi). The statement and proof modules in the proof map machine check:

- (i): existence and uniqueness, the criterion, and every instrument's readings with their one-sided forms;
- (ii): the no-trade test at the ETF-only optimum;
- (iii): the band holding (144), the bought and sold conditions with the box conditions, the zero-incumbent readings and the netting decomposition;
- (iv): the bracket (145), the re-hedge equalities and the converse's counterexample, the pinned-slope thresholds, and the sufficient conditions at the incumbent;
- (v): the shifted strip (146), the shifted exposure and the bracket with a binding budget;
- (vi): the thresholds (147) and their mirrors at the cap in the one-instrument model. The criterion in (i) and the parts that use it, (ii), (iv) and (v), take AX-13 as a hypothesis, so they are machine checked conditional on that citation; existence, uniqueness, (iii) and (vi) are unconditional. The reduction of the multi-review spanning problem to per-fund problems in (vi), the static ceiling's transfer, and the reading of the at-zero case are paper level. The proof imports the proof modules of claims 104 and 029, and the axiom audit reports only standard axioms. The independent paper-to-formal fidelity review matches the claim; it found the prose differing from the formal statement at two boundaries, the box conditions in (iii) and the sale side of (vi), which math has since corrected to the formal forms stated above.

Proposition 45 corresponds to `claims/104-m7-two-stage-exactness-loss.md`, the refile of claim 103, which red refuted because its exactness criterion in (iii), the ETF self-band alone, was necessary but not sufficient: a fund can leave the exposure drifting rather than re-hedge through a costly ETF. Red's review of the refile then required two corrections, the box multiplier μ_J in (149)'s upper bound (red's capped counterexample) and the fee and residual terms and the choice of shadow price in the binding-budget criterion; mathb made them, red's recheck passed them, and PM approved. It is the companion result that Propositions 35 and 39 build on. The statement and proof modules in the proof map machine check:

- the joint criterion, conditional on AX-13, through a lifted problem with one cost bound per instrument and the box signs in normal-cone form;
- (i): the transfer of Propositions 22 and 23, the reference case's vanishing cross moments, the concavity of the residual value, and the representation of every instance by finitely many scenarios;
- (ii): the explicit optimum and exactness of both procedures;
- (iii), conditional on AX-13: exactness iff (148), with general V and ETF residual risk, and the soft procedure's exactness;
- (iv): (149), with $\mu_J = 0$ at an interior holding;
- (v): (150) at b^* , with the operator norms entering as any bounds, and, conditional on AX-13, the binding-budget criterion and the bound with the fibre problem's cash multiplier. The measure-zero reading in (iii) is paper level. The proof imports the proof modules of claims 027, 028 and 031, and the axiom audit reports only standard axioms. The independent paper-to-formal fidelity review matches the claim, with the joint criterion, (iii) and the binding-budget case conditional on AX-13 and the rest unconditional.

Proposition 46 corresponds to `claims/040-m7-fund-decision-with-etfs-at-zero.md`, the companion result Propositions 40 and 41 build on. Red’s review required one correction, the drop-Z comparison in (iv) at a box bound, where both rules can clip to the same bound; after math added the premium-error formula in (iii) and the fold-in reading, PM withdrew the verdict, and red’s re-review required a second correction, the sign of the fold-in’s ETF premium. Math made both, red passed the result, and PM approved. The statement and proof modules in the proof map machine check:

- the coordinate change, the equivalence of the bound with $w \geq Qx^A$, and the transfer of the coordinate optimum to the review’s optimum;
- (i), conditional on AX-13: the ETF lines with their slacks and the fund lines with the at-zero term;
- (ii): existence and uniqueness of $w(q)$ and $\zeta(q)$, the Schur-block characterization (151), and the concavity and gradient of V_E ;
- (iii): the marginal (152), the optimization of the exposure, the hold test for any N , the readings at the optimum, the one-fund rule with its box conditions, and the premium-error shift;
- (iv): the pieces and their slopes, the unique roots, the optimum, the drop-Z comparison with its box-bound case, and the ordering when the at-zero set grows or shrinks along the trade;
- (v): the one-ETF test and marginal. Only (i) takes AX-13 as a hypothesis; the rest is unconditional. The identification of (152) with Proposition 28’s reduced moments, the reading that a held fund does not move until the premium shift leaves its band, the direction-dependence reading and the comparison with Jagannathan and Ma are paper level. The proof imports claim 104’s proof module, whose ETF-coordinate objects Proposition 40’s second formal stage now uses, and the axiom audit reports only standard axioms. The independent paper-to-formal fidelity review matches the claim, with (i) conditional on AX-13; it notes that (i) needs every ETF below its cap, which the text states.

Proposition 47 corresponds to `claims/110-m7-one-etf-two-scalars.md`. Red passed it on red’s own implementation of the two-scalar rule against a joint solve, and PM approved; red’s one nit, that the cash-price monotonicity in (iv) holds at a fixed exposure price, is made. Experiment 045 agrees with it and illustrates it. The statement and proof modules in the proof map machine check:

- (i): the clip as the unique maximizer of each fund’s one-fund problem, the monotonicity of $r_i x_i$ in m and of x_i in η , the properties of q and p , the unique roots m_0 , m_I and the traded prices, and the fund marginal as a coordinate derivative;
- (ii), for every $\eta \geq 0$ and $\sigma_E \geq 0$: the Lagrangian’s unique maximizer, the funds’ clips and the ETF’s holding at it, and the status table (155) with each regime’s price;
- (iii): the monotone cash slack, the slack-budget case, and that any budget-exhausting η gives the unique constrained optimum;
- (iv): (156) and its mirror, and both comparative statics. The existence of such an η when the budget binds is paper level, through AX-13, and (iii)’s constrained-optimum statement is conditional on it. The formal model is the claim’s own one-ETF coordinates; its link to the general review model is paper level, as PM’s scope note says. The proof uses Mathlib only, and the axiom audit reports only standard axioms. The independent paper-to-formal fidelity review matches the claim in these one-ETF coordinates, unconditional apart from the declared cash price; the auditor checked the link to the general model by hand.

Proposition 48 corresponds to `claims/111-m7-incumbent-aware-first-stage.md`. Lean B's prose points were made before red's review; red passed the claim on 400 random instances with fees, ETF rates, ETFs at zero and a tight budget on half, and PM approved, with part 4 reading exactness and fundability, not loss. Experiment 045 illustrates it: on the 594 draws where both first stages could fund their exposure, the incumbent-aware stage was exact on 122 against 15 for Proposition 40's, but lost more when inexact (median 32 against 15.5 basis points); experiment 044 found Proposition 40's first stage unfundable on 41% of its draws. The statement and proof modules in the proof map machine check:

- the marginals g_E and the fund identity $g_A = \tilde{\alpha} - \gamma Vx^A + Q^\top g_E$;
- (i), unconditionally: the objective bound at any feasible point, the lower bound by strong concavity (the paper proof cites AX-13 at x_J ; the formal one does not need it), (158), and $\Lambda = 0$ iff $x_2 = x_J$;
- (ii): the equivalence of $s = 0$ with Proposition 41's joint criterion, unconditionally, with its consequence $x_2 = x_J$; the converse, conditional on AX-13 through Proposition 45's joint optimality, with no binding ETF cap; and $s_j = 0$ at an interior traded ETF under a slack budget;
- (iii): the three exactness cases, the joint criterion's sufficiency proved directly. The existence of the fibre multipliers is AX-13 on the fibre problem, a paper step; the statements that hold for every valid choice rest on it. The binding-budget reading, part 4's comparison and the experiment readings are paper level. The proof imports the proof modules of claims 104, 109 and 040, and the axiom audit reports only standard axioms. An independent paper-to-formal fidelity review of this result has not yet been recorded.

Proposition 49 corresponds to `claims/043-m7-fine-cost-sandwich-with-one-costly-etf.md`. Red's review required one correction, applying AX-16 through a change of variables that turns the pair's gradient set into the source's cash-only box, with the cost, covariance, equation and support function checked to transform consistently; math made it, red's recheck passed it with a nit (that k^* lies in the sandwich while its position is open), and PM approved. Experiment 046 finds every cell it resolves inside the sandwich, an illustration. The statement and proof modules in the proof map machine check, for every number a that lies between the values of all classical subsolutions and supersolutions with the stated growth:

- (i): the separable test functions as sub- and supersolutions, the general lower bound with its maximum attained, (160), and the third lower bound;
- (ii): the uncorrelated case;
- (iii): the identity and the bound (by concavity);
- (iv): the effective rate, the band-width arithmetic of claim 042's reading, and both limits. None of these uses AX-16: it enters the formal statement only as a proposition asserting that such an a exists. That the pair's corrector eigenvalue is such an a , with the hypothesis check through the change of variables and the classical-to-viscosity step, is the cited AX-16 and a reviewed paper step. The first-order expansion of (iii), the identification with Proposition 38's re-hedge term, the non-ergodic reading of the frozen end and the band readings are paper level. The proof imports claim 042's proof module, and the axiom audit reports only standard axioms. An independent paper-to-formal fidelity review of this result has not yet been recorded.

Proposition 50 corresponds to `claims/044-m7-two-reviews-binding-budget.md`, the first result of the direction on dynamics with a binding budget. Red passed it and PM approved; experiment 047 agrees

with its proved parts and found the joint hoarding and front-loading reading failing by substitution in 12 of 82 cases, so math narrowed that reading to one instrument with the other held fixed, and this write-up follows PM's instruction to present the dynamic cash price and the incumbent values as the result. The statement and proof modules in the proof map machine check, in the claim's two-review coordinates for any finite instrument set and state set, with each covariance only positive semidefinite and finite caps:

- (i): existence, the concavity of the whole problem, of tomorrow's value and of the root objective, and monotonicity in cash;
- (ii): sufficiency of the lines by a direct Lagrangian bound, unconditionally; necessity, with one tomorrow slope in both sets of lines, conditional on AX-13; and the readings of $s_i(z')$;
- (iii): the slack-tomorrow case and the bounds (163) with the weak purchase and sale conditions;
- (iv): the myopic-optimality test, conditional on AX-13; the pinned multiplier; the slack-tomorrow condition and the held-today band; $\beta E[g; t_{1,i}] = 0$ for an interior trade with today's budget slack; the residual identity; and the tangent-line bounds of slope r along one instrument;
- (v): each review's holdings as its one-review Lagrangian's maximizer, the band in the marginal with its width, and the costless instrument's line. Paper level are the exact value of the one-sided derivative (equal to r only for a given choice of tomorrow's multipliers), strict improvement in the opposite direction, the interval case of the multiplier set, the joint comparative static, the width in holdings, the link to Proposition 47's coordinates, and the uncapped ETF. Two readings in the claim hold formally only as stated here: the zero condition in (iv) needs today's budget slack, and a purchase today with some state buying again tomorrow is not myopically optimal only if no other state sells. The proof imports claim 104's statement and proof module, with the two-review lifting proved here, and the axiom audit reports only standard axioms. An independent paper-to-formal fidelity review of this result has not yet been recorded.

Proposition 51 corresponds to `claims/046-m7-two-reviews-bounds-in-the-inputs.md`, the refile of claim 045, which red refuted because its (iii) failed when today's budget binds at the myopic root, where today's cash price is free, and for a costless instrument with a slack budget tomorrow, where the equation is an identity. The refile states (iii) with today's budget slack and names both cases; red's review required one more correction (the identity needs the instrument costless on both sides), which math made; red's recheck passed it and PM approved. Experiments 049 and 047 illustrate it. The statement and proof modules in the proof map machine check, reusing claim 044's objects:

- (i): the admissible selection below $\bar{\eta}$, the zero multiplier when the cash covers need, the bound for any tomorrow family in states that trade or keep cash, the bracket on $\hat{\eta}_0$, and $\hat{\eta}_0 = \eta_0$ when every state is covered and some cash is carried;
- (ii): (165), the residual's bracket and the two sufficient sign conditions;
- (iii): the root line's value at an interior trade with today's budget slack, the costless identity, and (166) with today's budget binding;
- (iv): the hedge-term identity at the reduced point and the one-variable concave step. Nonnegative off-diagonal covariances tomorrow are stated for any finite set of instruments. Paper level are the link from (iv)'s two formal ingredients to (167) (the reduced root objective's concavity and its one-sided derivatives as extremes over tomorrow's multipliers, which needs Danskin's theorem with AX-13 on tomorrow's value), the edge case of (i) with no cash carried, the joint admissibility of a per-state selection, and

the reading of (iii)'s equation as nongeneric. The proof imports claim 044's modules, and the axiom audit reports only standard axioms. The independent paper-to-formal fidelity review matches (i)-(iii), unconditionally given tomorrow's multipliers, and records (iv) as two formal ingredients whose link to (167) is paper level, as stated.

Proposition 52 corresponds to `claims/112-m8-worked-learning-example.md`, on the lab's M8 (the two-review worked model with both shock laws). Red's reviews required six corrections, among them restricting the Gaussian-law second-review statements to nonnegative marked holdings, the condition under which the two means differ, and citing the filter's two meanings through AX-17 and AX-18; mathb made them, red passed the claim and PM approved. Experiment 050 reproduces the example independently. The statement and proof modules in the proof map machine check, in the model's own coordinates with its finite tree mapped into Proposition 50's:

- (i) under the finite law: the posterior variance, the positive definite predictive covariance and unique target, marking, the nonempty feasible sets and unique review optima, and the existence of an optimal two-review policy (through claim 044's part 1);
- (ii)'s moment content for any finite laws with the stated moments: the innovation's mean and variance, the error variance, the zero cross-block moment, the posterior mean's average, the non-coincidence condition, red's two-point counterexample and the example's node;
- (iii)'s slack-budget threshold;
- (iv)'s first-review numbers exactly. Paper level are the filter's two meanings (cited, AX-17 and AX-18), the Gaussian-law statements, the transfer in (iii), the case with no cash today, and the example's later tables. The proof imports claim 044's modules, and the axiom audit reports only standard axioms. The independent paper-to-formal fidelity review matches this scope; it notes that the claim's phrase "the exact Bayes posterior only under Gaussian noise" should read as sufficiency after AX-17's fix, which this write-up follows.

Proposition 53 corresponds to `claims/047-m8-reserve-rule-one-fund-one-etf.md`, the first result of the second focused request's reserve direction. The reviews required corrections: red's, reading the cost-channel reserve through the ETF's expected marked slope including held marginals rather than the probability of a trade, and the factors on the threshold errors; Lean B's, taking the certificate at the dynamic root's own need; the claim was also revised after experiment 053 showed that its one-shot reading misleads, to state (169) as a fixed point. Red's rechecks passed and PM approved. The statement and proof modules in the proof map machine check, on Propositions 50-51's objects with the revision entering as a hypothesis on tomorrow's means bounded by its extremes:

- (ii): the budget channel's absence at any root with positive cash covering its need, the certificate, (168), the finite-law bounds over the uncovered states for every tomorrow family, and the threshold errors;
- (iii): the reserve premium as the root line less the one-review line, its expansion by state, (169), its bound, and the no-shift case;
- (iv): (170) and the need's slope in the purchase rate on the active set, and the ETF unit count;
- (v): (171), given that some tomorrow family exists and every family prices cash somewhere. Paper level are (i)'s definitions and readings, the direction of (iii)'s reading (Proposition 51's one-variable step), (169)'s uniqueness and its fixed-point computation, (ii)'s sufficient conditions for the cost channel and

the idleness bound, the Gaussian tail moments (which need Proposition 50's lines under the Gaussian law, proved only for the finite law), the reserve's non-monotonicity, and the experiments' readings. No cited result or hypothesis structure is used, and the axiom audit reports only standard axioms. An independent paper-to-formal fidelity review of this result has not yet been recorded.

Proposition 54 corresponds to `claims/113-m7-several-funds-one-etf-reserve.md`. Red passed it on red's own two-fund, one-ETF solver, Lean B's three prose points were made, and PM approved; red's two nits went to mathb. The statement and proof modules in the proof map machine check, on Propositions 50-51's objects for any finite set of instruments, with the model's factor structure as a hypothesis:

- (i): each fund's marginal and its root holding as Proposition 47's clip with S_i added, the clip shown equal to Proposition 47's formal one;
- (ii): the state check with the threshold clipped at zero, the every-state family with $\eta_1 = 0$ and Proposition 26's brackets (every family when cash is positive), and the pooling inequality;
- (iii): (172) on the trading pieces;
- (iv): (173), the clip's Lipschitz bound, and the relation to Proposition 53's reserve counted at the ETF's worst-case liquidation. Paper level are the common cash price as Proposition 47's root with N funds, the product-tree pooling remark, the fixed-point remark, and the carried-over cost channel. The proof imports claims 044's and 046's modules, and the axiom audit reports only standard axioms. An independent paper-to-formal fidelity review of this result has not yet been recorded.

Proposition 55 corresponds to `claims/048-m7-many-funds-many-etfs-two-reviews.md`. Red's review required the qualification that the separation is of the lines given the tomorrow numbers at the optimum, a fixed point of the joint root, not of two independent problems (red's test moved the ETFs 1% and changed the funds' incumbent values); math made it, red's recheck passed and PM approved. The statement and proof modules in the proof map machine check, in Proposition 50's formal model with funds and ETFs as the instruments and the root's moments from the factor model:

- (i): sufficiency of the lines without citation, necessity conditional on AX-13, and the root as the one-review Lagrangian's maximizer at $(\mu_0 + S, \hat{\eta}_0)$;
- (ii): necessity and sufficiency of (175) at the tilted inputs, the tomorrow numbers read from the same policy's multipliers;
- (iii): both identities in (176). The ETFs' rates are review-independent in the model, so frictionless today means frictionless tomorrow. (iv) is a reading of Proposition 41's obstruction, and the check's costly-ETF count illustrates it. The proof imports claims 044's, 104's and 110's modules, and the axiom audit reports only standard axioms. An independent paper-to-formal fidelity review of this result has not yet been recorded.

Proposition 56 corresponds to `claims/049-m7-quantile-flexibility-test-full-menu.md`. Red's review re-derived every step and tested (iii)'s bound on red's own solver at 300 of 300 instances (164 with a positive tail term, largest ratio of loss to bound 0.61). It required one correction: the one-ETF band term uses the ETF's variance net of the funds, not its own variance (30 of 300 instances exceeded the old form, by up to 5.9 times). Math made it, with red's point that under the test the tail term is the plain expectation, and PM approved. The statement and proof modules in the proof map machine check, in Proposition 55's model with the ETFs a set among the instruments:

- (i): need $\leq \text{liq}$ at a state makes $\eta_1 = 0$ admissible there, from any admissible multiplier, with the proceeds summed over the ETFs;
- (ii): the quantile equivalence, its state-free form, and the tail identity with its bound, proved directly on the finite law;
- (iii): the loss bound for every feasible policy, from today's strong concavity at the myopic root, the relaxed tomorrow's supergradient and Proposition 48's completed square, with the tail term from the budgeted tomorrow's Lagrangian bound; and the band term's bound in the inputs. The multipliers' existence (AX-13) enters as hypotheses; the one-ETF form of the band term is at paper level, and the ledger's tail measure (AX-19) is cited for its reading, not used formally. The proof imports claims 113's and 111's proof modules, and the axiom audit reports only standard axioms. An independent paper-to-formal fidelity review of this result has not yet been recorded.

Proposition 57 corresponds to `claims/114-m9-predictable-and-unpredictable-target-move.md`, on the lab's M9. Red's review required four corrections, made by mathb: the width's direction per instrument, with the positive semidefinite order only sufficient for all instruments together (red's counterexample at M8's inputs); the positive part kept around the whole need in (v); the variance recursion's convergence argued block by block, with AX-10 cited for the steady state's form only; and the half-open probability in (iii) when the innovation has atoms. PM approved. Red's recheck then found that the threshold reading in (iv) used the myopic tomorrow's incumbent value at the dynamic root; mathb restated it at the dynamic root with its own incumbent value, as (180), the direction $a_0^{\text{dyn}} \geq a_0^{\text{my}}$ unchanged. Experiment 057 illustrates it: in 24 of 24 cells meeting the myopic condition the order held, while the dynamic root's tomorrow bought in only 4 to 14 of 16 states and the threshold lay strictly between its ends. The statement and proof modules in the proof map machine check, on Proposition 52's objects with scalar blocks and Proposition 50's two-review objects for (iv):

- (i): the decomposition as an identity, and the innovation part's mean zero and covariance under any finite laws with the stated moments and independent blocks;
- (ii): the variance change, the per-instrument criterion, the cases where both blocks move the same way, the mixed-blocks instance, and the recursion's convergence to a fixed point, unique when $q > 0$ or $\phi^2 < 1$;
- (iii): $U - D$ as the half-open probability, the tilt's sign, and the full shift beyond the support;
- (iv): the tangent-plane bound, the pinned S_A , the strict decrease, $a_0^{\text{dyn}} \geq a_0^{\text{my}}$ (and its mirror) when the roots agree off the fund, the equality case of (180), and (180) itself with its equality condition, for a purchase. These are one-coordinate statements; nothing is said about the joint move, which experiment 047 showed can reverse. The case of a free ETF and (v) rest on Propositions 54 and 51 at paper level, and so does the mirror bound on the sale threshold; (vi) is a reading, and the transient filter's posterior meaning is not shown. The proof imports claim 044's module, and the axiom audit reports only standard axioms. An independent paper-to-formal fidelity review of this result has not yet been recorded.

Proposition 58 corresponds to `claims/115-m9-quantile-flexibility-test-time-varying.md`, on the lab's M9. Red rederived each part and tested (ii)'s bound on its own solver at 583 holdings (120 front-loaded points, 120 myopic roots and 343 random feasible roots; the loss at most 0.99 of the bound). Red required one correction: the root residual must minimize over the admissible slopes at tomorrow's bound states, since with one fixed choice the residual at the front-loaded point was nonzero at 8 of 120 points,

and with the free choice zero at all 120. Mathb made it, red's fresh verdict passed it, and PM approved; math's reply recorded how the claim composes with Proposition 56. The statement and proof modules in the proof map machine check:

- (i): the split test's equivalence and its state-free form, as exact rewritings, with the planned purchase from tomorrow's predictable mean and $g^{\min}$;
- (ii): the bound at any feasible holding, for every admissible choice of line values and relaxed slope families, including the interval slopes at bound states, and the corollary that a zero residual leaves the tail term alone;
- (iii): the pinned S_A at the myopic root's relaxed tomorrow and the block-inverse entry. At paper level: Proposition 56's instantiation on this tree (its formal statement is over any finite tree with state-dependent means); the zero residual at the front-loaded point, from the relaxed program's optimality conditions (AX-13); the tail-measure form of the bound; the assembly of (183) and the mirror for a fall; the reading of which point to hold; and the Gaussian quantile, which is not shown. The proof imports claim 049's proof module, and the axiom audit reports only standard axioms. An independent paper-to-formal fidelity review of this result has not yet been recorded.

The conditional parts in this draft rest on three cited theorems:

- the linear-programming vertex theorem: Proposition 24's face finiteness;
- the polyhedral Karush-Kuhn-Tucker theorem: Proposition 39's multiplier test in (i), Proposition 41 (i) and (iv)'s criterion, Proposition 44 (i), (ii), (iv) and (v), Proposition 45's joint criterion, (iii) and binding-budget case in (v), Proposition 46 (i), Proposition 55 (i)'s necessity, Proposition 56's multipliers, Proposition 58's zero residual at the front-loaded point (at paper level), Proposition 47 (iii) when the budget binds (at paper level), Proposition 48 (ii)'s converse, with the existence of its multipliers at paper level, and Proposition 50 (ii)'s necessity and (iv)'s test, and Proposition 51 (iv)'s link at paper level;
- Topkis's lattice theorems: Proposition 43 (ii).

The detailed statement comparison belongs in board/FIDELITY.md.

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